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MISO DPP 2016 August West Area Phase 1 Study

Prepared for

MISO

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Contents

Legal Notice.....	xi
Executive Summary	xiii
1.1 Project List	xiii
1.2 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)	xiv
1.3 Total Network Upgrades for all Projects	xv
1.4 Per Project Summary	xviii
1.4.1 J302 Summary	xviii
1.4.2 J414 Summary	xx
1.4.3 J415 Summary	xxi
1.4.4 J439 Summary	xxii
1.4.5 J457 Summary	xxiv
1.4.6 J459 Summary	xxv
1.4.7 J476 Summary	xxvii
1.4.8 J503 Summary	xxviii
1.4.9 J511 Summary	xxix
1.4.10 J512 Summary	xxxi
1.4.11 J541 Summary	xxxii
1.4.12 J555 Summary	xxxiii
1.4.13 J569 Summary	xxxiv
1.4.14 J575 Summary	xxxv
1.4.15 J577 Summary	xxxvi
1.4.16 J583 Summary	xxxvii
1.4.17 J587 Summary	xxxviii
1.4.18 J590 Summary	xxxix
1.4.19 J593 Summary	xl
1.4.20 J594 Summary	xlii
1.4.21 J596 Summary	xliii
1.4.22 J597 Summary	xliv
1.4.23 J598 Summary	xlvi
1.4.24 J599 Summary	xlvii

1.4.25	J607 Summary	xlvi
1.4.26	J611 Summary	l
1.4.27	J613 Summary	li
1.4.28	J614 Summary	liii
1.4.29	J615 Summary	liv
1.4.30	J637 Summary	lv
1.4.31	J638 Summary	lvi
1.5	Study Compliance with NERC FAC-002-2 Standard	lvii
Introduction		1
Section 2 – Model Development and Study Criteria.....		2-1
2.1	Model Development.....	2-1
2.1.1	Benchmark Cases	2-1
2.1.2	Study Cases	2-2
2.2	Contingency Criteria	2-2
2.3	Monitored Elements.....	2-3
2.4	Performance Criteria.....	2-4
2.5	Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)	2-5
Section 3 – Summer Peak Steady-State Analysis		3-1
3.1	Study Procedure.....	3-1
3.1.1	Computer Programs	3-1
3.1.2	Study Methodology.....	3-1
3.2	Summer Peak Contingency Analysis Results.....	3-1
3.2.1	System Intact Conditions.....	3-1
3.2.2	Post Contingency Conditions	3-1
3.3	Network Upgrades Identified in MISO ERIS Analysis for 2022 Summer Peak Scenario	3-2
Section 4 – Summer Shoulder Steady-State Analysis		4-1
4.1	Study Procedure.....	4-1
4.2	Step 1 – Stage-1 ACCC Analysis.....	4-1
4.2.1	Stage-1 System Intact Conditions	4-2
4.2.2	Stage-1 Post Contingency Conditions.....	4-2
4.2.3	Worst Thermal Constraints in the Stage-1 ACCC.....	4-2
4.3	Step 2 – Project Justification Analysis	4-31

4.3.1	Project Justification in Dakotas-Minnesota.....	4-31
4.3.2	Project Justification in Iowa Area.....	4-32
4.3.3	Project Justification of Reactive Power NUs	4-32
4.4	Initial Backbone Network Upgrades	4-33
4.5	Step 3 – Stage-2 ACCC Analysis.....	4-33
4.5.1	Stage-2 System Intact Conditions	4-33
4.5.2	Stage-2 Post Contingency Conditions.....	4-33
4.5.3	Summer Shoulder Worst Thermal Constraints in the Stage-2 ACCC.....	4-34
4.6	Step 4 – Justification of Additional Projects Based on the MWEX Voltage Stability Study and Stage-2 ACCC Analysis	4-55
4.6.1	MWEX Project Justification	4-55
4.6.2	Webster-Franklin Project Justification	4-64
4.7	Step 5 – Stage-3 ACCC Analysis.....	4-64
4.7.1	Stage-3 System Intact Conditions	4-65
4.7.2	Stage-3 Post Contingency Conditions.....	4-65
4.7.3	Summer Shoulder Worst Thermal Constraints in the Stage-3 ACCC.....	4-65
4.8	Network Upgrades Identified in MISO ERIS Analysis for 2022 Summer Shoulder Scenario	4-79
Section 5 – Local Planning Criteria Analysis		5-1
5.1	GRE Local Planning Criteria Analysis (J511, J596).....	5-1
5.1.1	Additional Network Upgrades Identified in J511 GRE LPC Analysis	5-1
5.1.2	Additional Network Upgrades Identified in J596 GRE LPC Analysis	5-3
5.2	OTP Local Planning Criteria Analysis.....	5-4
5.2.1	Additional Network Upgrades Identified in OTP LPC Analysis	5-4
5.3	MDU Local Planning Criteria Analysis.....	5-5
5.3.1	Additional Network Upgrades Identified in MDU LPC Analysis	5-5
5.4	DPC Local Planning Criteria Analysis	5-6
5.4.1	Additional Network Upgrades Identified in DPC LPC Analysis	5-6
5.5	Ameren Local Planning Criteria Analysis	5-7
5.5.1	Steady-State Results Summary	5-7
5.5.2	Stability Results Summary.....	5-7
5.5.3	Additional Network Upgrades Identified in Ameren LPC Analysis	5-8

Section 6 – Affected System Steady-State Analysis	6-1
6.1 Affected System Analysis for CIPCO Company.....	6-1
6.2 MPC Affected System Analysis.....	6-3
6.2.1 Study Results.....	6-3
6.2.2 Study Summary.....	6-3
6.3 PJM Affected System Analysis.....	6-4
6.3.1 Study Results.....	6-4
6.3.2 Study Summary.....	6-4
6.4 SPP Affected System AC Contingency Analysis	6-5
6.5 AECI Affected System Analysis	6-9
6.5.1 Study Summary.....	6-9
6.5.2 Network Upgrades Cost	6-9
Section 7 – Stability Analysis	7-1
7.1 Procedure	7-1
7.1.1 Computer Programs.....	7-1
7.1.2 Study Methodology.....	7-1
7.2 Case Development.....	7-1
7.2.1 Study Case	7-1
7.2.2 Benchmark Case.....	7-2
7.3 Disturbance Criteria.....	7-2
7.4 Performance Criteria.....	7-2
7.4.1 MISO Criteria.....	7-3
7.4.2 Local Planning Criteria.....	7-3
7.5 Stability Results	7-4
7.5.1 Slow Voltage Recovery at ITCM 345 kV Buses.....	7-4
7.5.2 Out of Step (OOS) Relay at Forbes on M602F Tripping.....	7-5
7.5.3 Coal Creek 41G GSU DS Zone 1 Relay Tripping.....	7-5
7.5.4 Tripping of Wind Farms by Low Voltage Protection.....	7-6
7.5.5 Transient Voltage Violations under CUDC Permanent Bipole Fault.....	7-6
7.6 Network Upgrades Identified in Stability Analysis	7-7
Section 8 – MWEX Voltage Stability Study	8-1
Section 9 – Short Circuit Analysis	9-1
9.1 Introduction	9-1

9.2	J302 Short Circuit Study Performed by Siemens PTI	9-1
9.3	J414 Short Circuit Study Performed by ITCM	9-1
9.4	J415 Short Circuit Study Performed by MEC	9-1
9.5	J439 Short Circuit Study Performed by MEC	9-1
9.6	J457 Short Circuit Study Performed by Siemens PTI	9-2
9.7	J459 Short Circuit Study Performed by OTP	9-2
9.8	J476 Short Circuit Study Performed by MEC	9-2
9.9	J503 Short Circuit Study Performed by Siemens PTI	9-2
9.10	J511 Short Circuit Study Performed by GRE	9-2
9.11	J512 Short Circuit Study Performed by Xcel	9-3
9.12	J541 Short Circuit Study Performed by Ameren	9-3
9.13	J555 Short Circuit Study Performed by MEC	9-3
9.14	J569 Short Circuit Study Performed by Xcel	9-3
9.15	J575, J577, J587, and J597 Short Circuit Study Performed by Xcel	9-3
9.16	J583 Short Circuit Study Performed by MEC	9-4
9.17	J590 Short Circuit Study Performed by MEC	9-4
9.18	J593 Short Circuit Study Performed by Siemens PTI	9-4
9.19	J594 Short Circuit Study Performed by ITCM	9-4
9.20	J596 Short Circuit Study Performed by GRE	9-4
9.21	J598 Short Circuit Study Performed by Ameren	9-5
9.22	J599 Short Circuit Study Performed by Siemens PTI	9-5
9.23	J607 Short Circuit Study Performed by Siemens PTI	9-5
9.24	J611 Short Circuit Study Performed by MEC	9-5
9.25	J613 Short Circuit Study Performed by OTP	9-6
9.26	J614 Short Circuit Study Performed by DPC	9-6
9.27	J615 Short Circuit Study Performed by MEC	9-6
9.28	J637 Short Circuit Study Performed by OTP	9-6
9.29	J638 Short Circuit Study Performed by OTP	9-6
Section 10	– Deliverability Study.....	10-1
10.1	Project Description.....	10-1
10.2	Introduction	10-1
10.3	Study Methodology.....	10-1
10.4	Determining the MW restriction	10-1
10.5	2022 Deliverability Study Result.....	10-2

10.5.1	J302	10-2
10.5.2	J414	10-2
10.5.3	J415	10-2
10.5.4	J439	10-3
10.5.5	J457	10-3
10.5.6	J459	10-3
10.5.7	J476	10-4
10.5.8	J503	10-4
10.5.9	J511	10-5
10.5.10	J512	10-5
10.5.11	J541	10-6
10.5.12	J555	10-6
10.5.13	J569	10-6
10.5.14	J575	10-6
10.5.15	J577	10-6
10.5.16	J583	10-6
10.5.17	J587	10-6
10.5.18	J590	10-7
10.5.19	J593	10-7
10.5.20	J594	10-8
10.5.21	J596	10-8
10.5.22	J597	10-8
10.5.23	J598	10-8
10.5.24	J599	10-9
10.5.25	J607	10-9
10.5.26	J611	10-10
10.5.27	J613	10-10
10.5.28	J614	10-10
10.5.29	J615	10-11
10.5.30	J637	10-11
10.5.31	J638	10-11
Section 11 – Shared Network Upgrades Analysis		11-1
Section 12 – Cost Allocation.....		12-1
12.1	Cost Assumptions for Network Upgrades	12-1

12.2	ERIS Network Upgrades Proposed for DPP West Area Projects.....	12-1
12.3	Cost Allocation Methodology	12-12
12.4	Cost Allocation.....	12-12
Appendix A – Model Development for Steady-State and Stability		
Analysis		A-1
A.1	DPP 2016 August Generation Projects	A-2
A.2	DPP 2016 February West Area Phase 2 Network Upgrades	A-9
A.3	Model Review Comments	A-11
A.4	MISO Classic as the Study Sink.....	A-16
A.5	PJM Market as PJM Projects Sink	A-17
A.6	SPP Market as SPP Projects Sink	A-18
A.7	Contingency Files used in Steady-State Analysis	A-19
Appendix B – Model Data		B-1
B.1	Power Flow Model Data	B-1
B.2	Dynamic Model Data	B-2
B.3	2022 Slider Diagrams	B-4
Appendix C – Reactive Power Requirement Analysis Results (FERC Order 827)		C-1
Appendix D – 2022 Summer Peak Contingency Analysis Results		D-1
	2022 Summer Peak (SPK) Constraints.....	D-1
Appendix E – 2022 Summer Shoulder Contingency Analysis Results		E-1
E.1	Thermal Constraints Identified in Stage-1 Contingency Analysis with Fictitious SVCs	E-1
E.2	Backbone Network Upgrades Justification Results.....	E-3
E.3	Constraints Identified in Stage-2 Contingency Analysis with Initial Backbone NUs.....	E-16
E.4	Step 4 - MWEX Project Justification.....	E-17
E.5	Constraints Identified in Stage-3 Contingency Analysis with Backbone and MWEX NUs	E-47
Appendix F – Local Planning Criteria Analysis Results		F-1
F.1	GRE Local Planning Criteria Analysis Results.....	F-1
F.2	OTP LPC Analysis.....	F-5
F.3	MDU LPC Analysis.....	F-7
F.4	DPC LPC Analysis.....	F-9

F.5	Ameren LPC Analysis.....	F-11
Appendix G – Affected System Contingency Analysis Results		G-1
G.1	CIPCO Company Affected System Analysis Results.....	G-1
G.2	MPC Affected System Study Results.....	G-3
G.3	PJM Affected System Study Results.....	G-5
G.4	SPP Affected System Study Results.....	G-7
G.5	AECI Affected System Study Results.....	G-9
Appendix H – Transient Stability Results		H-1
H.1	2022 Summer Shoulder Stability Results Summary	H-1
H.2	2022 Summer Shoulder Stability Plots.....	H-3
Appendix I – MWEX Voltage Study Details		I-1
Appendix J – Short Circuit Analysis.....		J-1
J.1	J302 Short Circuit Study Performed by Siemens PTI	J-1
J.2	J414 Short Circuit Study Performed by ITCM	J-1
J.3	J415 Short Circuit Study Performed by ITCM	J-1
J.4	J439 Short Circuit Study Performed by MEC.....	J-1
J.5	J457 Short Circuit Study Performed by Siemens PTI	J-1
J.6	J459, J613, J637, and J638 Short Circuit Study Performed by OTP.....	J-1
J.7	J476 Short Circuit Study Performed by MEC.....	J-1
J.8	J503 Short Circuit Study Performed by Siemens PTI	J-1
J.9	J511 Short Circuit Study Performed by GRE	J-1
J.10	J512, J569, J575, J577, J587, and J597 Short Circuit Studies Performed by Xcel	J-1
J.11	J541 and J598 Short Circuit Study Performed by Ameren	J-1
J.12	J555 Short Circuit Study Performed by MEC.....	J-1
J.13	J583 Short Circuit Study Performed by MEC.....	J-1
J.14	J590 Short Circuit Study Performed by MEC.....	J-1
J.15	J593 Short Circuit Study Performed by Siemens PTI	J-1
J.16	J594 Short Circuit Study Performed by ITCM	J-1
J.17	J596 Short Circuit Study Performed by GRE	J-1
J.18	J599 Short Circuit Study Performed by Siemens PTI	J-2
J.19	J607 Short Circuit Study Performed by Siemens PTI	J-2
J.20	J611 Short Circuit Study Performed by MEC.....	J-2

J.21	J614 Short Circuit Study Performed by SMMPA.....	J-2
J.22	J615 Short Circuit Study Performed by MEC.....	J-2
Appendix K – 2022 Cost Allocation Results.....		K-1
K.1	Distribution Factor (DF) and MW Contribution Results for Cost Allocation in 2022	K-1
K.2	Cost Allocation Details.....	K-3

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Executive Summary

This report presents results of a System Impact Study (SIS) performed to evaluate the interconnection of the DPP 2016 August Phase 1 West Area Group (DPP West Area) generating facilities. The results for 2022 scenario are summarized below.

1.1 Project List

The DPP West Area study group has thirty one (31) generation projects with a combined nameplate rating of 5618 MW. The DPP West Area generating facilities are listed in Table ES-1. The modeling details and projects' slider diagrams are shown in Appendix B.

Table ES-1: Generating Facilities in DPP 2016 August West Area Group

MISO Project #	Service Type	TO	County	State	Point Of Interconnection	Fuel Type	Max Output	SH MW	SPK MW	Stability MW
J302	NRIS	MDU	Emmons, Logan	ND	Heskett-Wishek 230 kV	Wind	101.2	101.20	15.79	101.20
J414	NRIS	ITCM	Freeborn	MN	Freeborn 161 kV Sub	Wind	120	120.00	18.72	120.00
J415	NRIS	ITCM	Franklin	IA	Quinn-Blackhawk 345 kV (7 mi south of Quinn)	Wind	200	200.00	31.20	200.00
J439	NRIS	MEC	Emmet	IA	Obrien-Kossuth 345 kV	Wind	500	500.00	78.00	500.00
J457	NRIS	MDU	Dickey	ND	Merricourt 230 kV Sub	Wind	150	150.00	23.40	150.00
J459	NRIS	OTP	Grant	SD	Big Stone South-Brookings Co 345 kV (11 miles from Big Stone South)	Wind	200	200.00	31.20	200.00
J476	NRIS	MEC	Atchison	MO	Atchison Co-Orient 345 kV (0.2 mi from Atchison)	Wind	246	246.00	38.38	246.00
J503	NRIS	MDU	Emmons, Logan	ND	Heskett-Wishek 230 kV (20 miles NW of Wishek)	Wind	98.8	98.80	15.41	98.80
J511	NRIS	GRE	Oliver	ND	Stanton 230 Sub	Wind	200	200.00	31.20	200.00
J512	NRIS	Xcel	Nobles	MN	Nobles-Fenton 115 kV	Wind	250	250.00	39.00	250.00
J541	NRIS	ATXI	Schuyler	MO	Zachary-Ottumwa 345 kV	Wind	400	400.00	62.40	400.00
J555	NRIS	MEC	Poweshiek	IA	Montezuma 345 kV	Wind	140	140.00	21.84	140.00
J569	NRIS	Xcel	Rock	MN	Rock County 161 kV	Wind	100	100.00	15.60	100.00
J575	NRIS	Xcel	Lincoln	MN	Brookings Co 345 kV	Wind	100	100.00	15.60	100.00
J577	NRIS	Xcel	Pipestone	MN	Brookings Co 345 kV	Wind	100	100.00	15.60	100.00
J583	NRIS	MEC	Audubon	IA	Fallow 345 kV	Wind	200	200.00	31.20	200.00
J587	NRIS	Xcel	Lincoln	MN	J460 Sub on the Brookings-	Wind	200	200.00	31.20	200.00

MISO Project #	Service Type	TO	County	State	Point Of Interconnection	Fuel Type	Max Output	SH MW	SPK MW	Stability MW
					H081 345 kV					
J590	NRIS	MEC	Palo Alto	IA	J529 POI	Wind	90	90.00	14.04	90.00
J593	NRIS	MDU	Williams	ND	Tioga 230 kV	Wind	224	224.00	34.94	224.00
J594	NRIS	ITCM	Jackson, Martin	MN	Jackson North 161 kV	Wind	300	300.00	46.80	300.00
J596	NRIS	GRE	Big Stone, Stevens	MN	Morris-Moro 115 kV	Wind	100	100.00	15.60	100.00
J597	NRIS	Xcel	Deuel, Hamlin	SD	Brookings Co 345 kV	Wind	300	300.00	46.80	300.00
J598	NRIS	ATXI	Adair, Schuyler	MO	Zachary–Ottumwa 345 kV	Wind	300	300.00	46.80	300.00
J599	NRIS	MDU	Walworth	SD	Glenham 230 kV	Wind	200	200.00	31.20	200.00
J607	NRIS	MDU	Logan	ND	Wishek-Heskett 230 kV	Wind	150	150.00	23.40	150.00
J611	NRIS	MEC	Nodaway	MO	Clarinda-Merryville 161 kV	Wind	110	110.00	17.16	110.00
J613	NRIS	OTP	Stutsman	ND	Jamestown 115 kV	Wind	100	100.00	15.60	100.00
J614	NRIS	SMMPA	Howard	IA	Rice 161 kV	Wind	66	66.00	10.30	66.00
J615	NRIS	MEC	Black Hawk	IA	Electric Farms-Shaulis 161 kV	Wind	70	70.00	10.92	70.00
J637	NRIS	OTP	Deuel	SD	Big Stone-Brookings 345 kV	Wind	98	98.00	15.29	98.00
J638	NRIS	OTP	Deuel	SD	Big Stone-Brookings 345 kV	Wind	204	204.00	31.82	204.00

1.2 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

For non-synchronous generation projects in the DPP 2016 August West Area study group, if they do not have signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

Except for generation project J590, all other non-synchronous generation projects in this study group are required to meet the reactive power requirements per FERC Order 827.

The reactive power requirement analysis results are summarized as following:

- J459, J587, and J615 do not meet the reactive power requirements per FERC Order 827.
- J590 is exempted from the FERC Order 827 reactive power requirements.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

1.3 Total Network Upgrades for all Projects

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection Service as of the System Impact Study report date. The total cost of network upgrades in the interconnection plan required for each generation project is listed in Table ES-2. The costs for Network Upgrades are planning level estimates and subject to be revised in the facility studies.

Table ES-2: Total Cost of Network Upgrades for DPP 2016 August West Area Generation Projects

Project Num	ERIS Network Upgrades (\$)																NRIS Network Upgrades (\$)	Inter - connection Substation TO NUs (\$)	TOIF ¹ (\$)	SNU (\$)	Total Network Upgrade Cost (Exclude TOIF & Affected System) (\$)	M2 Received (\$)	M3 (\$)
	Backbone NUs	MWEX Voltage Stability	MISO Thermal & Voltage	Transient Stability	Short Circuit	DPC LPC	GRE LPC	MDU LPC	OTP LPC	Ameren LPC	CIPCO AFS	MPC AFS	PJM AFS	SPP AFS	AECI AFS	Ameren AFS							
J302	\$17,873,404	\$24,656,882	\$6,895,959	\$2,501,512	\$0	\$0	\$56,686,704	\$206,092	\$0	\$0	\$0	\$0	\$0	\$1,432,746	\$0	\$0	\$3,167,955	\$3,250,000	NA	\$0	\$115,238,509	\$342,196	\$11,181,655
J414	\$7,683,456	\$5,954,168	\$46,210,422	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$42,861	\$0	\$74,080	\$43,292	\$0	\$0	\$0	\$500,000	NA	\$8,964,311	\$69,312,357	\$824,032	\$6,107,204
J415	\$117,925,881	\$5,420,738	\$30,604,293	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$15,610,615	\$0	\$0	\$72,182	\$0	\$0	\$0	\$10,000,000	NA	\$0	\$163,950,912	\$838,032	\$15,557,059
J439	\$125,309,755	\$37,640,581	\$10,539,805	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$131,096	\$0	\$179,768	\$323,684	\$0	\$0	\$0	\$10,000,000	NA	\$0	\$183,490,141	\$1,786,660	\$16,562,354
J457	\$31,024,608	\$34,944,619	\$5,708,632	\$1,440,235	\$0	\$0	\$32,637,138	\$520,306	\$0	\$0	\$0	\$0	\$0	\$1,241,966	\$0	\$0	\$7,680,635	\$30,000	NA	\$1,544,216	\$115,530,388	\$1,202,998	\$10,350,041
J459	\$79,728,828	\$40,015,498	\$3,345,870	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$1,779,209	\$0	\$0	\$0	\$12,000,000	NA	\$4,475,458	\$139,565,654	\$1,244,644	\$12,711,921
J476	\$18,063,866	\$5,285,404	\$2,700,804	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$81,907,429	\$0	\$4,171,856	\$0	\$11,000,000	NA	\$0	\$37,050,073	\$929,085	\$2,775,922
J503	\$17,449,528	\$24,072,134	\$6,732,419	\$2,442,188	\$0	\$0	\$55,342,355	\$201,205	\$0	\$0	\$0	\$0	\$0	\$1,418,560	\$0	\$0	\$3,092,826	\$3,250,000	NA	\$0	\$112,582,655	\$690,726	\$10,567,539
J511	\$6,608,574	\$53,021,976	\$58,496,722	\$10,615,610	\$0	\$0	\$240,560,054	\$68,223	\$0	\$0	\$0	\$4,658,600	\$0	\$6,474,269	\$0	\$0	\$3,329,109	\$816,000	NA	\$0	\$373,516,268	\$1,217,452	\$36,134,175
J512	\$28,093,673	\$37,475,943	\$3,026,073	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$58,760	\$0	\$90,555	\$136,670	\$0	\$0	\$0	\$10,000,000	NA	\$0	\$78,595,689	\$1,484,315	\$6,375,254
J541	\$80,824,273	\$0	\$37,888,245	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$140,959	\$0	\$3,380,000	\$0	\$0	\$8,500,000	\$500,000	\$57,277	\$127,269,795	\$1,513,856	\$11,213,124
J555	\$44,804,259	\$0	\$6,587,068	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	NA	\$53,075	\$51,444,402	\$820,523	\$3,763,917 ²
J569	\$6,416,408	\$12,343,806	\$1,079,090	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$64,893	\$0	\$0	\$0	\$0	\$325,800	\$0	\$19,839,304	\$1,027,079	\$956,851
J575	\$24,382,811	\$19,546,760	\$1,147,256	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$68,931	\$0	\$0	\$0	\$6,000,000	NA	\$0	\$51,076,827	\$737,000	\$4,370,683
J577	\$24,382,811	\$19,546,760	\$1,147,256	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$70,862	\$0	\$0	\$0	\$6,000,000	NA	\$0	\$51,076,827	\$1,001,610	\$4,106,073
J583	\$36,966,451	\$4,663,398	\$2,997,039	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$19,492,571	\$0	\$0	\$0	\$5,500,000	NA	\$0	\$50,126,889	\$1,045,535	\$3,967,154
J587	\$53,865,155	\$39,398,102	\$2,364,763	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$120,398	\$0	\$0	\$0	\$13,500,000	NA	\$0	\$109,128,021	\$1,382,159	\$9,530,643
J590	\$14,799,936	\$6,601,199	\$1,920,156	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$24,238	\$0	\$0	\$58,189	\$0	\$0	\$0	\$0	NA	\$0	\$23,321,291	\$819,471	\$1,512,658
J593	\$8,628,405	\$53,050,623	\$2,673,751	\$6,737,277	\$0	\$0	\$152,673,251	\$58,127	\$0	\$0	\$0	\$0	\$90,677	\$12,184,044	\$0	\$0	\$435,018	\$2,100,000	NA	\$0	\$226,356,452	\$1,380,198	\$21,255,447
J594	\$89,857,025	\$34,037,046	\$13,977,689	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$77,249	\$0	\$109,045	\$239,086	\$0	\$0	\$390,000	\$500,000	NA	\$13,277,230	\$152,038,990	\$1,618,238	\$13,585,661
J596	\$5,643,424	\$23,421,950	\$3,779,095	\$0	\$0	\$0	\$7,300,000	\$0	\$0	\$0	\$0	\$0	\$0	\$1,102,108	\$0	\$0	\$55,000	\$3,810,000	NA	\$0	\$44,009,469	\$1,408,198	\$2,992,749
J597	\$79,967,431	\$58,640,280	\$3,441,769	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$109,167	\$206,794	\$0	\$0	\$0	\$6,000,000	NA	\$0	\$148,049,480	\$1,723,238	\$13,081,710
J598	\$58,713,211	\$0	\$27,709,398	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$105,750	\$0	\$0	\$0	\$0	\$8,500,000	\$500,000	\$54,532	\$94,977,142	\$1,289,238	\$8,208,476
J599	\$6,489,659	\$39,443,378	\$2,047,798	\$3,901,652	\$0	\$0	\$88,415,235	\$44,207	\$0	\$0	\$0	\$0	\$0	\$26,390,414	\$0	\$0	\$0	\$15,000,000	NA	\$0	\$155,341,929	\$1,277,159	\$14,257,034
J607	\$25,222,335	\$35,920,104	\$10,667,804	\$2,816,340	\$0	\$0	\$63,821,004	\$389,933	\$0	\$0	\$0	\$0	\$0	\$1,779,676	\$0	\$0	\$5,954,225	\$6,500,000	NA	\$1,513,019	\$152,804,765	\$764,659	\$14,515,817
J611	\$6,513,748	\$0	\$2,144,431	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$5,828,144	\$0	\$7,000,000	NA	\$76,265	\$15,734,444	\$1,106,198	\$467,246
J613	\$5,841,923	\$28,299,379	\$5,197,813	\$575,185	\$0	\$0	\$13,034,259	\$11,907	\$0	\$0	\$0	\$3,191,400	\$0	\$1,067,305	\$0	\$0	\$1,390,231	\$900,000	NA	\$2,872,391	\$58,123,090	\$746,079	\$5,066,230
J614	\$5,964,330	\$1,852,682	\$6,935,361	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$25,355	\$0	\$0	\$0	\$0	\$0	\$0	\$3,682,249	\$852,822	\$0	\$18,434,622	\$496,212	\$1,347,250

Executive Summary

Project Num	ERIS Network Upgrades (\$)																NRIS Network Upgrades (\$)	Inter - connection Substation TO NUs (\$)	TOIF ¹ (\$)	SNU (\$)	Total Network Upgrade Cost (Exclude TOIF & Affected System) (\$)	M2 Received (\$)	M3 (\$)
	Backbone NUs	MWEX Voltage Stability	MISO Thermal & Voltage	Transient Stability	Short Circuit	DPC LPC	GRE LPC	MDU LPC	OTP LPC	Ameren LPC	CIPCO AFS	MPC AFS	PJM AFS	SPP AFS	AECI AFS	Ameren AFS							
J615	\$28,017,037	\$0	\$3,366,751	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$10,029,824	\$0	\$0	\$0	\$0	\$0	\$24,000,000	\$5,092,200	\$673,600	\$0	\$60,475,988	\$700,000	\$5,347,599
J637	\$38,131,258	\$19,712,469	\$1,702,496	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$472,944	\$0	\$0	\$0	\$12,000,000	NA	\$2,824,885	\$74,371,108	\$980,000	\$6,457,111
J638	\$89,938,037	\$41,034,120	\$3,543,972	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$984,496	\$0	\$0	\$0	\$1,500,000	NA	\$5,879,888	\$141,896,016	\$1,257,202	\$12,932,400
Total (\$)	\$1,185,131,500	\$706,000,000	\$316,580,000	\$31,030,000	\$0	\$0	\$710,470,000	\$1,500,000	\$0	\$0	\$26,000,000	\$7,850,000	\$900,000	\$159,132,718	\$3,380,000	\$10,000,000	\$49,495,000	\$172,930,449	NA	\$41,592,547	\$3,214,729,496	\$33,653,992	\$287,258,958

Note 1: TO's Interconnection Facilities
Note 2: J555 has previously paid \$560,000 PGIA M3 payment.

Executive Summary

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

1.4 Per Project Summary

This section provides estimated cost of Network Upgrades on a per project basis for the 2022 scenario. The shared cost of Network Upgrades for all the generation projects are listed below.

The Interconnection Customers are required to mitigate the constraints observed from the 2022 scenario.

1.4.1 J302 Summary

Network Upgrade	Cost	J302	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$6,089,492	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$8,685,487	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$2,653,646	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$336,534	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$108,244	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$24,656,882	MWEX Voltage Stability NU
Add breaker at Merricourt to eliminate overloads under one P2-3 contingency	\$1,500,000	\$231,104	MISO SH
J302&J503 POI-Heskett 230 kV	\$9,000,000	\$2,226,106	SH MISO and/or NRIS
J607 POI-Wishek 230 kV	\$1,100,000	\$318,057	MISO SH
Wing River-Inman 230 kV	\$50,000	\$5,898	MISO SH
Stanton-Square Butte 230 kV	\$10,875,000	\$544,811	SH MISO and/or NRIS
Raun-S3451 345 kV	\$200,000	\$3,912	MISO SH

Network Upgrade	Cost	J302	NUs Type
Whitlock-Glenham 230 kV	\$10,000	\$440	SH MISO and/or NRIS
Merricourt-Wishek 230 kV	\$6,000,000	\$1,586,480	SH MISO and/or NRIS
Merricourt-Ellendale 230 kV	\$4,500,000	\$708,597	MISO SH
New 345 kV line Center-Ellendale New 345 kV line Prairie-Bison 150 Mvar SVC at Jamestown 345 kV 150 Mvar SVC at Prairie 230 kV 2×50 Mvar MSC at Lake Park 230 kV	\$703,170,000	\$56,686,704	GRE LPC
Add a breaker on the existing 345 kV line reactor at Ellendale to make it switchable	\$1,500,000	\$206,092	MDU LPC
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$133,258	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$192,241	Reactive Power NU
2×20 Mvar switched cap bank at Fergus Falls 230 kV (658109)	\$1,750,000	\$133,569	Reactive Power NU
3×50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$231,909	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$8,130	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$41,922	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$65,130	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$7,275	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$153,600	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$97,809	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$104,517	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$101,195	Reactive Power NU
(0, 200) Mvar SVC at Winger 230 kV	\$31,030,000	\$2,501,512	Stability
New Tatanka-Ellendale 230 kV line (NRIS)	\$20,500,000	\$2,954,258	NRIS
J607 POI-Wishek 230 kV (NRIS)	\$400,000	\$86,855	NRIS
Merricourt-Tatanka North 230 kV (NRIS)	\$900,000	\$126,842	NRIS
Oahe-Sully Buttes 230 kV (SPP NRIS)	\$9,410,000	\$343,280	SPP AFS

Executive Summary

Network Upgrade	Cost	J302	NUs Type
Whitlock-Sully Buttes 230 kV (SPP NRIS)	\$10,780,000	\$392,917	SPP AFS
Whitlock-Glenham 230 kV (SPP NRIS)	\$18,165,000	\$662,091	SPP AFS
Low voltages near Summit 69 kV (SPP ERIS)	\$440,000	\$34,458	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$113,421,255	

1.4.2 J414 Summary

Network Upgrade	Cost	J414	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$427,559	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$746,725	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$2,440,689	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$3,787,012	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$281,471	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$5,954,168	MWEX Voltage Stability NU
Murphy Creek 161-69 kV xfmr	\$2,500,000	\$2,500,000	MISO SH
Austin-Murphy Creek 161 kV	\$650,000	\$479,534	MISO SH
Murphy Creek-Hayward 161 kV	\$4,510,000	\$3,327,124	MISO SH
Ellendale-Steel Center 69 kV	\$1,500,000	\$1,500,000	MISO SH
Pratt Tap-Steel Center 69 kV	\$5,250,000	\$5,250,000	MISO SH
Adams-Hayward 161 kV	\$36,500,000	\$31,878,720	MISO SH
Raun-S3451 345 kV	\$200,000	\$6,133	MISO SH
Hazleton-Dundee 161 kV	\$500,000	\$42,861	CIPCO AFS
Twin Branch–Argenta 345 kV	\$900,000	\$74,080	PJM AFS
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$72,677	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$84,774	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$5,789	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV	\$4,000,000	\$46,799	Reactive Power NU

Network Upgrade	Cost	J414	NUs Type
(636000)			
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$42,415	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$19,822	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$210,899	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$184,170	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$208,445	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$393,122	Reactive Power NU
Hayward-Freeborn 161 kV	\$10,259,820	\$5,841,841	SNU
NIW-Colby 161 kV	\$8,250,000	\$3,122,470	SNU
Low voltage near Mars 69 kV (SPP ERIS)	\$880,000	\$43,292	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$68,972,590	

1.4.3 J415 Summary

Network Upgrade	Cost	J415	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$1,211,427	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$2,794,676	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$102,451,947	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$9,346,355	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2x150 MVar MSC at Franklin 345kV	\$8,000,000	\$2,121,476	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$5,420,738	MWEX Voltage Stability NU
Adams 345-161-13.8 kV xfmr	\$4,000,000	\$1,591,852	MISO SH
Austin-Murphy Creek 161 kV	\$650,000	\$170,466	MISO SH
Murphy Creek-Hayward 161 kV	\$4,510,000	\$1,182,876	MISO SH
Emery-Floyd 161 kV	\$21,000,000	\$21,000,000	MISO SH

Executive Summary

Network Upgrade	Cost	J415	NUs Type
Adams-Hayward 161 kV	\$36,500,000	\$4,621,280	MISO SH
Raun-S3451 345 kV	\$200,000	\$7,894	MISO SH
Hazleton-Dundee 161 kV	\$500,000	\$82,624	CIPCO AFS
Liberty-Dundee 161 kV	\$25,000,000	\$15,221,671	CIPCO AFS
Liberty-Hickory Crk 161 kV	\$500,000	\$306,320	CIPCO AFS
3x50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$119,473	Reactive Power NU
2x50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$41,580	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$8,510	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$43,406	Reactive Power NU
1x25 Mvar switched cap bank at J615 POI 161 kV (86151)	\$2,500,000	\$320,881	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$19,029	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$8,262	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$509,817	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$393,313	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$565,655	Reactive Power NU
Low voltage near Mars 69 kV (SPP ERIS)	\$880,000	\$72,182	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$169,633,709	

1.4.4 J439 Summary

Network Upgrade	Cost	J439	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$14,379,730	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$25,113,992	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$58,008,737	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV

Network Upgrade	Cost	J439	NUs Type
Webster-Franklin 345 kV	\$117,000,000	\$26,709,313	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$1,097,984	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$37,640,581	MWEX Voltage Stability NU
J506 POI-Raun 345 kV	\$700,000	\$462,540	SH MISO and/or NRIS
J506 POI-Highland 345 kV	\$200,000	\$169,492	MISO SH
Raun-S3451 345 kV	\$200,000	\$48,809	MISO SH
Hazleton-Dundee 161 kV	\$500,000	\$131,096	CIPCO AFS
Twin Branch–Argenta 345 kV	\$900,000	\$179,768	PJM AFS
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$772,109	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$508,320	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$56,504	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$1,001,970	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$1,136,436	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$2,939,724	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$1,072,936	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$649,880	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$711,747	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$1,009,338	Reactive Power NU
Low voltage near Mars 69 kV (SPP ERIS)	\$880,000	\$323,684	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$174,124,689	

1.4.5 J457 Summary

Network Upgrade	Cost	J457	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$10,160,338	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$16,219,270	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$3,979,554	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$500,848	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$164,598	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$34,944,619	MWEX Voltage Stability NU
Add breaker at Merricourt to eliminate overloads under one P2-3 contingency	\$1,500,000	\$521,639	MISO SH
J302&J503 POI-Heskett 230 kV	\$9,000,000	\$1,301,018	SH MISO and/or NRIS
Wing River-Inman 230 kV	\$50,000	\$8,594	MISO SH
Raun-S3451 345 kV	\$200,000	\$6,543	MISO SH
Merricourt-Ellendale 230 kV	\$4,500,000	\$1,767,390	MISO SH
New 345 kV line Center-Ellendale New 345 kV line Prairie-Bison 150 Mvar SVC at Jamestown 345 kV 150 Mvar SVC at Prairie 230 kV 2×50 Mvar MSC at Lake Park 230 kV	\$703,170,000	\$32,637,138	GRE LPC
Add a breaker on the existing 345 kV line reactor at Ellendale to make it switchable	\$1,500,000	\$520,306	MDU LPC
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$225,005	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$299,247	Reactive Power NU
2×20 Mvar switched cap bank at Fergus Falls 230 kV (658109)	\$1,750,000	\$267,905	Reactive Power NU
3×50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$428,354	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$11,725	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$64,089	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV	\$4,000,000	\$101,641	Reactive Power NU

Network Upgrade	Cost	J457	NUs Type
(635200)			
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$12,911	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$228,663	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$147,777	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$159,677	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$156,456	Reactive Power NU
(0, 200) Mvar SVC at Winger 230 kV	\$31,030,000	\$1,440,235	Stability
Hankinson-Forman 230 kV	\$400,000	\$159,777	SNU
Oakes-Forman 230 kV (NU1)	\$300,000	\$115,765	SNU
Oakes-Forman 230 kV (NU2)	\$950,000	\$365,993	SNU
Oakes-Ellendale 230 kV (NU1)	\$700,000	\$269,295	SNU
Oakes-Ellendale 230 kV (NU2)	\$1,650,000	\$633,386	SNU
New Tatanka-Ellendale 230 kV line (NRIS)	\$20,500,000	\$7,364,828	NRIS
Merricourt-Tatanka North 230 kV (NRIS)	\$900,000	\$315,807	NRIS
Oahe-Sully Buttes 230 kV (SPP NRIS)	\$9,410,000	\$287,388	SPP AFS
Whitlock-Sully Buttes 230 kV (SPP NRIS)	\$10,780,000	\$329,381	SPP AFS
Whitlock-Glenham 230 kV (SPP NRIS)	\$18,165,000	\$555,029	SPP AFS
Low voltages near Summit 69 kV (SPP ERIS)	\$440,000	\$70,168	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$116,742,354	

1.4.6 J459 Summary

Network Upgrade	Cost	J459	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$27,893,425	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$45,360,114	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$5,543,304	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$699,014	Backbone NU Webster-

Executive Summary

Network Upgrade	Cost	J459	NUs Type
			Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$232,970	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$40,015,498	MWEX Voltage Stability NU
Wing River-Inman 230 kV	\$50,000	\$9,611	MISO SH
Raun-S3451 345 kV	\$200,000	\$9,253	MISO SH
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$399,003	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$434,050	Reactive Power NU
2×20 Mvar switched cap bank at Fergus Falls 230 kV (658109)	\$1,750,000	\$181,870	Reactive Power NU
3×50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$1,066,242	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$15,407	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$92,234	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$151,288	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$21,925	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$311,469	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$204,810	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$223,067	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$225,641	Reactive Power NU
J510 POI-Brookings Co 345 kV #2	\$31,400,000	\$4,475,458	SNU
Low voltages near Granite Falls 230 kV (SPP ERIS)	\$4,000,000	\$1,668,687	SPP AFS
Low voltage near Flandreau 69 kV (SPP ERIS)	\$880,000	\$110,522	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$129,344,863	

1.4.7 J476 Summary

Network Upgrade	Cost	J476	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$326,267	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$569,820	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$14,282,225	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$2,791,463	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$94,091	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$5,285,404	MWEX Voltage Stability NU
J530 POI-Hills 345 kV	\$27,000,000	\$1,407,262	MISO SH
Additional 2×25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$10,000,000	\$4,171,856	Ameren AFS
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$170,774	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$57,192	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$54,858	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$230,501	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$59,574	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$33,640	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$508,643	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$134,978	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$43,381	Reactive Power NU
Cooper South interface (SPP ERIS) ¹	\$101,400,000	\$81,907,429	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$112,129,358	

Note 1: If recommended mitigation is not valid, a new 345 kV line from Cooper to Stranger Creek may be an alternate solution. The project's total cost is \$301,000,000 and J476 cost allocation is \$243,137,437.

1.4.8 J503 Summary

Network Upgrade	Cost	J503	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$5,945,077	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$8,479,507	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$2,590,713	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$328,553	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$105,677	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$24,072,134	MWEX Voltage Stability NU
Add breaker at Merricourt to eliminate overloads under one P2-3 contingency	\$1,500,000	\$225,623	MISO SH
J302&J503 POI-Heskett 230 kV	\$9,000,000	\$2,173,313	SH MISO and/or NRIS
J607 POI-Wishek 230 kV	\$1,100,000	\$310,514	MISO SH
Wing River-Inman 230 kV	\$50,000	\$5,758	MISO SH
Stanton-Square Butte 230 kV	\$10,875,000	\$531,891	SH MISO and/or NRIS
Raun-S3451 345 kV	\$200,000	\$3,819	MISO SH
Whitlock-Glenham 230 kV	\$10,000	\$429	SH MISO and/or NRIS
Merricourt-Wishek 230 kV	\$6,000,000	\$1,548,856	SH MISO and/or NRIS
Merricourt-Ellendale 230 kV	\$4,500,000	\$691,793	MISO SH
New 345 kV line Center-Ellendale New 345 kV line Prairie-Bison 150 Mvar SVC at Jamestown 345 kV 150 Mvar SVC at Prairie 230 kV 2×50 Mvar MSC at Lake Park 230 kV	\$703,170,000	\$55,342,355	GRE LPC
Add a breaker on the existing 345 kV line reactor at Ellendale to make it switchable	\$1,500,000	\$201,205	MDU LPC
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$130,097	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$187,682	Reactive Power NU
2×20 Mvar switched cap bank at Fergus Falls 230 kV (658109)	\$1,750,000	\$130,401	Reactive Power NU
3×50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$226,409	Reactive Power NU

Network Upgrade	Cost	J503	NUs Type
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$7,937	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$40,928	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$63,586	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$7,102	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$149,957	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$95,489	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$102,038	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$98,795	Reactive Power NU
(0, 200) Mvar SVC at Winger 230 kV	\$31,030,000	\$2,442,188	Stability
New Tatanka-Ellendale 230 kV line (NRIS)	\$20,500,000	\$2,884,196	NRIS
J607 POI-Wishek 230 kV (NRIS)	\$400,000	\$84,796	NRIS
Merricourt-Tatanka North 230 kV (NRIS)	\$900,000	\$123,834	NRIS
Oahe-Sully Buttes 230 kV (SPP NRIS)	\$9,410,000	\$339,882	SPP AFS
Whitlock-Sully Buttes 230 kV (SPP NRIS)	\$10,780,000	\$389,027	SPP AFS
Whitlock-Glenham 230 kV (SPP NRIS)	\$18,165,000	\$655,535	SPP AFS
Low voltages near Summit 69 kV (SPP ERIS)	\$440,000	\$34,116	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$110,751,215	

1.4.9 J511 Summary

Network Upgrade	Cost	J511	NUs Type
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$617,249	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$5,132,155	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$652,002	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$207,169	Backbone NU Franklin

Executive Summary

Network Upgrade	Cost	J511	NUs Type
			Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$53,021,976	MWEX Voltage Stability NU
Sheyenne-Lake Park 230 kV	\$250,000	\$55,892	MISO SH
Sheyenne-Mapleton 115 kV	\$4,650,000	\$1,696,172	SH MISO and/or NRIS
Stanton-Square Butte 230 kV	\$10,875,000	\$8,185,569	SH MISO and/or NRIS
Ramsey-Prairie 230 kV	\$46,150,000	\$46,150,000	MISO SH
Whitlock-Glenham 230 kV	\$10,000	\$926	SH MISO and/or NRIS
Merricourt-Wishek 230 kV	\$6,000,000	\$262,765	SH MISO and/or NRIS
Voltage constraint at Bison 345 kV (MPC)	\$7,500,000	\$4,425,000	MPC AFS
Square Butte-Stanton 230 kV (MPC)	\$50,000	\$50,000	MPC AFS
Center-Roughrider (MPC)	\$300,000	\$183,600	MPC AFS
New 345 kV line Center-Ellendale New 345 kV line Prairie-Bison 150 Mvar SVC at Jamestown 345 kV 150 Mvar SVC at Prairie 230 kV 2×50 Mvar MSC at Lake Park 230 kV	\$703,170,000	\$240,560,054	GRE LPC
Add a breaker on the existing 345 kV line reactor at Ellendale to make it switchable	\$1,500,000	\$68,223	MDU LPC
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$213,798	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$329,486	Reactive Power NU
2×20 Mvar switched cap bank at Fergus Falls 230 kV (658109)	\$1,750,000	\$139,923	Reactive Power NU
3×50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$358,084	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$16,241	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$79,559	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$114,707	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$11,944	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$300,388	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$189,085	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345	\$5,500,000	\$200,987	Reactive Power NU

Network Upgrade	Cost	J511	NUs Type
kV (631148)			
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$191,196	Reactive Power NU
(0, 200) Mvar SVC at Winger 230 kV	\$31,030,000	\$10,615,610	Stability
New Tatanka-Ellendale 230 kV line (NRIS)	\$20,500,000	\$1,122,051	NRIS
J607 POI-Wishek 230 kV (NRIS)	\$400,000	\$33,382	NRIS
Merricourt-Tatanka North 230 kV (NRIS)	\$900,000	\$48,810	NRIS
Coyote 345-115 kV xfmr (NRIS)	\$3,000,000	\$1,964,002	NRIS
Coyote-Beulah 115 kV (NRIS)	\$250,000	\$160,864	NRIS
Oahe-Sully Buttes 230 kV (SPP NRIS)	\$9,410,000	\$688,769	SPP AFS
Whitlock-Sully Buttes 230 kV (SPP NRIS)	\$10,780,000	\$787,586	SPP AFS
Whitlock-Glenham 230 kV (SPP NRIS)	\$18,165,000	\$1,327,133	SPP AFS
Broadland 345-230-13.8 kV xfmr (SPP ERIS)	\$9,413,718	\$3,564,615	SPP AFS
Low voltages near Summit 69 kV (SPP ERIS)	\$440,000	\$106,166	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$383,833,137	

1.4.10 J512 Summary

Network Upgrade	Cost	J512	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$5,894,504	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$10,294,666	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$7,763,651	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$3,741,463	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2x150 MVar MSC at Franklin 345kV	\$8,000,000	\$399,389	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$37,475,943	MWEX Voltage Stability NU
J506 POI-Raun 345 kV	\$700,000	\$57,719	SH MISO and/or NRIS
Raun-S3451 345 kV	\$200,000	\$15,516	MISO SH
Hazleton-Dundee 161 kV	\$500,000	\$58,760	CIPCO AFS
Twin Branch-Argenta 345 kV	\$900,000	\$90,555	PJM AFS

Executive Summary

Network Upgrade	Cost	J512	NUs Type
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$536,686	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$424,230	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$24,861	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$199,524	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$270,523	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$71,282	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$458,674	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$283,138	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$298,276	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$385,643	Reactive Power NU
Low voltage near Flandreau 69 kV (SPP ERIS)	\$880,000	\$29,122	SPP AFS
Low voltage near Mars 69 kV (SPP ERIS)	\$880,000	\$107,548	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$68,881,675	

1.4.11 J541 Summary

Network Upgrade	Cost	J541	NUs Type
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$66,021,141	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$14,803,131	Backbone NU Webster-Franklin 345 kV (49.2 mi)
J530 POI-Montezuma 345 kV	\$125,000	\$56,209	MISO SH
J530 POI-Hills 345 kV	\$27,000,000	\$10,502,521	MISO SH
Zackary 345-161 kV xfmr	\$7,000,000	\$4,350,094	MISO SH
Adair-Zackary 161 kV	\$2,000,000	\$1,242,884	MISO SH
Henry Co-Jeff 161 kV	\$200,000	\$114,278	MISO SH
Wapello-Jeff 161 kV	\$23,000,000	\$13,141,260	MISO SH

Network Upgrade	Cost	J541	NUs Type
Ottumwa 345-161 kV xfmr	\$9,000,000	\$5,142,785	MISO SH
Twin Branch–Argenta 345 kV	\$900,000	\$140,959	PJM AFS
Novelty 161 -69 kV xfmr	\$3,270,000	\$3,270,000	AECI AFS
South River-Emerson 161 kV	\$110,000	\$110,000	AECI AFS
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$239,264	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$164,121	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$2,846,193	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$88,637	Reactive Power NU
J530 POI-Hills 345 kV	\$300,000	\$57,277	SNU
Total Cost Per Project for Actual NRIS Elections for each Project		\$122,290,754	

1.4.12 J555 Summary

Network Upgrade	Cost	J555	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$158,525	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$276,862	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$37,341,104	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$7,027,767	Backbone NU Webster-Franklin 345 kV (49.2 mi)
J530 POI-Montezuma 345 kV	\$125,000	\$26,630	MISO SH
J530 POI-Hills 345 kV	\$27,000,000	\$4,990,627	MISO SH
3x50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$50,985	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$75,564	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$138,721	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$4,210	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$1,223,242	Reactive Power NU

Executive Summary

Network Upgrade	Cost	J555	NUs Type
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$77,088	Reactive Power NU
J530 POI-Hills 345 kV	\$300,000	\$53,075	SNU
Total Cost Per Project for Actual NRIS Elections for each Project		\$51,444,402	

1.4.13 J569 Summary

Network Upgrade	Cost	J569	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$972,131	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$1,697,813	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$3,172,182	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$425,119	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$149,163	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$12,343,806	MWEX Voltage Stability NU
Raun-S3451 345 kV	\$200,000	\$6,471	MISO SH
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$187,988	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$150,053	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$11,049	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$63,493	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$109,382	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$13,657	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$186,728	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$109,571	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$110,388	Reactive Power NU

Network Upgrade	Cost	J569	NUs Type
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$130,307	Reactive Power NU
Low voltage near Flandreau 69 kV (SPP ERIS)	\$880,000	\$28,874	SPP AFS
Low voltage near Mars 69 kV (SPP ERIS)	\$880,000	\$36,019	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$19,904,197	

1.4.14 J575 Summary

Network Upgrade	Cost	J575	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$7,144,263	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$12,971,085	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$3,604,971	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$539,518	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2x150 MVar MSC at Franklin 345kV	\$8,000,000	\$122,973	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$19,546,760	MWEX Voltage Stability NU
Raun-S3451 345 kV	\$200,000	\$5,219	MISO SH
3x50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$248,075	Reactive Power NU
2x50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$216,111	Reactive Power NU
3x50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$29,237	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$8,409	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$49,818	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$84,656	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$10,870	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$162,365	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$104,379	Reactive Power NU

Executive Summary

Network Upgrade	Cost	J575	NUs Type
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$111,429	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$116,688	Reactive Power NU
Low voltage near Flandreau 69 kV (SPP ERIS)	\$880,000	\$68,931	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$45,145,758	

1.4.15 J577 Summary

Network Upgrade	Cost	J577	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$7,144,263	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$12,971,085	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$3,604,971	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$539,518	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2x150 MVar MSC at Franklin 345kV	\$8,000,000	\$122,973	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$19,546,760	MWEX Voltage Stability NU
Raun-S3451 345 kV	\$200,000	\$5,219	MISO SH
3x50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$248,075	Reactive Power NU
2x50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$216,111	Reactive Power NU
3x50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$29,237	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$8,409	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$49,818	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$84,656	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$10,870	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$162,365	Reactive Power NU

Network Upgrade	Cost	J577	NUs Type
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$104,379	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$111,429	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$116,688	Reactive Power NU
Low voltage near Flandreau 69 kV (SPP ERIS)	\$880,000	\$70,862	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$45,147,689	

1.4.16 J583 Summary

Network Upgrade	Cost	J583	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$283,433	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$2,500,170	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$23,570,564	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$10,523,435	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2x150 MVar MSC at Franklin 345kV	\$8,000,000	\$88,849	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$4,663,398	MWEX Voltage Stability NU
J530 POI-Hills 345 kV	\$27,000,000	\$1,611,535	MISO SH
3x50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$144,933	Reactive Power NU
2x50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$47,979	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$67,248	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$265,538	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$7,900	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$23,852	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$622,398	Reactive Power NU

Executive Summary

Network Upgrade	Cost	J583	NUs Type
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$157,018	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$48,637	Reactive Power NU
Cooper South interface (SPP ERIS) ¹	\$101,400,000	\$19,492,571	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$64,119,460	

Note 1: If recommended mitigation is not valid, a new 345 kV line from Cooper to Stranger Creek may be an alternate solution. The project's total cost is \$301,000,000 and J583 cost allocation is \$57,862,563.

1.4.17 J587 Summary

Network Upgrade	Cost	J587	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$16,276,681	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$29,012,306	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$7,635,919	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$694,341	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2x150 MVar MSC at Franklin 345kV	\$8,000,000	\$245,909	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$39,398,102	MWEX Voltage Stability NU
Raun-S3451 345 kV	\$200,000	\$10,433	MISO SH
3x50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$550,963	Reactive Power NU
2x50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$460,183	Reactive Power NU
3x50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$45,489	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$16,023	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$99,157	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$161,972	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$24,174	Reactive Power NU

Network Upgrade	Cost	J587	NUs Type
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$320,729	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$210,292	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$227,034	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$238,315	Reactive Power NU
Low voltage near Flandreau 69 kV (SPP ERIS)	\$880,000	\$120,398	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$95,748,419	

1.4.18 J590 Summary

Network Upgrade	Cost	J590	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$1,893,495	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$3,306,962	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$7,123,525	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$2,268,851	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2x150 MVar MSC at Franklin 345kV	\$8,000,000	\$207,103	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$6,601,199	MWEX Voltage Stability NU
J506 POI-Raun 345 kV	\$700,000	\$83,257	SH MISO and/or NRIS
J506 POI-Highland 345 kV	\$200,000	\$30,508	MISO SH
Raun-S3451 345 kV	\$200,000	\$8,640	MISO SH
Hazleton-Dundee 161 kV	\$500,000	\$24,238	CIPCO AFS
3x50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$127,102	Reactive Power NU
2x50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$86,377	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$10,040	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$200,450	Reactive Power NU

Executive Summary

Network Upgrade	Cost	J590	NUs Type
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$204,546	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$529,148	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$195,910	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$119,811	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$133,072	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$191,296	Reactive Power NU
Low voltage near Mars 69 kV (SPP ERIS)	\$880,000	\$58,189	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$23,403,718	

1.4.19 J593 Summary

Network Upgrade	Cost	J593	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$366,119	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$1,335,092	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$5,920,359	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$773,788	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2x150 MVar MSC at Franklin 345kV	\$8,000,000	\$233,046	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$53,050,623	MWEX Voltage Stability NU
Sheyenne-Lake Park 230 kV	\$250,000	\$50,197	MISO SH
Whitlock-Glenham 230 kV	\$10,000	\$918	SH MISO and/or NRIS
Merricourt-Wishek 230 kV	\$6,000,000	\$250,398	SH MISO and/or NRIS
Twin Branch–Argenta 345 kV	\$900,000	\$90,677	PJM AFS
New 345 kV line Center-Ellendale New 345 kV line Prairie-Bison 150 Mvar SVC at Jamestown 345 kV 150 Mvar SVC at Prairie 230 kV	\$703,170,000	\$152,673,251	GRE LPC

Network Upgrade	Cost	J593	NUs Type
2×50 Mvar MSC at Lake Park 230 kV			
Add a breaker on the existing 345 kV line reactor at Ellendale to make it switchable	\$1,500,000	\$58,127	MDU LPC
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$262,824	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$390,279	Reactive Power NU
2×20 Mvar switched cap bank at Fergus Falls 230 kV (658109)	\$1,750,000	\$142,798	Reactive Power NU
3×50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$338,040	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$19,648	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$90,149	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$139,971	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$9,573	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$343,332	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$210,512	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$216,957	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$208,157	Reactive Power NU
(0, 200) Mvar SVC at Winger 230 kV	\$31,030,000	\$6,737,277	Stability
New Tatanka-Ellendale 230 kV line (NRIS)	\$20,500,000	\$356,726	NRIS
J607 POI-Wishek 230 kV (NRIS)	\$400,000	\$31,811	NRIS
Merricourt-Tatanka North 230 kV (NRIS)	\$900,000	\$46,481	NRIS
Bismarck-Hilken 230 kV (SPP ERIS)	\$3,500,000	\$3,500,000	SPP AFS
Oahe-Sully Buttes 230 kV (SPP NRIS)	\$9,410,000	\$606,719	SPP AFS
Whitlock-Sully Buttes 230 kV (SPP NRIS)	\$10,780,000	\$690,390	SPP AFS
Whitlock-Glenham 230 kV (SPP NRIS)	\$18,165,000	\$1,163,352	SPP AFS
Broadland 345-230-13.8 kV xfmr (SPP ERIS)	\$9,413,718	\$5,849,103	SPP AFS
Low voltage near Belden 69 Kv (SPP ERIS)	\$264,000	\$264,000	SPP AFS

Executive Summary

Network Upgrade	Cost	J593	NUs Type
Low voltages near Summit 69 kV (SPP ERIS)	\$440,000	\$110,480	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$236,531,173	

1.4.20 J594 Summary

Network Upgrade	Cost	J594	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$10,192,037	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$17,800,247	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$48,863,667	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$12,421,446	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$579,626	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$34,037,046	MWEX Voltage Stability NU
J506 POI-Raun 345 kV	\$700,000	\$96,483	SH MISO and/or NRIS
Rutland-Fox Lake 161 kV	\$9,000,000	\$9,000,000	MISO SH
Rutland-Huntley 161 kV	\$730,000	\$730,000	MISO SH
Cayler-Wisdom 161 kV	\$500,000	\$500,000	MISO SH
Raun-S3451 345 kV	\$200,000	\$17,645	MISO SH
Hazleton-Dundee 161 kV	\$500,000	\$77,249	CIPCO AFS
Twin Branch–Argenta 345 kV	\$900,000	\$109,045	PJM AFS
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$582,389	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$426,610	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$29,074	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$258,185	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$280,571	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$121,602	Reactive Power NU

Network Upgrade	Cost	J594	NUs Type
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$579,840	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$369,488	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$400,192	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$585,609	Reactive Power NU
Rutland-Fox Lake 161 kV	\$11,900,000	\$6,638,615	SNU
Rutland-Huntley 161 kV	\$10,220,000	\$6,638,615	SNU
Lakefield-Dickinson 161 kV (NRIS)	\$390,000	\$390,000	NRIS
Low voltage near Mars 69 kV (SPP ERIS)	\$880,000	\$239,086	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$151,964,370	

1.4.21 J596 Summary

Network Upgrade	Cost	J596	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$1,619,649	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$1,084,118	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$2,530,212	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$298,663	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2x150 MVar MSC at Franklin 345kV	\$8,000,000	\$110,782	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$23,421,950	MWEX Voltage Stability NU
Grove Lake-Lake Johnason 69 kV	\$2,260,000	\$2,260,000	MISO SH
Wing River-Inman 230 kV	\$50,000	\$5,699	MISO SH
Raun-S3451 345 kV	\$200,000	\$4,201	MISO SH
Morris-Grant Co 115 kV	\$350,000	\$350,000	MISO SH
Kerkhoven Tap-Kerkhoven 115kV	\$7,300,000	\$7,300,000	GRE LPC
3x50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$163,278	Reactive Power NU

Executive Summary

Network Upgrade	Cost	J596	NUs Type
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$219,169	Reactive Power NU
2×20 Mvar switched cap bank at Fergus Falls 230 kV (658109)	\$1,750,000	\$55,281	Reactive Power NU
3×50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$121,107	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$6,947	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$42,826	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$66,889	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$11,480	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$149,138	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$100,651	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$112,230	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$110,200	Reactive Power NU
Hoot Lake-Fergus Falls 115 kV (NRIS)	\$55,000	\$55,000	NRIS
Low voltages near Granite Falls 230 kV (SPP ERIS)	\$4,000,000	\$1,068,727	SPP AFS
Low voltage near Flandreau 69 kV (SPP ERIS)	\$880,000	\$33,381	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$41,301,577	

1.4.22 J597 Summary

Network Upgrade	Cost	J597	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$23,752,475	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$43,412,568	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$10,814,913	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$1,618,555	Backbone NU Webster-

Network Upgrade	Cost	J597	NUs Type
			Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$368,920	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$58,640,280	MWEX Voltage Stability NU
Raun-S3451 345 kV	\$200,000	\$15,656	MISO SH
Twin Branch–Argenta 345 kV	\$900,000	\$109,167	PJM AFS
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$744,226	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$648,332	Reactive Power NU
3×50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$87,710	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$25,227	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$149,455	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$253,967	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$32,611	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$487,094	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$313,138	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$334,287	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$350,065	Reactive Power NU
Low voltage near Flandreau 69 kV (SPP ERIS)	\$880,000	\$206,794	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$142,365,440	

1.4.23 J598 Summary

Network Upgrade	Cost	J598	NUs Type
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$47,609,181	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$11,104,031	Backbone NU Webster-Franklin 345 kV (49.2 mi)
J530 POI-Montezuma 345 kV	\$125,000	\$42,161	MISO SH
J530 POI-Hills 345 kV	\$27,000,000	\$7,877,703	MISO SH
Zackary 345-161 kV xfmr	\$7,000,000	\$2,649,906	MISO SH
Adair-Zackary 161 kV	\$2,000,000	\$757,116	MISO SH
Henry Co-Jeff 161 kV	\$200,000	\$85,722	MISO SH
Wapello-Jeff 161 kV	\$23,000,000	\$9,858,740	MISO SH
Ottumwa 345-161 kV xfmr	\$9,000,000	\$3,857,215	MISO SH
Twin Branch–Argenta 345 kV	\$900,000	\$105,750	PJM AFS
3x50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$32,379	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$185,408	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$153,672	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$2,135,150	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$74,225	Reactive Power NU
J530 POI-Hills 345 kV	\$300,000	\$54,532	SNU
Total Cost Per Project for Actual NRIS Elections for each Project		\$86,582,891	

1.4.24 J599 Summary

Network Upgrade	Cost	J599	NUs Type
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$5,524,871	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$758,831	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$205,958	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$39,443,378	MWEX Voltage Stability NU
Sheyenne-Lake Park 230 kV	\$250,000	\$58,654	MISO SH
Whitlock-Glenham 230 kV	\$10,000	\$6,385	SH MISO and/or NRIS
New 345 kV line Center-Ellendale New 345 kV line Prairie-Bison 150 Mvar SVC at Jamestown 345 kV 150 Mvar SVC at Prairie 230 kV 2×50 Mvar MSC at Lake Park 230 kV	\$703,170,000	\$88,415,235	GRE LPC
Add a breaker on the existing 345 kV line reactor at Ellendale to make it switchable	\$1,500,000	\$44,207	MDU LPC
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$237,351	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$313,882	Reactive Power NU
2×20 Mvar switched cap bank at Fergus Falls 230 kV (658109)	\$1,750,000	\$106,467	Reactive Power NU
3×50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$237,168	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$20,112	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$80,130	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$138,374	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$317,290	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$183,920	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$177,783	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$170,283	Reactive Power NU

Executive Summary

Network Upgrade	Cost	J599	NUs Type
(0, 200) Mvar SVC at Winger 230 kV	\$31,030,000	\$3,901,652	Stability
Oahe-Sully Buttes 230 kV (SPP NRIS)	\$9,410,000	\$6,465,895	SPP AFS
Whitlock-Sully Buttes 230 kV (SPP NRIS)	\$10,780,000	\$7,414,441	SPP AFS
Whitlock-Glenham 230 kV (SPP NRIS)	\$18,165,000	\$12,493,816	SPP AFS
Low voltage near Flandreau 69 kV (SPP ERIS)	\$880,000	\$16,262	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$166,732,343	

1.4.25 J607 Summary

Network Upgrade	Cost	J607	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$8,176,930	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$12,432,059	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$3,951,603	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$499,656	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$162,087	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$35,920,104	MWEX Voltage Stability NU
Add breaker at Merricourt to eliminate overloads under one P2-3 contingency	\$1,500,000	\$521,634	MISO SH
J302&J503 POI-Heskett 230 kV	\$9,000,000	\$3,299,564	SH MISO and/or NRIS
J607 POI-Wishek 230 kV	\$1,100,000	\$471,429	MISO SH
Sheyenne-Lake Park 230 kV	\$250,000	\$42,024	MISO SH
Stanton-Square Butte 230 kV	\$10,875,000	\$666,936	SH MISO and/or NRIS
Raun-S3451 345 kV	\$200,000	\$5,947	MISO SH
Whitlock-Glenham 230 kV	\$10,000	\$558	SH MISO and/or NRIS
Merricourt-Wishek 230 kV	\$6,000,000	\$2,351,502	SH MISO and/or NRIS
Merricourt-Ellendale 230 kV	\$4,500,000	\$1,332,220	MISO SH
New 345 kV line Center-Ellendale New 345 kV line Prairie-Bison 150 Mvar SVC at Jamestown 345 kV 150 Mvar SVC at Prairie 230 kV 2×50 Mvar MSC at Lake Park 230 kV	\$703,170,000	\$63,821,004	GRE LPC

Network Upgrade	Cost	J607	NUs Type
Add a breaker on the existing 345 kV line reactor at Ellendale to make it switchable	\$1,500,000	\$389,933	MDU LPC
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$208,319	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$290,570	Reactive Power NU
2×20 Mvar switched cap bank at Fergus Falls 230 kV (658109)	\$1,750,000	\$225,468	Reactive Power NU
3×50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$383,170	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$11,923	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$62,911	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$98,537	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$11,621	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$228,073	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$146,071	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$156,803	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$152,525	Reactive Power NU
(0, 200) Mvar SVC at Winger 230 kV	\$31,030,000	\$2,816,340	Stability
Hankinson-Forman 230 kV	\$400,000	\$160,628	SNU
Oakes-Forman 230 kV (NU1)	\$300,000	\$113,202	SNU
Oakes-Forman 230 kV (NU2)	\$950,000	\$357,890	SNU
Oakes-Ellendale 230 kV (NU1)	\$700,000	\$262,916	SNU
Oakes-Ellendale 230 kV (NU2)	\$1,650,000	\$618,383	SNU
New Tatanka-Ellendale 230 kV line (NRIS)	\$20,500,000	\$5,552,843	NRIS
J607 POI-Wishek 230 kV (NRIS)	\$400,000	\$163,156	NRIS
Merricourt-Tatanka North 230 kV (NRIS)	\$900,000	\$238,226	NRIS
Oahe-Sully Buttes 230 kV (SPP NRIS)	\$9,410,000	\$422,385	SPP AFS
Whitlock-Sully Buttes 230 kV (SPP NRIS)	\$10,780,000	\$483,659	SPP AFS

Executive Summary

Network Upgrade	Cost	J607	NUs Type
Whitlock-Glenham 230 kV (SPP NRIS)	\$18,165,000	\$814,996	SPP AFS
Low voltages near Summit 69 kV (SPP ERIS)	\$440,000	\$58,636	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$148,084,441	

1.4.26 J611 Summary

Network Upgrade	Cost	J611	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$148,786	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$259,853	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$4,920,028	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$1,147,513	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$37,568	Backbone NU Franklin Reactive Support
J530 POI-Hills 345 kV	\$27,000,000	\$610,353	MISO SH
J611-Maryville 161 kV	\$1,000,000	\$1,000,000	MISO SH
Additional 2×25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$10,000,000	\$5,828,144	Ameren AFS
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$73,686	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$24,585	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$21,870	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$94,839	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$19,954	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$14,130	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$217,639	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$58,204	Reactive Power NU

Network Upgrade	Cost	J611	NUs Type
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$9,172	Reactive Power NU
J274 POI-Creston 161 kV	\$175,000	\$60,145	SNU
Clarinda-Brooks 161 kV	\$70,000	\$16,120	SNU
Total Cost Per Project for Actual NRIS Elections for each Project		\$14,562,588	

1.4.27 J613 Summary

Network Upgrade	Cost	J613	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$1,901,509	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$1,086,534	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$2,454,073	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$296,279	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2x150 MVar MSC at Franklin 345kV	\$8,000,000	\$103,528	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$28,299,379	MWEX Voltage Stability NU
Sheyenne-Lake Park 230 kV	\$250,000	\$43,233	MISO SH
Sheyenne-Mapleton 115 kV	\$4,650,000	\$2,953,828	SH MISO and/or NRIS
Stanton-Square Butte 230 kV	\$10,875,000	\$945,794	SH MISO and/or NRIS
Raun-S3451 345 kV	\$200,000	\$3,767	MISO SH
Whitlock-Glenham 230 kV	\$10,000	\$345	SH MISO and/or NRIS
Voltage constraint at Bison 345 kV (MPC)	\$7,500,000	\$3,075,000	MPC AFS
Center-Roughrider (MPC)	\$300,000	\$116,400	MPC AFS
New 345 kV line Center-Ellendale New 345 kV line Prairie-Bison 150 Mvar SVC at Jamestown 345 kV 150 Mvar SVC at Prairie 230 kV 2x50 Mvar MSC at Lake Park 230 kV	\$703,170,000	\$13,034,259	GRE LPC
Add a breaker on the existing 345 kV line reactor at Ellendale to make it switchable	\$1,500,000	\$11,907	MDU LPC
3x50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$103,843	Reactive Power NU

Executive Summary

Network Upgrade	Cost	J613	NUs Type
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$182,412	Reactive Power NU
2×20 Mvar switched cap bank at Fergus Falls 230 kV (658109)	\$1,750,000	\$91,936	Reactive Power NU
3×50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$308,399	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$7,225	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$39,375	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$61,116	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$8,500	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$146,295	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$95,891	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$105,860	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$99,995	Reactive Power NU
(0, 200) Mvar SVC at Winger 230 kV	\$31,030,000	\$575,185	Stability
Jamestown 345-115-41.6 kV #1	\$4,042,642	\$1,370,219	SNU
Jamestown 345-115-41.6 kV #2	\$4,042,642	\$1,381,802	SNU
Buffalo-Bison 345 kV	\$175,000	\$60,027	SNU
Jamestown-Buffalo 345 kV	\$175,000	\$60,343	SNU
New Tatanka-Ellendale 230 kV line (NRIS)	\$20,500,000	\$265,097	NRIS
Coyote 345-115 kV xfmr (NRIS)	\$3,000,000	\$1,035,998	NRIS
Coyote-Beulah 115 kV (NRIS)	\$250,000	\$89,136	NRIS
Oahe-Sully Buttes 230 kV (SPP NRIS)	\$9,410,000	\$255,682	SPP AFS
Whitlock-Sully Buttes 230 kV (SPP NRIS)	\$10,780,000	\$292,599	SPP AFS
Whitlock-Glenham 230 kV (SPP NRIS)	\$18,165,000	\$493,048	SPP AFS
Low voltages near Summit 69 kV (SPP ERIS)	\$440,000	\$25,976	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$61,481,795	

1.4.28 J614 Summary

Network Upgrade	Cost	J614	NUs Type
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$812,611	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$4,361,177	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$736,808	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$53,734	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$1,852,682	MWEX Voltage Stability NU
Adams 345-161-13.8 kV xfmr	\$4,000,000	\$2,408,148	MISO SH
Harmony-Cresco 69 kV	\$4,000,000	\$4,000,000	MISO SH
Hazleton-Dundee 161 kV	\$500,000	\$25,355	CIPCO AFS
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$16,559	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$67,556	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$894	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$8,768	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$13,697	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$5,981	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$105,729	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$96,337	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$112,743	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$98,949	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$14,777,729	

1.4.29 J615 Summary

Network Upgrade	Cost	J615	NUs Type
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$27,989,138	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$27,899	Backbone NU Franklin Reactive Support
Hazleton-Washburn 161 kV	\$370,000	\$370,000	MISO SH
Hazleton-Arnold 345 kV	\$200,000	\$200,000	MISO SH
Hazleton-Dundee 161 kV	\$500,000	\$57,815	CIPCO AFS
Liberty-Dundee 161 kV	\$25,000,000	\$9,778,329	CIPCO AFS
Liberty-Hickory Crk 161 kV	\$500,000	\$193,680	CIPCO AFS
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$17,672	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$35,627	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$1,120	Reactive Power NU
1×25 Mvar switched cap bank at J615 POI 161 kV (86151)	\$2,500,000	\$2,179,119	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$2,770	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$181,244	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$185,379	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$175,848	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$17,973	Reactive Power NU
Laurel-Jasper 161 kV (NRIS)	\$24,000,000	\$24,000,000	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$65,413,612	

1.4.30 J637 Summary

Network Upgrade	Cost	J637	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$13,987,734	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$19,967,724	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$3,527,766	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$529,003	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$119,031	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$19,712,469	MWEX Voltage Stability NU
Wing River-Inman 230 kV	\$50,000	\$4,686	MISO SH
Raun-S3451 345 kV	\$200,000	\$4,843	MISO SH
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$235,741	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$215,913	Reactive Power NU
2×20 Mvar switched cap bank at Fergus Falls 230 kV (658109)	\$1,750,000	\$89,038	Reactive Power NU
3×50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$522,271	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$7,978	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$47,910	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$80,060	Reactive Power NU
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$10,979	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$157,124	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$102,076	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$109,794	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$114,083	Reactive Power NU

Executive Summary

Network Upgrade	Cost	J637	NUs Type
J510 POI-Brookings Co 345 kV #2	\$31,400,000	\$2,824,885	SNU
Low voltages near Granite Falls 230 kV (SPP ERIS)	\$4,000,000	\$409,713	SPP AFS
Low voltage near Flandreau 69 kV (SPP ERIS)	\$880,000	\$63,231	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$62,844,052	

1.4.31 J638 Summary

Network Upgrade	Cost	J638	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$31,054,151	Backbone NU Big Stone South-Alexandria 345 kV
J510-Hazel Creek-Scott County 345 kV	\$330,000,000	\$50,191,404	Backbone NU J510-Hazel Creek-Scott County 345 kV
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	\$532,731,500	\$7,343,514	Backbone NU Franklin-Morgan Valley & Beverly-Sub 92 345 kV
Webster-Franklin 345 kV	\$117,000,000	\$1,101,189	Backbone NU Webster-Franklin 345 kV (49.2 mi)
2×150 MVar MSC at Franklin 345kV	\$8,000,000	\$247,779	Backbone NU Franklin Reactive Support
N Rochester-S Rocky Run MWEX NU	\$706,000,000	\$41,034,120	MWEX Voltage Stability NU
Wing River-Inman 230 kV	\$50,000	\$9,755	MISO SH
Raun-S3451 345 kV	\$200,000	\$10,081	MISO SH
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	\$7,500,000	\$490,727	Reactive Power NU
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	\$7,000,000	\$449,451	Reactive Power NU
2×20 Mvar switched cap bank at Fergus Falls 230 kV (658109)	\$1,750,000	\$185,344	Reactive Power NU
3×50 Mvar switched cap bank at Alexandria 345 kV (658047)	\$5,500,000	\$1,087,176	Reactive Power NU
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	\$1,000,000	\$16,607	Reactive Power NU
1x150 Mvar switched cap bank at Webster 345 kV (636000)	\$4,000,000	\$99,732	Reactive Power NU
1x150 Mvar switched cap bank at Raun 345 kV (635200)	\$4,000,000	\$166,655	Reactive Power NU

Network Upgrade	Cost	J638	NUs Type
1x100 Mvar switched cap bank at OBrien 345 kV (635368)	\$4,000,000	\$22,855	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$327,075	Reactive Power NU
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$5,500,000	\$212,484	Reactive Power NU
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	\$5,500,000	\$228,552	Reactive Power NU
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	\$5,500,000	\$237,479	Reactive Power NU
J510 POI-Brookings Co 345 kV #2	\$31,400,000	\$5,879,888	SNU
Low voltages near Granite Falls 230 kV (SPP ERIS)	\$4,000,000	\$852,873	SPP AFS
Low voltage near Flandreau 69 kV (SPP ERIS)	\$880,000	\$131,623	SPP AFS
Total Cost Per Project for Actual NRIS Elections for each Project		\$141,380,512	

1.5 Study Compliance with NERC FAC-002-2 Standard

This DPP 2016 August West Area study was completed in compliance with NERC FAC-002-2:

R1.1: The reliability impact of the new interconnection, or materially modified existing interconnection, on affected system(s).

Section 3 covers summer peak steady-state analysis results which include thermal and voltage constraints impacted by the DPP West Area generating facilities. Thermal and voltage upgrades required to interconnect the new generating facilities are also identified.

Section 4 covers summer shoulder steady-state analysis results which include thermal and voltage constraints impacted by the DPP West Area generating facilities. Thermal and voltage upgrades required to interconnect the new generating facilities are also identified.

Section 5.1 covers reliability impact of the J511 and J596 generating facilities per GRE Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 5.2 covers reliability impact of the generating facilities per OTP Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 5.3 covers reliability impact of the generating facilities per MDU Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Executive Summary

Section 5.4 covers reliability impact of the generating facilities per DPC Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 5.5 covers reliability impact of the generating facilities per Ameren Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 6.1 covers reliability impact of the new generating facilities in the CIPCO affected systems. Network Upgrades required to interconnect the new generating facilities are also identified.

Section 6.2 covers reliability impact of the new generating facilities in the MPC affected systems. Network Upgrades required to interconnect the new generating facilities are also identified.

Section 6.3 covers reliability impact of the new generating facilities in the PJM affected systems. Network Upgrades required to interconnect the new generating facilities are also identified.

Section 6.4 covers reliability impact of the new generating facilities in the SPP affected systems. Network Upgrades required to interconnect the new generating facilities are also identified.

Section 6.5 covers reliability impact of the new generating facilities in the AECI affected systems. Network Upgrades required to interconnect the new generating facilities are also identified.

Section 7 covers transient stability analysis results.

Section 8 covers voltage stability (PV) analysis on the MWEX System Operating Limit (SOL). Network Upgrades required for MWEX voltage stability are identified.

Section 9 covers short circuit reliability impact of the new generating facilities.

Section 10 covers Deliverability reliability impact of the new NRIS generating facilities.

R1.2: Adherence to applicable NERC Reliability Standards; regional and Transmission Owner planning criteria; and Facility interconnection requirements.

Sections 2.2-2.4, Section 5, Section 6, and Section 7 all cover NERC Reliability Standard TPL-001-4.

Section 5.1 covers GRE Local Planning Criteria (LPC).

Section 5.2 covers OTP LPC.

Section 5.3 covers MDU LPC.

Section 5.4 covers DPC LPC.

Section 5.5 covers Ameren LPC.

Section 6.1 covers CIPCO system planning criteria.

Section 6.2 covers MPC system planning criteria.

Section 6.3 covers PJM system planning criteria.

Section 6.4 covers SPP system planning criteria.

Section 6.5 covers AECI system planning criteria.

Section 8 (voltage stability analysis) covers individual system planning criteria (ATC).

Section 10 covers MISO system planning criteria.

R1.3: Steady-state, short-circuit, and dynamics studies, as necessary, to evaluate system performance under both normal and contingency conditions.

Section 3 and Section 4 cover MISO steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.1 covers GRE's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.2 covers OTP's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.3 covers MDU's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.4 covers DPC's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.5 covers Ameren's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6.1 covers CIPCO steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6.2 covers MPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6.3 covers PJM assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Executive Summary

Section 6.4 covers SPP steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6.5 covers AECI steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 7 covers transient stability studies under NERC category P0 to P7 contingencies (TPL-001-4).

Section 8 covers steady-state voltage stability assessment.

Section 9 covers short-circuit assessment.

Section 10 covers MISO deliverability study (steady-state assessment) including NERC category P0 to P1 contingencies (TPL-001-4).

R1.4: Study assumptions, system performance, alternatives considered, and coordinated recommendations. While these studies may be performed independently, the results shall be evaluated and coordinated by the entities involved.

Section 2.1, Section 2.2, Section 2.3, Section 2.4, Section 7.2, Section 7.3, and Section 7.4 cover study assumptions and system performance criteria.

Jointly coordinated recommendations can be found in Section 5.1 (MISO and GRE), Section 5.2 (MISO and OTP), Section 5.3 (MISO and MDU), Section 5.4 (MISO and DPC), Section 5.5 (MISO and Ameren), Sections 6.1 (MISO and CIPCO), Section 6.2 (MISO and MPC), Section 6.3 (MISO and PJM), Section 6.4 (MISO and SPP), Section 6.5 (MISO and AECI), and Section 8 (MISO and ATC). Results in Section 3, 4, 5, 6, 7, 9, and 10 have also been reviewed by PJM, SPP, CIPCO, MPC, and AECI.

Introduction

Thirty one (31) generation projects, listed in Table A-1 (Appendix A.1), have requested to interconnect to the MISO transmission network in the West Area and have advanced to the Definitive Planning Phase (DPP) 2016 August Phase 1 study (DPP West Area). All generating facilities have requested Energy Resource Interconnection Service (ERIS) and Network Resource Interconnection Service (NRIS).

This report presents the study results of a System Impact Study (SIS) performed to evaluate the interconnection of the generating facilities in the DPP West Area Phase 1 study.

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

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Model Development and Study Criteria

2.1 Model Development

2.1.1 Benchmark Cases

DPP 2016 August West area power flow benchmark cases representing 2022 summer shoulder and summer peak conditions were developed from the DPP 2016 February West Phase 2 steady state models, which were originally developed from MTEP17 models with LBA dispatch.

The benchmark cases for DPP 2016 August study were created as follows:

- MISO Prior queued generation projects and their associated Network Upgrades (NU) were modeled. Appendix A.2 lists all DPP 2016 February West Area Phase 2 MISO Network Upgrades included in the models.
- DPP 2016 August generation projects in the West Area (DPP West Area, Table A-1) were modeled with offline status.
- DPP 2016 August generation projects in the Central Area (Table A-4), Michigan Area (Table A-5), and ATC Area (Table A-6) were modeled and dispatched.
- For MISO generation projects, their output was sunk to the MISO Classic (Appendix A.4, Table A-9), where generation was scaled uniformly;
- PJM generation projects were modeled and dispatched. The generation output was sunk to the PJM market (Appendix A.5, Table A-10), where generation was scaled uniformly.
- SPP generation projects were modeled and dispatched. The generation output was sunk to the SPP market (Appendix A.6, Table A-11), where generation was scaled uniformly. The following Network Upgrades identified in the SPP DIS2016-001 study were also modeled:
 - Advance Gentleman–Thedford–Holt 345 kV project
 - Build approximately 140 miles of new 345 kV from Banner County–Keystone
 - Build approximately 30 miles of second 345 kV circuit from Keystone–Gentleman
- The Hickory Creek–Cardinal 345 kV project (MVP project 3127) was included in the 2022 models; the Hickory Creek–Cardinal 345 kV project has an in-service date of 12/31/2023.
- Models were further reviewed by the Ad Hoc study members (transmission owners and customers). Model corrections and changes were made based on the comments and feedback. These modeling changes are listed in Appendix A.2.
- Adjusted Square Butte DC to match the total output of the Bison wind farms.
- Adjusted CU DC to match the total output of Coal Creek generation units #1 and #2.
- Set MHEX interface transfer level at 492 MW in summer shoulder and 1406 MW in summer peak cases.

2.1.2 Study Cases

Summer peak study case was created by dispatching the DPP West Area generating facilities at the specified summer peak level (Table ES-1) from the benchmark cases.

Summer shoulder study case was created by dispatching the DPP West Area generating facilities at the specified summer shoulder level (Table ES-1) from the benchmark cases. Due to voltage collapse under system intact condition, three (3) fictitious large size SVCs (Table 2-1) were added to the summer shoulder study case to achieve converged power flow thermal solutions.

Table 2-1: Fictitious SVCs Added Only in Summer Shoulder Case

Location	Bus #	SVC Mvar
Mingo 345 kV	531451	500
Helena 345 kV	601050	800
Cedar Mountain 345 kV	635730	500

The MISO Classic was used for power balance, where generation was scaled uniformly.

Both study and benchmark power flow cases were solved with transformer tap adjustment enabled, area interchange disabled, phase shifter adjustment enabled and switched shunt adjustment enabled.

The interface transfer levels in the 2022 study cases are summarized in Table 2-2.

Table 2-2: Interface Transfer Levels in 2022 Steady State Study Cases

Interface	2022 SH Case (MW)	2022 SPK Case (MW)
MHEX	492	1406
MWEX	1587	696
Arrowhead – Stone Lake 345 kV	613	259

2.2 Contingency Criteria

A variety of contingencies were considered for steady-state analysis:

- NERC Category P0 with system intact (no contingencies)
- NERC Category P1 contingencies
 - NERC Category P1 contingencies, at buses with a nominal voltage of 69 kV and above, in the following areas: CWLD (area 333), AMMO (area 356), AMIL (area 357), CWLP (area 360), SIPC (area 361), WEC (area 295), WEC MI (area 296), XCEL (area 600), MP (area 608), SMMPA (area 613), GRE (area 615), OTP (area 620), ITCM (area 627), MPW (area 633), MEC (area 635), MDU (area 661), BEPC-MISO (area 663), MHEB (area 667), DPC

- (area 680), ALTE (area 694), WPS (area 696), MGE (area 697), UPPC (area 698), CE (area 222), NPPD (area 640), OPPD (area 645), LES (area 650), WAPA (area 652), BEPC-SPP (area 659), AECI (area 330), MIPU (area 540), KCPL (area 541), KACY (area 542), INDN (area 545).
- o Multiple-element NERC Category P1 contingencies, in Dakotas, Illinois, Iowa, Minnesota, Missouri, and Wisconsin. These specified Category P1 contingencies are listed in Appendix A.7.
- NERC Category P2-P7 contingencies
 - o Selected NERC Category P2-P7 contingencies provided by the Ad Hoc Study Group, in the study region of Dakotas, Illinois, Iowa, Minnesota, Missouri, and Wisconsin. These specified Category P2-P7 contingencies are listed in Appendix A.7.

For all contingency and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

2.3 Monitored Elements

The study area is defined in Table 2-3. Facilities in the study area were monitored for system intact and contingency conditions. Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal (PSS[®]E rate A) rating. Under NERC category P1-P7 conditions, branches were monitored for loading as shown in the column labeled "Post-Disturbance Thermal Limits".

Table 2-3: Monitored Elements

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
AECI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
AMIL	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.05/0.95
AMMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.05/0.95
ATCLLC	69 kV and above	95% of Rate A	95% of Rate B	1.05/0.95	1.10/0.90
BEPC-MISO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
BEPC-SPP	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CWLD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CWLP	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.10/0.95
CE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
DPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
GMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
GRE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.92
INDN	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
ITCM	69 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.95	1.10/0.93

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
KACY	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
KCPL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
LES	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
MDU	57 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
MEC	69 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.96/0.95	1.05/0.96/0.95 ³
MHEB	69 kV and above	100% of Rate A	100% of Rate B	1.12/1.1/1.07/1.05/1.04/0.99/0.97/0.96/0.95	1.15/1.10/0.94/0.90
MP	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.10/0.95
MPW	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.06/0.92
NPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
OPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
OTP	40 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.97	1.10/0.92
PPI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.05/0.95
SIPC	69 kV and above	100% of Rate A	100% of Rate B	1.07/0.95	1.09/0.91
SMMPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
WAPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
XEL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.05/0.92

Notes

- 1: PSS®E Rate A, Rate B or Rate C
- 2: Limits dependent on nominal bus voltage
- 3: For facilities in Cedar Falls Utilities or Ames Municipal Utilities, post-contingency voltage limits are 1.05/0.94 for >200 kV, and 1.05/0.93 for others.

2.4 Performance Criteria

A branch is considered as a thermal injection constraint if the branch is loaded above its applicable normal or emergency rating for the post-change case, and any of the following conditions are met:

- 1) the generator (NR/ER) has a larger than 20% DF on the overloaded facility under post contingent condition or 5% DF under system intact condition, or
- 2) the megawatt impact due to the generator is greater than or equal to 20% of the applicable rating (normal or emergency) of the overloaded facility, or
- 3) the overloaded facility or the overload-causing contingency is at generator's outlet, or
- 4) for any other constrained facility, where none of the study generators meet one of the above criteria in 1), 2), or 3), however, the cumulative megawatt impact of the group of study generators (NR/ER) is greater than 20% of the applicable rating, then only those study generators whose individual MW impact is greater than 5% of the

applicable rating and has DF greater than 5% (OTDF or PTDF) will be responsible for mitigating the cumulative MW impact constraint.

A bus is considered a voltage constraint if both of the following conditions are met. All voltage constraints must be resolved before a project can receive interconnection service.

- 1) the bus voltage is outside of applicable normal or emergency limits for the post-change case, and
- 2) the change in bus voltage is greater than 0.01 per unit.

All DPP 2016 August West Area study generators must mitigate thermal injection constraints and voltage constraints in order to obtain unconditional Interconnection Service.

Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the deliverability algorithm, to meet the system performance criteria for NERC category P0-P1 events, if the constraint demonstrates an incremental flow caused by the generator equal to or greater than 5% of the generator's maximum dispatch level in each case.

2.5 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

For non-synchronous generation projects in the DPP 2016 August West Area study group, if they do not have signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

Except for generation project J590, all other non-synchronous generation projects in this study group are required to meet the reactive power requirements per FERC Order 827.

Collector system and shunt compensation of DPP West projects are modeled, which are listed in Appendix A.1, Table A-3. An analysis was performed to study the reactive power requirements (FERC Order 827) for the non-synchronous generation projects in the DPP 2016 August West study group. The analysis was performed as follows:

Step 1: Verify whether total dynamic reactive power (reactive power from generators and dynamic compensation devices) in the plant can meet the dynamic reactive power range of 0.95 leading to 0.95 lagging at the generator terminal bus. The verification in Step 1 was performed when generator data was submitted and modeled.

Step 2: Verify whether total reactive power (reactive power from generators, dynamic compensation devices, and static compensation devices) in the plant can meet the dynamic reactive power range of 0.95 leading to 0.95 lagging at the high-side of the generator substation. The testing procedure in Step 2 is described in the following:

- Lock the high-side of the generator substation at 1.0 pu voltage by adding a fictitious SVC. This is to ensure that the test result is not affected by the system condition.
- Lock the reactive power output of the generator to the maximum limit (Qmax). Make sure all shunt compensation devices within the substation are at the maximum capacitive output. Adjust transformer tap to ensure bus voltages within the substation are within 0.95 – 1.05 pu range. Measure real power and reactive power from the

- generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 lagging requirement.
- Lock the reactive power output of the generator to the minimum limit (Q_{min}). Make sure all shunt compensation devices within the substation are at the maximum inductive output. Adjust transformer tap to ensure bus voltages within the substation are within 0.95 – 1.05 pu range. Measure real power and reactive power from the generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 leading requirement.

Appendix C lists reactive power requirement analysis results for the DPP West generation projects. The results are summarized as following:

- J459, J587, and J615 do not meet the reactive power requirements per FERC Order 827.
- J590 is exempted from the FERC Order 827 reactive power requirements.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

Summer Peak Steady-State Analysis

Summer peak steady-state analysis was performed in summer peak scenario to identify thermal and voltage upgrades required interconnecting the generating facilities in the DPP 2016 August West Area group to the transmission system.

3.1 Study Procedure

3.1.1 Computer Programs

Steady-state analyses were performed using PSS[®]E version 33.11 and PSS[®]MUST version 12.0.

3.1.2 Study Methodology

Summer peak power flow case was created in the procedure as described in Section 2.1. The summer peak case can converge under post-contingency conditions. Therefore fictitious SVCs are not modeled. Nonlinear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP West Area generating facilities was evaluated by comparing the steady-state performance of the transmission system in the benchmark and study cases. Network upgrades were identified to mitigate any summer peak constraints.

3.2 Summer Peak Contingency Analysis Results

The incremental impact of the proposed interconnection on individual facilities was evaluated by comparing flows and voltages between benchmark case (without DPP West Area projects) and study case (with DPP West Area projects). Analysis was performed in the 2022 summer peak scenario using PSS[®]E and PSS[®]MUST.

3.2.1 System Intact Conditions

For NERC category P0 (system intact) conditions, no thermal or voltage constraints were identified (Table D-1, Table D-2).

3.2.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All category P1 contingencies were converged.

For P1 contingencies in the 2022 summer peak scenario, no thermal or voltage constraints were identified (Table D-3, Table D-4).

Three category P2-P7 contingencies (Table D-7) were not converged, and their dc thermal results are listed in Table D-8. The contingency was not converged in the benchmark or study cases. No mitigation plan is required for the study projects for this contingency.

For P2-P7 contingencies in the 2022 summer peak scenario, no thermal or voltage constraints were identified (Table D-5, Table D-6).

3.3 Network Upgrades Identified in MISO ERIS Analysis for 2022 Summer Peak Scenario

Table 3-1 lists worst thermal constraints, Network Upgrades and their costs in the 2022 summer peak scenario.

Table 3-1: 2022 Summer Peak Thermal Constraints, Network Upgrades and Costs, Maximum Screened Loading

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation	Cost (\$)
			(MVA)	(%)				
Jamestown Pike 115-41.6 kV xfmr	70.0	OTP	94.6	135.1	CEII Redacted	P2-P7	This contingency leaves the entire Jamestown 115 kV loop on a radial being fed from a long 41.6 kV circuit. The under voltage tripping will disconnect the 115 kV and part of the 41.6 kV network. If the under voltage tripping is simulated this constraint would not occur.	\$0

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Summer Shoulder Steady-State Analysis

Summer shoulder steady-state analysis was performed in summer shoulder scenario to identify thermal and voltage upgrades required interconnecting the generating facilities in the DPP 2016 August West Area group to the transmission system.

4.1 Study Procedure

Summer shoulder power flow case was created in the procedure as described in Section 2.1. Due to voltage collapse and severe thermal overloads in the initial power flow case, steady-state analysis was performed in the following five-step procedure:

- Step 1: With 3 fictitious large size SVCs (Table 2-1) added to the case, non-linear (AC) thermal analysis (Stage-1 ACCC) was performed to identify thermal constraints.
- Step 2: Based on the identified thermal constraints in the Stage-1 ACCC, and based on voltage collapse under system intact condition and post critical contingencies conditions, perform project justification analysis to determine Network Upgrades (NUs) required for interconnection of DPP West projects. These selected NUs are called Backbone NUs.
- Step 3: The fictitious SVCs in MISO (Helena 345 kV, Cedar Mountain 345 kV) were removed and the Backbone NUs were added to the summer shoulder case. Stage-2 ACCC was performed to identify any remaining thermal and voltage constraints.
- Step 4: Based on the Stage-2 ACCC results and MWEX voltage stability analysis performed by ATC (Section 8), additional NUs were justified and determined for resolving MWEX stability constraint and additional significant thermal/voltage constraints.
- Step 5: The additional NUs identified in Step 4 were added to the Stage-2 models to create Stage-3 models. Stage-3 ACCC was performed to identify any remaining thermal and voltage constraints.

4.2 Step 1 – Stage-1 ACCC Analysis

Three fictitious large size SVCs (Table 2-1) were added to the summer shoulder case. The incremental impact of the proposed interconnection on individual facilities was evaluated by comparing flows between benchmark case (without DPP West Area projects or fictitious SVCs) and study case (with DPP West Area projects and fictitious SVCs). Analysis was performed in the 2022 summer shoulder scenario using PSS[®]E and PSS[®]MUST.

4.2.1 Stage-1 System Intact Conditions

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table E-1.

4.2.2 Stage-1 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies.

For P1 contingencies in the 2022 summer shoulder scenario, thermal constraints are listed in Table E-2.

For P2-P7 contingencies in the 2022 summer shoulder scenario, thermal constraints are listed in Table E-3.

Non-converged contingencies are listed in Table E-4, and their dc thermal results are listed in Table E-5.

4.2.3 Worst Thermal Constraints in the Stage-1 ACCC

Table 4-1 lists worst thermal constraints identified in the Stage-1 ACCC for 2022 summer shoulder scenario.

Table 4-1: 2022 Shoulder Thermal Constraints, Maximum Screened Loading, Stage-1 ACCC

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
J506 POI-Raun 345 kV	864.0	MEC	1050.3	121.6	CEII Redacted	P1	J590,J439
J506 POI-Raun 345 kV	864.0	MEC	1057.2	122.4	CEII Redacted	P2-P7	J590,J439
J506 POI-Highland 345 kV	864.0	MEC	873.3	101.1	CEII Redacted	P1	J590,J439
J506 POI-Highland 345 kV	864.0	MEC	881.6	102.0	CEII Redacted	P2-P7	J590,J439
J530 POI-Montezuma 345 kV	1010.0	MEC	1335.5	132.2	CEII Redacted	P1	J541,J555,J598
J530 POI-Montezuma 345 kV	1010.0	MEC	1392.8	137.9	CEII Redacted	P2-P7	J541,J555,J598
J530 POI-Hills 345 kV	956.0	MEC	1158.4	121.2	CEII Redacted	P0	J476,J541,J555,J583,J598,J611
J530 POI-Hills 345 kV	1010.0	MEC	1585.0	156.9	CEII Redacted	P1	J541,J555,J598

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
J530 POI-Hills 345 kV	1010.0	MEC	1646.0	163.0	CEII Redacted	P2-P7	J541,J555,J598
J302&J503 POI-J607 POI 230 kV	257.0	MDU	298.2	116.0	CEII Redacted	P2-P7	J457,J607
J302&J503 POI-Heskett 230 kV	234.0	MDU	239.6	102.4	CEII Redacted	P0	J302,J457,J459,J503,J607
J302&J503 POI-Heskett 230 kV	257.0	MDU	412.0	160.3	CEII Redacted	P1	J302,J457,J503,J607
J302&J503 POI-Heskett 230 kV	257.0	MDU	499.5	194.3	CEII Redacted	P2-P7	J302,J457,J503,J607
J607 POI-Wishek 230 kV	257.0	MDU	362.7	141.1	CEII Redacted	P1	J302,J503,J607
J607 POI-Wishek 230 kV	257.0	MDU	362.9	141.2	CEII Redacted	P2-P7	J302,J503,J607
J611-Maryville 161 kV	199.0	MEC GMO	199.2	100.1	CEII Redacted	P0	J611
J611-Maryville 161 kV	199.0	MEC	245.4	123.3	CEII Redacted	P1	J611

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
		GMO					
J611-Maryville 161 kV	199.0	MEC GMO	236.6	118.9	CEII Redacted	P2-P7	J611
J615 POI-Shaulis Road 161 kV	215.0	MEC	285.6	132.8	CEII Redacted	P2-P7	J615
Novelty 161 -69 kV xfmr	56.0	AECI	62.0	110.7	CEII Redacted	P1	J541
South River-Emerson 161 kV	167.0	AECI	189.9	113.7	CEII Redacted	P1	J541
South River-Palmyra 161 kV	167.0	AECI	173.5	103.9	CEII Redacted	P1	J541
Zackary 345-161 kV xfmr	560.0	AMMO	569.1	101.6	CEII Redacted	P1	J541,J598,J541
Adair-Zackary 161 kV	557.0	AMMO	580.0	104.1	CEII Redacted	P1	J541,J598,J541
St. Joseph-Cooper 345 kV	1195.0	NPPD GMO	1580.5	132.3	CEII Redacted	P1	J476
Adams 345-161-13.8 kV xfmr	334.6	ITCM	372.6	111.4	CEII Redacted	P1	J614,J415

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Adams 345-161-13.8 kV xfmr	334.6	ITCM	395.3	118.1	CEII Redacted	P2-P7	J614
Wilmarth-Sheas Lake 345 kV	1515.8	XEL	1818.3	120.0	CEII Redacted	P1	J594
Wilmarth-Sheas Lake 345 kV	1515.8	XEL	2070.6	136.6	CEII Redacted	P2-P7	J594
Split Rock 345-115-13.8 kV xfmr #11	358.5	XEL	420.9	117.4	CEII Redacted	P2-P7	J512,J587,J597,J638
Split Rock-White 345 kV	717.1	XEL WAPA	921.1	128.5	CEII Redacted	P0	J302,J457,J459,J503,J511,J575, J577,J587,J593,J596,J597,J599, J607,J613,J637,J638
Split Rock-White 345 kV	1075.0	XEL WAPA	1675.9	155.9	CEII Redacted	P1	J457,J459,J575,J577,J587,J597, J637,J638
Split Rock-White 345 kV	1075.0	XEL WAPA	1669.3	155.3	CEII Redacted	P2-P7	J457,J459,J575,J577,J587,J597, J637,J638
Split Rock-Sioux City 345 kV	717.0	XEL WAPA	827.7	115.4	CEII Redacted	P1	J512
Split Rock-Sioux City 345 kV	717.0	XEL WAPA	865.9	120.8	CEII Redacted	P2-P7	J512,J569

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Lakefield Gen-Lakefield 345 kV	956.1	XEL ITCM	1101.1	115.2	CEII Redacted	P2-P7	J439,J594
Brookings Co-White 345 kV #1	788.8	XEL WAPA	1139.5	144.5	CEII Redacted	P1	J459,J575,J577,J587,J597,J637, J638
Brookings Co-White 345 kV #1	788.8	XEL WAPA	1137.3	144.2	CEII Redacted	P2-P7	J459,J575,J577,J587,J597,J637, J638
Brookings Co-White 345 kV #2	788.8	XEL WAPA	1264.7	160.3	CEII Redacted	P1	J459,J575,J577,J587,J597,J637, J638
Brookings Co-White 345 kV #2	788.8	XEL WAPA	1698.2	215.3	CEII Redacted	P2-P7	J459,J575,J577,J597,J637,J638
Field South-Crandal 345 kV	1332.6	XEL	1409.5	105.8	CEII Redacted	P1	J594
Field South-Crandal 345 kV	1332.6	XEL	1487.0	111.6	CEII Redacted	P2-P7	J594
Lyon Co-Hawknest 345 kV	1792.7	XEL	1837.7	102.5	CEII Redacted	P1	J459,J575,J577,J587,J597,J637, J638
Lyon Co-Hawknest 345 kV	1792.7	XEL	1838.6	102.6	CEII Redacted	P2-P7	J459,J575,J577,J587,J597,J637, J638
Helena-Scott Co 345 kV	1378.0	XEL WAPA	1601.3	116.2	CEII Redacted	P0	J590,J439,J459,J512,J569,J575, J577,J587,J594,J597,J637,J638

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Helena-Scott Co 345 kV	1515.8	XEL WAPA	2401.5	158.4	CEII Redacted	P2-P7	J439
Helena-Sheas Lake 345 kV	1513.0	XEL	1795.1	118.6	CEII Redacted	P1	J594
Helena-Sheas Lake 345 kV	1513.0	XEL	2069.6	136.8	CEII Redacted	P2-P7	J594
Hickory Creek 345-230-13.8 kV xfmr #6	386.0	XEL	571.2	148.0	CEII Redacted	P2-P7	J587,J597
Sheyenne-Lake Pike 230 kV	294.0	XEL MPC	326.3	111.0	CEII Redacted	P0	J613
Sheyenne-Lake Pike 230 kV	294.0	XEL MPC	332.1	113.0	CEII Redacted	P1	J599
Minn Valley-Granite Falls 230 kV	259.0	XEL WAPA	391.8	151.3	CEII Redacted	P1	J459,J511,J593,J596
Minn Valley-Granite Falls 230 kV	259.0	XEL WAPA	391.8	151.3	CEII Redacted	P2-P7	J459,J511,J593,J596
Minn Valley Tap-Granite Falls 230 kV	413.0	XEL WAPA	429.7	104.0	CEII Redacted	P1	J587
Lawrence-Sioux Falls 115 kV	239.0	XEL WAPA	265.4	111.0	CEII Redacted	P2-P7	J459,J569,J587,J597,J638
Sheyenne-Mapleton 115 kV	162.0	XEL OTP	223.3	137.8	CEII Redacted	P1	J613

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Sheyenne-Mapleton 115 kV	162.0	XEL OTP	205.6	126.9	CEII Redacted	P2-P7	J511,J613
Grove Lake-Lake Johnson 69 kV	38.6	XEL GRE	48.9	126.7	CEII Redacted	P1	J596
Murphy Creek 161-69 kV xfmr	94.0	SMMPA	120.5	128.2	CEII Redacted	P1	J414
Murphy Creek 161-69 kV xfmr	94.0	SMMPA	120.5	128.2	CEII Redacted	P2-P7	J414
Austin-Murphy Creek 161 kV	279.0	SMMPA	290.9	104.3	CEII Redacted	P0	J414
Austin-Murphy Creek 161 kV	307.0	SMMPA	381.6	124.3	CEII Redacted	P1	J414
Austin-Murphy Creek 161 kV	307.0	SMMPA	427.5	139.3	CEII Redacted	P2-P7	J414
Murphy Creek-Hayward 161 kV	276.0	SMMPA ITCM	313.1	113.4	CEII Redacted	P0	J414

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Murphy Creek-Hayward 161 kV	276.0	SMMPA ITCM	407.6	147.7	CEII Redacted	P1	J414
Murphy Creek-Hayward 161 kV	276.0	SMMPA ITCM	451.2	163.5	CEII Redacted	P2-P7	J414
Rice 161-69 kV xfmr	106.0	SMMPA ITCM	141.4	133.4	CEII Redacted	P1	J614
Rice 161-69 kV xfmr	106.0	SMMPA ITCM	141.5	133.5	CEII Redacted	P2-P7	J614
Rutland-Fox Lake 161 kV	168.0	SMMPA ITCM	254.0	151.2	CEII Redacted	P1	J594

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Rutland-Fox Lake 161 kV	168.0	SMMPA ITCM	268.7	159.9	CEII Redacted	P2-P7	J594
Rutland-Huntley 161 kV	168.0	SMMPA ITCM	206.3	122.8	CEII Redacted	P1	J594
Rutland-Huntley 161 kV	168.0	SMMPA ITCM	224.4	133.6	CEII Redacted	P2-P7	J594
Panther-McLeod 230 kV	321.9	GRE	386.4	120.0	CEII Redacted	P0	J596
Panther-McLeod 230 kV	321.9	GRE	418.1	129.9	CEII Redacted	P1	J597
Panther-McLeod 230 kV	321.9	GRE	597.8	185.7	CEII Redacted	P2-P7	J597
Wing River-Inman 230 kV	252.5	GRE	303.8	120.3	CEII Redacted	P0	J457,J596

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Wing River-Inman 230 kV	252.5	GRE	318.0	125.9	CEII Redacted	P1	J459
Wing River-Inman 230 kV	252.5	GRE	324.4	128.5	CEII Redacted	P2-P7	J459,J638
Silver Lake-Henning 230 kV	334.0	GRE	335.1	100.3	CEII Redacted	P0	J457,J607
Silver Lake-Henning 230 kV	334.0	GRE	353.8	105.9	CEII Redacted	P1	J459
Silver Lake-Fergus Falls 230 kV	334.0	GRE OTP	342.4	102.5	CEII Redacted	P0	J457,J607
Silver Lake-Fergus Falls 230 kV	334.0	GRE OTP	361.4	108.2	CEII Redacted	P1	J459
Stanton-Square Butte 230 kV	430.2	GRE MPC	437.5	101.7	CEII Redacted	P1	J511
Ramsey-Prairie 230 kV	320.9	GRE MPC	348.7	108.7	CEII Redacted	P1	J511
Ramsey-Prairie 230 kV	320.9	GRE MPC	350.5	109.2	CEII Redacted	P2-P7	J511
Johnson Jct-Ortonville 115 kV	108.0	MRES GRE OTP	175.2	162.2	CEII Redacted	P1	J459,J637,J638

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Johnson Jct-Morris 115 kV	92.0	GRE WAPA	149.2	162.2	CEII Redacted	P1	J637,J638
Wilmarth-Granite Falls 230 kV	264.9	GRE WAPA	296.7	112.0	CEII Redacted	P1	J596
Oakes 230-41.6 kV xfmr #1	42.0	OTP	135.8	323.3	CEII Redacted	P2-P7	J613
Oakes 230-41.6 kV xfmr #2	41.2	OTP	127.1	308.5	CEII Redacted	P2-P7	J613
Jamestown Pike 115-41.6 kV xfmr	70.0	OTP	264.2	377.4	CEII Redacted	P2-P7	J613
Canby-Granite Falls 115 kV	113.0	OTP WAPA	121.0	107.1	CEII Redacted	P1	J459,J637,J638
Canby-Granite Falls 115 kV	113.0	OTP WAPA	130.4	115.4	CEII Redacted	P2-P7	J459,J637,J638
Big Stone-Highway 12 115 kV	230.0	OTP	267.1	116.1	CEII Redacted	P1	J459,J638
Big Stone-Highway 12 115 kV	230.0	OTP	277.3	120.6	CEII Redacted	P2-P7	J459,J638
Big Stone 115-230-13.8 kV xfmr	233.0	OTP	234.1	100.5	CEII Redacted	P0	J459
Highway 12-Ortonville 115 kV	230.0	OTP	258.2	112.3	CEII Redacted	P1	J459,J638

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Highway 12-Ortonville 115 kV	230.0	OTP	268.5	116.7	CEII Redacted	P2-P7	J459,J638
Jamestown-North Jamestown 115 kV	130.0	OTP	178.3	137.2	CEII Redacted	P2-P7	J613
Big Stone 230-115-13.8 kV xfmr	233.0	OTP	240.7	103.3	CEII Redacted	P0	J459
Big Stone 230-115-13.8 kV xfmr	291.2	OTP	376.0	129.1	CEII Redacted	P1	J459,J638
Big Stone 230-115-13.8 kV xfmr	291.2	OTP	391.7	134.5	CEII Redacted	P2-P7	J459,J638
Big Stone-Big Stone South 230 kV #1	526.0	OTP	762.8	145.0	CEII Redacted	P1	J459
Big Stone-Big Stone South 230 kV #1	526.0	OTP	730.8	138.9	CEII Redacted	P2-P7	J459,J637,J638
Big Stone-Big Stone South 230 kV #2	605.0	OTP	788.7	130.4	CEII Redacted	P1	J459,J637,J638
Big Stone-Big Stone South 230 kV #2	605.0	OTP	790.8	130.7	CEII Redacted	P2-P7	J459,J637,J638
Big Stone-Browns Valley 230 kV	285.0	OTP	376.9	132.3	CEII Redacted	P1	J459

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Big Stone-Browns Valley 230 kV	285.0	OTP	384.4	134.9	CEII Redacted	P2-P7	J459,J638
Big Stone-Blair 230 kV	630.0	OTP WAPA	786.2	124.8	CEII Redacted	P1	J457,J459,J637,J638
Big Stone-Blair 230 kV	630.0	OTP WAPA	771.6	122.5	CEII Redacted	P2-P7	J457,J459,J637,J638
Hankinson-Wahpeton 230 kV	319.0	OTP	512.5	160.7	CEII Redacted	P0	J302,J457,J459,J503,J607,J637,J638
Hankinson-Wahpeton 230 kV	351.0	OTP	647.8	184.6	CEII Redacted	P1	J457,J459,J607,J637,J638
Hankinson-Wahpeton 230 kV	351.0	OTP	643.3	183.3	CEII Redacted	P2-P7	J457,J459,J607
Hankinson-Forman 230 kV	257.0	OTP	331.9	129.2	CEII Redacted	P0	J302,J457,J503,J607
Hankinson-Forman 230 kV	257.0	OTP	536.6	208.8	CEII Redacted	P1	J302,J457,J503,J607
Hankinson-Forman 230 kV	257.0	OTP	539.9	210.1	CEII Redacted	P2-P7	J302,J457,J503,J607
Wahpeton-Fergus Falls 230 kV	319.0	OTP MRES	393.3	123.3	CEII Redacted	P0	J302,J457,J503,J607
Wahpeton-Fergus Falls 230 kV	351.0	OTP MRES	441.1	125.7	CEII Redacted	P1	J459,J638

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Wahpeton-Fergus Falls 230 kV	351.0	OTP MRES	451.8	128.7	CEII Redacted	P2-P7	J459,J638
Audubon-Lake Pike 230 kV	294.0	OTP MPC	320.6	109.0	CEII Redacted	P0	J613
Audubon-Lake Pike 230 kV	294.0	OTP MPC	326.4	111.0	CEII Redacted	P1	J599
Buffalo 345-115-41.6 kV xfmr	140.0	OTP	214.1	153.0	CEII Redacted	P2-P7	J511,J613
Buffalo 345-115-41.6 kV xfmr	140.0	OTP	212.9	152.0	CEII Redacted	P2-P7	J511,J613
Oakes-Forman 230 kV	319.0	OTP	373.7	117.1	CEII Redacted	P0	J302,J457,J503,J607
Oakes-Forman 230 kV	331.0	OTP	649.1	196.1	CEII Redacted	P1	J302,J457,J503,J607
Oakes-Forman 230 kV	331.0	OTP	651.1	196.7	CEII Redacted	P2-P7	J302,J457,J503,J607
Oakes-Ellendale 230 kV	319.0	OTP MDU	377.3	118.3	CEII Redacted	P0	J302,J457,J503,J607
Oakes-Ellendale 230 kV	351.0	OTP MDU	666.4	189.9	CEII Redacted	P1	J302,J457,J503,J607
Oakes-Ellendale 230 kV	351.0	OTP MDU	668.4	190.4	CEII Redacted	P2-P7	J302,J457,J503,J607
Big Stone South 345-230-34.5 kV xfmr #1	525.8	OTP	583.8	111.0	CEII Redacted	P1	J459,J637,J638

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Big Stone South 345-230-34.5 kV xfmr #1	525.8	OTP	585.9	111.4	CEII Redacted	P2-P7	J459,J637,J638
Big Stone South 345-230-34.5 kV xfmr #2	525.8	OTP	584.7	111.2	CEII Redacted	P1	J459,J637,J638
Big Stone South 345-230-34.5 kV xfmr #2	525.8	OTP	586.7	111.6	CEII Redacted	P2-P7	J459,J637,J638
Ellendale-Steel Center 69 kV	45.0	ITCM	56.3	125.2	CEII Redacted	P1	J414
Triboji-Fly Cloud 69 kV	48.0	ITCM	54.0	112.5	CEII Redacted	P2-P7	J569
Walters-Alden 69 kV	45.0	ITCM	49.2	109.3	CEII Redacted	P1	J414
Pratt Tap-Steel Center 69 kV	45.0	ITCM	55.8	124.1	CEII Redacted	P1	J414
Alden-Alea West 69 kV	45.0	ITCM	49.1	109.1	CEII Redacted	P1	J414
Rice-Cresco 69 kV	69.0	ITCM	69.7	101.0	CEII Redacted	P1	J614

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Rice-Cresco 69 kV	69.0	ITCM	70.6	102.3	CEII Redacted	P2-P7	J614
DECORAH-MADISNT 69 kV	60.0	ITCM	62.6	104.3	CEII Redacted	P1	J614
DECORAH-MADISNT 69 kV	60.0	ITCM	62.2	103.7	CEII Redacted	P2-P7	J614
Fox Lake-Jackson 161 kV #2	276.0	ITCM	292.1	105.8	CEII Redacted	P1	J594
Fox Lake-Jackson 161 kV #2	276.0	ITCM	296.7	107.5	CEII Redacted	P2-P7	J594
Hayward-Glenworth 161 kV	325.0	ITCM	331.1	101.9	CEII Redacted	P1	J415

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Lime Creek 161 kV bus tie	335.0	ITCM	521.8	155.8	CEII Redacted	P2-P7	J439
Emery-Colby 161 kV	327.0	ITCM	396.4	121.2	CEII Redacted	P1	J439
Emery-Floyd 161 kV	272.0	ITCM MEC	352.0	129.4	CEII Redacted	P1	J415
Emery-Floyd 161 kV	272.0	ITCM MEC	406.7	149.5	CEII Redacted	P2-P7	J415
Hazleton-Blackhawk 161 kV	325.0	ITCM MEC	456.5	140.5	CEII Redacted	P1	J415
Hazleton-Blackhawk 161 kV	325.0	ITCM MEC	438.7	135.0	CEII Redacted	P2-P7	J415
Hazleton-Washburn 161 kV	325.0	ITCM MEC	442.6	136.2	CEII Redacted	P2-P7	J615
Lansing-Harmony 161 kV	153.0	ITCM DPC	162.6	106.3	CEII Redacted	P1	J614
Lansing-Harmony 161 kV	153.0	ITCM DPC	176.7	115.5	CEII Redacted	P2-P7	J614

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Parnell-J438 POI 161 kV	282.0	ITCM	310.5	110.1	CEII Redacted	P1	J555
Parnell-J438 POI 161 kV	282.0	ITCM	288.8	102.4	CEII Redacted	P2-P7	J598
Parnell-Hills 161 kV	325.0	MEC ITCM	357.1	109.9	CEII Redacted	P1	J555
Henry Co-Denmark 161 kV	169.0	ITCM	201.2	119.1	CEII Redacted	P2-P7	J541,J598
Henry Co-Jeff 161 kV	197.0	ITCM	229.8	116.6	CEII Redacted	P2-P7	J541,J598
Wapello-Jeff 161 kV	223.0	ITCM	230.2	103.2	CEII Redacted	P1	J541
Wapello-Jeff 161 kV	223.0	ITCM	275.0	123.3	CEII Redacted	P2-P7	J541,J598
Wapello-Appanoose 161 kV	171.0	ITCM	196.9	115.1	CEII Redacted	P1	J541
Wapello-Appanoose 161 kV	171.0	ITCM	172.9	101.1	CEII Redacted	P2-P7	J541,J598

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Ottumwa 345-161 kV xfmr	400.0	ITCM	535.9	134.0	CEII Redacted	P1	J541,J598
Ottumwa 345-161 kV xfmr	400.0	ITCM	518.8	129.7	CEII Redacted	P2-P7	J541,J598
Adams-Hayward 161 kV	216.0	ITCM	305.0	141.2	CEII Redacted	P1	J414
Adams-Hayward 161 kV	216.0	ITCM	305.0	141.2	CEII Redacted	P2-P7	J414
Hayward-Freeborn 161 kV	446.0	ITCM	449.3	100.7	CEII Redacted	P1	J414
Hayward-Freeborn 161 kV	446.0	ITCM	468.3	105.0	CEII Redacted	P2-P7	J414
Hazleton-Arnold 345 kV	1036.0	ITCM MEC	1207.9	116.6	CEII Redacted	P2-P7	J415,J615

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Hazleton-Mitchell Co 345 kV	995.0	ITCM	1003.5	100.9	CEII Redacted	P1	J415
Hazleton-Hickory Crk 345 kV	1195.0	ITCM	1207.9	101.1	CEII Redacted	P0	J590,J302,J414,J415,J439,J457,J459,J503,J511,J512,J569,J575,J577,J587,J593,J594,J596,J597,J599,J607,J613,J614,J615,J637,J638
Hazleton-Hickory Crk 345 kV	1195.0	ITCM	1411.5	118.1	CEII Redacted	P1	J415,J615
Hazleton-Hickory Crk 345 kV	1195.0	ITCM	1426.4	119.4	CEII Redacted	P2-P7	J415,J615
Hazleton-Blackhawk 345 kV	1195.0	ITCM MEC	1300.9	108.9	CEII Redacted	P1	J415
Hazleton-Blackhawk 345 kV	1195.0	ITCM MEC	1289.6	107.9	CEII Redacted	P2-P7	J415
Barton-Lime Creek 161 kV	240.0	ITCM	254.5	106.1	CEII Redacted	P2-P7	J415
Cayler-Wisdom 161 kV	215.0	ITCM CBPC	217.4	101.1	CEII Redacted	P1	J594

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Cayler-Wisdom 161 kV	215.0	ITCM CBPC	227.2	105.7	CEII Redacted	P2-P7	J594
Colby 345-161 kV xfmr	450.0	ITCM	459.3	102.1	CEII Redacted	P1	J439
Raun-S3451 345 kV	956.0	MEC OPPD	990.3	103.6	CEII Redacted	P0	J590,J302,J414,J439,J457,J459, J503,J511,J512,J569,J575,J577, J587,J593,J594,J596,J597,J599, J607,J613,J637,J638
Raun-S3451 345 kV	956.0	MEC OPPD	1267.5	132.6	CEII Redacted	P1	J439
Raun-S3451 345 kV	956.0	MEC OPPD	1264.9	132.3	CEII Redacted	P2-P7	J439
Grimes-Sycamore 345 kV #1	1209.0	MEC	1255.9	103.9	CEII Redacted	P1	J583
Grimes-Sycamore 345 kV #2	1209.0	MEC	1248.4	103.3	CEII Redacted	P1	J583
Grimes-Sycamore 345 kV #2	1209.0	MEC	1438.9	119.0	CEII Redacted	P2-P7	J583
Grimes-Beaver Creek 345 kV	1196.0	MEC	1341.4	112.2	CEII Redacted	P1	J439
Grimes-Beaver Creek 345 kV	1196.0	MEC	1403.9	117.4	CEII Redacted	P2-P7	J439
Bondurant-Sycamore 345 kV	1152.0	MEC	1217.7	105.7	CEII Redacted	P1	J583

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Bondurant-Sycamore 345 kV	1152.0	MEC	1304.0	113.2	CEII Redacted	P2-P7	J583
Bondurant-Montezuma 345 kV	926.0	MEC	1100.1	118.8	CEII Redacted	P0	J590,J302,J439,J457,J459,J476,J503,J511,J512,J569,J575,J577,J583,J587,J593,J594,J596,J597,J599,J607,J611,J613,J637,J638
Bondurant-Montezuma 345 kV	1010.0	MEC	1216.9	120.5	CEII Redacted	P1	J583
Bondurant-Montezuma 345 kV	1010.0	MEC	1250.0	123.8	CEII Redacted	P2-P7	J583
Webster 345-161-13.8 kV xfmr	560.0	MEC	572.0	102.1	CEII Redacted	P1	J439

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Webster 345-161-13.8 kV xfmr	560.0	MEC	578.5	103.3	CEII Redacted	P2-P7	J439
Blackhawk-Midport 161 kV	272.0	MEC	280.0	102.9	CEII Redacted	P2-P7	J415
Blackhawk-Deere Foundry 161 kV	257.0	MEC	269.8	105.0	CEII Redacted	P2-P7	J415
Blackhawk-Bremer Co 161 kV	258.0	MEC CBPC	275.2	106.7	CEII Redacted	P1	J415
Blackhawk-Bremer Co 161 kV	258.0	MEC CBPC	334.0	129.5	CEII Redacted	P2-P7	J415
Midport-Lundquist 161 kV	231.0	MEC	270.4	117.0	CEII Redacted	P2-P7	J415
Lundquist-Deere Foundry 161 kV	271.0	MEC	354.2	130.7	CEII Redacted	P2-P7	J415

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Deere Foundry-Washburn 161 kV	272.0	MEC	326.9	120.2	CEII Redacted	P2-P7	J415
Washburn-Shaulis Road 161 kV	261.0	MEC	278.1	106.6	CEII Redacted	P2-P7	J615
Union Tap-Butler 161 kV	410.0	MEC	411.9	100.5	CEII Redacted	P1	J415
Franklin-Butler 161 kV	410.0	MEC	417.1	101.7	CEII Redacted	P1	J415
Floyd-Bremer Co 161 kV	258.0	MEC CBPC	276.1	107.0	CEII Redacted	P1	J415
Floyd-Bremer Co 161 kV	258.0	MEC CBPC	290.1	112.5	CEII Redacted	P2-P7	J415
Hills-Sub 92 345 kV	1304.0	MEC	1403.1	107.6	CEII Redacted	P1	J555
Hills-Sub 92 345 kV	1304.0	MEC	1402.4	107.5	CEII Redacted	P2-P7	J555

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Red Willow-Mingo 345 kV	717.0	NPPD SUNC	922.3	128.6	CEII Redacted	P0	J599
Red Willow-Mingo 345 kV	828.5	NPPD SUNC	931.6	112.4	CEII Redacted	P1	J587
Blair-Granite Falls 230 kV	410.0	NPPD SUNC	410.0	100.0	CEII Redacted	P0	J457,J459
Blair-Granite Falls 230 kV	410.0	NPPD SUNC	695.4	169.6	CEII Redacted	P1	J597
Blair-Granite Falls 230 kV	410.0	NPPD SUNC	692.5	168.9	CEII Redacted	P2-P7	J597
Sioux Falls-Pahoja 230 kV	341.8	WAPA	349.5	102.3	CEII Redacted	P1	J597
Sioux Falls-Pahoja 230 kV	341.8	WAPA	383.5	112.2	CEII Redacted	P2-P7	J597
Whitlock-Glenham 230 kV	260.0	MDU WAPA	261.0	100.4	CEII Redacted	P1	J599
Watertown 345-230-13.8 kV xfmr	500.0	WAPA	689.0	137.8	CEII Redacted	P1	J587,J597,J638
Watertown 345-230-13.8 kV xfmr	500.0	WAPA	688.2	137.6	CEII Redacted	P2-P7	J587,J597,J638
Watertown-Appledorn 230 kV	341.8	WAPA	534.6	156.4	CEII Redacted	P1	J597

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Watertown-Appledorn 230 kV	341.8	WAPA	531.9	155.6	CEII Redacted	P2-P7	J597
Granite Falls-Appledorn 230 kV	409.0	WAPA	465.5	113.8	CEII Redacted	P1	J587
Morris-Grant Co 115 kV	103.0	MRES WAPA	107.1	104.0	CEII Redacted	P0	J596
Morris-Grant Co 115 kV	103.0	MRES WAPA	155.5	151.0	CEII Redacted	P1	J596
Morris-Grant Co 115 kV	103.0	MRES WAPA	165.2	160.4	CEII Redacted	P2-P7	J596
Abdeen Jct-Ellendale 115 kV	128.0	MDU WAPA	160.6	125.5	CEII Redacted	P1	J302,J457,J503,J607
Abdeen Jct-Ellendale 115 kV	128.0	MDU WAPA	159.8	124.8	CEII Redacted	P2-P7	J302,J457,J503,J607
Heskett-Mandan 230 kV	383.0	MDU	409.5	106.9	CEII Redacted	P2-P7	J302,J457,J503,J607
Foxtail-Tatanka North 230 kV	343.0	MDU	350.0	102.0	CEII Redacted	P1	J302,J457,J503,J607

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Foxtail-Ellendale 230 kV	419.0	MDU	461.1	110.0	CEII Redacted	P1	J302,J457,J503,J607
Foxtail-Ellendale 230 kV	419.0	MDU	446.1	106.5	CEII Redacted	P2-P7	J302,J457,J503,J607
Merricourt-Wishek 230 kV	257.0	MDU	359.3	139.8	CEII Redacted	P1	J302,J503,J607
Merricourt-Wishek 230 kV	257.0	MDU	359.4	139.9	CEII Redacted	P2-P7	J302,J503,J607
Merricourt-Ellendale 230 kV	383.0	MDU	446.2	116.5	CEII Redacted	P1	J302,J457,J503,J607
Merricourt-Ellendale 230 kV	383.0	MDU	431.9	112.8	CEII Redacted	P2-P7	J302,J457,J503,J607
Harmony-Cresco 69 kV	37.0	DPC	42.7	115.3	CEII Redacted	P1	J614

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Harmony-Cresco 69 kV	37.0	DPC	39.2	105.9	CEII Redacted	P2-P7	J614
Beaver Creek-Harmony 161 kV	378.0	DPC	391.0	103.4	CEII Redacted	P1	J614
Beaver Creek-Harmony 161 kV	378.0	DPC	391.6	103.6	CEII Redacted	P2-P7	J614
Wabaco-Rochester 161 kV	221.1	DPC	269.1	121.7	CEII Redacted	P1	J414

4.3 Step 2 – Project Justification Analysis

Based on the identified thermal constraints in the Stage-1 ACCC, and based on voltage collapse under system intact condition and post critical contingencies conditions, the Ad Hoc group proposed twelve potential transmission Network Upgrades (NUs). To justify the needs of the NUs only required for interconnection of the DPP West projects, project justification analysis was performed considering the following aspects:

- 1) Evaluate the Stage-1 ACCC identified constraints to be relieved by the NU.
- 2) Evaluate the Stage-1 ACCC identified constraints to be aggravated by the NU
- 3) Evaluate the new constraints caused by the NU.
- 4) Evaluate effectiveness of reactive power loss reduction by the NU.
- 5) Select the transmission NUs required and justified for interconnection of DPP West projects.
- 6) Model the justified transmission NUs in the summer shoulder case; Remove the fictitious large size SVCs at Helena 345 kV and Cedar Mountain 345 kV buses (Table 2-1). Justify the needs of additional MISO reactive power NUs to resolve the voltage collapse issue under system intact condition and post critical contingencies conditions. As the higher queued SPP generators are still under study, and the higher queued SPP generators were included without associated network upgrades, the fictitious SVC at Mingo 345 kV bus (SPP area) is still kept in the case and its need will be further verified in the SPP affected system study.

Project justification analysis was performed in two study areas: Iowa and Dakotas-Minnesota.

4.3.1 Project Justification in Dakotas-Minnesota

Seven transmission NUs (Table 4-2) in Dakotas-Minnesota area were proposed and evaluated. Within these 7 transmission NUs, the Big Stone South-Alexandria and J510-Hazel Creek-Scott County 345 kV lines were justified and selected as required NUs based on cost and effectiveness. Dakotas-Minnesota area project justification details are in Appendix E.2.1.

Table 4-2: Transmission NUs Evaluated in Dakotas-Minnesota Area

Transmission NUs	Miles
Big Stone South-Alexandria 345 kV	95
J510-Hazel Creek-Scott County 345 kV	165
Ellendale-Alexandria 345 kV	180
Big Stone South-Alexandria 345 kV + 2nd Alexandria-Monticello 345 kV	NA
Ellendale-Jamestown 345 kV	76
Ellendale-Buffalo 345 kV	118
Big Stone South-Hazel Creek 345 kV	80

4.3.2 Project Justification in Iowa Area

Five transmission NUs (Table 4-3) in Iowa area were proposed and evaluated. Within these 5 transmission NUs, the Franklin-Morgan Valley & Beverly-Sub 92 345 kV line was justified and selected as required NU. Iowa area project justification details are in Appendix E.2.2.

Table 4-3: Transmission NUs Evaluated in Central-East Iowa Area

Transmission NUs	Miles
Franklin-Morgan Valley & Beverly-Sub 92 345 kV 1 - Franklin-Morgan Valley 345 kV; 2 - Beverly-Sub 92 345 kV	198
New Hartford-Morgan Valley & Beverly-Sub 92 345 kV 1 - New Hartford-Morgan Valley 345 kV; 2 - Beverly-Sub 92 345 kV	168
Ida Co-Avoca 345 kV	75.6
Webster-Franklin 345 kV	49.2
Ottumwa-Mt Pleasant 345 kV	67.2

4.3.3 Project Justification of Reactive Power NUs

The transmission NUs required and justified for interconnection of DPP West projects are listed in Table 4-4.

Table 4-4: Justified and Required Transmission NUs

Area	NUs	Miles	Cost (\$)
Dakotas-Minnesota	Big Stone South-Alexandria 345 kV	95	\$197,400,000
Dakotas-Minnesota	J510-Hazel Creek-Scott County 345 kV	165	\$330,000,000
Iowa	Franklin-Morgan Valley 345 kV	138	\$384,056,500
	Beverly-Sub 92 345 kV	60	\$148,675,000

These transmission NUs were modeled in the summer shoulder case as backbone NUs. The 2 fictitious SVCs in MISO (Helena 345 kV, Cedar Mountain 345 kV) were removed. Summer shoulder power flow case under system intact condition was solved without voltage collapse. Since voltage at Franklin 345 kV bus is only at 0.915 pu under system intact condition, and the Franklin – J415 POI 345 kV contingency still cause voltage collapse, reactive power NUs were justified as follows:

- 1×150 Mvar switched capacitor bank and ±150 Mvar SVC at Franklin 345 kV bus were required.

4.4 Initial Backbone Network Upgrades

Based on the project justifications in Section 4.3, a minimal set of transmission and reactive power NUs as initial Backbone NUs was required and justified for interconnecting of the DPP West projects. The initial Backbone NUs are listed in Table 4-5.

Table 4-5: Initial Backbone NUs Required and Justified

Area	NUs	Miles	Cost (\$)
Dakotas-Minnesota	Big Stone South-Alexandria 345 kV	95	\$197,400,000
Dakotas-Minnesota	J510-Hazel Creek-Scott County 345 kV	165	\$330,000,000
Iowa	Franklin-Morgan Valley 345 kV	138	\$384,056,500
	Beverly-Sub 92 345 kV	60	\$148,675,000
Iowa	150 MVar MSC at Franklin 345kV		\$4,000,000
	150 MVar SVC at Franklin 345kV		\$60,000,000

4.5 Step 3 – Stage-2 ACCC Analysis

With the initial Backbone NUs added and the fictitious SVCs in MISO removed, Stage-2 AC contingency analysis was performed to identify any remaining thermal and voltage constraints not mitigated by the initial Backbone NUs.

4.5.1 Stage-2 System Intact Conditions

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table E-12, and voltage constraints are listed in Table E-13.

4.5.2 Stage-2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies.

All category P1 contingencies were converged. For P1 contingencies in the 2022 summer shoulder scenario, thermal constraints are listed in Table E-14, and worst voltage constraints are listed in Table E-15.

Three category P2-P7 contingencies (Table E-18) were not converged in the benchmark or study cases. No mitigation plan is required for the study projects for these non-converged contingencies. The dc thermal results for non-converged contingencies are listed in Table E-19.

For P2-P7 contingencies in the 2022 summer shoulder scenario, thermal constraints are listed in Table E-16, and worst voltage constraints are listed in Table E-17.

4.5.3 Summer Shoulder Worst Thermal Constraints in the Stage-2 ACCC

Table 4-6 lists worst thermal constraints identified in the Stage-2 ACCC for 2022 summer shoulder scenario.

Table 4-6: 2022 Summer Shoulder Thermal Constraints, Maximum Screened Loading, Stage-2 ACCC

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
J506 POI-Raun 345 kV	864.0	MEC	1048.0	121.3	CEII Redacted	P1	J590,J439
J506 POI-Raun 345 kV	864.0	MEC	1057.3	122.4	CEII Redacted	P2-P7	J590,J439
J506 POI-Highland 345 kV	864.0	MEC	871.2	100.8	CEII Redacted	P1	J590,J439
J506 POI-Highland 345 kV	864.0	MEC	881.6	102.0	CEII Redacted	P2-P7	J590,J439
J530 POI-Montezuma 345 kV	1010.0	MEC	1288.3	127.6	CEII Redacted	P1	J555,J541,J598,J415
J530 POI-Montezuma 345 kV	1010.0	MEC	1376.6	136.3	CEII Redacted	P2-P7	J555,J541,J598
J530 POI-Hills 345 kV	956.0	MEC	1082.7	113.3	CEII Redacted	P0	J555,J476,J541,J583,J598,J611
J530 POI-Hills 345 kV	1152.0	MEC	1540.3	133.7	CEII Redacted	P1	J555,J541,J598
J530 POI-Hills 345 kV	1152.0	MEC	1658.4	144.0	CEII Redacted	P2-P7	J555,J541,J598

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
J302&J503 POI-J607 POI 230 kV	257.0	MDU	297.9	115.9	CEII Redacted	P2-P7	J457,J607,J302,J503
J302&J503 POI-Heskett 230 kV	257.0	MDU	344.5	134.0	CEII Redacted	P1	J302,J607,J503
J302&J503 POI-Heskett 230 kV	257.0	MDU	498.8	194.1	CEII Redacted	P2-P7	J302,J457,J607,J503
J607 POI-Wishek 230 kV	257.0	MDU	350.4	136.3	CEII Redacted	P1	J302,J607,J503
J607 POI-Wishek 230 kV	257.0	MDU	350.5	136.4	CEII Redacted	P2-P7	J302,J607,J503
J611-Maryville 161 kV	199.0	MEC GMO	212.4	106.7	CEII Redacted	P0	J611
J611-Maryville 161 kV	199.0	MEC GMO	274.7	138.1	CEII Redacted	P1	J611
J611-Maryville 161 kV	199.0	MEC GMO	265.8	133.6	CEII Redacted	P2-P7	J611
Gentry-Fairport 161 kV	227.0	AECI	239.3	105.4	CEII Redacted	P1	J611

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Gentry-Nodaway 161 kV	227.0	AECI	234.1	103.1	CEII Redacted	P1	J611
Novelty 161 -69 kV xfmr	56.0	AECI	64.7	115.5	CEII Redacted	P1	J541
South River-Emerson 161 kV	167.0	AECI	194.9	116.7	CEII Redacted	P1	J541
South River-Palmyra 161 kV	167.0	AECI	175.5	105.1	CEII Redacted	P1	J541
Zackary 345-161 kV xfmr	560.0	AMMO	601.1	107.3	CEII Redacted	P1	J541,J598
Adair-Zackary 161 kV	557.0	AMMO	608.2	109.2	CEII Redacted	P1	J541,J598
St. Joseph-Cooper 345 kV	1195.0	NPPD GMO	1216.9	101.8	CEII Redacted	P0	J476,J583
St. Joseph-Cooper 345 kV	1195.0	NPPD GMO	1704.4	142.6	CEII Redacted	P1	J302,J503,J414,J415,J439,J459, J476,J512,J555,J583,J590,J599, J611,J637,J638
Adams 345-161-13.8 kV xfmr	334.6	ITCM	363.3	108.6	CEII Redacted	P1	J614,J415
Adams 345-161-13.8 kV xfmr	334.6	ITCM	386.0	115.4	CEII Redacted	P2-P7	J614
Split Rock-Sioux City 345 kV	717.0	XEL WAPA	735.9	102.6	CEII Redacted	P2-P7	J597
Blue Lake-Scott Co 345 kV	1515.8	XEL	1613.7	106.5	CEII Redacted	P2-P7	J575,J577,J587,J597,J637,J638

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Field South-Crandal 345 kV	1332.6	XEL	1332.6	100.0	CEII Redacted	P2-P7	J512,J594
Sheyenne-Lake Pike 230 kV	294.0	XEL MPC	324.7	110.4	CEII Redacted	P0	J613
Sheyenne-Lake Pike 230 kV	294.0	XEL MPC	332.3	113.0	CEII Redacted	P1	J511,J593,J599,J607
Minn Valley-Granite Falls 230 kV	259.0	XEL WAPA	326.8	126.2	CEII Redacted	P1	J593,J596
Minn Valley-Granite Falls 230 kV	259.0	XEL WAPA	326.8	126.2	CEII Redacted	P2-P7	J593,J596
Sheyenne-Mapleton 115 kV	162.0	XEL OTP	206.9	127.7	CEII Redacted	P1	J613
Sheyenne-Mapleton 115 kV	162.0	XEL OTP	199.9	123.4	CEII Redacted	P2-P7	J511,J613
Grove Lake-Lake Johnson 69 kV	38.6	XEL GRE	40.5	105.0	CEII Redacted	P1	J596
Murphy Creek 161-69 kV xfmr	94.0	SMMPA	109.2	116.2	CEII Redacted	P1	J414
Murphy Creek 161-69 kV xfmr	94.0	SMMPA	109.2	116.2	CEII Redacted	P2-P7	J414
Austin-Murphy Creek 161 kV	307.0	SMMPA	358.2	116.7	CEII Redacted	P1	J414,J415

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Austin-Murphy Creek 161 kV	307.0	SMMPA	390.6	127.2	CEII Redacted	P2-P7	J414
Murphy Creek-Hayward 161 kV	276.0	SMMPA ITCM	281.0	101.8	CEII Redacted	P0	J414
Murphy Creek-Hayward 161 kV	276.0	SMMPA ITCM	386.4	140.0	CEII Redacted	P1	J414,J415
Murphy Creek-Hayward 161 kV	276.0	SMMPA ITCM	422.3	153.0	CEII Redacted	P2-P7	J414
Rice 161-69 kV xfmr	106.0	SMMPA ITCM	140.6	132.7	CEII Redacted	P1	J614
Rice 161-69 kV xfmr	106.0	SMMPA ITCM	141.0	133.0	CEII Redacted	P2-P7	J614
Rutland-Fox Lake 161 kV	168.0	SMMPA ITCM	239.0	142.2	CEII Redacted	P1	J594

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Rutland-Fox Lake 161 kV	168.0	SMMPA ITCM	267.8	159.4	CEII Redacted	P2-P7	J594
Rutland-Huntley 161 kV	168.0	SMMPA ITCM	192.9	114.8	CEII Redacted	P1	J594
Rutland-Huntley 161 kV	168.0	SMMPA ITCM	223.8	133.2	CEII Redacted	P2-P7	J594
Wing River-Inman 230 kV	252.5	GRE	291.2	115.3	CEII Redacted	P0	J596
Wing River-Inman 230 kV	252.5	GRE	324.6	128.6	CEII Redacted	P1	J637,J638,J457,J459,J596

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Stanton-Square Butte 230 kV	430.2	GRE MPC	455.1	105.8	CEII Redacted	P1	J511
Stanton-Square Butte 230 kV	430.2	GRE MPC	587.8	136.6	CEII Redacted	P2-P7	J511
Ramsey-Prairie 230 kV	320.9	GRE MPC	340.7	106.2	CEII Redacted	P1	J511
Ramsey-Prairie 230 kV	320.9	GRE MPC	346.2	107.9	CEII Redacted	P2-P7	J511

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Oakes 230-41.6 kV xfmr #1	42.0	OTP	136.0	323.8	CEII Redacted	P2-P7	J613
Oakes 230-41.6 kV xfmr #2	41.2	OTP	127.3	308.6	CEII Redacted	P2-P7	J613
Jamestown Pike 115-41.6 kV xfmr	70.0	OTP	264.0	377.2	CEII Redacted	P2-P7	J613
Jamestown-North Jamestown 115 kV	130.0	OTP	178.3	137.2	CEII Redacted	P2-P7	J613
Audubon-Lake Pike 230 kV	294.0	OTP MPC	319.1	108.5	CEII Redacted	P0	J613
Audubon-Lake Pike 230 kV	294.0	OTP MPC	326.5	111.1	CEII Redacted	P1	J511,J599,J607
Buffalo 345-115-41.6 kV xfmr #1	140.0	OTP	201.6	144.0	CEII Redacted	P2-P7	J613
Buffalo 345-115-41.6 kV xfmr #2	140.0	OTP	200.3	143.1	CEII Redacted	P2-P7	J613
Ellendale-Steel Center 69 kV	45.0	ITCM	50.1	111.4	CEII Redacted	P1	J414

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Triboji-Fly Cloud 69 kV	48.0	ITCM	49.4	102.8	CEII Redacted	P2-P7	J569
Walters-Alden 69 kV	45.0	ITCM	50.7	112.7	CEII Redacted	P1	J414
Pratt Tap-Steel Center 69 kV	45.0	ITCM	49.7	110.4	CEII Redacted	P1	J414
Alden-Alea West 69 kV	45.0	ITCM	50.6	112.5	CEII Redacted	P1	J414
DECORAH-MADISNT 69 kV	60.0	ITCM	61.3	102.2	CEII Redacted	P1	J614
DECORAH-MADISNT 69 kV	60.0	ITCM	61.7	102.8	CEII Redacted	P2-P7	J614
Fox Lake-Jackson 161 kV #2	276.0	ITCM	291.6	105.7	CEII Redacted	P1	J594

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Fox Lake-Jackson 161 kV #2	276.0	ITCM	296.6	107.5	CEII Redacted	P2-P7	J594
Emery-Killdeer 345 kV	450.0	ITCM	713.1	158.5	CEII Redacted	P1	J439
Emery-Colby 161 kV	327.0	ITCM	504.5	154.3	CEII Redacted	P1	J439
Emery-Floyd 161 kV	272.0	ITCM MEC	382.4	140.6	CEII Redacted	P1	J415
Emery-Floyd 161 kV	272.0	ITCM MEC	440.1	161.8	CEII Redacted	P2-P7	J415
Hazleton-Blackhawk 161 kV	325.0	ITCM MEC	343.4	105.7	CEII Redacted	P1	J415
Hazleton-Blackhawk 161 kV	325.0	ITCM MEC	329.3	101.3	CEII Redacted	P2-P7	J415
Hazleton-Washburn 161 kV	325.0	ITCM MEC	333.6	102.6	CEII Redacted	P2-P7	J615
Lansing-Harmony 161 kV	153.0	ITCM DPC	163.2	106.7	CEII Redacted	P1	J614

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Lansing-Harmony 161 kV	153.0	ITCM DPC	178.3	116.5	CEII Redacted	P2-P7	J614
Parnell-J438 POI 161 kV	274.0	MEC ITCM	289.9	105.8	CEII Redacted	P1	J555
Parnell-J438 POI 161 kV	274.0	MEC ITCM	277.9	101.4	CEII Redacted	P2-P7	J598
Parnell-Hills 161 kV	325.0	MEC ITCM	330.8	101.8	CEII Redacted	P1	J555
Henry Co-Denmark 161 kV	169.0	ITCM	211.0	124.9	CEII Redacted	P2-P7	J541,J598
Henry Co-Jeff 161 kV	197.0	ITCM	240.3	122.0	CEII Redacted	P2-P7	J541,J598
Wapello-Jeff 161 kV	223.0	ITCM	235.9	105.8	CEII Redacted	P1	J541
Wapello-Jeff 161 kV	223.0	ITCM	285.3	127.9	CEII Redacted	P2-P7	J541,J598

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Wapello-Appanoose 161 kV	171.0	ITCM	213.1	124.6	CEII Redacted	P1	J541
Wapello-Appanoose 161 kV	171.0	ITCM	201.6	117.9	CEII Redacted	P2-P7	J541,J598
Ottumwa 345-161 kV xfmr	400.0	ITCM	557.6	139.4	CEII Redacted	P1	J541,J598
Ottumwa 345-161 kV xfmr	400.0	ITCM	525.9	131.5	CEII Redacted	P2-P7	J541,J598
Adams-Hayward 161 kV	216.0	ITCM	279.6	129.5	CEII Redacted	P1	J414,J415
Adams-Hayward 161 kV	216.0	ITCM	279.6	129.5	CEII Redacted	P2-P7	J414
Hayward-Freeborn 161 kV	446.0	ITCM	468.5	105.1	CEII Redacted	P1	J414
Hayward-Freeborn 161 kV	446.0	ITCM	478.3	107.2	CEII Redacted	P2-P7	J414
Hazleton-Arnold 345 kV	923.0	ITCM MEC	1062.4	115.1	CEII Redacted	P2-P7	J615
Hazleton-Mitchell Co 345 kV	995.0	ITCM	1178.9	118.5	CEII Redacted	P1	J415

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Hazleton-Hickory Crk 345 kV	1195.0	ITCM	1313.2	109.9	CEII Redacted	P1	J541,J555J615
Hazleton-Hickory Crk 345 kV	1195.0	ITCM	1326.3	111.0	CEII Redacted	P2-P7	J615
Ottumwa-Montezuma 345 kV	1139.0	ITCM MEC	1154.4	101.4	CEII Redacted	P2-P7	J555
Morgan Valley-Tiffin 345 kV	846.0	ITCM MEC	929.9	109.9	CEII Redacted	P1	J415
Barton-Lime Creek 161 kV	240.0	ITCM	263.0	109.6	CEII Redacted	P2-P7	J415
Cayler-Wisdom 161 kV	215.0	ITCM CBPC	222.1	103.3	CEII Redacted	P1	J594
Cayler-Wisdom 161 kV	215.0	ITCM CBPC	231.2	107.5	CEII Redacted	P2-P7	J594
Colby 345-161 kV xfmr	450.0	ITCM	529.6	117.7	CEII Redacted	P1	J439
Raun-S3451 345 kV	956.0	MEC OPPD	1035.5	108.3	CEII Redacted	P0	J590,J302,J414,J439,J457,J459, J511,J512,J569,J575,J577,J587, J593,J594,J596,J597,J599,J607, J613,J637,J638,J503

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Raun-S3451 345 kV	956.0	MEC OPPD	1314.5	137.5	CEII Redacted	P1	J439,J415
Raun-S3451 345 kV	956.0	MEC OPPD	1311.9	137.2	CEII Redacted	P2-P7	J439
Grimes-Sycamore 345 kV #1	1209.0	MEC	1255.6	103.9	CEII Redacted	P1	J583
Grimes-Sycamore 345 kV #2	1319.0	MEC	1437.4	109.0	CEII Redacted	P2-P7	J583
Grimes-Beaver Creek 345 kV	1222.0	MEC	1360.5	111.3	CEII Redacted	P1	J439,J415,J590
Grimes-Beaver Creek 345 kV	1222.0	MEC	1398.9	114.5	CEII Redacted	P2-P7	J439,J590
Bondurant-Sycamore 345 kV	1152.0	MEC	1214.0	105.4	CEII Redacted	P1	J583
Bondurant-Sycamore 345 kV	1152.0	MEC	1300.4	112.9	CEII Redacted	P2-P7	J583
Bondurant-Montezuma 345 kV	1067.0	MEC	1089.8	102.1	CEII Redacted	P0	J590,J302,J439,J457,J459,J476, J511,J512,J569,J575,J577,J583, J587,J593,J594,J596,J597,J599, J607,J611,J613,J637,J638,J503
Bondurant-Montezuma 345 kV	1067.0	MEC	1270.5	119.1	CEII Redacted	P1	J583,J415,J476

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Bondurant-Montezuma 345 kV	1067.0	MEC	1348.6	126.4	CEII Redacted	P2-P7	J583,J555
Webster 345-161-13.8 kV xfmr	560.0	MEC	604.7	108.0	CEII Redacted	P1	J439
Webster 345-161-13.8 kV xfmr	560.0	MEC	635.9	113.5	CEII Redacted	P2-P7	J439
Webster-Wright 161 kV	212.0	MEC	218.3	103.0	CEII Redacted	P2-P7	J439
Blackhawk-Bremer Co 161 kV	258.0	MEC CBPC	305.0	118.2	CEII Redacted	P1	J415
Blackhawk-Bremer Co 161 kV	258.0	MEC CBPC	369.1	143.1	CEII Redacted	P2-P7	J415
Union Tap-Butler 161 kV	410.0	MEC	439.7	107.2	CEII Redacted	P1	J415
Franklin-Butler 161 kV	410.0	MEC	444.8	108.5	CEII Redacted	P1	J415
Floyd-Bremer Co 161 kV	258.0	MEC CBPC	302.0	117.1	CEII Redacted	P1	J415
Floyd-Bremer Co 161 kV	258.0	MEC CBPC	356.7	138.3	CEII Redacted	P2-P7	J415
Red Willow-Mingo 345 kV	717.0	NPPD SUNC	1007.2	140.5	CEII Redacted	P0	J599

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Red Willow-Mingo 345 kV	828.5	NPPD SUNC	1010.4	122.0	CEII Redacted	P1	J575,J577,J587,J597,J599
Oahe-Sully Buttes 230 kV	260.0	WAPA	261.0	100.4	CEII Redacted	P2-P7	J599
Whitlock-Sully Buttes 230 kV	264.0	WAPA	272.6	103.3	CEII Redacted	P1	J599

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Whitlock-Sully Buttes 230 kV	264.0	WAPA	279.7	106.0	CEII Redacted	P2-P7	J599
Whitlock-Glenham 230 kV	260.0	MDU WAPA	279.4	107.5	CEII Redacted	P1	J599

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Whitlock-Glenham 230 kV	260.0	MDU WAPA	286.5	110.2	CEII Redacted	P2-P7	J599
Morris-Grant Co 115 kV	103.0	MRES WAPA	104.5	101.5	CEII Redacted	P1	J596
Morris-Grant Co 115 kV	103.0	MRES WAPA	106.0	102.9	CEII Redacted	P2-P7	J596

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Sheffield-Hampton Tap 161 kV	215.0	CBPC	229.4	106.7	CEII Redacted	P2-P7	J415
Heskett-Mandan 230 kV	383.0	MDU	409.0	106.8	CEII Redacted	P2-P7	J302,J457,J607,J503
Foxtail-Tatanka North 230 kV	343.0	MDU	436.5	127.3	CEII Redacted	P1	J302,J457,J607,J503
Foxtail-Tatanka North 230 kV	343.0	MDU	361.1	105.3	CEII Redacted	P2-P7	J302,J457,J607,J503
Foxtail-Ellendale 230 kV	419.0	MDU	547.2	130.6	CEII Redacted	P1	J302,J457,J607,J503
Foxtail-Ellendale 230 kV	419.0	MDU	473.9	113.1	CEII Redacted	P2-P7	J302,J457,J607,J503
Merricourt-Wishek 230 kV	257.0	MDU	346.9	135.0	CEII Redacted	P1	J302,J607,J503
Merricourt-Wishek 230 kV	257.0	MDU	347.1	135.1	CEII Redacted	P2-P7	J302,J607,J503,J457
Merricourt-Ellendale 230 kV	383.0	MDU	531.2	138.7	CEII Redacted	P1	J302,J457,J607,J503
Merricourt-Ellendale 230 kV	383.0	MDU	521.7	136.2	CEII Redacted	P2-P7	J302,J457,J607,J503
Harmony-Cresco 69 kV	37.0	DPC	39.8	107.5	CEII Redacted	P1	J614

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Beaver Creek-Harmony 161 kV	378.0	DPC	393.0	104.0	CEII Redacted	P1	J614
Beaver Creek-Harmony 161 kV	378.0	DPC	390.7	103.3	CEII Redacted	P2-P7	J614
Wabaco-Rochester 161 kV	221.1	DPC	268.0	121.2	CEII Redacted	P1	J414

4.6 Step 4 – Justification of Additional Projects Based on the MWEX Voltage Stability Study and Stage-2 ACCC Analysis

Based on the MWEX voltage stability study and Stage-2 ACCC analysis, additional new transmission NUs are determined to be required. Project justification analysis for these additional NUs was performed based on the MWEX study and Stage-2 ACCC analysis.

4.6.1 MWEX Project Justification

4.6.1.1 MWEX Voltage Stability Study

ATC performed MWEX voltage stability study based on the Stage-2 model (Appendix I). Seven potential options were studied in detail, and eight additional alternative solutions were considered. It was concluded that the below two NU alternatives can ensure MWEX voltage stability and meet ATC's voltage stability criteria. Both alternatives are recommended based on the MWEX voltage stability study.

Alternative 1: (North Rochester-South Rocky Run or NRH-SRR alternative)

1. Construct a new South Rocky Run 345 kV switching station with one 75 MVAR shunt capacitor.
2. Construct a new North Rochester – South Rocky Run 345 kV line.
3. Construct a new 161/69 kV substation in Minnesota connecting the Wabaco – Alma 161 kV line to the Dairyland 69 kV system.
4. Install redundant bus differential relays at the Rocky Run 345 kV Substation.
5. Install 150 MVAR of shunt capacitance at the North Rochester 345 kV Substation.
6. Install a 35% series capacitor at the Briggs Road 345 kV Substation on the Briggs Road – North Madison 345 kV line.
7. Install one breaker at the Briggs Road 345 kV Substation.
8. Additional facilities, e.g. Forest Junction – Elkhart Lake 138 kV, are expected to be overloaded by this alternative and will need to be assigned Network Upgrades.

Alternative 2: (J510-Sub 91 alternative)

1. Remove the J510 – Hazel Creek – Scott County 345 kV line from the Stage-2 Backbone NUs
2. Construct a new J510 – Huntley 345 kV line.
3. Construct a new Huntley – Adams 345 kV line.
4. Construct a new Adams – Hickory Creek 345 kV line.
5. Construct a new Hickory Creek – Sub 91 345 kV line.

6. Install one 50 MVAR shunt capacitor at Hazleton 345 kV Substation.
7. Install one 75 MVAR shunt capacitor at Adams 345 kV Substation.
8. Install one 50 MVAR shunt capacitor at Hickory Creek 345 kV Substation.

The MWEX voltage stability study results for these two alternatives are summarized below:

1. Both alternatives can ensure MWEX voltage stability and meet ATC's voltage stability criteria.
2. Alternative 1 (NRH-SRR) has 14% voltage stability margin, and Alternative 2 (J510-Sub 91) has 10% voltage stability margin (ATC Planning Criteria require a 10% voltage stability margin).
3. Alternative 1 (NRH-SRR) has nose voltage less than post-contingent minimum voltage, and Alternative 2 (J510-Sub 91) has nose voltage 0.959 pu, which is greater than post-contingent minimum voltage (0.950 pu) but with voltage aggravation less than 0.01 pu (ATC Planning Criteria requires $V_{nose} < V_{min}$).

4.6.1.2 Comparison of Flows on New NUs and System Losses

Between Alternative 1 and Alternative 2 NUs, the effectiveness of new transmission NUs can be reflected on power flows carried by the new transmission lines (Table 4-7) and system losses reduced due to addition of new transmission NUs (Table 4-8).

Table 4-7: Flows Carried by New Transmission NUs under System Intact Condition

New Transmission NU	Flow (MVA)	
	Alternative 1	Alternative 2
J510 - Hazel Creek	935.6	NA
Hazel Creek - Scott County	830.6	NA
North Rochester - South Rocky Run	714.8	NA
J510 - Huntley	NA	552.8
Huntley - Adams	NA	847.2
Adams - Hickory Creek	NA	691.2
Hickory Creek - Sub 91	NA	347.3

Table 4-8: MISO/SPP Areas with Significant Loss Changes by MWEX Alternatives

Area #	Area Name	Alternative 1		Alternative 2		Difference (Alternative 1 - Alternative 2)	
		P losses (MW)	Q losses (MVAR)	P losses (MW)	Q losses (MVAR)	P losses (MW)	Q losses (MVAR)
295	WEC	90.8	1454.7	79	1321.2	11.8	133.5
356	AMMO	168.8	2229.3	171.9	2258.4	-3.1	-29.1

Area #	Area Name	Alternative 1		Alternative 2		Difference (Alternative 1 - Alternative 2)	
		P losses (MW)	Q losses (MVAR)	P losses (MW)	Q losses (MVAR)	P losses (MW)	Q losses (MVAR)
357	AMIL	183.2	2562.1	187.5	2603.6	-4.3	-41.5
600	XEL	727.6	6653.1	735.5	6763.1	-7.9	-110
608	MP	128.3	1416.6	134.2	1453.4	-5.9	-36.8
620	OTP	294.7	2492	310.5	2621.7	-15.8	-129.7
627	ALTW	417.6	3600.8	454.6	4094.4	-37	-493.6
635	MEC	473.3	4962.5	460.5	4868.4	12.8	94.1
680	DPC	89.6	468.6	103.5	530.1	-13.9	-61.5
694	ALTE	101.3	984.8	116.9	1105.8	-15.6	-121
696	WPS	117.6	1371	74.8	888.6	42.8	482.4
697	MGE	53.9	675.3	44.5	561.8	9.4	113.5
Total (MISO Classic)		3966.4	41998.8	3992.8	42193.9	-26.4	-195.1
640	NPPD	274.6	2595	273	2582.4	1.6	12.6
645	OPPD	82.6	990.1	80.9	973.4	1.7	16.7
652	WAPA	413.8	3931.3	416.1	3965.1	-2.3	-33.8
Total (SPP)		1888.2	20449.4	1885.8	20440.7	2.4	8.7

Based on the comparison of flows and loss changes, below are some observations:

1. Alternative 1 (NRH-SRR) new transmission lines carry much higher flows, compared with Alternative 2 (J510-Sub 91).
2. Compared with Alternative 2, Alternative 1 has significant reactive power loss reductions in ITCM, OTP, Xcel, ALTE, DPC, MP, Ameren, and WAPA, which means voltage profile in these areas are better with Alternative 1.
3. Compared with Alternative 2, Alternative 1 has significant reactive power loss increase in ATC (WPS, WEC, MGE) and MEC.

4.6.1.3 Comparison of Contingent Voltages

1) Voltage Comparison Based on Stage-2 Worst Voltage Results

The Stage-2 ACCC analysis identified worst voltage constraints (Appendix E.3) under NERC Category P0-P7 contingencies. Performance of the two alternatives was also compared based on their improvement on the Stage-2 worst voltage constraints. The comparison results are in Appendix E.4.1.

A summary of the comparison results is listed below:

1. The below contingency is converged and has good voltage performance with Alternative 1 (NRH-SRR), but is not converged with Alternative 2 (J510-Sub 91).

CEII Redacted

2. With voltage difference threshold of more than 0.01 pu, Alternative 1 has better voltage performance for almost all buses except one bus (Hazel Creek 345 kV).

2) Voltage Comparison under Three Critical Contingencies

Bus voltages were also compared between the two alternatives under three critical contingencies (Table 4-9). Based on the comparison results (Table E-21, Table E-22, Table E-23), with voltage difference threshold of more than 0.01 pu, Alternative 1 (NRH-SRR) has better voltage performance than Alternative 2 (J510-Sub 91).

Table 4-9: Three Critical Contingencies

Three Critical Contingencies
CEII Redacted
CEII Redacted
CEII Redacted

4.6.1.4 Comparison of Potential Thermal Constraints

To compare performance of the two alternatives based on potential thermal constraints (>95% of applicable rating) per MISO ERIS criteria (Section 2.4), full-blown linear (DC) contingency analysis was performed on the Stage-2 cases with addition of Alternative 1 (NRH-SRR) or Alternative 2 (J510-Sub 91). Comparison results of potential thermal constraints with maximum screened loading are listed in Table E-24 (Appendix E.4.2).

Potential thermal constraints which were only identified with Alternative 1 are listed in Table 4-10, and potential thermal constraints which were only identified with Alternative 2 are listed in Table 4-11. For potential common thermal constraints, Alternative 1 and Alternative 2 have similar maximum loadings. Overall, the two alternatives have similar and comparable potential thermal constraints.

Table 4-10: Potential Thermal Constraints Only Identified with Alternative 1, Maximum Screened Loading

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type
			(MVA)	(%)		
Parnell-J438 POI 161 kV	274.0	MEC ITCM	269.9	98.5	CEII Redacted	P1
J506 POI-Highland 345 kV	864.0	MEC	-840.3	97.3	CEII Redacted	P1
J506 POI-Highland 345 kV	864.0	MEC	-840.7	97.3	CEII Redacted	P2-P7
Red Willow-Mingo 345 kV	717.0	NPPD SUNC	-937.9	130.8	CEII Redacted	P0
Red Willow-Mingo 345 kV	828.5	NPPD SUNC	-939.6	113.4	CEII Redacted	P1
Adams 345-161-13.8 kV xfmr	334.6	ITCM	-353.2	105.6	CEII Redacted	P1
Adams 345-161-13.8 kV xfmr	334.6	ITCM	-377.8	112.9	CEII Redacted	P2-P7
Blue Lake-Scott Co 345 kV	1515.8	XEL	-1488.4	98.2	CEII Redacted	P2-P7
Murphy Creek 161-69 kV xfmr	94.0	SMMPA	-110.0	117.0	CEII Redacted	P1
Murphy Creek 161-69 kV xfmr	94.0	SMMPA	-110.0	117.0	CEII Redacted	P2-P7
Alden-Alea West 69 kV	45.0	ITCM	50.5	112.2	CEII Redacted	P1

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type
			(MVA)	(%)		
Hayward-Glenworth 161 kV	325.0	ITCM	-325.4	100.1	CEII Redacted	P1
Emery-Colby 161 kV	327.0	ITCM	-457.9	140.0	CEII Redacted	P1
Hazleton-Blackhawk 161 kV	325.0	ITCM MEC	-316.3	97.3	CEII Redacted	P1
Parnell-Hills 161 kV	325.0	MEC ITCM	310.9	95.7	CEII Redacted	P1
Hayward-Freeborn 161 kV	446.0	ITCM	-470.8	105.6	CEII Redacted	P1
Hayward-Freeborn 161 kV	446.0	ITCM	-479.8	107.6	CEII Redacted	P2-P7
Hazleton-Arnold 345 kV	923.0	ITCM MEC	880.7	95.4	CEII Redacted	P2-P7
Barton-Lime Creek 161 kV	240.0	ITCM	-235.5	98.1	CEII Redacted	P1
Barton-Lime Creek 161 kV	240.0	ITCM	-258.6	107.8	CEII Redacted	P2-P7
Colby 345-161 kV xfmr	450.0	ITCM	468.9	104.2	CEII Redacted	P1
Webster-Wright 161 kV	212.0	MEC	207.2	97.7	CEII Redacted	P2-P7

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type
			(MVA)	(%)		
Union Tap-Butler 161 kV	410.0	MEC	-401.6	98.0	CEII Redacted	P1
Franklin-Butler 161 kV	410.0	MEC	406.3	99.1	CEII Redacted	P1

Table 4-11: Potential Thermal Constraints Only Identified with Alternative 2, Maximum Screened Loading

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type
			(MVA)	(%)		
Brookings Co-White 345 kV #2	788.8	XEL WAPA	905.2	114.8	CEII Redacted	P2-P7
Helena-Scott Co 345 kV	1378.0	XEL	1311.9	95.2	CEII Redacted	P0
Kerkhoven Tap-Kerkhoven 115 kV	103.0	GRE XEL	-97.9	95.0	CEII Redacted	P2-P7
Lakefield-Obrien 345 kV	864.0	ITCM MEC	-834.8	96.6	CEII Redacted	P1
Lakefield-Obrien 345 kV	864.0	ITCM MEC	-848.1	98.2	CEII Redacted	P2-P7

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type
			(MVA)	(%)		
Hazleton-Hickory Crk 345 kV	1195.0	ITCM	1323.2	110.7	CEII Redacted	P2-P7
Blair-Granite Falls 230 kV	410.0	WAPA	443.2	108.1	CEII Redacted	P1
Watertown 345-230-13.8 kV xfmr	500.0	WAPA	510.2	102.0	CEII Redacted	P1
Watertown 345-230-13.8 kV xfmr	500.0	WAPA	510.2	102.0	CEII Redacted	P2-P7
Watertown-Appledorn 230 kV	341.8	WAPA	374.5	109.6	CEII Redacted	P1

4.6.1.5 Cost Comparison

Detailed estimated costs of the two alternatives are in Appendix E.4.3. Total estimated cost for Alternative 1 (NRH-SRR) is \$1,062,400,000 (including cost of J510 POI-Hazel Creek-Scott County), and total estimated cost for Alternative 2 (J510-Sub 91) is \$1,549,810,000.

4.6.1.6 Recommended MWEX Project

Alternative 1 (NRH-SRR) is recommended as MWEX project based on its better reliability and economical performance compared with Alternative 2 (J510-Sub 91).

4.6.2 Webster-Franklin Project Justification

The Webster-Franklin project justification is described in Appendix E.2.2. It was further confirmed in the Stage-2 ACCC analysis that this Webster-Franklin project was required for eliminating or reducing post-contingent overloads and improving post-contingent voltages. With a 4th 345 kV line into new Franklin substation, the previously required 150 Mvar SVC can be replaced by a 150 Mvar mechanically switched capacitor. Therefore the Webster-Franklin 345 kV project is justified as Backbone NU.

4.7 Step 5 – Stage-3 ACCC Analysis

In summary, the following projects have been justified based on the studies in Steps 1 to 4:

MISO Backbone Network Upgrades:

- Big Stone South-Alexandria 345 kV line
- J510-Hazel Creek-Scott County 345 kV line
- Franklin-Morgan Valley 345 kV line
- Beverly-Sub 92 345 kV line
- Webster-Franklin 345 kV line
- 2×150 MVar MSC at Franklin 345 kV

MWEX Voltage Stability Network Upgrades (NRH-SRR alternative):

- Construct a new South Rocky Run 345 kV switching station with one 75 MVAR shunt capacitor.
- Construct a new North Rochester – South Rocky Run 345 kV line.
- Construct a new 161/69 kV substation in Minnesota connecting the Wabaco – Alma 161 kV line to the Dairyland 69 kV system.
- Install redundant bus differential relays at the Rocky Run 345 kV Substation.
- Install 150 MVAR of shunt capacitance at the North Rochester 345 kV Substation.
- Install a 35% series capacitor at the Briggs Road 345 kV Substation on the Briggs Road – North Madison 345 kV line.
- Install one breaker at the Briggs Road 345 kV Substation.
- Additional facilities, e.g. Forest Junction – Elkhart Lake 138 kV, are expected to be overloaded by this alternative and will need to be assigned Network Upgrades.

Both MISO Backbone NUs and MWEX NUs were added to the Stage-2 models to create Stage-3 models. AC contingency analysis was performed in the Stage-3 models to identify any remaining thermal and voltage constraints.

4.7.1 Stage-3 System Intact Conditions

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table E-27, and voltage constraints are listed in Table E-28.

4.7.2 Stage-3 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies.

All category P1 contingencies were converged. For P1 contingencies in the 2022 summer shoulder scenario, thermal constraints are listed in Table E-29, and voltage constraints are listed in Table E-30.

Two category P2-P7 contingencies (Table E-33) were not converged in the benchmark or study cases. No mitigation plan is required for the study projects for these non-converged contingencies. The dc thermal results for non-converged contingencies are listed in Table E-34.

For P2-P7 contingencies in the 2022 summer shoulder scenario, thermal constraints are listed in Table E-31, and voltage constraints are listed in Table E-32.

4.7.3 Summer Shoulder Worst Thermal Constraints in the Stage-3 ACCC

Table 4-13 lists worst thermal constraints identified in the Stage-3 ACCC for 2022 summer shoulder scenario.

Table 4-12: 2022 Summer Shoulder Thermal Constraints, Maximum Screened Loading, Stage-3 ACCC

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
J506 POI-Raun 345 kV	864.0	MEC	1055.0	122.1	CEII Redacted	P1	J590,J439
J506 POI-Raun 345 kV	864.0	MEC	1098.8	127.2	CEII Redacted	P2-P7	J590,J439
J506 POI-Highland 345 kV	864.0	MEC	878.7	101.7	CEII Redacted	P1	J590,J439
J506 POI-Highland 345 kV	864.0	MEC	921.6	106.7	CEII Redacted	P2-P7	J590,J439
J530 POI-Montezuma 345 kV	1010.0	MEC	1145.5	113.4	CEII Redacted	P1	J555,J541,J598
J530 POI-Montezuma 345 kV	1010.0	MEC	1200.6	118.9	CEII Redacted	P2-P7	J555,J541,J598
J530 POI-Hills 345 kV	956.0	MEC	976.7	102.2	CEII Redacted	P0	J555,J476,J541,J583,J598,J611
J530 POI-Hills 345 kV	1152.0	MEC	1391.3	120.8	CEII Redacted	P1	J555,J541,J598

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
J530 POI-Hills 345 kV	1152.0	MEC	1449.0	125.8	CEII Redacted	P2-P7	J555,J541,J598
J302&J503 POI-J607 POI 230 kV	257.0	MDU	296.5	115.4	CEII Redacted	P2-P7	J457,J607
J302&J503 POI-Heskett 230 kV	257.0	MDU	346.2	134.7	CEII Redacted	P1	J302,J607,J503
J302&J503 POI-Heskett 230 kV	257.0	MDU	499.4	194.3	CEII Redacted	P2-P7	J302,J457,J607,J503
J607 POI-Wishek 230 kV	257.0	MDU	352.0	137.0	CEII Redacted	P1	J302,J607,J503
J607 POI-Wishek 230 kV	257.0	MDU	352.0	137.0	CEII Redacted	P2-P7	J302,J607,J503
J611-Maryville 161 kV	199.0	MEC GMO	206.0	103.5	CEII Redacted	P0	J611
J611-Maryville 161 kV	199.0	MEC GMO	263.3	132.3	CEII Redacted	P1	J611
J611-Maryville 161 kV	199.0	MEC GMO	245.6	123.4	CEII Redacted	P2-P7	J611

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
J615 POI-Shaulis Rd 161 kV	215.0	MEC	228.0	106.0	CEII Redacted	P2-P7	J615
Gentry-Fairport 161 kV	227.0	AECI	230.2	101.4	CEII Redacted	P1	J611
Novelty 161 -69 kV xfmr	56.0	AECI	64.8	115.7	CEII Redacted	P1	J541
South River-Emerson 161 kV	167.0	AECI	186.6	111.7	CEII Redacted	P1	J541
Zackary 345-161 kV xfmr	560.0	AMMO	592.8	105.9	CEII Redacted	P1	J541,J598
Adair-Zackary 161 kV	557.0	AMMO	597.9	107.3	CEII Redacted	P1	J541,J598
St. Joseph-Cooper 345 kV	1195.0	NPPD GMO	1647.9	137.9	CEII Redacted	P1	J476,J555,J598,J611
Adams 345-161-13.8 kV xfmr	334.6	XEL	344.5	102.9	CEII Redacted	P1	J614,J415
Adams 345-161-13.8 kV xfmr	334.6	XEL	380.9	113.8	CEII Redacted	P2-P7	J614
Split Rock-Sioux City 345 kV	717.0	XEL WAPA	723.9	101.0	CEII Redacted	P2-P7	J512,J569
Sheyenne-Lake Park 230 kV	294.0	XEL OTP	322.1	109.6	CEII Redacted	P0	J613
Sheyenne-Lake Park 230 kV	294.0	XEL	329.6	112.1	CEII Redacted	P1	J511,J593,J599,J607

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
		OTP					
Minn Valley-Granite Falls 230 kV	259.0	XEL WAPA	336.5	129.9	CEII Redacted	P1	J593,J596
Sheyenne-Mapleton 115 kV	162.0	XEL OTP	206.5	127.5	CEII Redacted	P1	J613
Sheyenne-Mapleton 115 kV	162.0	XEL OTP	198.9	122.8	CEII Redacted	P2-P7	J511,J613
Grove Lake-Lake Johnason 69 kV	38.6	XEL GRE	39.7	102.8	CEII Redacted	P1	J596
Murphy Creek 161-69 kV xfmr	94.0	SMMPA	117.2	124.7	CEII Redacted	P1	J414
Austin-Murphy Creek 161 kV	279.0	SMMPA	281.8	101.0	CEII Redacted	P0	J414
Austin-Murphy Creek 161 kV	307.0	SMMPA	360.5	117.4	CEII Redacted	P1	J414,J415
Austin-Murphy Creek 161 kV	307.0	SMMPA	408.0	132.9	CEII Redacted	P2-P7	J414
Murphy Creek-Hayward 161 kV	276.0	SMMPA ITCM	304.3	110.3	CEII Redacted	P0	J414
Murphy Creek-Hayward 161 kV	276.0	SMMPA ITCM	384.5	139.3	CEII Redacted	P1	J414,J415

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Murphy Creek-Hayward 161 kV	276.0	SMPA ITCM	432.5	156.7	CEII Redacted	P2-P7	J414
Rice 161-69 kV xfmr	106.0	SMPA	140.5	132.5	CEII Redacted	P1	J614
Rice 161-69 kV xfmr	106.0	SMPA	140.8	132.8	CEII Redacted	P2-P7	J614
Rutland-Fox Lake 161 kV	168.0	ITCM	234.0	139.3	CEII Redacted	P1	J594
Rutland-Fox Lake 161 kV	168.0	ITCM	267.6	159.3	CEII Redacted	P2-P7	J594
Rutland-Huntley 161 kV	168.0	ITCM	187.6	111.7	CEII Redacted	P1	J594

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Rutland-Huntley 161 kV	168.0	ITCM	223.3	132.9	CEII Redacted	P2-P7	J594
Wing River-Inman 230 kV	252.5	GRE	287.3	113.8	CEII Redacted	P0	J596
Wing River-Inman 230 kV	252.5	GRE	321.0	127.1	CEII Redacted	P1	J637,J638,J457,J459,J596,J302,J503
Stanton-Square Butte 230 kV	430.2	GRE MPC	454.5	105.6	CEII Redacted	P1	J511

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Stanton-Square Butte 230 kV	430.2	GRE MPC	586.3	136.3	CEII Redacted	P2-P7	J511
Ramsey-Prairie 230 kV	320.9	GRE MPC	335.8	104.7	CEII Redacted	P1	J511
Ramsey-Prairie 230 kV	320.9	GRE MPC	337.4	105.2	CEII Redacted	P2-P7	J511
Audubon-Lake Park 230 kV	294.0	OTP MPC	316.6	107.7	CEII Redacted	P0	J613
Audubon-Lake Park 230 kV	294.0	OTP MPC	324.0	110.2	CEII Redacted	P1	J511,J599,J607
Ellendale-Steel Center 69 kV	45.0	ITCM	53.8	119.5	CEII Redacted	P1	J414

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Triboji-Fly Cloud 69 kV	48.0	ITCM	49.9	103.9	CEII Redacted	P2-P7	J569
Walters-Alden 69 kV	45.0	ITCM	47.1	104.7	CEII Redacted	P1	J414
Pratt Tap-Steel Center 69 kV	45.0	ITCM	53.3	118.4	CEII Redacted	P1	J414
Alden-Alea West 69 kV	45.0	ITCM	47.0	104.5	CEII Redacted	P1	J414
Fox Lake-Jackson 161 kV #2	276.0	ITCM	292.3	105.9	CEII Redacted	P1	J594
Fox Lake-Jackson 161 kV #2	276.0	ITCM	296.7	107.5	CEII Redacted	P2-P7	J594
Emery-Floyd 161 kV	272.0	ITCM MEC	285.8	105.1	CEII Redacted	P1	J415
Emery-Floyd 161 kV	272.0	ITCM MEC	323.9	119.1	CEII Redacted	P2-P7	J415

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Hazleton-Washburn 161 kV	325.0	ITCM MEC IP&L	361.9	111.4	CEII Redacted	P2-P7	J615
Lansing-Harmony 161 kV	153.0	ITCM DPC	168.1	109.9	CEII Redacted	P2-P7	J614
Henry Co-Denmark 161 kV	169.0	ITCM	195.3	115.6	CEII Redacted	P2-P7	J541,J598
Henry Co-Jeff 161 kV	197.0	ITCM NMEC	224.9	114.2	CEII Redacted	P2-P7	J541,J598
Wapello-Jeff 161 kV	223.0	ITCM	268.3	120.3	CEII Redacted	P2-P7	J541,J598
Wapello-Appanoose 161 kV	171.0	ITCM	204.6	119.6	CEII Redacted	P1	J541
Ottumwa 345-161 kV xfmr	400.0	ITCM	555.5	138.9	CEII Redacted	P1	J541,J598

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Ottumwa 345-161 kV xfmr	400.0	ITCM	545.6	136.4	CEII Redacted	P2-P7	J541,J598
Adams-Hayward 161 kV	216.0	ITCM	292.7	135.5	CEII Redacted	P1	J414,J415
Adams-Hayward 161 kV	216.0	ITCM	292.7	135.5	CEII Redacted	P2-P7	J414
Hazleton-Arnold 345 kV	923.0	ITCM MEC IP&L	1017.4	110.2	CEII Redacted	P2-P7	J615
Hazleton-Blackhawk 345 kV	1195.0	ITCM MEC	1293.6	108.3	CEII Redacted	P1	J415
Hazleton-Blackhawk 345 kV	1195.0	ITCM MEC	1275.6	106.7	CEII Redacted	P2-P7	J415
Cayler-Wisdom 161 kV	215.0	MEC ITCM CBPC	223.5	104.0	CEII Redacted	P1	J594
Cayler-Wisdom 161 kV	215.0	MEC ITCM CBPC	230.2	107.1	CEII Redacted	P2-P7	J594
Raun-S3451 345 kV	956.0	MEC OPPD	989.8	103.5	CEII Redacted	P0	J590,J302,J439,J457,J459,J512, J569,J575,J577,J587,J594,J596, J597,J607,J613,J637,J638,J503

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Raun-S3451 345 kV	956.0	MEC OPPD	1189.0	124.4	CEII Redacted	P1	J439,J414,J415,J457,J587,J590, J594,J555,J577,J597,J569,J607
Raun-S3451 345 kV	956.0	MEC OPPD	1213.2	126.9	CEII Redacted	P2-P7	J439
Grimes-Beaver Creek 345 kV	1222.0	MEC	1302.4	106.6	CEII Redacted	P2-P7	J439
Bondurant-Sycamore 345 kV	1152.0	MEC	1158.8	100.6	CEII Redacted	P2-P7	J583
Red Willow-Mingo 345 kV	828.5	NPPD SUNC	993.8	120.0	CEII Redacted	P1	J599
Whitlock-Sully Buttes 230 kV	264.0	WAPA	273.5	103.6	CEII Redacted	P1	J599
Whitlock-Glenham 230 kV	260.0	MDU WAPA	280.3	107.8	CEII Redacted	P1	J599
Morris-Grant Co 115 kV	103.0	OTP MRES WAPA	104.5	101.5	CEII Redacted	P2-P7	J596
Heskett-Mandan 230 kV	383.0	MDU	409.1	106.8	CEII Redacted	P2-P7	J302,J457,J607,J503
Foxtail-Tatanka North 230 kV	343.0	MDU	440.1	128.3	CEII Redacted	P1	J302,J457,J607,J503

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Foxtail-Tatanka North 230 kV	343.0	MDU	363.4	106.0	CEII Redacted	P2-P7	J302,J457,J607,J503
Foxtail-Ellendale 230 kV	419.0	MDU	551.9	131.7	CEII Redacted	P1	J302,J457,J607,J503
Foxtail-Ellendale 230 kV	419.0	MDU	477.4	113.9	CEII Redacted	P2-P7	J302,J457,J607,J503
Merricourt-Wishek 230 kV	257.0	MDU	348.6	135.6	CEII Redacted	P1	J302,J607,J503
Merricourt-Wishek 230 kV	257.0	MDU	348.7	135.7	CEII Redacted	P2-P7	J302,J607,J503
Merricourt-Ellendale 230 kV	383.0	MDU	534.9	139.7	CEII Redacted	P1	J302,J457,J607,J503
Merricourt-Ellendale 230 kV	383.0	MDU	524.9	137.0	CEII Redacted	P2-P7	J302,J457,J607,J503
Harmony-Cresco 69 kV	37.0	DPC	40.1	108.4	CEII Redacted	P1	J614

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Harmony-Cresco 69 kV	37.0	DPC	37.5	101.2	CEII Redacted	P2-P7	J614

4.8 Network Upgrades Identified in MISO ERS Analysis for 2022 Summer Shoulder Scenario

Based on the MISO 2022 summer shoulder steady state analyses, the MISO Backbone thermal NUs and cost are listed in Table 4-13, and MWEX stability NUs and cost are listed in Table 4-14. Additional thermal NUs and cost are listed in Table 4-15, and additional reactive power NUs and cost are listed in Table 4-16.

Table 4-13: MISO Backbone NUs

NUs	Miles	Cost (\$)
Big Stone South-Alexandria 345 kV	95	\$197,400,000
J510-Hazel Creek-Scott County 345 kV	165	\$330,000,000
Franklin-Morgan Valley 345 kV	138	\$384,056,500
Beverly-Sub 92 345 kV	60	\$148,675,000
Webster-Franklin 345 kV	49.2	\$117,000,000
2×150 MVar MSC at Franklin 345 kV		\$8,000,000
Total		\$1,185,131,500

Table 4-14: MWEX Stability NUs

NUs	Miles	Cost (\$)
Construct a new South Rocky Run 345 kV switching station with one 75 MVAR shunt capacitor.		\$53,500,000
Construct a new North Rochester – South Rocky Run 345 kV line	180	\$621,000,000
Construct a new 161/69 kV substation in Minnesota connecting the Wabaco – Alma 161 kV line to the Dairyland 69 kV system		\$5,500,000
Install 150 MVAR of shunt capacitance at the North Rochester 345 kV Substation		\$6,000,000
Install a 35% series capacitor at the Briggs Road 345 kV Substation on the Briggs Road – North Madison 345 kV line		\$18,000,000
Install one breaker at the Briggs Road 345 kV Substation		\$2,000,000
Additional facilities, e.g. Forest Junction – Elkhart Lake 138 kV, are expected to be overloaded by this alternative and will need to be assigned Network Upgrades		NA
Total		\$706,000,000

Table 4-15: Additional Thermal NUs

Constraint	Owner	Mitigation	Cost (\$)
J506 POI-Raun 345 kV	MEC	Structure Replacements.	\$700,000
J506 POI-Highland 345 kV	MEC	Structure Replacements.	\$200,000
J530 POI-Montezuma 345 kV	MEC	Structure Replacements.	\$125,000
J530 POI-Hills 345 kV	MEC	Reconductor with structure replacements as needed. Substation terminal equipment upgrades.	\$27,000,000
J302&J503 POI-J607 POI 230 kV	MDU	Add breaker at Merricourt	\$1,500,000
J302&J503 POI-Heskett 230 kV	MDU	Line rebuild	\$9,000,000
J607 POI-Wishek 230 kV	MDU	Rebuild with high temp. conductor. New Rating: 383 MVA	\$1,100,000
J611-Maryville 161 kV	MEC GMO	MEC: Reconductor with structure replacements as needed. Assumes J611 POI is one mile from Missouri border. \$1,000,000 GMO: NU is not required unless it is identified as constraint in affected system study.	\$1,000,000
J615 POI-Shaulis Rd 161 kV	MEC	MEC: No mitigation required. Expected rating will be 272 MVA.	\$0
Gentry-Fairport 161 kV	AECI	NU is not required unless it is identified as constraint in affected system study.	\$0
Novelty 161 -69 kV xfmr	AECI	NU is not required unless it is identified as constraint in affected system study.	\$0
South River-Emerson 161 kV	AECI	NU is not required unless it is identified as constraint in affected system study.	\$0
Zackary 345-161 kV xfmr	AMMO	Add Second 560 MVA 345-161 kV transformer.	\$7,000,000
Adair-Zackary 161 kV	AMMO	Add second 161 kV line between Adair and Zachary	\$2,000,000
St. Joseph-Cooper 345 kV	NPPD GMO	NU is not required unless it is identified as constraint in affected system study.	\$0
Adams 345-161-13.8 kV xfmr	XEL	Upgrade Transformer Nameplate to 336 MVA and terminal equipment	\$4,000,000
Split Rock-Sioux City 345 kV	XEL WAPA	XEL: Rating should be 980 MVA WAPA: NU is not required unless it is identified as constraint in affected system study.	\$0
Sheyenne-Lake Park 230 kV	XEL OTP	XEL: OTP conductor / MPS equipment is limiter. \$0 OTP: Line clearance mitigation required. \$250,000	\$250,000
Minn Valley-Granite Falls 230 kV	XEL WAPA	XEL: Rating should be 407 MVA WAPA: NU is not required unless it is identified as constraint in affected system study.	\$0

Summer Shoulder Steady-State Analysis

Constraint	Owner	Mitigation	Cost (\$)
Sheyenne-Mapleton 115 kV	XEL OTP	OTP: Complete rebuild of OTP's 477 ACSR portion of the line. \$3,150,000 Xcel: Complete rebuild of Xcel's portion of the line \$1,500,000	\$4,650,000
Grove Lake-Lake Johnason 69 kV	XEL GRE	XEL: GRE Equipment is limiter. \$0 GRE: Upgrade 4 miles of 69 kV line, \$2.26 million	\$2,260,000
Murphy Creek 161-69 kV xfmr	SMMPA	Install new 112 MVA Transformer at Murphy Creek Sub.	\$2,500,000
Austin-Murphy Creek 161 kV	SMMPA	Re-conductoring of the Austin to Murphy Creek Line.	\$650,000
Murphy Creek-Hayward 161 kV	SMMPA ITCM	ITCM: Apply RF 2 on CT's, replace jumpers and transducer, \$10k SMMPA: Re-conductoring of the Hayward to Murphy Creek Line, \$4.5 M	\$4,510,000
Rice 161-69 kV xfmr	SMMPA	SMMPA: MOD project # 110359 to increase the Rice 161/69kV transformer to 190MVA rating as per the GIA J614	\$0
Rutland-Fox Lake 161 kV	ITCM	Rebuild 9 miles of 161 kV line, \$9,000,000	\$9,000,000
Rutland-Huntley 161 kV	ITCM	Rebuild 0.73 miles of 161 kV, \$730,000	\$730,000
Wing River-Inman 230 kV	GRE	Replace jumpers at Wing River	\$50,000
Stanton-Square Butte 230 kV	GRE MPC	GRE: Rebuild 16.5 miles of 230 kV line > 590 MVA; 2) replace hookstick switches; 3) replace jumpers to >590 MVA. \$10,875,000 MPC: NU is not required unless it is identified as constraint in affected system study.	\$10,875,000
Ramsey-Prairie 230 kV	GRE MPC	GRE: rebuild 71 miles of 230 kV transmission >350 MVA. \$46,150,000 MPC: NU is not required unless it is identified as constraint in affected system study.	\$46,150,000
Audubon-Lake Park 230 kV	OTP MPC	OTP: OTP's equipment will be capable of handling the flow seen in this study. MPC: NU is not required unless it is identified as constraint in affected system study.	\$0
Ellendale-Steel Center 69 kV	ITCM	Rebuild 2 miles of 69 kV with T2-477 ACSR	\$1,500,000
Triboji-Fly Cloud 69 kV	ITCM	RF 2 on CTs - not a constraint, rating 62 MVA	\$0
Walters-Alden 69 kV	ITCM	New ratings removes constraint = 48 MVA	\$0
Pratt Tap-Steel Center 69 kV	ITCM	Rebuild 7 miles of 69 kV with T2-477 ACSR	\$5,250,000
Alden-Alea West 69 kV	ITCM	New ratings removes constraint = 48 MVA	\$0
Fox Lake-Jackson 161 kV #2	ITCM	New ratings removes constraint, 308/328 MVA SN/SE	\$0

Summer Shoulder Steady-State Analysis

Constraint	Owner	Mitigation	Cost (\$)
Emery-Floyd 161 kV	ITCM MEC	MEC: Reconductor / Terminal Equipment Upgrades. \$21,000,000 ITCM: New ratings removes constraint, 407/418 MVA SN/SE \$0	\$21,000,000
Hazleton-Washburn 161 kV	ITCM MEC IP&L	MEC: No additional mitigation required. Existing MEC emergency rating expected to be 410 MVA after DPP2016 FEB reconductor mitigation. ITCM: ITCM limit is wavetrap - \$200k. IP&L: Install 3 new 161 kV CT/PT combo units, \$170,000	\$370,000
Lansing-Harmony 161 kV	ITCM DPC	ITCM: Not a constraint. ITC rating is 180 MVA. \$0 DPC: ITC's line rating. \$0	\$0
Henry Co-Denmark 161 kV	ITCM	Ratings change has line rating at 207 MVA summer at 162 degrees F	\$0
Henry Co-Jeff 161 kV	ITCM NMEC	Northeast Missouri Electric Coop: NU is not required unless it is identified as constraint in affected system study. ITCM: Relay upgrade. \$200,000	\$200,000
Wapello-Jeff 161 kV	ITCM	Rebuild of 23 miles of 161 kV required. Future rating of 515 MVA	\$23,000,000
Wapello-Appanoose 161 kV	ITCM	New ITCM rating is 209 MVA - not a constraint	\$0
Ottumwa 345-161 kV xfmr	ITCM	Add 2nd 450 MVA transformer.	\$9,000,000
Adams-Hayward 161 kV	ITCM	Rebuild 36.5 miles of 161 kV	\$36,500,000
Hazleton-Arnold 345 kV	ITCM MEC IP&L	MEC: Structure Replacements. \$200,000 ITCM: Pending ITCM ratings methodology changes will increase the rating of the line to 1174 MVA. \$0 IP&L: Alliant does not require meter data for flows on this line at either end. \$0	\$200,000
Hazleton-Blackhawk 345 kV	ITCM MEC	MEC: Existing MEC rating is 1,740 MVA. No mitigation required. ITCM: Not a constraint. ITCM limit at 1324/1463 MVA SN/SE. The 1195 rating does not account for ITCM CT's RF of 2.	\$0
Cayler-Wisdom 161 kV	MEC ITCM CBPC	MEC: Structure Replacements. \$500,000 CBPC: NU is not required unless it is identified as constraint in affected system study. ITCM: Not a constraint. ITCM equipment at 240 MVA. \$0	\$500,000
Raun-S3451 345 kV	MEC OPPD	MEC: Terminal Equipment Upgrades. \$200,000 OPPD: NU is not required unless it is identified as constraint in affected system study.	\$200,000
Grimes-Beaver Creek 345 kV	MEC	MEC: No mitigation required. The proposed Franklin substation will be constructed so that these two lines do not share a common breaker. There are no other contingencies reported as requiring mitigation.	\$0
Bondurant-Sycamore 345 kV	MEC	MEC: No mitigation required. Rating expected to be 1202 MVA after planned 2018 relay upgrade project.	\$0
Red Willow-Mingo 345 kV	NPPD	NU is not required unless it is identified as constraint in affected	\$0

Summer Shoulder Steady-State Analysis

Constraint	Owner	Mitigation	Cost (\$)
	SUNC	system study.	
Whitlock-Sully Buttes 230 kV	WAPA	NU is not required unless it is identified as constraint in affected system study.	\$0
Whitlock-Glenham 230 kV	MDU WAPA	MDU: Relay Setting WAPA: NU is not required unless it is identified as constraint in affected system study.	\$10,000
Morris-Grant Co 115 kV	OTP MRES WAPA	OTP: All OTP equipment is good for the flow of 110.6 MVA seen in the study. MRES: Phase raisers, \$350,000 WAPA: NU is not required unless it is identified as constraint in affected system study.	\$350,000
Heskett-Mandan 230 kV	MDU	Add breaker at Merricourt	Incl.
Foxtail-Tatanka North 230 kV	MDU	Line Rating is 478 MVA	\$0
Foxtail-Ellendale 230 kV	MDU	Line Rating is 623 MVA	\$0
Merricourt-Wishek 230 kV	MDU	Rebuild Line with high temp. conductor. New Rating: 478 MVA	\$6,000,000
Merricourt-Ellendale 230 kV	MDU	Rebuild Line with high temp. conductor New Rating: 597 MVA	\$4,500,000
Harmony-Cresco 69 kV	DPC	Rebuild line with 477 ACSR	\$4,000,000

Table 4-16: Additional Reactive Power NUs Required for Voltage Constraints

Network Upgrades	Owner	Cost (\$)
Additional 2×25 Mvar switched cap bank at Lathrop 161 kV (344923) ¹	Ameren AFS	\$10,000,000
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	Xcel	\$7,500,000
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	Xcel	\$7,000,000
2×20 Mvar switched cap bank at Fergus Falls 230 kV (658109)	MRES	\$1,750,000
3×50 Mvar switched cap bank at Alexandria 345 kV (658047)	MRES	\$5,500,000
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	MEC	\$1,000,000
1x150 Mvar switched cap bank at Webster 345 kV (636000)	MEC	\$4,000,000
1×25 Mvar switched cap bank at J615 POI 161 kV (86151)	MEC	\$2,500,000
1x150 Mvar switched cap bank at Raun 345 kV (635200)	MEC	\$4,000,000
1x100 Mvar switched cap bank at O'Brien 345 kV (635368)	MEC	\$4,000,000
2x150 Mvar switched cap bank at Hills 345 kV (636400)	MEC	\$15,000,000
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	ITCM	\$5,500,000
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	ITCM	\$5,500,000

Summer Shoulder Steady-State Analysis

Network Upgrades	Owner	Cost (\$)
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	ITCM	\$5,500,000
Reset transformer taps back to the midpoint at Hazleton, Hickory Creek, Killdeer and Colby. Taps are not locked, but were reset to midpoint.	ITCM	\$0

Note 1: Lathrop 161 kV is not MISO facility and voltage violation was flagged and identified in MISO ERIS analysis. Per Ameren's comments Ameren agreed with MISO's ERIS analysis results and no additional Ameren affected system impact study was required.

Section

5

Local Planning Criteria Analysis

Local Planning Criteria (LPC) analyses were performed to identify additional constraints per Transmission Owning Companies' LPC.

5.1 GRE Local Planning Criteria Analysis (J511, J596)

Siemens PTI performed the local planning criteria analysis for J511 and J596 based on GRE's LPC. The J511 GRE local planning criteria analysis details can be found in Appendix F.1.1, and the J596 GRE local planning criteria analysis details can be found in Appendix F.1.2.

5.1.1 Additional Network Upgrades Identified in J511 GRE LPC Analysis

Except the thermal constraints or NUs identified in MISO ACCC analysis, no additional thermal constraints or NUs were required in the J511 GRE LPC analysis.

With the MISO reactive power NUs applied, all voltage constraints except those under one CUDC bipole contingencies (Table 5-1) can be mitigated.

The MISO reactive power NUs will not mitigate severe voltage constraints identified under the CUDC bipole contingencies (Table 5-1). Neither will the MISO reactive power NUs resolve the system instability identified under the permanent CUDC bipole fault (Table 5-2).

Table 5-1: CUDC Bipole Contingency

CEII Redacted

Table 5-2: System Unstable under Permanent CUDC Bipole Faults

CEII Redacted

To mitigate the transient stability violations and voltage constraints under the CUDC bipole contingency, the following network upgrades are required:

Table 5-3: Network Upgrades Required in J511 LPC Study

Network Upgrades	Mileage	Cost (Million)	TO
New 345 kV line Center-Ellendale	200.0	\$406.75	OTP MDU
New 345 kV line Prairie-Bison	90.08	\$243.2	MPC

Local Planning Criteria Analysis

Network Upgrades	Mileage	Cost (Million)	TO
			XEL
150 Mvar SVC at Jamestown 345 kV (620369)		\$24.37	OTP
150 Mvar SVC at Prairie 230 kV (657755)		\$24.37	OTP
2*50 Mvar MSC at Lake Park 230 kV (bus #: 657780)		\$4.48	OTP

5.1.2 Additional Network Upgrades Identified in J596 GRE LPC Analysis

Except the thermal constraints or NUs identified in MISO ACCC analysis, additional thermal constraints or NUs identified in the J596 GRE LPC analysis are listed in Table 5-4.

With the MISO reactive power NUs applied, all voltage constraints identified in the J596 GRE LPC analysis can be mitigated.

Additional Network Upgrades required in the J596 GRE LPC study are listed in Table 5-4.

When J596 POI – Morris 115 kV is out of service, Short Circuit Ratio (SCR) of the J596 generation is around 2.79. Further study is recommended to provide evidence that the J596 wind farm can operate at the SCR of 2.79.

Table 5-4: Additional Network Upgrades for Constraints Identified in J596 GRE LPC Analysis

Constraint	Owner	Mitigation	Cost (\$)
Kerkhoven Tap-Kerkhoven 115kV	XEL GRE	GRE: Upgrade 11 miles of 115 kV line (Kerkhoven to Kerkhoven tap); replace jumpers at Kerkhoven; upgrade CTs at Kerkhoven, \$7.3 M. XEL: NU is not required for GRE LPC.	\$7,300,000

5.2 OTP Local Planning Criteria Analysis

Siemens PTI performed the local planning criteria analysis based on OTP's LPC. The OTP local planning criteria analysis details can be found in Appendix F.2.

5.2.1 Additional Network Upgrades Identified in OTP LPC Analysis

One category P2-P7 contingency (Table 5-5) was solved in the benchmark case but not solved in the study case. Voltage collapse was observed in the study case. With the GRE LPC network upgrades (Table 5-3) included, this contingency is converged without voltage constraints.

Except the thermal constraints or NUs identified in MISO ACCC analysis, no additional thermal constraints or NUs were required in the OTP LPC analysis.

With the MISO reactive power NUs applied, there are still a few OTP buses with low voltage violations. Except low voltage violations at Viking 115 kV bus, all other remaining low voltage constraints can be mitigated by adding a breaker on the existing 345 kV line reactor at Ellendale to make it switchable. This mitigation is also required for resolving reactive power and voltage oscillations identified in the MDU LPC stability analysis. The low voltage violations at Viking 115 kV bus can be mitigated by 85 MVar switched capacitor at Bison 345 kV bus, which is required in MPC affected system study.

No high voltage constraints were identified on OTP buses under the light load no wind (LLNW) condition. Under a few contingencies, several bus voltages are above 1.05 p.u. but below their applicable high voltage limits.

In summary, no additional network upgrades were identified in the OTP LPC study.

Table 5-5: Non-Converged P7 Contingency in the Study Case

Contingency	Contingency Details	Type	Benchmark	Study
P7:400X2:GRE:CCR-DKN BIPOLE	CEII Redacted	P2-P7	Converged	Blown up

5.3 MDU Local Planning Criteria Analysis

Siemens PTI performed the local planning criteria analysis based on MDU's LPC. The MDU local planning criteria analysis details can be found in Appendix F.3.

5.3.1 Additional Network Upgrades Identified in MDU LPC Analysis

Except the thermal constraints or NUs identified in MISO ACCC analysis, no additional thermal constraints or NUs were required in the MDU LPC analysis.

With the MISO reactive power NUs applied, all voltage constraints identified in the MDU LPC analysis can be mitigated.

Additional Network Upgrades required in the MDU LPC study is listed in Table 5-6.

Table 5-6: Additional Network Upgrades for Constraints Identified in MDU LPC Analysis

Constraint	Owner	Mitigation	Cost (\$)
Reactive power and voltage oscillations (J302, J503, J457, J526, J607, Foxtail, etc.)	MDU	Add a breaker on the existing 345 kV line reactor at Ellendale to make it switchable	\$1,500,000

5.4 DPC Local Planning Criteria Analysis

Siemens PTI performed the local planning criteria analysis based on DPC's LPC. The DPC local planning criteria analysis details can be found in Appendix F.4.

5.4.1 Additional Network Upgrades Identified in DPC LPC Analysis

Except the thermal constraints or NUs identified in MISO ACCC analysis, no additional thermal constraints or NUs were identified in the DPC LPC analysis.

No voltage constraints were identified in the DPC LPC analysis.

No additional Network Upgrades are required in the DPC LPC study.

5.5 Ameren Local Planning Criteria Analysis

Ameren performed the local planning criteria analysis based on Ameren's LPC for generation projects J541 and J598. The Ameren local planning criteria analysis details can be found in Appendix F.5.

5.5.1 Steady-State Results Summary

The following constraints were identified under shoulder peak conditions for Line + Generator contingencies:

Monitored Element	Balancing Area	First Level Contingency Second Level Contingency	Summer Emergency Rating (MVA)	MVA Flow	%Load
Zachary 345/161 kV transformer	AMMO	CEII Redacted	560	602.6	108%
Zachary-Adair 161 kV line			557	593.8	107%

The following constraints were identified under shoulder peak conditions for 345 kV Line + Line contingencies:

Monitored Element	Area	First Level Contingency Second Level Contingency	Summer Emergency Rating (MVA)	MVA Flow	%Load
Zachary 345/161 kV transformer	AMMO	CEII Redacted	560	684.8	122.28
Adair 161 kV bus tie 2-3			335	370.3	110.53
Adair-Zachary 161 kV line			557	699.2	125.52

5.5.2 Stability Results Summary

Mitigation is required for the scenarios where all the J541 and J598 generation flows through the Zachary 345/161 kV or Ottumwa 345/161 kV transformer. In these scenarios the J541 and J598 electrical power output becomes unstable with tripping due to overvoltage. These scenarios also result in large thermal overloads. The thermal overload mitigation may also resolve the generator tripping observed.

5.5.3 Additional Network Upgrades Identified in Ameren LPC Analysis

Except the constraints or NUs identified in MISO ACCC analysis, no additional constraints or NUs were identified in the Ameren LPC analysis.

Affected System Steady-State Analysis

Steady state analyses were performed to identify constraints on affected systems.

6.1 Affected System Analysis for CIPCO Company

Per CIPCO Affected System Planning Criteria, transmission facilities which meet all of the following three (3) conditions are considered as constraints:

- 1) the branch is loaded above its applicable normal or emergency rating for the post-change case, and
- 2) the generator has a larger than 3% DF on the overloaded facility under post contingent condition or 5% DF under system intact condition, and
- 3) the loading increase of the overloaded facility is greater than 1 MVA compared with that in the pre-change case under system intact or contingency conditions.

AC contingency analysis was performed for this CIPCO affected system analysis, using the following benchmark and study cases:

- Summer peak benchmark and study cases (without any network upgrades)
- Summer shoulder benchmark and study cases (with Backbone and MWEX NUs)

All NERC category P0-P7 contingencies described in Section 2.2 were simulated. The CIPCO affected system was monitored.

The additional thermal constraints in the affected systems identified in this CIPCO affected system analysis are listed in Appendix G.1. These include the thermal constraints for the 2022 summer peak and shoulder scenarios. Whether these constraints will be mitigated or not will be determined by the affected transmission owning company's planning criteria.

The highest loading of CIPCO local planning criteria thermal constraints and potential network upgrades are listed in Table 6-1 for the 2022 summer shoulder scenario. No CIPCO's thermal constraints were identified in the 2022 summer peak scenario.

Table 6-1: Additional Thermal Constraints Identified in CIPCO Affected System Analysis and their Highest Loadings

Generator	Constraint	Rating	Owner	Worst Loading		Cont Type	Mitigation	Cost (\$)
				(MVA)	(%)			
J415,J615	Hazleton-Dundee 161 kV	327.0	CIPCO ITCM	366.5	112.1	P1	CIPCO: Upgrade Dundee terminal to 3000 Amps ITCM: CIPCO LPC. \$0	\$500,000
J590,J414,J415,J439, J512,J594,J614,J615	Hazleton-Dundee 161 kV	327.0	CIPCO ITCM	518.3	158.5	P2-P7	CIPCO: Upgrade Dundee terminal to 3000 Amps ITCM: CIPCO LPC. \$0	
J415,J615	Liberty-Dundee 161 kV	327.0	CIPCO ITCM	384.1	117.5	P1	CIPCO: Rebuild 25 miles of 161 kV line with bundled conductor. ITCM: CIPCO LPC	\$25,000,000
J415,J615	Liberty-Dundee 161 kV	327.0	CIPCO ITCM	431.9	132.1	P2-P7	CIPCO: Rebuild 25 miles of 161 kV line with bundled conductor. ITCM: CIPCO LPC	
J415,J615	Liberty-Hickory Crk 161 kV	327.0	CIPCO ITCM	376.0	115.0	P1	CIPCO: Upgrade Liberty terminal to 3000 Amps. ITCM: CIPCO LPC	\$500,000
J415,J615	Liberty-Hickory Crk 161 kV	327.0	CIPCO ITCM	424.5	129.8	P2-P7	CIPCO: Upgrade Liberty terminal to 3000 Amps. ITCM: CIPCO LPC	

6.2 MPC Affected System Analysis

The MPC affected system analysis details (dated on 06/19/2018) can be found in Appendix G.2.

6.2.1 Study Results

Table 6-2: Voltage and Thermal Constraints Impacting MPC System and Attributable to MISO Projects

Scenario	Constraint	Limit	Result	Contingency	Project Impact	
					J511	J613
2024SuOp	Bison 345 kV Bus Voltage	0.92 pu	0.9005 pu Blown Case	CEII Redacted	0.038 pu	0.026 pu
2024SuOp	Square Butte–Stanton 230 kV	478 MVA	528.9 MVA	CEII Redacted	43.8%	-17.3%
2024SuOp	Center-Roughrider	614 MVA	625.8 MVA	CEII Redacted	27.1%	34.3%

6.2.2 Study Summary

The voltage constraint at Bison 345 kV can be mitigated by construction of two 55 MVAR switched capacitor banks at the Jamestown 345 kV substation and two 85 MVAR switched capacitor banks at the Bison 345 kV substation. A non-binding, planning level estimate of the cost of these is \$7,500,000. The cost of this upgrade would be borne by J511 and J613 at 59% and 41%, respectively.

The thermal constraint on Square Butte–Stanton 230 kV can be mitigated by replacing bus work at Square Butte. A non-binding, planning level estimate of the cost of this is \$50,000 and would be borne solely by J511.

The thermal constraint on Center-Roughrider 230 kV can be mitigated by raising structures to attain a higher conductor rating. A non-binding, planning level estimate of the cost of this is \$300,000 and would be borne by both J511 and J613 at 61.2% and 38.8%, respectively.

6.3 PJM Affected System Analysis

The PJM affected system analysis details (dated on 07/18/2018) can be found in Appendix G.2.

6.3.1 Study Results

6.3.1.1 Overload on Twin Branch–Argenta 345 kV line

To relieve the Twin Branch–Argenta 345 kV line overload:

- a. A sag check will be required for the ACSR ~ 954 ~ 45/7 ~ RAIL - Conductor Section 1 to determine if the line section can be operated above its emergency rating of 1409 MVA. Existing PJM Network Upgrade N5240.
- b. Additional Reinforcement on AEP end: Replace Twin Branch Wavetrap. PJM Network Upgrade: N5310.1. \$400,000
- c. Additional Reinforcement on MISO end: Replace substation structure at Argenta substation to accommodate larger wire: N5310.2. \$500,000

The cost allocation is as follows:

Queue	MW contribution	% of Cost	Cost (\$900 K)
J512	14.84	10.06	90,555
J541	23.10	15.66	140,959
J593	14.86	10.08	90,677
J594	17.87	12.12	109,045
J597	17.89	12.13	109,167
J414	12.14	8.23	74,080
J439	29.46	19.97	179,768
J598	17.33	11.75	105,750

6.3.2 Study Summary

Multiple projects in the MISO DPP 2016 August West Area group contribute loading to the overloads in the PJM system. These projects are responsible for the cost of Network Upgrades per PJM cost allocation rules.

6.4 SPP Affected System AC Contingency Analysis

Southwest Power Pool (SPP) conducted an Affected System Interconnection System Impact Study (ASISIS) to evaluate potential impacts to the SPP Transmission System related to the interconnection of generators on the Mid-Continent Independent System Operation (MISO) Transmission System.

Power flow analysis has indicated that for the power flow cases studied, the SPP Transmission System is impacted by the interconnection of the MISO DPP 2016 August West Phase 1 generation. These impacts are described in Appendix G.4. Steady state thermal and voltage constraintss can be mitigated by determinations listed in Section 5 of Appendix G.4. Within the scope of this ASISIS, SPP coordinated with associated SPP Transmission Owners and MISO Transmission Owners for planning level estimates for mitigation of constraints requiring transmission reinforcements and will require Affected System Facilities Study (ASFS) if the MISO Interconnection Request(s) impacting the SPP transmission system proceed forward.

The study results presented in this ASISIS are contingent upon completion of all SPP planned projects through 2017 ITPNT and DISIS-2016-001-1.

Potential mitigations for constraints identified in the SPP affected system are listed in Table 6-3.

The SPP affected system analysis results (Final REP-0328) for this study are in Appendix G.4.

Table 6-3: SPP Suggested Mitigations for DPP 2016 August West Area Study

Mitigation Required	TO	Constraint	Type	Cost Allocation	Cost Estimate
Reroute of Bismark to Washburn 230 kV circuit through Hilken 230kV station	WAPA	Bismark-Hilken 230 kV	ERIS	J593: \$3,500,000	\$3,500,000
Reconductor line	WAPA	Oahe-Sully Buttes 230 kV	NRIS	J302: \$343,280 J457: \$287,388 J503: \$339,882 J511: \$688,769 J593: \$606,719 J599: \$6,465,895 J607: \$422,385 J613: \$255,682	\$9,410,000
Reconductor line	WAPA	Whitlock-Sully Buttes 230 kV	NRIS	J302: \$392,917 J457: \$329,381 J503: \$389,027 J511: \$787,586 J593: \$690,390 J599: \$7,414,441 J607: \$483,659 J613: \$292,599	\$10,780,000
Reconductor line	WAPA	Whitlock-Glenham 230 kV	NRIS	J302: \$662,091 J457: \$555,029 J503: \$655,535 J511: \$1,327,133 J593: \$1,163,352 J599: \$12,493,816 J607: \$814,996 J613: \$493,048	\$18,165,000
New transformer to achieve higher MVA rating	BEPC	Broadland 345-230-13.8 kV xfmr	ERIS	J511: \$3,564,615 J593: \$5,849,103	\$9,413,718

Affected System Steady-State Analysis

Mitigation Required	TO	Constraint	Type	Cost Allocation	Cost Estimate
Build new 345 kV substations, reroute transmission lines, and reconductor line to achieve higher MVA rating	NPPD AECI KCPL	Cooper South interface ¹	ERIS	J476: \$81,907,429 J583: \$19,492,571	\$101,400,000
or Build new Cooper–Stranger Creek 345kV circuit				or J476: \$243,137,437 J583: \$57,862,563	or \$301,000,000
Increase Granite Falls 115 kV Capacitor from 2 steps of 30 Mvar to 5 steps of 30 Mvar (150 Mvar total)	WAPA	Low voltages near Granite Falls 230 kV	ERIS	J459: \$1,668,687 J596: \$1,068,727 J637: \$409,713 J638: \$852,873	\$4,000,000
Install 20 Mvar capacitor at Flandreau 69 kV	WAPA	Low voltage near Flandreau 69 kV radial lines serving load	ERIS	J459: \$110,522 J512: \$29,122 J569: \$28,874 J575: \$68,931 J577: \$70,862 J587: \$120,398 J596: \$33,381 J597: \$206,794 J599: \$16,262 J637: \$63,231 J638: \$131,623	\$880,000
Install 20 Mvar capacitor at Mars 69 kV	WAPA	Low voltage near Mars 69 kV radial lines serving load	ERIS	J414: \$43,292 J415: \$72,182 J439: \$323,684 J512: \$107,548 J569: \$36,019 J590: \$58,189 J594: \$239,086	\$880,000
Increase Belden 69 kV capacitor by 6 Mvar (1 step)	WAPA	Low voltage near Belden 69 kV	ERIS	J593: \$264,000	\$264,000

Affected System Steady-State Analysis

Mitigation Required	TO	Constraint	Type	Cost Allocation	Cost Estimate
Install 10 Mvar capacitor at Summit 69 kV	WAPA	Low voltages near Summit 69 kV	ERIS	J302: \$34,458 J457: \$70,168 J503: \$34,116 J511: \$106,166 J593: \$110,480 J607: \$58,636 J613: \$25,976	\$440,000

Note 1: If recommended mitigation is not valid, a new 345 kV line from Cooper to Stranger Creek may be an alternate solution.

6.5 AECl Affected System Analysis

Associated Electric Cooperative Inc. (AECl) performed this affected system analysis.

6.5.1 Study Summary

The AECl affected system analysis details (dated on 08/27/2018) can be found in Appendix G.5. The results are summarized below:

1. Based on the information received from MISO and results reviewed, the following AECl facilities were impacted by project J541:
 - 300106 5NOVELY_SW 161 300364 2NOVLTY_SW 69.0 1
 - 300113 5SRIVER 161 300339 5EMERSN 161 1
2. The AECl analysis did not determine additional impacts to those identified by MISO on the AECl system due to J541 or other select Study Cycle projects.

6.5.2 Network Upgrades Cost

The table below provides a non-binding, good faith estimate of the timing and cost estimates for upgrades needed as a result of J541.

Table 6-4: AECl Network Upgrades and Cost

Option / Description	Current Cost	Need Date	Year In Service
Replace Novelty 161/69 kV transformer to 84 MVA unit	\$2,500,000	2020	2020
Upgrade 600A disconnect switches at South River	\$100,000	2020	2020
Engineering	\$520,000	2020	2020
Contingencies	\$260,000	2020	2020
Total Cost	\$3,380,000		

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Stability Analysis

Stability analysis was performed to evaluate the transient stability and impact on the region of the generating facilities in the DPP 2016 August West Area group.

7.1 Procedure

7.1.1 Computer Programs

Stability analysis was performed using PSS[®]E revision 33.11.

7.1.2 Study Methodology

Study stability package representing 2022 summer shoulder (SH) conditions with generating facilities in the DPP 2016 August West Area group was created from the stability package used for the DPP 2016 February West Phase 2 study, which was originally developed from MTEP17 stability package. Benchmark case was created by removing the DPP West Area generating facilities from the study case. Disturbances were simulated to evaluate the transient stability and impact on the region of the generating facilities. If a simulation for the study case violates MISO transient stability criteria or local TOs' planning criteria, the simulation was repeated on the benchmark case to assess the impact of the generating facilities on the violation.

7.2 Case Development

7.2.1 Study Case

Study case representing 2022 shoulder (SH) condition was developed from the DPP 2016 February West Phase 2 stability package, which was originally developed from MTEP17 stability package.

The stability study case for DPP 2016 August study was created in the same procedure as the steady state models, as described in Section 2.1.1. Stage-3 stability case with MISO Backbone NUs and MWEX voltage stability NUs included was used in the stability study.

The interface transfer levels are summarized in Table 7-1.

Table 7-1: Interface Transfer Levels in Stage-3 2022 Stability Study Case

Interface	2022 SH Case (MW)
MHEX	492
MWEX	1397
Arrowhead – Stone Lake 345 kV	542

7.2.2 Benchmark Case

The DPP West Area generating facilities as described in Table A-1 (Appendix A.1) were removed from the study case. MISO Classic was used for power balance, where generation was scaled uniformly.

7.3 Disturbance Criteria

The stability simulations performed as part of this study considered all the regional and local contingencies listed in Table 7-2. Regional contingencies with the pre-defined switching sequences were selected from the MISO MTEP17 study; switching sequences for local contingencies were developed based on the generic clearing times shown in Table 7-3. The admittance for local single line-to-ground (SLG) faults were estimated by assuming that the Thevenin impedance of the positive, negative and zero sequence networks at the fault point are equal.

Table 7-2: Regional and Local Disturbance Descriptions

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Table 7-3: Generic Clearing Time Assumption

Voltage Level (kV)	Primary Clearing Time (cycle)	Backup Clearing Time (cycle)
345 kV	4	11
230 kV	5	13
161/138 kV	6	18
115 kV	6	20
69 kV	8	24

7.4 Performance Criteria

All generators must mitigate the stability constraints listed below in order to obtain any type of Interconnection Service:

- System instability
- Transient voltage constraint
- Damping violation

7.4.1 MISO Criteria

Stability simulation results are evaluated based on the following MISO criteria:

- All on-line generating units are stable
- No unexpected generator tripping
- Post-fault transient voltage limits: 1.2 per unit maximum, 0.7 per unit minimum.
- Per local TOs' planning criteria, specific transient voltage limits are applied to specific buses, areas or companies that have different requirements.
- All machine rotor angle oscillations must be positively damped with a minimum damping ratio of 0.81633% for disturbances with a fault or 1.6766% for line trips without a fault.

A bus is considered a transient voltage constraint if both of the following conditions are met. All transient voltage constraints must be resolved before a project can receive interconnection service.

1. the bus transient voltage is outside of specified transient voltage limits during transient period, and
2. the bus voltage is at least 0.01 per unit worse than the benchmark case voltage for the same contingency.

7.4.2 Local Planning Criteria

7.4.2.1 ATC Local Planning Criteria

ATC has the following local transient voltage recovery criteria. For facilities in the ATC footprint, transient voltage recovery is evaluated based on ATC's local planning criteria.

- Voltage recovery within 80 percent and 120 percent of nominal for between 2 and 20 seconds following the clearing of a disturbance.

7.4.2.2 ITCM Local Planning Criteria

ITCM has the following local transient voltage and damping criteria. For facilities in the ITCM footprint, transient voltages and dampings are evaluated based on ITCM's local planning criteria.

- Voltages at all busses on the Transmission Systems should not drop below 0.70 per unit after the first swing for more than 5 cycles. The duration for the minimum voltage dip starts after the first swing post clearing of fault.
- Voltage at all Transmission System buses should recover to the applicable post-contingency steady-state voltage level (ITCM post-disturbance limits in Table 2-3), within 1.0 second of the clearing of the fault.
- Rotor angle oscillation damping ratios are not to be less than 0.03.

7.4.2.3 MEC Local Planning Criteria

MEC has the following local transient voltage and damping criteria. For facilities in the MEC footprint, transient voltages and dampings are evaluated based on MEC's local planning criteria.

- Generator bus transient voltage limits shall adhere to the high voltage duration and low voltage duration curve in Attachment 2 of NERC PRC-024, which is:
 - o Generator bus transient over voltage limits (after fault clearing): 1.2 pu voltage from 0.0 to and including 0.2 s; 1.175 pu voltage from 0.2 to and including 0.5 s; 1.15 pu voltage from 0.5 to and including 1.0 s; 1.1 pu voltage for greater than 1.0 s.
 - o Generator bus transient low voltage limits (after fault clearing): may be less than 0.45 pu voltage from 0 to 0.15 seconds; Voltage shall remain above 0.45 pu from 0.15 to 0.3 s; Voltage shall remain above 0.65 pu from 0.3 to 2.0 s; Voltage shall remain above 0.75 pu from 2.0 to 3.0 s; Voltage shall recover to 0.9 pu after 3 s.
- Load bus transient voltage limits:
 - o Load bus transient over voltage limits (after fault clearing): 1.6 pu voltage from 0.01 to and including 0.04 s; 1.2 pu voltage from 0.04 to and including 0.5 s; 1.1 pu voltage from 0.5 to and including 5 s; and 1.05 pu voltage for greater than 5 s. These voltage limits also apply to buses without loads or generators.
 - o Load bus transient low voltage limits (after fault clearing): may be less than 0.7 pu voltage from 0 to 2 s; Voltage shall remain above 0.7 pu from 2 to 20 s; Voltage shall recover to 0.9 pu after 20 s.
- Angular transient stability minimum damping ratio (ζ) should not be less than 0.03.

7.5 Stability Results

The contingencies listed in Table 7-2 were simulated using the 2022 summer shoulder Stage-3 study case. If a transient stability criteria violation was identified, the same disturbance was repeated in the benchmark case.

Appendix H.2 contains plots of generator rotor angles, generator power output, generator terminal voltages, bus voltages, and branch flows for each simulation. Simulations were performed with a 2.0 seconds steady-state run followed by the appropriate disturbance. Simulations were run for a 12-second duration.

Stability study results summary is in Appendix H, Table H-1. The following stability related issues were identified.

7.5.1 Slow Voltage Recovery at ITCM 345 kV Buses

Per ITCM LPC, voltage at all ITCM transmission system buses should recover to the applicable post-contingency steady-state voltage level (ITCM post-disturbance limits in Table 2-3) within 1.0 second of the clearing of the fault. Under disturbances in Table 7-4, voltages at

several ITCM buses are still below post-contingency steady-state voltage limit (0.93 pu) after 1.0 second with the fault cleared.

Voltage constraints at ITCM buses were also identified in the steady state analysis. With addition of the reactive power NUs (Table 4-16) required in the steady state analysis, no stability violations were identified under these disturbances (Table 7-4).

Table 7-4: Disturbances Causing Slow Voltage Recovery at ITCM Buses

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7.5.2 Out of Step (OOS) Relay at Forbes on M602F Tripping

The OOS relay at Forbes on M602F tripped for 3-phase fault '0890_w_xel_p12_pc3_at_king-eauclaire' (Table 7-5) in both study case and benchmark case. This OOS trip is initiated prior to fault clearing. During the fault OOS relay can operate if the apparent impedance initially remains between blinders long enough for the timer to time out and then moves into the trip area before the fault is cleared (Figure 7-1). M602F OOS relay settings may need to be adjusted when the GNTL 500 kV project goes into service.

The 3-phase fault '0890_w_xel_p12_pc3_at_king-eauclaire' was simulated again by blocking the OOS relay at Forbes on M602F to keep it from operating during the fault. No stability violations were identified.

Table 7-5: Disturbance Causing OOS Relay at Forbes on M602F Tripping

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Figure 7-1: Forbes M602F OOS Relay Apparent Impedance under Disturbance "0890_w_xel_p12_pc3_at_king-eauclaire"

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7.5.3 Coal Creek 41G GSU DS Zone 1 Relay Tripping

Distance Zone 1 relay (DISTR model) on the Coal Creek 41G generator step up transformer (GSU) took tripping action at 1 cycle after the 3PH fault at Stanton 230 kV bus (Table 7-6) was initiated. The relay cross-tripped the Coal Creek – Stanton 230 kV and Coal Creek – Underwood 230 kV lines. The Coal Creek 41G DS Zone 1 relay tripping occurs in both study and benchmark cases. GRE reviewed and confirmed that the Coal Creek Isolation SPS is no longer valid. The faults in Table 7-6 were simulated again by disabling the Coal Creek Isolation SPS. No stability violations were identified.

Table 7-6: Coal Creek 41G GSU DS Zone 1 Relay Tripping

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7.5.4 Tripping of Wind Farms by Low Voltage Protection

Under the disturbances listed in Table 7-7, wind farms were tripped by their low voltage protections. The fault “1672_w_otp_p55_jamestown” is cleared after 5.1 second, which causes Court WTG and J613 WTG (in OTP area) tripping during the fault by their low voltage protection. Network Upgrade is not required for DPP West project.

The Tatanka WTG was tripped after the fault “2155_w_MDU_P42_Heskett-Wishek_Bkr7149” was cleared due to post-disturbance voltage below 0.9 pu. With addition of the reactive power NUs (Table 4-16) required in the steady state analysis, the Tatanka WTG was not tripped any more.

Table 7-7: Disturbances Causing Wind Farm Low Voltage Protection Tripping

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7.5.5 Transient Voltage Violations under CUDC Permanent Bipole Fault

MISO reactive power NUs (Table 4-16) required in the steady state analysis were added in the model. Under the CUDC permanent bipole fault listed in Table 7-8, stability simulation is stable but with 37 transient voltage violations (Table 7-9).

Table 7-8: CUDC Permanent Bipole Fault

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Table 7-9: Transient Voltage Violations under CUDC Permanent Bipole Fault

Bus #	Bus Name	Voltage Level	Worst Voltage	Area	Voltage Limit
620229	WAHPETN7	115	0.756	OTP	0.8
620238	WINGER 7	115	0.689	OTP	0.7
620239	BAGLEY 7	115	0.686	OTP	0.7
620240	MN PIPE7	115	0.678	OTP	0.7
620241	CLEARBR7	115	0.678	OTP	0.7
620242	CLBKPIP7	115	0.678	OTP	0.7
620249	CRKSTON7	115	0.684	OTP	0.7
620250	FERTILE7	115	0.684	OTP	0.7
620251	PLUMTAP7	115	0.666	OTP	0.7
620252	PLUMMER7	115	0.666	OTP	0.7
620253	PLUMPIP7	115	0.665	OTP	0.7
620254	VIKING 7	115	0.667	OTP	0.7

Bus #	Bus Name	Voltage Level	Worst Voltage	Area	Voltage Limit
620284	OSLO TN7	115	0.678	OTP	0.7
657713	TRFALLS7	115	0.665	OTP	0.7
657714	WARSAW 7	115	0.682	OTP	0.7
657722	OSLO 7	115	0.679	OTP	0.7
657723	GREENWD7	115	0.664	OTP	0.7
657724	ALVARAD7	115	0.675	OTP	0.7
657725	WARREN 7	115	0.669	OTP	0.7
657726	ANDERNW7	115	0.663	OTP	0.7
657752	DRAYTON4	230	0.740	OTP	0.8
657758	WINGER 4	230	0.670	OTP	0.7
657761	BAGLEYW4	230	0.691	OTP	0.7
657946	PRAIRIE3	345	0.668	OTP	0.7
620149	KARLSTA7	115	0.696	OTP	0.7
620244	ITASCA 7	115	0.695	OTP	0.7
620336	AUDUBON4	230	0.696	MP	0.7
652430	EGF IND7	115	0.698	WAPA	0.7
652437	GRNDFKS4	230	0.693	WAPA	0.7
652443	GRNDFKS7	115	0.695	WAPA	0.7
657706	FALCONR7	115	0.695	OTP	0.7
657718	HALMA 7	115	0.699	OTP	0.7
657727	RICELKT7	115	0.698	OTP	0.7
657728	MPLLPRT7	115	0.692	OTP	0.7
657729	RICELAK7	115	0.698	OTP	0.7
657755	PRAIRIE4	230	0.695	OTP	0.7
657780	LAKEPK 4	230	0.695	OTP	0.7

Per MPC affected system study, network upgrades of two 55 MVAR switched capacitor banks at the Jamestown 345 kV substation and two 85 MVAR switched capacitor banks at the Bison 345 kV substation are required. With the addition of the MPC NUs, and (0, 200) Mvar SVC at Winger 230 kV bus, all the transient voltage violations (Table 7-9) can be mitigated.

7.6 Network Upgrades Identified in Stability Analysis

Additional Network Upgrades required for mitigating constraints identified in the stability analysis are listed in Table 7-10.

Table 7-10: Additional Stability NUs

Network Upgrades	Owner	Cost (\$)
(0, 200) Mvar SVC at Winger 230 kV (bus #: 657758)	OTP	\$31,030,000

MWEX Voltage Stability Study

ATC performed steady state voltage stability analysis. Voltage stability analysis is required to determine if the initial conditions of the DPP system models under study are in a stable state as defined by Power-Voltage (PV) curves of the Minnesota Wisconsin Export Interface (MWEX) for loss of the worst contingency.

The voltage stability analysis used 2022 shoulder load cases to compare the Pre-DPP without Network Upgrades and Post-DPP with the Backbone Network Upgrades. The MISO Backbone Network Upgrades are the following projects.

1. Construction of a J510 – Hazel Creek – Scott County 345 kV line;
2. Construction of a Big Stone South – Alexandria 345 kV line;
3. Construction of a Beverly – Sub 92 345 kV line;
4. Construction of a Franklin – Morgan Valley 345 kV line;
5. Construction of a Franklin – Webster 345 kV line; and
6. Installation of 2x150 MVAR capacitors at Franklin 345 kV.

As shown in Table 8-1, the Post-DPP + Backbone Network Upgrades scenario in the 2022SH case is not voltage stable.

ATC recommends that either of the following Network Upgrades be installed. Either alternative will result in the Post-DPP case that meets ATC's voltage stability criteria.

MWEX Network Upgrade Alternative #1 (NU #1), North Rochester – South Rocky Run 345 kV line:

1. Construct all Backbone Network Upgrades.
2. Construct a new South Rocky Run 345 kV switching station with one 75 MVAR shunt capacitor. The station will be configured to eliminate P2 contingencies that result in the loss of two or more 345 kV lines.
3. Construct a new North Rochester – South Rocky Run 345 kV line. This line will be constructed such that no common mode failures will exist with it and any other 345 kV lines (i.e. no P2-P7, excluding P6, contingencies are allowed to be created).
4. Construct a new 161/69 kV substation in Minnesota connecting the Wabaco – Alma 161 kV line to the Dairyland 69 kV system. This is required for the North Rochester – South Rocky Run 345 kV line to utilize the existing 161 kV position on CapX Alma transmission structures crossing the Mississippi. The final configuration will result in the 345 kV circuits being separated by the 161 kV circuit such that there won't be a P7.1 contingency that causes an outage of both 345 kV circuits.
5. Install redundant bus differential relays at the Rocky Run 345 kV Substation.
6. Install 150 MVAR of shunt capacitance at the North Rochester 345 kV Substation.

7. Install a 35% series capacitor at the Briggs Road 345 kV Substation on the Briggs Road – North Madison 345 kV line.
8. Install one breaker at the Briggs Road 345 kV Substation.
9. Additional facilities, e.g. Forest Junction – Elkhart Lake 138 kV, are expected to be overloaded by this alternative and will need to be assigned Network Upgrades.

MWEX Network Upgrade Alternative #2 (NU #2), J510 – Huntley – Adams – Hickory Creek – Sub 91 345 kV line:

1. Construct all Backbone Network Upgrades, except the J510 – Hazel Creek – Scott County 345 kV line.
2. Construct a new J510 – Huntley 345 kV line.
3. Construct a new Huntley – Adams 345 kV line.
4. Construct a new Adams – Hickory Creek 345 kV line. This line will be constructed such that no common mode failures will exist with it and any portion of Hickory Creek – Hazleton 345 kV (i.e. no P2-P7, excluding P6, contingencies are allowed to be created).
5. Construct a new Hickory Creek – Sub 91 345 kV line.
6. Install one 50 MVAR shunt capacitor at Hazleton 345 kV Substation.
7. Install one 75 MVAR shunt capacitor at Adams 345 kV Substation.
8. Install one 50 MVAR shunt capacitor at Hickory Creek 345 kV Substation.

Table 8-1: MWEX Margins to Collapse in the 2022SH Cases

	Real Power Flow (MW)							
	AHD-SLK ¹	MWEX				Margin to Nose ²		
Case	N-0 Initial Condition	N-0 I.C. ³	N-1 I.C. ³	N- 1 I.C. After Phase Shift ⁴	N-1 Nose	(MW)	(%)	Notes
Pre-DPP without Backbone NUs	453	1201	579	N/A	697	118	17	Voltage Stable Sufficient Margin ⁵ Vnose > Vmin ⁶ (0.963 > 0.950)
Post-DPP with Backbone NUs	640	1634	N/A: N-1 Does Not Converge ⁷					Not Voltage Stable
Post-DPP with MWEX NU #1	537	1389	630	N/A	732	102	14	Voltage Stable Sufficient Margin ⁵
Post-DPP with MWEX NU #2	597	1525	739	699	777	78	10	Voltage Stable Sufficient Margin ⁵ Vnose > Vmin ⁶ (0.959 > 0.950)

Notes:

1. As described in the active MWEX Operating Guide, the AHD-SLK interface is a single element PTDF interface measured at the Minnesota Power 230 kV side of the Arrowhead 230 kV phase shifter.
2. Margin to Nose (MW) is defined as the MWEX N-1 Nose minus the N-1 Initial Condition.
3. Initial Condition flows were measured in the base cases with the worst contingency plus operation of various control systems as needed with all transformer taps, switched shunts, and PARs locked.
4. Arrowhead PAR modeled as changing from neutral tap to a maximum of the 14th tap in the retard direction.
5. ATC Planning Criteria requires a 10% voltage stability margin.
6. ATC Planning Criteria requires $V_{nose} < V_{min}$.
7. N-1 measurements can't be made because the worst contingency doesn't converge.

The results of this analysis show that the Post-DPP scenario in the 2022 shoulder load cases is not voltage stable because the initial condition doesn't converge with the worst contingency. With either of the proposed MWEX Network Upgrade Alternatives, the Post-DPP case is voltage stable and meets ATC's voltage stability criteria.

The MWEX voltage stability study details can be found in Appendix I.

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Short Circuit Analysis

9.1 Introduction

Siemens PTI and several transmission owning companies performed short circuit analysis for the DPP 2016 August West Area generation projects.

9.2 J302 Short Circuit Study Performed by Siemens PTI

The J302 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 6,492 A (increased by 469 A) and SLG fault current is 4,376 A (increased by 136 A) at the J302&503 POI 230 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J302 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.1.

9.3 J414 Short Circuit Study Performed by ITCM

The J414 short circuit study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J414. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.2.

9.4 J415 Short Circuit Study Performed by MEC

The J415 short circuit study was performed by MEC. The study results show that the changes in fault current at buses more than a couple of buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project.

Study details can be found in Appendix J.3.

9.5 J439 Short Circuit Study Performed by MEC

The J439 short circuit study was performed by MEC. The study results show that the changes in fault current at buses more than a couple of buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project.

Study details can be found in Appendix J.4.

9.6 J457 Short Circuit Study Performed by Siemens PTI

The J457 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 7,292 A (increased by 1,024 A) and SLG fault current is 7,865 A (increased by 1,634 A) at the Merricourt 230 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J457 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.5.

9.7 J459 Short Circuit Study Performed by OTP

The J459 short circuit study was performed by OTP.

With the proposed project J459 addition, the fault currents are estimated to be roughly 15.3 kA at the Big Stone South 345 kV bus, 22.7 kA at the Big Stone South 230 kV bus, and 13 kA at the Big Stone South 34.5 kV bus. Based on the short circuit analysis performed, the fault current ratings of the Transmission Owner's equipment at the Big Stone South Substation are not exceeded and there are no upgrades required. The Transmission Owner did not evaluate any impacts on the fault current levels at substations owned by other Transmission Owners.

Study details can be found in Appendix J.6.

9.8 J476 Short Circuit Study Performed by MEC

The J476 short circuit study was performed by MEC. The study results show that the changes in fault current at buses more than a couple of buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project.

Study details can be found in Appendix J.7.

9.9 J503 Short Circuit Study Performed by Siemens PTI

The J503 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 6,492 A (increased by 456 A) and SLG fault current is 4,376 A (increased by 132 A) at the J302&503 POI 230 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J503 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.8.

9.10 J511 Short Circuit Study Performed by GRE

The J511 short circuit study was performed by GRE. Fault currents were calculated before and after the addition of J511's 200 MW of wind generation. The results show that none of the circuit breaker interrupting capabilities at Stanton or Coal Creek Substations will be exceeded after the addition of J511.

Study details can be found in Appendix J.9.

9.11 J512 Short Circuit Study Performed by Xcel

The J512 short circuit study was performed by Xcel. Based on the results of the study, the fault current ratings of the Transmission Owner's equipment in the area will not be exceeded. With the proposed J512 generation modeled the fault currents in the area are below 10 kA for 69 kV busses and approximately 12 kA for 115 kV and 345 kV busses. All of the Transmission Owner equipment is rated at 31.5kA or greater. No short circuit upgrades are required in this area due to the proposed additional generation.

Study details can be found in Appendix J.10.

9.12 J541 Short Circuit Study Performed by Ameren

The J541 short circuit study was performed by Ameren. The study results show that no circuit breaker upgrades are required on the Ameren System based on the expected fault contribution of the J541 and J598 generation. Breaker ratings for MEC and AECI would be required in order to determine if ratings are exceeded for their facilities in this study. However, since the incremental increase in fault current is greater than 5% for only the Ottumwa 345 kV bus, this would be the only facility where the customers would be required to provide mitigation if necessary.

Study details can be found in Appendix J.11.

9.13 J555 Short Circuit Study Performed by MEC

The J555 short circuit study was performed by MEC. The study results show that the changes in fault current at buses more than a couple of buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project.

Study details can be found in Appendix J.12.

9.14 J569 Short Circuit Study Performed by Xcel

The J569 short circuit study was performed by Xcel. Based on the results of the study, the fault current ratings of the Transmission Owner's equipment in the area will not be exceeded. With the proposed J569 generation modeled the fault currents in the area are below 15 kA for 161 kV and 115 kV busses with the exception of the Anson 115 kV bus which was approximately 37 kA. All of the Transmission Owner equipment is rated at 40 kA or greater and the 115 kV equipment at Anson is rated at 63 kA. No short circuit upgrades are required in this area due to the proposed additional generation.

Study details can be found in Appendix J.10.

9.15 J575, J577, J587, and J597 Short Circuit Study Performed by Xcel

The J575 (100 MW wind), J577 (102.8 MW Wind), J587 (200 MW wind) and J597 (300MW wind) short circuit study was performed by Xcel to determine the impact of the cumulative proposed 702.8 MW wind generation interconnecting at the 345 kV Brookings County substation and on the 345 kV Transmission Line between Brookings County and Hawks Nest Lake substations. Based on the results of the study, the fault current ratings of the

Transmission Owner's equipment in the area will not be exceeded. With the proposed J575, J577, J587 and J597 generation modeled the fault currents in the area are below 30 kA for 115 kV and 345 kV busses. All of the Transmission Owner equipment is rated at 40 kA or greater. No short circuit upgrades are required in this area due to the proposed additional generation.

Study details can be found in Appendix J.10.

9.16 J583 Short Circuit Study Performed by MEC

The J583 short circuit facilities study was performed by MEC. The study results show that the changes in fault current at buses more than a couple of buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project.

Study details can be found in Appendix J.13.

9.17 J590 Short Circuit Study Performed by MEC

The J590 short circuit facilities study was performed by MEC. The study results show that the changes in fault current at buses more than a couple of buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project.

Study details can be found in Appendix J.14.

9.18 J593 Short Circuit Study Performed by Siemens PTI

The J593 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 6,933 A (increased by 1,324 A) and SLG fault current is 7,086 A (increased by 1,497 A) at the Tioga 230 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J593 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.15.

9.19 J594 Short Circuit Study Performed by ITCM

The J594 short circuit facilities study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J594. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.16.

9.20 J596 Short Circuit Study Performed by GRE

The J596 short circuit facilities study was performed by GRE. Fault currents were calculated before and after the addition of J596's 100 MW of wind generation. The results show that none of the circuit breaker interrupting capabilities at nearby substations will be exceeded after the addition of J596.

Based on the results shown in the below table, the short-circuit ratio at the POI for Project J596 (100 MW wind) could be as low as 2.14.

Min Fault Current (Tap Bus-Morris 115kV Open, All Local Generation Offline)

Location	Fault Type	Short Circuit Current (A)	ANSI X/R	MVA
Morris-Walden Tap 115kV Bus	1LG	1,310	3.90	261.1
	3LG	1,075	3.82	214.3

Study details can be found in Appendix J.17.

9.21 J598 Short Circuit Study Performed by Ameren

The J598 short circuit facilities study was performed by Ameren. The study results show that no circuit breaker upgrades are required on the Ameren System based on the expected fault contribution of the J541 and J598 generation. Breaker ratings for MEC and AECI would be required in order to determine if ratings are exceeded for their facilities in this study. However, since the incremental increase in fault current is greater than 5% for only the Ottumwa 345 kV bus, this would be the only facility where the customers would be required to provide mitigation if necessary.

Study details can be found in Appendix J.11.

9.22 J599 Short Circuit Study Performed by Siemens PTI

The J599 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 5,099 A (increased by 1,256 A) and SLG fault current is 5,751 A (increased by 1,767 A) at the Glenham 230 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J599 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.18.

9.23 J607 Short Circuit Study Performed by Siemens PTI

The J607 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 6,543 A (increased by 985 A) and SLG fault current is 6,114 A (increased by 1,796 A) at the J607 POI 230 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J607 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.19.

9.24 J611 Short Circuit Study Performed by MEC

The J611 short circuit facilities study was performed by MEC. The study results show that the changes in fault current at buses more than a couple of buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project.

Study details can be found in Appendix J.20.

9.25 J613 Short Circuit Study Performed by OTP

The J613 short circuit study was performed by OTP.

With the proposed projects additions, the fault currents are roughly 15.6 kA at the Jamestown 115 kV bus, 6.4 kA at the Jamestown 345 kV bus, 9.3 kA at the Jamestown 41.6 kV bus. Based on the short circuit analysis performed, the fault current ratings of the Transmission Owner's equipment in the area are not exceeded and there are no upgrades required. The Transmission Owner did not evaluate any impacts on the fault current levels at substations owned by other Transmission Owners.

Study details can be found in Appendix J.6.

9.26 J614 Short Circuit Study Performed by DPC

The J614 short circuit facilities study was performed by DPC. The short circuit study concludes the fault current rating of the existing Rice Substation equipment has the capabilities to withstand the additional generation.

Study details can be found in Appendix J.21.

9.27 J615 Short Circuit Study Performed by MEC

The J615 short circuit facilities study was performed by MEC. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J615 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.22.

9.28 J637 Short Circuit Study Performed by OTP

The J637 short circuit study was performed by OTP.

With the proposed projects additions, the fault currents are roughly 15.5 kA at the Big Stone South 345 kV bus, 22.9 kA at the Big Stone South 230 kV bus, 13 kA at the Big Stone South 34.5 kV bus. Based on the short circuit analysis performed, the fault current ratings of the Transmission Owner's equipment at the Big Stone South Substation are not exceeded and there are no upgrades required. The Transmission Owner did not evaluate any impacts on the fault current levels at substations owned by other Transmission Owners.

Study details can be found in Appendix J.6.

9.29 J638 Short Circuit Study Performed by OTP

The J638 short circuit study was performed by OTP.

With the proposed projects additions, the fault currents are roughly 15.7 kA at the Big Stone South 345 kV bus, 23.1 kA at the Big Stone South 230 kV bus, 13.2 kA at the Big

Stone South 34.5 kV bus. Based on the short circuit analysis performed, the fault current ratings of the Big Stone South Substation are not exceeded and there are no upgrades required. The Transmission Owner did not evaluate any impacts on the fault current levels at substations owned by other Transmission Owners.

Study details can be found in Appendix J.6.

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Deliverability Study

10.1 Project Description

Interconnection requests requesting Network Resource Interconnection Services (NRIS) were considered for deliverability analysis.

10.2 Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted Network Resource Interconnection Services (NRIS).

If the generator is determined as not fully deliverable, the customer can choose either to change his project to an Energy Resource (ER) project or proceed with the system upgrades that will make the generator fully deliverable.

Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottlenecked up. The wind generators are tested at 100 % of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

10.3 Study Methodology

MISO Generator Deliverability Study whitepaper describing the algorithm can be found at https://cdn.misoenergy.org/Generator_Deliverability_Study_Methodology108139.pdf.

10.4 Determining the MW restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study is in the “Top 30 DF List” (see white paper for detail), part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

The generator is also responsible for any NEW base case (pre-shift) overload or NEW “severe yet credible dispatch overload” where the generator is not in the “Top 30 DF List”, if the generator’s DF is greater than 5%. Please see white paper for detail. The formula above also applies to these situations.

10.5 2022 Deliverability Study Result

10.5.1 J302

J302 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Stanton-Square Butte 230 kV	4.30	6.95%	Yes	J511	J302,J503,J607, J613		\$10,875,000
Merricourt-Ellendale 230 kV	11.00	39.56%	Yes ¹	J302,J457,J503 ,J607	J302,J457,J503, J511,J593,J607		\$20,500,000
Foxtail-Ellendale 230 kV	25.10	40.34%	No		J302,J457,J503, J511,J607,J613		Incl.
Foxtail-Tatanka North 230 kV	38.60	40.35%	No		J302,J457,J503, J511,J607,J613		Incl.
J607 POI-Wishek 230 kV	44.50	43.09%	Yes ¹	J302, J503,J607	J302,J503,J511, J593,J607		\$400,000
Merricourt-Wishek 230 kV	54.20	43.09%	Yes	J302, J503,J607	J302,J503,J511, J593,J607		\$6,000,000
J302&J503 POI-Heskett 230 kV	77.10	100.00%	Yes	J302,J457,J503 ,J607	J302,J457,J503, J607		\$9,000,000
Merricourt-Tatanka North 230 kV	93.60	40.47%	No		J302,J457,J503, J511,J593,J607		\$900,000
Whitlock-Glenham 230 kV	101.20	7.43%	Yes	J599	J302,J503,J511, J593,J599,J607, J613		\$10,000

Note 1: ERIS NU is not sufficient.

10.5.2 J414

J414 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	120 MW (100%)
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10.5.3 J415

J415 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	200 MW (100%)
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10.5.4 J439

J439 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	333.8 MW (66.75%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
J506 POI-Raun 345 kV	500.00	41.19%	Yes	J439,J590	J439,J512,J594, J590		\$700,000

10.5.5 J457

J457 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	6.4 MW (4.24%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Merricourt-Ellendale 230 kV	16.30	66.54%	Yes ¹	J302,J457,J503 ,J607	J302,J457,J503, J511,J593,J607		\$20,500,000
Foxtail-Ellendale 230 kV	37.10	67.85%	No		J302,J457,J503, J511,J607,J613		Incl.
Foxtail-Tatanka North 230 kV	80.40	67.86%	No		J302,J457,J503, J511,J607,J613		Incl.
Merricourt-Tatanka North 230 kV	126.80	67.98%	No		J302,J457,J503, J511,J593,J607		\$900,000
J302&J503 POI-Heskett 230 kV	150.00	39.43%	Yes	J302,J457,J503 ,J607	J302,J457,J503, J607		\$9,000,000

Note 1: ERIS NU is not sufficient..

10.5.6 J459

J459 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	200 MW (100%)
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10.5.7 J476

J476 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	246 MW (100%)
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10.5.8 J503

J503 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Stanton-Square Butte 230 kV	4.20	6.95%	Yes	J511	J302,J503,J607,J613		\$10,875,000
Merricourt-Ellendale 230 kV	10.70	39.56%	Yes ¹	J302,J457,J503,J607	J302,J457,J503,J511,J593,J607		\$20,500,000
Foxtail-Ellendale 230 kV	24.50	40.34%	No		J302,J457,J503,J511,J607,J613		Incl.
Foxtail-Tatanka North 230 kV	37.50	40.35%	No		J302,J457,J503,J511,J607,J613		Incl.
J607 POI-Wishek 230 kV	43.40	43.09%	Yes ¹	J302,J503,J607	J302,J503,J511,J593,J607		\$400,000
Merricourt-Wishek 230 kV	53.00	43.09%	Yes	J302,J503,J607	J302,J503,J511,J593,J607		\$6,000,000
J302&J503 POI-Heskett 230 kV	75.30	100.00%	Yes	J302,J457,J503,J607	J302,J457,J503,J607		\$9,000,000
Merricourt-Tatanka North 230 kV	91.40	40.47%	No		J302,J457,J503,J511,J593,J607		\$900,000
Whitlock-Glenham 230 kV	98.80	7.43%	Yes	J599	J302,J503,J511,J593,J599,J607,J613		\$10,000

Note 1: ERIS NU is not sufficient.

10.5.9 J511

J511 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)		0 MW (0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Sheyenne-Mapleton 115 kV	8.50	5.86%	Yes	J511,J613	J511,J613		\$4,650,000
Merricourt-Ellendale 230 kV	21.70	7.60%	Yes ¹	J302,J457,J503,J607	J302,J457,J503,J511,J593,J607		\$20,500,000
Foxtail-Ellendale 230 kV	39.80	7.75%	No		J302,J457,J503,J511,J607,J613		Incl.
Foxtail-Tatanka North 230 kV	76.00	7.76%	No		J302,J457,J503,J511,J607,J613		Incl.
J607 POI-Wishek 230 kV	87.90	8.38%	Yes ¹	J302,J503,J607	J302,J503,J511,J593,J607		\$400,000
Merricourt-Wishek 230 kV	103.80	8.38%	Yes	J302,J503,J607	J302,J503,J511,J593,J607		\$6,000,000
Coyote 345-115 kV xfmr	107.20	7.82%	No		J511,J613		\$3,000,000
Coyote-Beulah 115 kV	152.40	5.36%	No		J511,J613		\$250,000
Merricourt-Tatanka North 230 kV	185.00	7.88%	No		J302,J457,J503,J511,J593,J607		\$900,000
Whitlock-Glenham 230 kV	200.00	7.92%	Yes	J599	J302,J503,J511,J593,J599,J607,J613		\$10,000

Note 1: ERIS NU is not sufficient..

10.5.10 J512

J512 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)		208.4 MW (83.38%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
J506 POI-Raun 345 kV	250.00	10.28%	Yes	J439,J590	J439,J512,J594,J590		\$700,000

10.5.11 J541

J541 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	400 MW (100%)
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10.5.12 J555

J555 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	140 MW (100%)
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10.5.13 J569

J569 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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10.5.14 J575

J575 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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10.5.15 J577

J577 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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10.5.16 J583

J583 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	200 MW (100%)
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10.5.17 J587

J587 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	200 MW (100%)
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10.5.18 J590

J590 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	60.1 MW (66.76%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
J506 POI-Raun 345 kV	90.00	41.19%	Yes	J439,J590	J439,J512,J594,J590		\$700,000

10.5.19 J593

J593 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	85.1 MW (38.00%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
J607 POI-Wishek 230 kV	98.40	7.13%	Yes ¹	J302,J503,J607	J302,J503,J511,J593,J607		\$400,000
Merricourt-Wishek 230 kV	143.00	7.13%	Yes	J302,J503,J607	J302,J503,J511,J593,J607		\$6,000,000
Merricourt-Tatanka North 230 kV	207.20	6.70%	No		J302,J457,J503,J511,J593,J607		\$900,000
Whitlock-Glenham 230 kV	215.20	7.01%	Yes	J599	J302,J503,J511,J593,J599,J607,J613		\$10,000
Merricourt-Ellendale 230 kV	224.00	6.56%	Yes ¹	J302,J457,J503,J607	J302,J457,J503,J511,J593,J607		\$20,500,000

Note 1: ERIS NU is not sufficient..

10.5.20 J594

J594 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Lakefield-Dickinson 161 kV	200.30	11.26%	No		J594		\$390,000
J506 POI-Raun 345 kV	300.00	14.32%	Yes	J439,J590	J439,J512,J594, J590		\$700,000

10.5.21 J596

J596 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Hoot Lake-Fergus Falls 115 kV	100.00	9.67%	No		J596		\$55,000

10.5.22 J597

J597 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	300 MW (100%)
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10.5.23 J598

J598 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	300 MW (100%)
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10.5.24 J599

J599 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	185.0 MW (92.50%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Whitlock-Glenham 230 kV	200.00	54.62%	Yes	J599	J302,J503,J511,J593,J599,J607,J613		\$10,000

10.5.25 J607

J607 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Stanton-Square Butte 230 kV	6.40	5.74%	Yes	J511	J302,J503,J607,J613		\$10,875,000
Merricourt-Ellendale 230 kV	16.30	50.17%	Yes ¹	J302,J457,J503,J607	J302,J457,J503,J511,J593,J607		\$20,500,000
Foxtail-Ellendale 230 kV	37.10	51.15%	No		J302,J457,J503,J511,J607,J613		Incl.
Foxtail-Tatanka North 230 kV	57.00	51.17%	No		J302,J457,J503,J511,J607,J613		Incl.
J607 POI-Wishek 230 kV	65.90	54.61%	Yes ¹	J302, J503,J607	J302,J503,J511,J593,J607		\$400,000
Merricourt-Wishek 230 kV	80.40	54.61%	Yes	J302, J503,J607	J302,J503,J511,J593,J607		\$6,000,000
J302&J503 POI-Heskett 230 kV	114.30	100.00%	Yes	J302,J457,J503,J607	J302,J457,J503,J607		\$9,000,000
Merricourt-Tatanka North 230 kV	138.70	51.28%	No		J302,J457,J503,J511,J593,J607		\$900,000
Whitlock-Glenham 230 kV	150.00	6.36%	Yes	J599	J302,J503,J511,J593,J599,J607,J613		\$10,000

Note 1: ERIS NU is not sufficient..

10.5.26 J611

J611 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	110 MW (100%)
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10.5.27 J613

J613 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Stanton-Square Butte 230 kV	0.00	12.21%	Yes	J511	J302,J503,J607, J613		\$10,875,000
Sheyenne-Mapleton 115 kV	19.90	20.41%	Yes	J511,J613	J511,J613		\$4,650,000
Coyote 345-115 kV xfmr	72.90	8.25%	No		J511,J613		\$3,000,000
Coyote-Beulah 115 kV	77.50	5.94%	No		J511,J613		\$250,000
Foxtail-Ellendale 230 kV	92.50	5.45%	No		J302,J457,J503, J511,J607,J613		Incl.
Whitlock-Glenham 230 kV	96.00	5.90%	Yes	J599	J302,J503,J511, J593,J599,J607, J613		\$10,000
Foxtail-Tatanka North 230 kV	100.00	5.47%	No		J302,J457,J503, J511,J607,J613		Incl.

10.5.28 J614

J614 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	66 MW (100%)
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10.5.29 J615

J615 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Laurel-Jasper 161 kV	70.00	5.71%	No		J615		\$24,000,000

10.5.30 J637

J637 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	98 MW (100%)
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10.5.31 J638

J638 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and IC upgrades and case assumptions)	204 MW (100%)
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Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) test for Network Upgrades driven by higher queued interconnection project was performed for this System Impact Study.

The maximum MW impacts and Shared Network Upgrade (SNU) cost allocations are listed in Table 11-1.

Table 11-1: Maximum MW Impact and SNU Cost Allocations

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility
Rutland-Fox Lake 161 kV	CS-5	G614	43.36	\$11,900,000	\$5,261,385
	DPP-2016-AUG	J594	54.71		\$6,638,615
Rutland-Huntley 161 kV	CS-5	G614	43.36	\$10,220,000	\$4,518,601
	DPP-2016-AUG	J594	54.71		\$6,638,615
Hayward-Freeborn 161 kV	DPP-2012-AUG	G870	101.40	\$10,259,820	\$4,417,979
	DPP-2016-AUG	J414	134.08		\$5,841,841
Jamestown 345-115-41.6 kV #1	DPP-2013-FEB	J262	47.15	\$4,042,642	\$1,336,211
	DPP-2013-FEB	J263	47.15		\$1,336,211
	DPP-2016-AUG	J613	48.35		\$1,370,219
Jamestown 345-115-41.6 kV #2	DPP-2013-FEB	J262	48.16	\$4,042,642	\$1,330,420
	DPP-2013-FEB	J263	48.16		\$1,330,420
	DPP-2016-AUG	J613	50.02		\$1,381,802
Buffalo-Bison 345 kV	DPP-2013-FEB	J262	52.71	\$175,000	\$57,486
	DPP-2013-FEB	J263	52.71		\$57,486
	DPP-2016-AUG	J613	55.04		\$60,027
Jamestown-Buffalo 345 kV	DPP-2013-FEB	J262	55.71	\$175,000	\$57,328
	DPP-2013-FEB	J263	55.71		\$57,328
	DPP-2016-AUG	J613	58.64		\$60,343
J274 POI-Creston 161 kV	DPP-2013-AUG	J274	28.74	\$175,000	\$114,855
	DPP-2016-AUG	J611	15.05		\$60,145

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility
Clarinda-Brooks 161 kV	DPP-2014-AUG	J343	61.70	\$70,000	\$53,880
	DPP-2016-AUG	J611	18.46		\$16,120
NIW-Colby 161 kV	DPP-2015-FEB	J407	77.39	\$8,250,000	\$5,127,530
	DPP-2016-AUG	J414	47.13		\$3,122,470
Hankinson-Forman 230 kV	DPP-2015-AUG	J436	7.48	\$400,000	\$39,798
	DPP-2015-AUG	J437	7.48		\$39,798
	DPP-2016-AUG	J457	30.03		\$159,777
	DPP-2016-AUG	J607	30.19		\$160,628
Oakes-Forman 230 kV	DPP-2015-AUG	J436	11.64	\$300,000	\$35,517
	DPP-2015-AUG	J437	11.64		\$35,517
	DPP-2016-AUG	J457	37.94		\$115,765
	DPP-2016-AUG	J607	37.10		\$113,202
Oakes-Forman 230 kV	DPP-2016-FEB	J488	11.72	\$950,000	\$113,058
	DPP-2016-FEB	J489	11.72		\$113,058
	DPP-2016-AUG	J457	37.94		\$365,993
	DPP-2016-AUG	J607	37.10		\$357,890
Oakes-Ellendale 230 kV	DPP-2015-AUG	J436	12.10	\$700,000	\$83,895
	DPP-2015-AUG	J437	12.10		\$83,895
	DPP-2016-AUG	J457	38.84		\$269,295
	DPP-2016-AUG	J607	37.92		\$262,916
Oakes-Ellendale 230 kV	DPP-2016-FEB	J488	12.21	\$1,650,000	\$199,115
	DPP-2016-FEB	J489	12.21		\$199,115
	DPP-2016-AUG	J457	38.84		\$633,386
	DPP-2016-AUG	J607	37.92		\$618,383
J530 POI-Hills 345 kV	DPP-2016-FEB	J475	66.60	\$300,000	\$63,461
	DPP-2016-FEB	J530	75.20		\$71,655
	DPP-2016-AUG	J541	60.11		\$57,277
	DPP-2016-AUG	J555	55.70		\$53,075
	DPP-2016-AUG	J598	57.23		\$54,532
J510 POI-Brookings Co 345 kV #2	DPP-2016-FEB	J510	90.56	\$31,400,000	\$5,804,891
	DPP-2016-FEB	J493	47.75		\$3,060,772
	DPP-2016-FEB	J488	27.79		\$1,781,338

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility
	DPP-2016-FEB	J489	27.79		\$1,781,338
	DPP-2016-FEB	J526	90.35		\$5,791,430
	DPP-2016-AUG	J459	69.82		\$4,475,458
	DPP-2016-AUG	J637	44.07		\$2,824,885
	DPP-2016-AUG	J638	91.73		\$5,879,888

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Section

12

Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

12.1 Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade was provided by the corresponding transmission owning company.

12.2 ERIS Network Upgrades Proposed for DPP West Area Projects

Network upgrades for Energy Resource Interconnection Service (ERIS) were identified in the MISO ERIS analysis and the affected system analysis. The ERIS network upgrades include voltage support network upgrades and thermal network upgrades identified in the MISO steady-state analysis, network upgrades identified in the Local Planning Criteria analysis and affected system analysis, voltage support network upgrades identified in the MWEX voltage stability analysis, stability network upgrades identified in the MISO transient stability analysis, and short circuit network upgrades identified in the MISO short circuit analysis. The total costs of ERIS network upgrades for the 2022 scenario are summarized in Table 12-1.

Table 12-1: Summary of ERIS Network Upgrades

Category of Network Upgrades	Cost (\$)
Backbone Network Upgrades	\$1,185,131,500
Network Upgrades Identified in MWEX Voltage Stability analysis	\$706,000,000
Additional Thermal Network Upgrades Identified in MISO Steady-State Analysis	\$247,830,000
Additional Reactive Power Network Upgrades for Voltage Constraints	\$68,750,000
Network Upgrades Identified in Transient Stability Analysis	\$31,030,000
Network Upgrades Identified in Short Circuit Analysis	\$0
Network Upgrades Identified in DPC LPC Analysis	\$0
Network Upgrades Identified in GRE LPC Analysis	\$710,470,000
Network Upgrades Identified in MDU LPC Analysis	\$1,500,000
Network Upgrades Identified in OTP LPC Analysis	\$0
Network Upgrades Identified in Ameren LPC Analysis	\$0

Cost Allocation

Category of Network Upgrades	Cost (\$)
Network Upgrades Identified in CIPCO Affected Systems	\$26,000,000
Network Upgrades Identified in MPC Affected Systems	\$7,850,000
Network Upgrades Identified in PJM Affected Systems	\$900,000
Network Upgrades Identified in SPP Affected Systems	\$159,132,718
Network Upgrades Identified in AECI Affected Systems	\$3,380,000
Network Upgrades Identified in Ameren Affected Systems	\$10,000,000
Shared Network Upgrades	\$41,592,548
Total	\$3,199,566,766

ERIS network upgrades are listed below.

Table 12-2: Backbone Network Upgrades

NUs	Miles	Cost (\$)
Big Stone South-Alexandria 345 kV	95	\$197,400,000
J510-Hazel Creek-Scott County 345 kV	165	\$330,000,000
Franklin-Morgan Valley 345 kV	138	\$384,056,500
Beverly-Sub 92 345 kV	60	\$148,675,000
Webster-Franklin 345 kV	49.2	\$117,000,000
2×150 MVar MSC at Franklin 345 kV		\$8,000,000
Total		\$1,185,131,500

Table 12-3: Network Upgrades Required for MWEX Voltage Stability

NUs	Miles	Cost (\$)
Construct a new South Rocky Run 345 kV switching station with one 75 MVAR shunt capacitor.		\$53,500,000
Construct a new North Rochester – South Rocky Run 345 kV line	180	\$621,000,000
Construct a new 161/69 kV substation in Minnesota connecting the Wabaco – Alma 161 kV line to the Dairyland 69 kV system		\$5,500,000
Install 150 MVAR of shunt capacitance at the North Rochester 345 kV Substation		\$6,000,000
Install a 35% series capacitor at the Briggs Road 345 kV Substation on the Briggs Road – North Madison 345 kV line		\$18,000,000
Install one breaker at the Briggs Road 345 kV Substation		\$2,000,000
Additional facilities, e.g. Forest Junction – Elkhart Lake 138 kV, are expected to be overloaded by this alternative and will need to be assigned Network Upgrades		NA

NUs	Miles	Cost (\$)
Total		\$706,000,000

**Table 12-4: Additional Thermal Network Upgrades in MISO
Steady-State Analysis**

Constraint	Owner	Mitigation	Cost (\$)
J506 POI-Raun 345 kV	MEC	Structure Replacements.	\$700,000
J506 POI-Highland 345 kV	MEC	Structure Replacements.	\$200,000
J530 POI-Montezuma 345 kV	MEC	Structure Replacements.	\$125,000
J530 POI-Hills 345 kV	MEC	Reconductor with structure replacements as needed. Substation terminal equipment upgrades.	\$27,000,000
J302&J503 POI-J607 POI 230 kV	MDU	Add breaker at Merricourt	\$1,500,000
J302&J503 POI-Heskett 230 kV	MDU	Line rebuild	\$9,000,000
J607 POI-Wishek 230 kV	MDU	Rebuild with high temp. conductor. New Rating: 383 MVA	\$1,100,000
J611-Maryville 161 kV	MEC GMO	MEC: Reconductor with structure replacements as needed. Assumes J611 POI is one mile from Missouri border. \$1,000,000 GMO: NU is not required unless it is identified as constraint in affected system study.	\$1,000,000
Zackary 345-161 kV xfmr	AMMO	Add Second 560 MVA 345-161 kV transformer.	\$7,000,000
Adair-Zackary 161 kV	AMMO	Add second 161 kV line between Adair and Zachary	\$2,000,000
Adams 345-161-13.8 kV xfmr	XEL	Upgrade Transformer Nameplate to 336 MVA and terminal equipment	\$4,000,000
Sheyenne-Lake Park 230 kV	XEL OTP	XEL: OTP conductor / MPS equipment is limiter. \$0 OTP: Line clearance mitigation required. \$250,000	\$250,000
Sheyenne-Mapleton 115 kV	XEL OTP	OTP: Complete rebuild of OTP's 477 ACSR portion of the line. \$3,150,000 Xcel: Complete rebuild of Xcel's portion of the line \$1,500,000	\$4,650,000
Grove Lake-Lake Johnason 69 kV	XEL GRE	XEL: GRE Equipment is limiter. \$0 GRE: Upgrade 4 miles of 69 kV line, \$2.26 million	\$2,260,000
Murphy Creek 161-69 kV xfmr	SMMPA	Install new 112 MVA Transformer at Murphy Creek Sub.	\$2,500,000
Austin-Murphy Creek 161 kV	SMMPA	Re-conductoring of the Austin to Murphy Creek Line.	\$650,000

Cost Allocation

Constraint	Owner	Mitigation	Cost (\$)
Murphy Creek-Hayward 161 kV	SMMPA ITCM	ITCM: Apply RF 2 on CT's, replace jumpers and transducer, \$10k SMMPA: Re-conductoring of the Hayward to Murphy Creek Line, \$4.5 M	\$4,510,000
Rice 161-69 kV xfmr	SMMPA	SMMPA: MOD project # 110359 to increase the Rice 161/69kV transformer to 190MVA rating as per the GIA J614	\$0
Rutland-Fox Lake 161 kV	ITCM	Rebuild 9 miles of 161 kV line, \$9,000,000	\$9,000,000
Rutland-Huntley 161 kV	ITCM	Rebuild 0.73 miles of 161 kV, \$730,000	\$730,000
Wing River-Inman 230 kV	GRE	Replace jumpers at Wing River	\$50,000
Stanton-Square Butte 230 kV	GRE MPC	GRE: Rebuild 16.5 miles of 230 kV line > 590 MVA; 2) replace hookstick switches; 3) replace jumpers to >590 MVA. \$10,875,000 MPC: NU is not required unless it is identified as constraint in affected system study.	\$10,875,000
Ramsey-Prairie 230 kV	GRE MPC	GRE: rebuild 71 miles of 230 kV transmission >350 MVA. \$46,150,000 MPC: NU is not required unless it is identified as constraint in affected system study.	\$46,150,000
Ellendale-Steel Center 69 kV	ITCM	Rebuild 2 miles of 69 kV with T2-477 ACSR	\$1,500,000
Pratt Tap-Steel Center 69 kV	ITCM	Rebuild 7 miles of 69 kV with T2-477 ACSR	\$5,250,000
Emery-Floyd 161 kV	ITCM MEC	MEC: Reconductor / Terminal Equipment Upgrades. \$21,000,000 ITCM: New ratings removes constraint, 407/418 MVA SN/SE. \$0	\$21,000,000
Hazleton-Washburn 161 kV	ITCM MEC IP&L	MEC: No additional mitigation required. Existing MEC emergency rating expected to be 410 MVA after DPP2016 FEB reconductor mitigation. ITCM: ITCM limit is wavetrap - \$200k. IP&L: Install 3 new 161 kV CT/PT combo units, \$170,000	\$370,000
Henry Co-Jeff 161 kV	ITCM NMEC	Northeast Missouri Electric Coop: NU is not required unless it is identified as constraint in affected system study. ITCM: Relay upgrade. \$200,000	\$200,000
Wapello-Jeff 161 kV	ITCM	Rebuild of 23 miles of 161 kV required. Future rating of 515 MVA	\$23,000,000
Ottumwa 345-161 kV xfmr	ITCM	Add 2nd 450 MVA transformer.	\$9,000,000
Adams-Hayward 161 kV	ITCM	Rebuild 36.5 miles of 161 kV	\$36,500,000

Cost Allocation

Constraint	Owner	Mitigation	Cost (\$)
Hazleton-Arnold 345 kV	ITCM MEC IP&L	MEC: Structure Replacements. \$200,000 ITCM: Pending ITCM ratings methodology changes will increase the rating of the line to 1174 MVA. \$0 IP&L: Alliant does not require meter data for flows on this line at either end. \$0	\$200,000
Cayler-Wisdom 161 kV	MEC ITCM CBPC	MEC: Structure Replacements. \$500,000 CBPC: NU is not required unless it is identified as constraint in affected system study. ITCM: Not a constraint. ITCM equipment at 240 MVA. \$0	\$500,000
Raun-S3451 345 kV	MEC OPPD	MEC: Terminal Equipment Upgrades. \$200,000 OPPD: NU is not required unless it is identified as constraint in affected system study.	\$200,000
Whitlock-Glenham 230 kV	MDU WAPA	MDU: Relay Setting. \$10,000 WAPA: NU is not required unless it is identified as constraint in affected system study.	\$10,000
Morris-Grant Co 115 kV	OTP MRES WAPA	OTP: All OTP equipment is good for the flow of 110.6 MVA seen in the study. MRES: Phase raisers, \$350,000 WAPA: NU is not required unless it is identified as constraint in affected system study.	\$350,000
Heskett-Mandan 230 kV	MDU	Add breaker at Merricourt	Incl.
Merricourt-Wishek 230 kV	MDU	Rebuild Line with high temp. conductor. New Rating: 478 MVA	\$6,000,000
Merricourt-Ellendale 230 kV	MDU	Rebuild Line with high temp. conductorNew Rating: 597 MVA	\$4,500,000
Harmony-Cresco 69 kV	DPC	Rebuild line with 477 ACSR	\$4,000,000

Table 12-5: Additional Reactive Power NUs Required for Voltage Constraints

Network Upgrades	Owner	Cost (\$)
3×50 Mvar switched cap bank at Blue Lake 230 kV (602014)	Xcel	\$7,500,000
2×50 Mvar switched cap bank at Prairie Island 161 kV (602001)	Xcel	\$7,000,000
2×20 Mvar switched cap bank at Fergus Falls 230 kV (658109)	MRES	\$1,750,000
3×50 Mvar switched cap bank at Alexandria 345 kV (658047)	MRES	\$5,500,000
Increase 1x100 Mvar switched cap bank to 1x150 Mvar at Montezuma 345 kV (635730)	MEC	\$1,000,000
1x150 Mvar switched cap bank at Webster 345 kV (636000)	MEC	\$4,000,000
1×25 Mvar switched cap bank at J615 POI 161 kV (86151)	MEC	\$2,500,000
1x150 Mvar switched cap bank at Raun 345 kV (635200)	MEC	\$4,000,000

Cost Allocation

Network Upgrades	Owner	Cost (\$)
1x100 Mvar switched cap bank at O'Brien 345 kV (635368)	MEC	\$4,000,000
2x150 Mvar switched cap bank at Hills 345 kV (636400)	MEC	\$15,000,000
2x150 Mvar switched cap bank at Hickory Creek 345 kV (631191)	ITCM	\$5,500,000
2x150 Mvar switched cap bank at Morgan Valley 345 kV (631148)	ITCM	\$5,500,000
1x150 Mvar switched cap bank at Killdeer 345 kV (631199)	ITCM	\$5,500,000
Reset transformer taps back to the midpoint at Hazleton, Hickory Creek, Killdeer and Colby. Taps are not locked, but were reset to midpoint.	ITCM	\$0

Table 12-6: Additional Network Upgrades in Transient Stability Analysis

Network Upgrades	Owner	Cost (\$)
(0, 200) Mvar SVC at Winger 230 kV	OTP	\$31,030,000

Table 12-7: Network Upgrades in Short Circuit Analysis

Constraint	Owner	Mitigation	Cost (\$)
No Constraints			\$0

Table 12-8: DPC Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
No additional NUs			\$0

Table 12-9: GRE Local Planning Criteria Network Upgrades

Network Upgrades	Owner	Mileage	Cost (Million)
New 345 kV line Center-Ellendale	OTP MDU	200.0	\$406.75
New 345 kV line Prairie-Bison	MPC XEL	90.08	\$243.2
(0, 150) Mvar SVC at Jamestown 345 kV	OTP		\$24.37
(0, 150) Mvar SVC at Prairie 230 kV	OTP		\$24.37
2*50 Mvar MSC at Lake Park 230 kV	OTP		\$4.48
Upgrade Kerkhoven Tap-Kerkhoven 115 kV line; replace jumpers at Kerkhoven; upgrade CTs at Kerkhoven	XEL GRE	11.0	\$7.3

Table 12-10: MDU Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
Reactive power and voltage oscillations (J302, J503, J457, J526, J607, Foxtail, etc.)	MDU	Add a breaker on the existing 345 kV line reactor at Ellendale to make it switchable	\$1,500,000

Table 12-11: OTP Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
No additional NUs			\$0

Table 12-12: Ameren Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
No additional NUs			\$0

Table 12-13: CIPCO Affected System Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
Hazleton-Dundee 161 kV	CIPCO ITCM	CIPCO: Upgrade Dundee terminal to 3000 Amps. \$500,000 ITCM: CIPCO LPC. \$0	\$500,000
Liberty-Dundee 161 kV	CIPCO ITCM	CIPCO: Rebuild 25 miles of 161 kV line with bundled conductor. If feasible, Re-conductor the line with high temp conductor for \$5,000,000 ITCM: CIPCO LPC. \$0	\$25,000,000
Liberty-Hickory Crk 161 kV	CIPCO ITCM	CIPCO: Upgrade Liberty terminal to 3000 Amps. \$500,000 ITCM: CIPCO LPC. \$0	\$500,000

Table 12-14: MPC Affected System Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)	DPP Projects
Bison 345 kV Bus Voltage	MPC	Install two 55 MVAR switched capacitor banks at the Jamestown 345 kV substation and two 85 MVAR switched capacitor banks at the Bison 345 kV substation.	\$7,500,000	J511,J613
Square Butte–Stanton 230 kV	MPC	Replace bus work at Square Butte.	\$50,000	J511
Center-Roughrider 230 kV	MPC	Raise structures to attain a higher conductor rating	\$300,000	J511,J613

Table 12-15: PJM Affected System Network Upgrades

Constraint	Owner	Mitigation	Total Cost (\$)	DPP Projects
Twin Branch–Argenta 345 kV	PJM	Replace Twin Branch Wavetrap; Replace substation structure at Argenta substation to accommodate larger wire.	\$900,000	J414,J439,J512,J541,J593,J594,J597,J598

Table 12-16: SPP Affected System Network Upgrades

Mitigation Required	TO	Constraint	Type	Cost Allocation	Cost Estimate
Reroute of Bismark to Washburn 230 kV circuit through Hilken 230kV station	WAPA	Bismark-Hilken 230 kV	ERIS	J593: \$3,500,000	\$3,500,000
Reconductor line	WAPA	Oahe-Sully Buttes 230 kV	NRIS	J302: \$343,280 J457: \$287,388 J503: \$339,882 J511: \$688,769 J593: \$606,719 J599: \$6,465,895 J607: \$422,385 J613: \$255,682	\$9,410,000
Reconductor line	WAPA	Whitlock-Sully Buttes 230 kV	NRIS	J302: \$392,917 J457: \$329,381 J503: \$389,027 J511: \$787,586 J593: \$690,390 J599: \$7,414,441 J607: \$483,659 J613: \$292,599	\$10,780,000
Reconductor line	WAPA	Whitlock-Glenham 230 kV	NRIS	J302: \$662,091 J457: \$555,029 J503: \$655,535 J511: \$1,327,133 J593: \$1,163,352 J599: \$12,493,816 J607: \$814,996 J613: \$493,048	\$18,165,000
New transformer to achieve higher MVA rating	BEPC	Broadland 345-230-13.8 kV xfmr	ERIS	J511: \$3,564,615 J593: \$5,849,103	\$9,413,718
Build new 345 kV substations, reroute transmission lines, and reconductor line to achieve higher MVA rating	NPPD AECI KCPL	Cooper South interface ¹	ERIS	J476: \$81,907,429 J583: \$19,492,571	\$101,400,000
or Build new Cooper–Stranger Creek 345kV circuit				or J476: 243,137,437 J583: \$57,862,563	or \$301,000,000

Cost Allocation

Mitigation Required	TO	Constraint	Type	Cost Allocation	Cost Estimate
Increase Granite Falls 115 kV Capacitor from 2 steps of 30 Mvar to 5 steps of 30 Mvar (150 Mvar total)	WAPA	Low voltages near Granite Falls 230 kV	ERIS	J459: \$1,668,687 J596: \$1,068,727 J637: \$409,713 J638: \$852,873	\$4,000,000
Install 20 Mvar capacitor at Flandreau 69 kV	WAPA	Low voltage near Flandreau 69 kV radial lines serving load	ERIS	J459: \$110,522 J512: \$29,122 J569: \$28,874 J575: \$68,931 J577: \$70,862 J587: \$120,398 J596: \$33,381 J597: \$206,794 J599: \$16,262 J637: \$63,231 J638: \$131,623	\$880,000
Install 20 Mvar capacitor at Mars 69 kV	WAPA	Low voltage near Mars 69 kV radial lines serving load	ERIS	J414: \$43,292 J415: \$72,182 J439: \$323,684 J512: \$107,548 J569: \$36,019 J590: \$58,189 J594: \$239,086	\$880,000
Increase Belden 69 kV capacitor by 6 Mvar (1 step)	WAPA	Low voltage near Belden 69 kV	ERIS	J593: \$264,000	\$264,000
Install 10 Mvar capacitor at Summit 69 kV	WAPA	Low voltages near Summit 69 kV	ERIS	J302: \$34,458 J457: \$70,168 J503: \$34,116 J511: \$106,166 J593: \$110,480 J607: \$58,636 J613: \$25,976	\$440,000

Note 1: If recommended mitigation is not valid, a new 345 kV line from Cooper to Stranger Creek may be an alternate solution.

Table 12-17: AECI Affected System Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
Novelty 161 -69 kV xfmr	AECI	Replace with 84 MVA.	\$3,270,000
South River-Emerson 161 kV	AECI	Upgrade 600 A disconnect switches at South River.	\$110,000

Table 12-18: Ameren Affected System Network Upgrades

Network Upgrades	Owner	Cost (\$)
Additional 2×25 Mvar switched cap bank at Lathrop 161 kV (344923)	Ameren AFS	\$10,000,000

Table 12-19: Shared Network Upgrades

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility
Rutland-Fox Lake 161 kV	CS-5	G614	43.36	\$11,900,000	\$5,261,385
	DPP-2016-AUG	J594	54.71		\$6,638,615
Rutland-Huntley 161 kV	CS-5	G614	43.36	\$10,220,000	\$4,518,601
	DPP-2016-AUG	J594	54.71		\$6,638,615
Hayward-Freeborn 161 kV	DPP-2012-AUG	G870	101.40	\$10,259,820	\$4,417,979
	DPP-2016-AUG	J414	134.08		\$5,841,841
Jamestown 345-115-41.6 kV #1	DPP-2013-FEB	J262	47.15	\$4,042,642	\$1,336,211
	DPP-2013-FEB	J263	47.15		\$1,336,211
	DPP-2016-AUG	J613	48.35		\$1,370,219
Jamestown 345-115-41.6 kV #2	DPP-2013-FEB	J262	48.16	\$4,042,642	\$1,330,420
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J274 POI-Creston 161 kV	DPP-2013-AUG	J274	28.74	\$175,000	\$114,855
	DPP-2016-AUG	J611	15.05		\$60,145
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	DPP-2016-AUG	J611	18.46		\$16,120
NIW-Colby 161 kV	DPP-2015-FEB	J407	77.39	\$8,250,000	\$5,127,530
	DPP-2016-AUG	J414	47.13		\$3,122,470
Hankinson-Forman 230 kV	DPP-2015-AUG	J436	7.48	\$400,000	\$39,798

Cost Allocation

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility
	DPP-2015-AUG	J437	7.48		\$39,798
	DPP-2016-AUG	J457	30.03		\$159,777
	DPP-2016-AUG	J607	30.19		\$160,628
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	DPP-2016-AUG	J457	37.94		\$115,765
	DPP-2016-AUG	J607	37.10		\$113,202
Oakes-Forman 230 kV	DPP-2016-FEB	J488	11.72	\$950,000	\$113,058
	DPP-2016-FEB	J489	11.72		\$113,058
	DPP-2016-AUG	J457	37.94		\$365,993
	DPP-2016-AUG	J607	37.10		\$357,890
Oakes-Ellendale 230 kV	DPP-2015-AUG	J436	12.10	\$700,000	\$83,895
	DPP-2015-AUG	J437	12.10		\$83,895
	DPP-2016-AUG	J457	38.84		\$269,295
	DPP-2016-AUG	J607	37.92		\$262,916
Oakes-Ellendale 230 kV	DPP-2016-FEB	J488	12.21	\$1,650,000	\$199,115
	DPP-2016-FEB	J489	12.21		\$199,115
	DPP-2016-AUG	J457	38.84		\$633,386
	DPP-2016-AUG	J607	37.92		\$618,383
J530 POI-Hills 345 kV	DPP-2016-FEB	J475	66.60	\$300,000	\$63,461
	DPP-2016-FEB	J530	75.20		\$71,655
	DPP-2016-AUG	J541	60.11		\$57,277
	DPP-2016-AUG	J555	55.70		\$53,075
	DPP-2016-AUG	J598	57.23		\$54,532
J510 POI-Brookings Co 345 kV #2	DPP-2016-FEB	J510	90.56	\$31,400,000	\$5,804,891
	DPP-2016-FEB	J493	47.75		\$3,060,772
	DPP-2016-FEB	J488	27.79		\$1,781,338
	DPP-2016-FEB	J489	27.79		\$1,781,338
	DPP-2016-FEB	J526	90.35		\$5,791,430
	DPP-2016-AUG	J459	69.82		\$4,475,458
	DPP-2016-AUG	J637	44.07		\$2,824,885

Cost Allocation

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility
	DPP-2016-AUG	J638	91.73		\$5,879,888

12.3 Cost Allocation Methodology

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Post Case. For constraints identified in the shoulder peak scenario, the MW impact is calculated using the shoulder peak post-DPP case. The MW impact on constraints identified in the summer peak scenario is calculated using the summer peak post-DPP case. With all Group Study generation projects dispatched in the Post Case, all thermal and voltage constraints will be identified and a distribution factor from each project on each constraint will be obtained.

Constraints which are mitigated by one or a subset of NU are identified. The MW contribution on these constraints from each generating facility is calculated in the Post Case without any network upgrades. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each generating facility on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of NU is:

$$\text{Project A cost portion of NU} = \text{Cost of NU} \times \left(\frac{\text{Max}(\text{Proj.A MW contribution on constraint})}{\sum_i \text{Max}(\text{Proj.i MW contribution on constraint})} \right)$$

12.4 Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

The Distribution Factor (DF) from each generating facility is calculated on the constraints identified in the steady-state analysis in the Post Case without any network upgrades. For a voltage support network upgrade required for mitigating voltage constraints identified in the steady-state AC contingency analysis, DFs are calculated under the most critical contingency on all branches (proxy branches for voltage support network upgrade) connecting to the high voltage side of the transformer, where the voltage support network upgrade is located. For a voltage support network upgrade required for mitigating MWEX voltage stability constraints identified in the voltage stability analysis, DFs are calculated under the most critical contingency on all branches (proxy branches) connecting to the high voltage side of the transformer, where the voltage collapse occurs. For network upgrades required for mitigating transient stability constraints under the GRE P7 CUDC bipole contingency, DFs are calculated under the GRE P7 CUDC bipole contingency on all branches (proxy branches) connecting to the Center 345 kV bus.

For each thermal constraint, the maximum MW contribution (increasing flow) from each generating facility is calculated. MW contribution from one generating facility is set as zero if

the MW contribution is less than 1 MW, or the constraint is not categorized as MISO ERIS constraint or affected system constraint for that specific generating facility.

For voltage support network upgrades, or MWEX network upgrades and other voltage stability network upgrades, or transient stability network upgrades required for the GRE P7 CUDC bipole contingency, generators with positive MW impact will be responsible for mitigating these constraints. DF cutoff of 1% is applied to the MW contribution from each generating facility.

The calculated DF results and the MW contribution on each constraint are in Appendix K.1 for the 2022 scenario.

Finally, the cost allocation for each NU is calculated based on the MW contribution of each generating facility, as detailed in Appendix K.2 for the 2022 scenario.

Assuming all generating facilities in the DPP 2016 August West Area group advance, a summary of the costs for total NUs (NUs for ERIS, NRIS, and Interconnection Facilities) allocated to each generating facility is listed in Table 12-20.

Table 12-20: Summary of Total NU Costs Allocated to Each Generation Project

Project	Max Output (MW)	Total Cost of NU per Project (\$)	\$/MW	Share %
J302	101.2	\$116,671,255	\$1,152,878	3.41%
J414	120	\$69,472,590	\$578,938	2.03%
J415	200	\$179,633,709	\$898,169	5.25%
J439	500	\$184,124,689	\$368,249	5.38%
J457	150	\$116,772,354	\$778,482	3.41%
J459	200	\$141,344,863	\$706,724	4.13%
J476	246	\$123,129,358	\$500,526	3.60%
J503	98.8	\$114,001,215	\$1,153,858	3.33%
J511	200	\$384,649,137	\$1,923,246	11.24%
J512	250	\$78,881,675	\$315,527	2.31%
J541	400	\$130,790,754	\$326,977	3.82%
J555	140	\$51,444,402	\$367,460	1.50%
J569	100	\$19,904,197	\$199,042	0.58%
J575	100	\$51,145,758	\$511,458	1.49%
J577	100	\$51,147,689	\$511,477	1.49%
J583	200	\$69,619,460	\$348,097	2.03%
J587	200	\$109,248,419	\$546,242	3.19%
J590	90	\$23,403,718	\$260,041	0.68%

Cost Allocation

Project	Max Output (MW)	Total Cost of NU per Project (\$)	\$/MW	Share %
J593	224	\$238,631,173	\$1,065,318	6.97%
J594	300	\$152,464,370	\$508,215	4.46%
J596	100	\$45,111,577	\$451,116	1.32%
J597	300	\$148,365,440	\$494,551	4.34%
J598	300	\$95,082,891	\$316,943	2.78%
J599	200	\$181,732,343	\$908,662	5.31%
J607	150	\$154,584,441	\$1,030,563	4.52%
J611	110	\$21,562,588	\$196,024	0.63%
J613	100	\$62,381,795	\$623,818	1.82%
J614	66	\$18,459,978	\$279,697	0.54%
J615	70	\$70,505,812	\$1,007,226	2.06%
J637	98	\$74,844,052	\$763,715	2.19%
J638	204	\$142,880,512	\$700,395	4.18%
Total/Average	5618.0	\$3,421,992,214	\$638,504	100.00%

Model Development for Steady-State and Stability Analysis

A.1 DPP 2016 August Generation Projects

Table A-1: DPP 2016 August West Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J302	ND	Emmons, Logan	MDU	Heskett-Wishek 230 kV	101.2	Wind	NRIS
J414	MN	Freeborn	ITCM	Freeborn 161 kV Sub	120	Wind	NRIS
J415	IA	Franklin	ITCM	Quinn-Blackhawk 345 kV (7 mi south of Quinn)	200	Wind	NRIS
J439	IA	Emmet	MEC	Obrien-Kossuth 345 kV	500	Wind	NRIS
J457	ND	Dickey	MDU	Merricourt 230 kV Sub	150	Wind	NRIS
J459	SD	Grant	OTP	Big Stone South-Brookings Co 345 kV (11 miles from Big Stone South)	200	Wind	NRIS
J476	MO	Atchison	MEC	Atchison Co-Orient 345 kV (0.2 mi from Atchison)	246	Wind	NRIS
J503	ND	Emmons, Logan	MDU	Heskett-Wishek 230 kV (20 miles NW of Wishek)	98.8	Wind	NRIS
J511	ND	Oliver	GRE	Stanton 230 Sub	200	Wind	NRIS
J512	MN	Nobles	Xcel	Nobles-Fenton 115 kV	250	Wind	NRIS
J541	MO	Schuyler	ATXI	Zachary–Ottumwa 345 kV	400	Wind	NRIS
J555	IA	Poweshiek	MEC	Montezuma 345 kV	140	Wind	NRIS
J569	MN	Rock	Xcel	Rock County 161 kV	100	Wind	NRIS
J575	MN	Lincoln	Xcel	Brookings Co 345 kV	100	Wind	NRIS
J577	MN	Pipestone	Xcel	Brookings Co 345 kV	100	Wind	NRIS
J583	IA	Audubon	MEC	Fallow 345 kV	200	Wind	NRIS
J587	MN	Lincoln	Xcel	J460 Sub on the Brookings-H081 345 kV	200	Wind	NRIS
J590	IA	Palo Alto	MEC	J529 POI	90	Wind	NRIS
J593	ND	Williams	MDU	Tioga 230 kV	224	Wind	NRIS
J594	MN	Jackson, Martin	ITCM	Jackson North 161 kV	300	Wind	NRIS
J596	MN	Big Stone, Stevens	GRE	Morris-Moro 115 kV	100	Wind	NRIS
J597	SD	Deuel, Hamlin	Xcel	Brookings Co 345 kV	300	Wind	NRIS
J598	MO	Adair, Schuyler	ATXI	Zachary–Ottumwa 345 kV	300	Wind	NRIS
J599	SD	Walworth	MDU	Glenham 230 kV	200	Wind	NRIS

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J607	ND	Logan	MDU	Wishek-Heskett 230 kV	150	Wind	NRIS
J611	MO	Nodaway	MEC	Clarinda-Merryville 161 kV	110	Wind	NRIS
J613	ND	Stutsman	OTP	Jamestown 115 kV	100	Wind	NRIS
J614	IA	Howard	SMMPA	Rice 161 kV	66	Wind	NRIS
J615	IA	Black Hawk	MEC	Electric Farms-Shaulis 161 kV	70	Wind	NRIS
J637	SD	Deuel	OTP	Big Stone-Brookings 345 kV	98	Wind	NRIS
J638	SD	Deuel	OTP	Big Stone-Brookings 345 kV	204	Wind	NRIS

Table A-2: Dynamic Modeling for DPP West Area Projects

MISO Project #	Dynamic Modeling	Generator Reactive Power Capability (power factor)
J302	44 GE 2.3 MW (GEWTG2)	± 0.9
J414	6 GE 2.3 MW (GEWTG2) and 43 GE 2.5 MW (GEWTG2)	± 0.9
J415	10 GE 2.3 MW (GEWTG2) and 71 GE 2.5 MW (GEWTG2)	± 0.9
J439	250 Vestas V110 2.0 MW (VS3103)	± 0.95
J457	75 Vestas V110 2.0 MW (VS3103)	± 0.95
J459	102 Vestas V100 2.0 MW (VS3103)	0.98 lagging 0.96 leading
J476	123 Vestas V110 2.0MW (VS3103)	± 0.95
J503	43 GE 2.3MW (GEWTG2)	± 0.9
J511	9 Vestas V110 2.0 MW (VS3103) & 51 Vestas V136 3.6 MW (CP17083200)	V136 3.6 MW: '-0.93 (leading), 0.913 (lagging) V110 2.0 MW: ± 0.95
J512	10 Vestas V110 2.0 MW (VS3103) & 64 Vestas V136 3.6MW (CP17083200)	V136 3.6 MW: '-0.93 (leading), 0.913 (lagging) V110 2.0 MW: ± 0.95
J541	200 Vestas V110 2.0 MW (VS3103)	± 0.95
J555	70 Vestas V110 2.0 MW (VS3103)	± 0.95
J569	42 Siemens SWT 2.5-120 2.5 MW (SWTGU2)	± 0.9
J575	40 GE 2.5 MW (GEWTG2)	± 0.9
J577	40 GE 2.5 MW (GEWTG2)	± 0.9
J583	80 GE 2.5 MW (GEWTG2)	± 0.9
J587	101 Vestas V110 2.0 MW (VS3103)	0.98 lagging 0.96 leading

MISO Project #	Dynamic Modeling	Generator Reactive Power Capability (power factor)
J590	45 Vestas V110 2.0 MW (VS3103)	± 0.95
J593	112 Vestas V110 2.0 MW (VS3103)	± 0.95
J594	150 Vestas V110 2.0 MW (VS3103)	± 0.95
J596	50 Vestas V110 2.0 MW (VS3103)	± 0.95
J597	150 Vestas V110 2.0 MW (VS3103)	± 0.95
J598	150 Vestas V110 2.0 MW (VS3103)	± 0.95
J599	100 Vestas V110 2.0 MW (VS3103)	± 0.95
J607	75 Vestas V110 2.0 MW (VS3103)	± 0.95
J611	55 Vestas V110 2.0 MW (VS3103)	± 0.95
J613	44 GE 116 - 2.3 MW (GEWTG2)	± 0.95
J614	33 Vestas V116 2.0 MW (VCS218)	± 0.95
J615	35 Vestas V110 2.0 MW (VS3103)	± 0.95
J637	49 Gamesa 2.0 MW (GXX048)	± 0.95
J638	102 Gamesa 2.0 MW (GXX048)	± 0.95

Table A-3: Collector System and Shunt Compensation Modeling for DPP West Area Non-Synchronous Projects

MISO Project #	Generator Modeling	Collector System Modeling	Shunt Compensation
J302	One 101.2 MW unit	Req = 0.00387 pu Xeq = 0.00639 pu Beq = 0.0 pu	3×8 MVAR capacitor bank on 34.5kV system
J414	One 13.8 MW unit and one 106.2 MW unit	R=0.007409 pu X=0.0495 pu B=0.07312 pu	15 MVAR capacitor bank on 34.5 kV system
J415	One 11.5 MW unit, one 11 MW unit, one 87.5 MW unit, one 90 MW unit	Circuit #1: R=0.007409 pu X=0.0495 pu B=0.07312 pu Circuit #2: R=0.007409 pu X=0.0495 pu B=0.07312 pu	5 MVAR capacitor bank on each of the two 34.5 kV collector system

MISO Project #	Generator Modeling	Collector System Modeling	Shunt Compensation
J439	One 238 MW unit and one 262 MW unit	Circuit #1: R=0.00314 pu X=0.00502 pu B=0.1688 pu Circuit #2: R=0.00225 pu X=0.0036 pu B=0.14322 pu	45 MVAR capacitor bank on one 34.5 kV system, 50 MVAR capacitor bank on another 34.5 kV system
J457	One 150 MW unit	R=0.006251 pu X=0.006792 pu B=0.0675 pu	2×17 MVAR capacitor bank on 34.5 kV system
J459	One 200 MW unit	R=0.007 pu X=0.025 pu B=0.047 pu	2×14 MVAR capacitor bank on 34.5 kV system
J476	One 246 MW unit	R=0.0062 pu X=0.0099 pu B=0.25631 pu	3×15 MVAR capacitor bank and 1×5 MVAR reactor bank on 34.5 kV system
J503	One 98.9 MW unit	R=0.003865 pu X=0.006385 pu B=0.0 pu	1×24 MVAR capacitor bank on 34.5 kV system
J511	One 18 MW unit and one 182 MW unit	Circuit #1: R=0.0065 pu X=0.01062 B=0.012 Circuit #2: R=0.00409 X=0.00653 B=0.10606	18 MVAR capacitor bank on 34.5 kV collector system
J512	One 20 MW unit and one 230 MW unit	Circuit #1: R=0.0065 pu X=0.01062 B=0.012 Circuit #2: R=0.00409 X=0.00653 B=0.10606	16 MVAR capacitor bank on 34.5 kV collector system
J541	One 400 MW unit	R=0.003382 pu X=0.003246 pu B=0.00197 pu	6×18 MVAR capacitor bank on 34.5 kV collector system
J555	One 140 MW unit	R=0.00716 pu X=0.01143 pu B=0.1694 pu	1×5, 1×10, and 2×15 MVAR capacitor banks on 34.5 kV collector system

MISO Project #	Generator Modeling	Collector System Modeling	Shunt Compensation
J569	One 100 MW unit	R=0.002352 pu X=0.003697 pu B=0.03911 pu	None
J575	One 100 MW unit	R=0.0074 pu X=0.0112 pu B=0.00583 pu	2×6 MVAR capacitor bank on 34.5 kV system
J577	One 100 MW unit	R=0.0074 pu X=0.0112 pu B=0.005832 pu	2×6 MVAR capacitor bank on 34.5 kV system
J583	One 200 MW unit	R=0.00267 pu X=0.00863 pu B=0.0 pu	3×8 MVAR capacitor bank on 34.5 kV system
J587	One 99 MW unit and one 101 MW unit	Circuit #1: R=0.0136 pu X=0.0448 pu B=0.0061 pu Circuit #2: R=0.0136 pu X=0.0448 pu B=0.0061 pu	20 MVAR capacitor bank on each 34.5 kV collector system
J590	Two 45 MW units	None	None
J593	One 224 MW unit	R=0.00297 pu X=0.00474 pu B=0.1771 pu	2×15 and 1×10 MVAR capacitor bank on 34.5 kV system
J594	Two 150 MW units	Circuit #1: R=0.00578 pu X=0.00923 pu B=0.08983 pu Circuit #2: R=0.00416 pu X=0.00663 pu B=0.09177 pu	25 MVAR capacitor bank on each 34.5 kV collector system
J596	One 100 MW unit	R=0.00606 pu X=0.00968 pu B=0.05929 pu	1×15 MVAR capacitor bank on 34.5 kV system
J597	Two 150 MW units	Circuit #1: R=0.00578 pu X=0.00923 pu B=0.08983 pu Circuit #2: R=0.00416 pu X=0.00663 pu	2×20 MVAR capacitor bank on one 34.5 kV collector system, and (1×20,1×5) MVAR capacitor bank on another 34.5 kV collector system

MISO Project #	Generator Modeling	Collector System Modeling	Shunt Compensation
		B=0.09177 pu	
J598	Two 150 MW units	Circuit #1: R=0.00486 pu X=0.00776 pu B=0.08187 pu Circuit #2: R=0.00481 pu X=0.00769 pu B=0.09973 pu	2x20 MVAR capacitor bank on one 34.5 kV collector system, and 1x20 MVAR capacitor bank on another 34.5 kV collector system
J599	One 200 MW unit	R=0.0031 pu X=0.00495 pu B=0.11505 pu	1x25 and 1x10 MVAR capacitor bank on 34.5 kV system
J607	One 150 MW unit	R=0.00313 pu X=0.00499 pu B=0.09245 pu	1x20 and 1x10 MVAR capacitor bank on 34.5 kV system
J611	One 110 MW unit	R=0.00416 pu X=0.00664 pu B=0.05313 pu	2x20 MVAR capacitor bank on 34.5 kV system
J613	One 100 MW unit	R=0.005 pu X=0.019 pu B=0.035 pu	1x21.5 MVAR capacitor bank on 34.5 kV system
J614	One 66 MW unit	R=0.02227 pu X=0.127 pu B=0.00182 pu	2x6.3 and 1x6.3 MVAR capacitor bank on 34.5 kV system
J615	One 70 MW unit	R=0.00219 pu X=0.12014 pu B=0.01795 pu	2x6.3 and 1x6.3 MVAR capacitor bank on 34.5 kV system
J637	One 98 MW unit	R=0.0105 pu X=0.069128 pu B=0.0181 pu	3x10 MVAR capacitor bank on 34.5 kV system
J638	One 100 MW unit and one 104 MW unit	Circuit #1: R=0.034201 pu X=0.2703 pu B=0.02068 pu Circuit #2: R=0.034201 pu X=0.26983 pu B=0.019805 pu	4x15 MVAR capacitor bank on each of the two 34.5 kV collector system

Table A-4: DPP 2016 August Central Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J446	IN	Clinton	DEI	Frankfort-New London 230 kV	200	Wind	NRIS
J456	IL	McDonough	AMIL	Niota-Macomb 138 kV	150	Wind	NRIS
J474	IL	DeWitt	AMIL	North Clinton 138 kV	144	Wind	NRIS
J513	IN	Jasper	NIPS	Reynolds 138 kV	100.05	Wind	NRIS
J641	IL	Morgan, Scott	AMIL	Meredosia East-Jacksonville Industrial Park 138 kV	140	Solar	NRIS
J643	IL	Jasper	NIPS	RM Schahfer-Starke 138 kV	175	Solar	NRIS
J644	IL	Greene, Scott	AMIL	Jerseyville 138 kV	110	Solar	NRIS
J648	IL	Cook	ComEd	SCEP switchyard 138 kV	296	Gas	External NRIS

Table A-5: DPP 2016 August Michigan Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J571	MI	Kalamazoo	METC	Morrow 138 kV	6.4	CT	NRIS
J572	MI	Manistee	METC	Filler City JCT 138 kV	150	CC	NRIS
J589	MI	Gratiot	METC	Regal-Summerton 138 kV	148.8	Wind	NRIS
J601	MI	Lapeer, Tuscola	ITCT	Kilgore 120 kV	100	Wind	NRIS
J602	MI	Shiawassee	METC	Goss 138 kV	200	Wind	NRIS
J608	MI	Montcalm	METC	Deja 138 kV	100	Wind	NRIS

Table A-6: DPP 2016 August ATC Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J652	WI	Lafayette	ATC	Darlington-Hillman 138 kV	98	Wind	NRIS Only

A.2 DPP 2016 February West Area Phase 2 Network Upgrades

Table A-7: DPP 2016 February West Phase 2 NUs

Constraint	Owner	Mitigation
J530 POI-Hills 345 kV	MEC	MEC: Structure replacements
Hazel Creek 345-230-13.8 kV xfmr #6	XEL	Upgrade Hazel Creek TR9 to 672 MVA
Panther-McLeod 230 kV	GRE	Remediate 4 Structures
Johnson Jct-Ortonville 115 kV	MRES GRE OTP	OTP: Ortonville terminal equipment requires replacement. OTP's portion of the 0.56 mile line will be required to be fully rebuilt with a larger conductor. GRE: GRE owned switch at Johnson Jct needs to be replaced. MRES: The MRES owned section (majority of the line) will have to be rebuilt for rating increases.
Johnson Jct-Morris 115 kV	GRE WAPA	GRE: Rebuild 15 miles of 115 kV line from Johnson Junction to Morris. WAPA: NU is not required unless it is identified as constraint in affected system study.
Big Stone 230-115-13.8 kV xfmr	OTP	Transformer and Terminal equipment will be required to be replaced at the substation.
Big Stone-Big Stone South 230 kV #1	OTP	Line will be required to be fully rebuilt with a larger conductor as well as replacement of terminal equipment at Big Stone.
Big Stone-Browns Valley 230 kV	OTP MDU	OTP: owns Big Stone and Browns Valley terminal equipment which is sufficient to handle the flow seen in the study. \$0 MDU: Conductor clearance mitigation.
Big Stone-Blair 230 kV	OTP BEPC	OTP: Some OTP terminal equipment will be required to be replaced at the Big Stone 230 kV sub. BEPC: NU is not required unless it is identified as constraint in affected system study.
Hankinson-Wahpeton 230 kV	OTP	Terminal equipment at both Hankinson and Wahpeton substations will be required to be replaced. In addition, conductor clearance mitigations will be required for the line.
Wahpeton-Fergus Falls 230 kV	OTP MRES	Terminal equipment at Wahpeton substation will be required to be replaced. In addition, conductor clearance mitigations will be required for the line.
Oakes-Forman 230 kV	OTP	Terminal equipment at Oakes and Forman substations will be required to be replaced. In addition, conductor clearance mitigations will be required for the line.
Oakes-Ellendale 230 kV	OTP MDU	OTP: Terminal equipment at Oakes will be required to be replaced. In addition, conductor clearance mitigation will be required for the line. MDU: The Ellendale MVP 230 kV bus will be built to 796 MVA.
Marshalltown-Blairtown 115 kV	ITCM	Sag limit at 162F. Raise/replace structures and replace bus conductor at Blairtown. New rating of 120 MVA.

Constraint	Owner	Mitigation
Blairstown-Prairie Crk 115 kV	CIPCO ITCM	CIPCO: No Mitigation Needed. \$0 ITCM: Clearance limit at 153F. Raise structures to reach a minimum rating of 114 MVA
Hazleton-Washburn 161 kV	ITCM MEC	MEC: Reconductor with structure replacements as appropriate and substation terminal equipment upgrades. ITCM: Terminal limit is 325 MVA.
Boone Jct-Sub T Fort Dodge 161 kV	MEC CIPCO	CIPCO: No Mitigation Needed. \$0 MEC: Reconductor with structure replacements as appropriate and substation terminal equipment upgrades.
Parnell-J438 POI 161 kV	MEC ITCM	MEC: Structure replacements. ITCM: ITCM Limit is 335 MVA at Parnell
Grimes-Sycamore 345 kV #2	MEC	Sycamore substation terminal equipment upgrades.
Grimes-Beaver Creek 345 kV	MEC	Structure replacements.
Granger Tap-108th & 54th 161 kV	MEC	Reconductor short section of line and terminal equipment upgrades.
Bondurant-Montezuma 345 kV	MEC	Structure replacements
Webster-Sub T Ft Dodge 161 kV	MEC	Reconductor with structure replacements as appropriate.
Voltage constraint identified at Lathrop 161 kV bus	AMMO	25 Mvar switched cap bank at Lathrop 161 kV (344923)
Voltage constraints in the areas of Hickory Creek, Salem, and Rock Creek	ITCM	Hickory Creek-Cardinal 345 kV project (MVP project 3127); XFMR tap should be locked at neutral position for the two 345-161 kV xfms at Salem (631140)
Voltage constraint in the area of Blackhawk	MEC	150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 1×50 Mvar switched cap bank at Midport 161 kV (636202)
Voltage constraint in the area of Montezuma, J530 POI	MEC	100 Mvar switched cap bank at Montezuma 345 kV (635730)
Voltage constraint in the area of J438 POI	MEC	1×25 Mvar switched cap bank at J438 POI 161 kV (51113)
Stability constraint	J488 J489	2nd J488&489 POI-Big Stone South 345 kV
Stability constraint	OTP GRE	2nd J510 POI-Brookings Co 345 kV

A.3 Model Review Comments

Table A-8: Model Review Comments

Company	Python/ Idev File Name	2022 SH	2022 SPK	2022 Stability
ATC	20210917_StoneLake_345kV_STATCOM.idv	x	x	x
ATC	Increase NED xfmr Rate B to 334 MVA.py	x	x	x
ATC	20231231_Z3_SB_EEN_345-138kV.IDV	x	x	x
ATC	20231231_Z3_SB_HCK-EEN_345kV.idv	x	x	x
ATC	20231231_Z3_SB_EEN-CDL_345kV.idv	x	x	x
ATC	20170808_CHC-NLL_Eden2.con	x	x	
BEPC	BEPC_MISO_2016_FEB_West_Phase2_Modelupdate_R1.idv	x	x	x
BEPC	BEPC_MISO_2016_FEB_West_Phase2_Gen_update_SH22-SPTI.idv	x		x
BEPC	BEPC_MISO_2016_FEB_West_Phase2_Gen_update_SP22-SPTI.idv		x	
CIPCO	CIPCO_Gen_SH.idv	x		x
CIPCO	CIPCO_Gen_PK.idv		x	
CIPCO	CIPCO DPP-2016-FEB Phase 2 Additional P6.con	x	x	
DPC	Update DPC Ratings.py	x	x	x
GRE	py-GRE-add-G667-at-RoundLakeTap.py	x	x	x
GRE	py-GRE-add-G549-at-Williams.py	x	x	x
GRE	py-open4X55atRockRiver.py	x	x	x
GRE	GRE_CCR-DKN.CON	x	x	
GRE	Adding H081 Info.idv	x	x	x
ITCM	ITCM DPP 16 Feb Phase 2 Comments.idv	x	x	x
MDU	MDU_Updates-MISO17_2022SUM_TA_SPTI.idv		x	
MDU	MDU_Updates-MISO17_2022SH90_TA_SPTI.idv	x		x
MEC	Adjust MEC Wind output_SH.idv	x		x
MEC	DPP2016FEB Ph2 MEC SH Updates 09.15.2017.py	x		x
MEC	DPP2016FEB Ph2 MEC SUM Updates 09.15.2017.py		x	
MEC	Turn off MEC Reactors.py	x		x
MEC	Adjust J416 SH output.py	x		x
MRES	MRES Summer Model Updates.idv	x	x	x
OTP	1) OTP_Buffalo_disconnect_xfmr2_416.idv	x	x	x
OTP	2) Jamestown345Transformers_10-25-16.idv	x	x	x
OTP	3) BSS XFMR updates.idv	x	x	x
OTP	4) OTP_unlock_BSS_reactors.idv	x	x	x

Company	Python/ Idev File Name	2022 SH	2022 SPK	2022 Stability
OTP	5) BigStone-BigStoneSouth230kV_LineImedanceUpdates_9-13-17.idv	x	x	x
OTP	6) SummerRatingUpdates_9-13-17.idv	x	x	x
OTP	7) BigStoneSouth-Brookings345kV_LineImedanceUpdates_9-13-17.idv	x	x	x
OTP	8) BigStoneSouth-Ellendale345kV_LineImedanceUpdates_9-13-17.idv	x	x	x
SPP	Turn off Ft Calhoun.py	x	x	x
MISO	Adjust J171 Gen output.py	x	x	x
MISO	Update MDU WTG outputs.py	x		x
MISO	Update J320 Pmax.py	x	x	x
MISO	Turn off J391_SH.py	x		x
IC	Address J493 Comments.idv	x	x	x
MISO	SystemModelingShoulderFinalPython.py	x		x
MISO	SystemModelingSummerFinalPython.py		x	
SPTI	Update Area Numbers.idv	x	x	x
GRE	Adjust J171 Gen output.py	x	x	x
MISO	Update MDU WTG outputs.py	x		x
MISO	Update J320 Pmax.py	x	x	x
MISO	Remove Duplicated DPP Projects.idv	x	x	x
MISO	Remove retired gens.idv	x	x	x
IC	J475_SH.idv	x		x
IC	J475_PK.idv		x	
IC	J493&J510_SH.idv	x		x
IC	J493&J510_PK.idv		x	
IC	J514_SH.idv	x		x
BEPC	SH-Turn Off BSS SSL.py	x		x
MISO	RMV_J264.py	x	x	x
MISO	RMV_J298.py	x	x	x
MEC	Cap_ShaulisRd 161.py	x	x	x
MISO	J514_SH.idv	x		x
MISO	J514_PK.idv		x	
MEC	Correct_J475_SH.idv	x		x
MEC	Correct_J475_PK.idv		x	
MISO	Adjust length of J436 J437 POI line.py	x	x	x
MISO	Turn off BSSE Reactor.py	x	x	x

Company	Python/ Idev File Name	2022 SH	2022 SPK	2022 Stability
SPTI	Update DPP Caps.py	x	x	x
OTP	OTPSummerRatingUpdates_10-24-17.idv	x	x	x
MDU	MDU_Updates-Bench_MISO17_2022SH90_NetInt.idv	x		x
MDU	MDU_Updates-MISO17_2022SH90.idv	x		x
MDU	MDU_Updates-MISO17_2022SUM.idv		x	
SPP	RMV_SPP-531449.py	x	x	x
IC	Update J525 GSU.idv	x	x	x
MISO	Correct J442-G736.py	x	x	x
MISO	Add 2015AugDPP changes_171031.py	x	x	x
MEC	MEC Stability Updates.py			x
MDU	MDU-ModelUpdates_StudyCase-2022_SH90_DS_171020.idv			x
MDU	MDU-OtherUpdates_StudyCase-2022_SH90_DS_171020.idv	x	x	x
AMRN	Gateway.idv	x	x	x
AMRN	Beehive.idv	x	x	x
AMRN	Watson Area.idv	x	x	x
AMRN	Tabor.idv	x	x	x
AMRN	AMRN_VSchedule.py	x	x	x
AMRN	AMRN_Rating Chng.py	x	x	x
AMRN	AMRN_Impedance Chng.py	x	x	x
BEPC	BEPC_Proj-Area Chng.py	x	x	x
MEC	MEC DPP2016AUG West Ph1 SH90 Updates 10.12.2017.py	x		x
MEC	MEC DPP2016AUG West Ph1 SUM Updates 10.12.2017.py		x	
J414	J414_DPP1_SH.py	x		x
J414	J414_DPP1_SPK.py		x	
J415	J415_DPP1_SH.py	x		x
J415	J415_DPP1_SPK.py		x	
SPP	RMV_GEN-2016-034.py	x	x	x
SPP	RMV_GEN-2016-055.py	x	x	x
SPP	RMV_GEN-2016-064.py	x	x	x
J457	J457_Xfmr_Rating.py	x	x	x
J457	Correct_Modeling errors.py	x	x	x
J459	Correct J459.py	x	x	x
J511	Correct_J511_Cap.py	x	x	x

Company	Python/ Idev File Name	2022 SH	2022 SPK	2022 Stability
J512	Correct_J512_SH.py	x		x
J512	Correct_J512_PK.py		x	
J569	J569_SH.idv	x		x
J569	J569_PK.idv		x	
J587	Correct_J587.py	x	x	x
J590	Correct_J590_SH.py	x		x
J590	Correct_J590_PK.py		x	
J613	Correct_J613_SH.py	x		x
J613	Correct_J613_PK.py		x	
J614	J614_Caps.py	x	x	x
J614	Correct_J614_0.95_SH.idv	x		x
J614	Correct_J614_0.95_PK.idv		x	
J615	J615_Caps.py	x	x	x
J637, J638	Correct_J637-J638.py	x	x	x
J439	J439.idv	x	x	x
J476	J476.idv	x	x	x
J555	J555.idv	x	x	x
J598	J598.idv	x	x	x
J599	J599.idv	x	x	x
MISO	Remove the original NUs required for DPP 2015 Aug West group	x	x	x
ITCM	Aug_15_DPP_RestudyMitigation_ITCM.idv	x	x	x
OTP	OTP_Mitigations_11-22-17.idv	x	x	x
MEC	M Ave-New Sharon 69 kV Reconductor.py	x	x	x
MEC	Poweshiek-New Sharon 69 kV Reconductor.py	x	x	x
J493	Add-J493-Cap.py	x	x	x
MISO	RMV_J601.py	x	x	x
MISO	RMV_J608.py	x	x	x
MISO	Order827_Updates.py	x	x	x
SPP	15.DIS1502PQ_ERIS_BUILD_RPLAN-PROJECT[16WP-17WP].idv	x	x	x
SPP	16.DIS1601_ERIS_BUILD_KEYSTONE-GGS-345kV-CKT2.idv	x	x	x
SPP	BANNERCO-KEYSTONE-345kV-CKT1.idv	x	x	x
MISO	correct J503 collector system.py	x	x	x
Xcel	G621_SH.py	x		x

Company	Python/ Idev File Name	2022 SH	2022 SPK	2022 Stability
Xcel	G621_PK.py		x	
J475	J475_update.py	x	x	x
MISO	Order827_Updates.py	x		x
MISO	Order827_Updates_SPK.py		x	
MDU	remove duplicate Ellendale shunt reactor.py	x	x	x

A.4 MISO Classic as the Study Sink

Table A-9: MISO Classic as the Study Sink

Area #	Area Name	Area #	Area Name
207	HE	600	Xcel
208	DEI	608	MP
210	SIGE	613	SMMPA
216	IPL	615	GRE
217	NIPS	620	OTP
218	METC	627	ALTW
219	ITC	633	MPW
295	WEC	635	MEC
296	MIUP	661	MDU
314	BREC	663	BEPC-MISO
333	CWLD	680	DPC
356	AMMO	694	ALTE
357	AMIL	696	WPS
360	CWLP	697	MGE
361	SIPC	698	UPPC

A.5 PJM Market as PJM Projects Sink

Table A-10: PJM Market as PJM Projects Sink

Area #	Area Name	Area #	Area Name
201	AP	229	PPL
202	ATSI	230	PECO
205	AEP	231	PSE&G
209	DAY	232	BGE
212	DEO&K	233	PEPCO
215	DLCO	234	AE
222	CE	235	DP&L
225	PJM	236	UGI
226	PENELEC	237	RECO
227	METED	320	EKPC
228	JCP&L	345	DVP

A.6 SPP Market as SPP Projects Sink

Table A-11: SPP Market as SPP Projects Sink

Area #	Area Name	Area #	Area Name
515	SWPA	541	KCPL
520	AEPW	542	KACY
523	GRDA	544	EMDE
524	OKGE	545	INDN
525	WFEC	546	SPRM
526	SPS	640	NPPD
527	OMPA	645	OPPD
531	MIDW	650	LES
534	SUNC	652	WAPA
536	WERE	659	BEPC-SPP
540	GMO		

A.7 Contingency Files used in Steady-State Analysis

Table A-12: List of Contingencies used in Steady-State Analysis

Contingency File Name	Description	2022
Automatic single element contingencies	Single element outages at buses 60 kV and above in the study region	x
20170808_CHC-NLL_Eden2.con	Specified category P1, P2 contingencies in ATC, ITCM	x
CIPCO DPP-2016-FEB Phase 2 Additional P6.con	Specified category P6 contingencies in CIPCO	x
CC Bipole Events.con	Specified category P1, P7 contingencies in GRE Coal Creek	x
MEC-DPP2016AUG West Ph1 2022 Cat P1 02.06.2018.con	Specified category P1 contingencies in MEC	x
MEC-DPP2016AUG West Ph1 2022 Cat P2 02.06.2018.con	Specified category P2 contingencies in MEC	x
MEC-DPP2016AUG West Ph1 2022 Cat P5 02.06.2018.con	Specified category P5 contingencies in MEC	x
MEC-DPP2016AUG West Ph1 2022 Cat P7 02.06.2018.con	Specified category P7 contingencies in MEC	x
P1_AMES_MTEP17-2022TA.CON	Specified category P1 contingencies in AMES	x
P2-P7_AMES_MTEP17-2022TA.con	Specified category P2-P7 contingencies in AMES	x
P1_ATC_MTEP17-2022TA.con	Specified category P1 contingencies in ATC	x
P2-P7_ATC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in ATC	x
P1_BEPC_MTEP17-2022TA.con	Specified category P1 contingencies in BEPC	x
P2-P7_BEPC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in BEPC	x
P1_CBPC_MTEP17-2022TA.con	Specified category P1 contingencies in CBPC	x
P2-P7_CBPC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in CBPC	x
P1_CFU_MTEP17-2022TA.con	Specified category P1 contingencies in CFU	x
P2-P7_CFU_MTEP17-2022TA.con	Specified category P2-P7 contingencies in CFU	x
P1_CIPCO_MTEP17-2022TA.con	Specified category P1 contingencies in CIPCO	x
P2-P7_CIPCO_MTEP17-2022TA.con	Specified category P2-P7 contingencies in CIPCO	x
P1_DPC_MTEP17-2022TA.con	Specified category P1 contingencies in DPC	x
P2-P7_DPC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in DPC	x
P1_GRE_MTEP17-2022TA.con	Specified category P1 contingencies in GRE	x
P2-P7_GRE_MTEP17-2022TA.con	Specified category P2-P7 contingencies in GRE	x

Contingency File Name	Description	2022
P1_ITCM_MTEP17-2022TA.con	Specified category P1 contingencies in ITCM	x
P2-P7_ITCM_MTEP17-2022TA.con	Specified category P2-P7 contingencies in ITCM	x
P1_MDU_MTEP17-2022TA.con	Specified category P1 contingencies in MDU	x
P2-P7_MDU_MTEP17-2022TA.con	Specified category P2-P7 contingencies in MDU	x
P1_MP_MTEP17-2022TA.con	Specified category P1 contingencies in MP	x
P2-P7_MP_MTEP17-2022TA.CON	Specified category P2-P7 contingencies in MP	x
P1_MPC_MTEP17-2022TA.con	Specified category P1 contingencies in MPC	x
P2-P7_MPC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in MPC	x
P1_MPW_MTEP17-2022TA.con	Specified category P1 contingencies in MPW	x
P2-P7_MPW_MTEP17-2022TA.CON	Specified category P2-P7 contingencies in MPW	x
P1_MRES_MTEP17-2022TA.con	Specified category P1 contingencies in MRES	x
P2-P7_MRES_MTEP17-2022TA.CON	Specified category P2-P7 contingencies in MRES	x
P1_OTP_MTEP17-2022TA.con	Specified category P1 contingencies in OTP	x
P2-P7_OTP_MTEP17-2022TA.con	Specified category P2-P7 contingencies in OTP	x
P1_RPU_MTEP17-2022TA.con	Specified category P1 contingencies in RPU	x
P2-P7_RPU_MTEP17-2022TA.con	Specified category P2-P7 contingencies in RPU	x
P1_SMPMA_MTEP17-2022TA.con	Specified category P1 contingencies in SMPMA	x
P2-P7_SMPMA_MTEP17-2022TA.con	Specified category P2-P7 contingencies in SMPMA	x
P1_XEL_MTEP17-2022TA.con	Specified category P1 contingencies in XEL	x
P2-P7_XEL_MTEP17-2022TA.con	Specified category P2-P7 contingencies in XEL	x
P1-4_ATC_MTEP17-2022TA.con	Specified category P1-4 contingencies in ATC	x
P1-4_DPC_MTEP17-2022TA.con	Specified category P1-4 contingencies in DPC	x
P1-4_GRE_MTEP17-2022TA.con	Specified category P1-4 contingencies in GRE	x
P1-4_ITCM_MTEP17-2022TA.con	Specified category P1-4 contingencies in ITCM	x
P1-4_MDU_MTEP17-2022TA.con	Specified category P1-4 contingencies in MDU	x
P1-4_MP_MTEP17-2022TA.con	Specified category P1-4 contingencies in MP	x
P1-4_MPC_MTEP17-2022TA.con	Specified category P1-4 contingencies in MPC	x
P1-4_OTP_MTEP17-2022TA.con	Specified category P1-4 contingencies in OTP	x
P1-4_SMPMA_MTEP17-2022TA.con	Specified category P1-4 contingencies in SMPMA	x
P1-4_XEL_MTEP17-2022TA.con	Specified category P1-4 contingencies in XEL	x

Contingency File Name	Description	2022
P1_AMRN_MTEP17-2022TA.con	Specified category P1 contingencies in Ameren	x
P2-P7_AMRN_MTEP17-2022TA.con	Specified category P2-P7 contingencies in Ameren	x
Ameren_N-2.con	Specified category P6 345 kV contingencies in Ameren	x
P1_CWLD_MTEP17-2022TA.con	Specified category P1 contingencies in CWLD	x
P2-P7_CWLD_MTEP17-2022TA.con	Specified category P2-P7 contingencies in CWLD	x
P1_CWLP_MTEP17-2022TA.con	Specified category P1 contingencies in CWLP	x
P2-P7_CWLP_MTEP17-2022TA.con	Specified category P2-P7 contingencies in CWLP	x
P1_PPI_MTEP17-2022TA.con	Specified category P1 contingencies in PPI	x
P2-P7_PPI_MTEP17-2022TA.con	Specified category P2-P7 contingencies in PPI	x
P1_SIPC_MTEP17-2022TA.con	Specified category P1 contingencies in SIPC	x
P2-P7_SIPC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in SIPC	x
ComEd_RTEP_Cat_P1.con	Specified category P1 contingencies in ComEd	x
ComEd_RTEP_Cat_P2-P7.con	Specified category P2-P7 contingencies in ComEd	x
P1_AECI_MTEP17-2022TA.con	Specified category P1 contingencies in AECI	x
P2-P7_AECI_MTEP17-2022TA.con	Specified category P2-P7 contingencies in AECI	x
P1_CE_MTEP17-2022TA.con	Specified category P1 contingencies in ComEd	x
P2-P7_CE_MTEP17-2022TA.con	Specified category P2-P7 contingencies in ComEd	x
P1_WAPA_MTEP17-2022TA.con	Specified category P1 contingencies in WAPA	x
P2-P7_WAPA_MTEP17-2022TA.con	Specified category P2-P7 contingencies in WAPA	x
P1-4_AMRN_MTEP17-2022TA.con	Specified category P1-4 contingencies in Ameren	x
P1-4_CWLP_MTEP17-2022TA.con	Specified category P1-4 contingencies in CWLP	x
P1-4_SIPC_MTEP17-2022TA.con	Specified category P1-4 contingencies in SIPC	x
P1-4_WAPA_MTEP17-2022TA.con	Specified category P1-4 contingencies in WAPA	x

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Model Data

B.1 Power Flow Model Data

CEII Redacted

B.2 Dynamic Model Data

CEII Redacted

B.3 2022 Slider Diagrams

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Appendix

C

Reactive Power Requirement Analysis Results (FERC Order 827)

Table C-1: Reactive Power Requirement Analysis Results

Project #	HV Side Bus #	Lagging Power Factor Results				Leading Power Factor Results				Turbine Inherent Power Factor	Shunt Compensation
		MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Lagging Power Factor at HV Side	Meet Lagging Power Factor Req.?	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Leading Power Factor at HV Side	Meet Leading Power Factor Req.?		
J302	83024	99.67	57.09	0.868	Yes	99.49	-65.85	-0.834	Yes	± 0.9	3×8 MVAR capacitor bank on 34.5kV system
J414	84141	117.43	45.69	0.932	Yes	116.30	-103.70	-0.746	Yes	± 0.9	15 MVAR capacitor bank on 34.5 kV system
J415	84152	196.17	70.50	0.941	Yes	194.90	-157.30	-0.778	Yes	± 0.9	5 Mvar capacitor bank on each of the two 34.5 kV collector system
J439	84391	490.60	185.60	0.935	Yes	489.30	-256.00	-0.886	Yes	± 0.95	45 MVAR capacitor bank on one 34.5 kV system, 50 MVAR capacitor bank on another 34.5 kV system
J457	84571	146.40	55.80	0.934	Yes	145.60	-90.20	-0.850	Yes	± 0.95	2×17 Mvar capacitor bank on 34.5 kV system
J459	84591	196.64	38.67	0.981	No	195.79	-98.07	-0.894	Yes	0.98 lagging 0.96 leading	2×14 Mvar capacitor bank on 34.5 kV system
J476	84761	239.28	84.83	0.943	Yes	238.31	-127.25	-0.882	Yes	± 0.95	3×15 Mvar capacitor bank and 1×5 Mvar reactor bank on 34.5 kV system
J503	83024	93.25	51.39	0.876	Yes	91.67	-73.99	-0.778	Yes	± 0.9	1×24 Mvar capacitor bank on 34.5 kV system
J511	85111	196.67	74.16	0.936	Yes	196.13	-118.49	-0.856	Yes	V136 3.6 MW: -0.93 (leading), 0.913 (lagging) V110 2.0 MW: ± 0.95	18 Mvar capacitor bank on 34.5 kV collector system
J512	85122	245.50	82.50	0.948	Yes	244.22	-161.93	-0.833	Yes	V136 3.6 MW: -0.93 (leading), 0.913 (lagging) V110 2.0 MW: ± 0.95	16 Mvar capacitor bank on 34.5 kV collector system
J541	85412	389.60	142.40	0.939	Yes	386.80	-246.80	-0.843	Yes	± 0.95	6×18 Mvar capacitor bank on 34.5 kV collector system
J555	65730	165.56	57.04	0.945	Yes	165.56	-92.80	-0.872	Yes	± 0.95	1×5, 1×10, and 2×15 Mvar capacitor banks on 34.5 kV collector system
J569	85691	98.92	33.81	0.946	Yes	98.70	-68.72	-0.821	Yes	± 0.9	None
J575	85751	97.99	40.83	0.923	Yes	97.42	-74.55	-0.794	Yes	± 0.9	2×6 MVAR capacitor bank on 34.5 kV system
J577	85771	97.99	40.83	0.923	Yes	97.42	-74.55	-0.794	Yes	± 0.9	2×6 MVAR capacitor bank on 34.5 kV system
J583	85831	197.10	77.02	0.931	Yes	196.68	-154.26	-0.787	Yes	± 0.9	3×8 MVAR capacitor bank on 34.5 kV system
J587	85873	196.57	43.58	0.976	No	195.90	-102.70	-0.886	Yes	0.98 lagging 0.96 leading	20 Mvar capacitor bank on each 34.5 kV collector system
J593	85931	220.75	84.35	0.934	Yes	220.13	-112.23	-0.891	Yes	± 0.95	2×15 and 1×10 MVAR capacitor bank on 34.5 kV system
J594	85941	295.49	107.90	0.939	Yes	294.52	-152.79	-0.888	Yes	± 0.95	25 Mvar capacitor bank on each 34.5 kV collector system
J596	85962	98.34	36.63	0.937	Yes	98.05	-47.37	-0.900	Yes	± 0.95	1×15 MVAR capacitor bank on 34.5 kV system
J597	85971	293.20	111.70	0.934	Yes	293.20	-164.50	-0.872	Yes	± 0.95	2×20 Mvar capacitor bank on one 34.5 kV collector system, and (1×20,1×5) Mvar capacitor bank on another 34.5 kV collector system
J598	85982	295.30	115.30	0.932	Yes	294.40	-150.80	-0.890	Yes	± 0.95	2×20 Mvar capacitor bank on one 34.5 kV collector system, and 1×20 Mvar capacitor bank on another 34.5 kV collector system
J599	85991	196.60	74.40	0.935	Yes	195.80	-102.20	-0.887	Yes	± 0.95	1×25 and 1×10 Mvar capacitor bank on 34.5 kV system
J607	86072	147.54	57.39	0.932	Yes	147.11	-77.36	-0.885	Yes	± 0.95	1×20 and 1×10 Mvar capacitor bank on 34.5 kV system
J611	86112	108.41	45.18	0.923	Yes	108.15	-50.18	-0.907	Yes	± 0.95	2×20 Mvar capacitor bank on 34.5 kV system
J613	86131	99.17	42.73	0.918	Yes	99.09	-45.77	-0.908	Yes	± 0.95	1×21.5 Mvar capacitor bank on 34.5 kV system
J614	86141	64.20	21.80	0.947	Yes	63.63	-48.68	-0.794	Yes	± 0.95	2×6.3 and 1×6.3 Mvar capacitor bank on 34.5 kV system
J615	86152	68.96	20.14	0.960	No	68.83	-46.92	-0.826	Yes	± 0.95	2×6.3 and 1×6.3 Mvar capacitor bank on 34.5 kV system
J637	86372	95.80	33.70	0.943	Yes	95.36	-65.75	-0.823	Yes	± 0.95	3×10 Mvar capacitor bank on 34.5 kV system
J638	86381	194.40	74.80	0.933	Yes	193.40	-130.40	-0.829	Yes	± 0.95	4×15 Mvar capacitor bank on each of the two 34.5 kV collector system

2022 Summer Peak Contingency Analysis Results

2022 Summer Peak (SPK) Constraints

Table D-1: 2022 SPK System Intact Thermal Constraints

Table D-2: 2022 SPK System Intact Voltage Constraints

Table D-3: 2022 SPK Category P1 Thermal Constraints

Table D-4: 2022 SPK Category P1 Voltage Constraints

Table D-5: 2022 SPK Category P2-P7 Thermal Constraints

Table D-6: 2022 SPK Category P2-P7 Voltage Constraints

Table D-7: 2022 SPK Non-Converged Contingencies

Table D-8: 2022 SPK Non-Converged Contingencies DCCC Results

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2022 Summer Shoulder Contingency Analysis Results

E.1 Thermal Constraints Identified in Stage-1 Contingency Analysis with Fictitious SVCs

Table E-1: 2022 SH System Intact Thermal Constraints in Stage-1 ACCC

Table E-2: 2022 SH Category P1 Thermal Constraints in Stage-1 ACCC

Table E-3: 2022 SH Category P2-P7 Thermal Constraints in Stage-1 ACCC

Table E-4: 2022 SH Non-Converged Contingencies in Stage-1 ACCC

Table E-5: 2022 SH Non-Converged Contingencies DCCC Results in Stage-1 ACCC

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E.2 Backbone Network Upgrades Justification Results

E.2.1 Project Justification Details in Dakotas-Minnesota

Table E-6: Project Evaluation Details in Dakotas-Minnesota

Transmission NUs	Miles	Constraints to be Relieved (MW)	Constraints to be aggravated (MW)	Potential New Constraints Caused by the NU (MW)
Big Stone South-Alexandria 345 kV	95	J302&J503 POI-Heskett 230 kV: -134 Wilmarth-Sheas Lake 345 kV: -257 Split Rock-White 345 kV: -446 Split Rock-Sioux City 345 kV: -73 Lakefield Gen-Lakefield 345 kV: -185 Brookings Co-White 345 kV #1: -339 Brookings Co-White 345 kV #2: -377 Field South-Crandal 345 kV: -111 Lyon Co-Hawksnest 345 kV: -277 Helena-Scott Co 345 kV: -192 Helena-Sheas Lake 345 kV: -258 Hazel Crk 345-230-13.8 kV xfmr: -70 Minn Valley Tap-Granite Falls 230 kV: -65 Silver Lk-Henning 230 kV: -64 Silver Lk-Fergus Falls 230 kV: -65 Johnson Jct-Ortonville 115 kV: -86 Johnson Jct-Morris 115 kV: -71 Lakemina-Brandon 115 kV: -112 Canby-Granite Falls 115 kV: -54 Big Stone-Highway 12 115 kV: -116 Big Stone 230-115-13.8 kV xfmr: -152 Highway 12-Ortonville 115 kV: -115 Big Stone-Big Stone South 230 kV #1: -352 Big Stone-Big Stone South 230 kV #2: -381 Big Stone-Brownsville 230 kV: -193 Big Stone-Blair 230 kV: -384 Hankinson-Wahpeton 230 kV: -244 Hankinson-Forman 230 kV: -117 Wahpeton-Fergus Falls 230 kV: -114 Oakes-Forman 230 kV: -128 Oakes-Ellendale 230 kV: -131	J530 POI-Hills 345 kV: 137 Hazleton-Blackhawk 345 kV: 104	MVA Loading increase: 1) Quarry 345-115-34.5 kV xfmr: 186 2) River Rd 115-69-34.5 kV xfmr: 50 3) ALEXMRES-Alex S.W. Station 115 kV: 94

2022 Summer Shoulder Contingency Analysis Results

Transmission NUs	Miles	Constraints to be Relieved (MW)	Constraints to be aggravated (MW)	Potential New Constraints Caused by the NU (MW)
		Big Stone South 345-230-34.5 kV xfmr #1: -367 Big Stone South 345-230-34.5 kV xfmr #2: -368 Raun-S3451 345 kV: -75 Grimes-Beaver Crk 345 kV: -117 Blair-Granite Falls 230 kV: -180 Watertown 345-230-13.8 kV xfmr: -113 Watertown-Appledorn 230 kV: -111 Morris-Grant Co 115 kV: -68 Elbow Lk-Brandon 115 kV: -111 Ellendale-Aberdeen Jct 115 kV: -53		
J510-Hazel Creek-Scott County 345 kV	165	J302&J503 POI-Heskett 230 kV: -140 Adams 345-161-13.8 kV xfmr: -94 Wilmarth-Sheas Lake 345 kV: -492 Split Rock-White 345 kV: -752 Split Rock-Sioux City 345 kV: -146 Lakefield Gen-Lakefield 345 kV: -331 Brookings Co-White 345 kV #1: -612 Brookings Co-White 345 kV #2: -817 Field South-Crandal 345 kV: -251 Lyon Co-Hawksnest 345 kV: -575 Helena-Scott Co 345 kV: -414 Helena-Sheas Lake 345 kV: -497 Hazel Crk 345-230-13.8 kV xfmr: -242 Minn Valley Tap-Granite Falls 230 kV #1: -182 Minn Valley Tap-Granite Falls 230 kV #2: -198 Panther-McLeod 230 kV: -150 Johnson Jct-Ortonville 115 kV: -83 Canby-Granite Falls 115 kV: -74 Big Stone-Highway 12 115 kV: -126 Big Stone 230-115-13.8 kV xfmr: -180 Highway 12-Ortonville 115 kV: -125 Big Stone-Big Stone South 230 kV #1: -420	J530 POI-Hills 345 kV: 136 Hazleton-Arnold 345 kV: 116 Hazleton-Blackhawk 345 kV: 110	MVA Loading increase: 1) Blue Lake-Scott Co 345 kV: 376 2) Scott Tap-Scott Co 115 kV: 43 3) Scott Co 345-115-34.5 kV xfmr: 100 4) Scott Co-Minn River 115 kV: 52 5) Minn River-CHANHAS 115 kV: 52

2022 Summer Shoulder Contingency Analysis Results

Transmission NUs	Miles	Constraints to be Relieved (MW)	Constraints to be aggravated (MW)	Potential New Constraints Caused by the NU (MW)
		Big Stone-Big Stone South 230 kV #2: -454 Big Stone-Brownsville 230 kV: -167 Big Stone-Blair 230 kV: -508 Hankinson-Wahpeton 230 kV: -190 Hankinson-Forman 230 kV: -69 Wahpeton-Fergus Falls 230 kV: -87 Oakes-Forman 230 kV: -107 Oakes-Ellendale 230 kV: -120 Big Stone South 345-230-34.5 kV xfmr #1: -437 Big Stone South 345-230-34.5 kV xfmr #2: -437 Raun-S3451 345 kV: -76 Grimes-Beaver Crk 345 kV: -110 Bondurant-Montezuma 345 kV: -89 Blair-Granite Falls 230 kV: -290 Watertown 345-230-13.8 kV xfmr: -259 Watertown-Appledorn 230 kV: -223 Granite Falls-Appledorn 230 kV: -170 Ellendale-Aberdeen Jct 115 kV: -69		
Big Stone South-Hazel Creek 345 kV	80	J302&J503 POI-Heskett 230 kV: -120 Wilmarth-Sheas Lake 345 kV: -299 Split Rock-White 345 kV: -490 Lakefield Gen-Lakefield 345 kV: -208 Brookings Co-White 345 kV #1: -370 Brookings Co-White 345 kV #2: -411 Field South-Crandal 345 kV: -127 Lyon Co-Hawksnest 345 kV: -387 Helena-Sheas Lake 345 kV: -301 Minn Valley Tap-Granite Falls 230 kV: -213 Johnson Jct-Ortonville 115 kV: -68 Johnson Jct-Morris 115 kV: -59 Canby-Granite Falls 115 kV: -70 Big Stone-Highway 12 115 kV: -110	J530 POI-Hills 345 kV: 143 Helena-Scott Co 345 kV: 120 Hazel Crk 345-230-13.8 kV xfmr: 162 Hazleton-Arnold 345 kV: 101 Hazleton-Blackhawk 345 kV: 106	1) Hazel Crk 345-230-13.8 kV xfmr is loaded @ 122.6% flo Lyon Co-Hazel Crk 345 kV

2022 Summer Shoulder Contingency Analysis Results

Transmission NUs	Miles	Constraints to be Relieved (MW)	Constraints to be aggravated (MW)	Potential New Constraints Caused by the NU (MW)
		Big Stone 230-115-13.8 kV xfmr: -164 Highway 12-Ortonville 115 kV: -110 Big Stone-Big Stone South 230 kV #1: -363 Big Stone-Big Stone South 230 kV #2: -393 Big Stone-Brownsville 230 kV: -128 Big Stone-Blair 230 kV: -447 Hankinson-Wahpeton 230 kV: -150 Oakes-Ellendale 230 kV: -97 Big Stone South 345-230-34.5 kV xfmr #1: -387 Big Stone South 345-230-34.5 kV xfmr #2: -388 Raun-S3451 345 kV: -71 Grimes-Beaver Crk 345 kV: -117 Blair-Granite Falls 230 kV: -279 Watertown 345-230-13.8 kV xfmr: -107 Watertown-Appledorn 230 kV: -184 Ellendale-Aberdeen Jct 115 kV: -57		

**Table E-7: Projects Not Evaluated in Details in Dakotas-
Minnesota Area**

Transmission NUs	Miles	Reasons for Not Testing
Ellendale-Alexandria 345 kV	180	1) Loading on Ellendale-Alexandria 345 kV line is 562 MVA with Ellendale-Alexandria NU, compared with 808 MVA with Big Stone South-Alexandria NU. 2) Ellendale-Alexandria NU is more expensive than Big Stone South-Alexandria NU (180 miles vs 95 miles)
Big Stone South-Alexandria 345 kV + 2nd Alexandria-Monticello 345 kV	NA	1) Loading on Big Stone South-Alexandria 345 kV line is 930 MVA with 2nd Alexandria-Monticello line, compared with 808 MVA without the 2nd line. The loading increase is not significant. 2) Without 2nd Alexandria-Monticello line, loadings are not very heavy on existing lines: 2a) Alexandria-River Road: 883 MVA, 49.3% 2b) River Road-Quarry: 857 MVA, 47.8% 2c) Quarry-Monticello: 636 MVA, 32.9%
Ellendale-Jamestown 345 kV	76	1) Loading on Ellendale-Jamestown 345 kV line is 305 MVA with Ellendale-Jamestown NU, which is low.
Ellendale-Buffalo 345 kV	118	1) Loading on Ellendale-Buffalo 345 kV line is 413 MVA with Ellendale-Buffalo NU, compared with 808 MVA with Big Stone South-Alexandria NU. 2) Ellendale-Buffalo NU is more expensive than Big Stone South-Alexandria NU (118 miles vs 95 miles).

Table E-8: Loss Reduction Comparison by Projects in Dakotas-Minnesota Area

Area #	Area Name	Without BSS-Alex NU		With BSS-Alex NU		Loss Change			Without J510-Hazel-Scott NU		With J510-Hazel-Scott NU		Loss Change	
		P losses (MW)	Q losses (MVAR)	P losses (MW)	Q losses (MVAR)	P losses (MW)	Q losses (MVAR)		P losses (MW)	Q losses (MVAR)	P losses (MW)	Q losses (MVAR)	P losses (MW)	Q losses (MVAR)
600	XEL	786.9	7122.9	739.8	6714.0	-47.1	-408.9		786.9	7122.9	760.2	7099.6	-26.7	-23.3
608	MP	127.2	1432.2	133.6	1448.7	6.4	16.5		127.2	1432.2	121.1	1399.0	-6.1	-33.2
615	GRE	107.2	1748.4	103.7	1724.4	-3.5	-24.0		107.2	1748.4	102.9	1722.3	-4.3	-26.1
620	OTP	339.3	2592.7	316.2	2659.7	-23.1	67.0		339.3	2592.7	294.0	2301.2	-45.3	-291.5
696	WPS	75.2	891.7	79.3	945.9	4.1	54.2		75.2	891.7	79.3	943.4	4.1	51.7
697	MGE	44.7	563.9	46.9	590.2	2.2	26.3		44.7	563.9	48.2	605.6	3.5	41.7
640	NPPD	249.6	2304.5	247.9	2293.9	-1.7	-10.6		249.6	2304.5	244.8	2272.4	-4.8	-32.1
652	WAPA	427.3	3898.3	399.3	3692.7	-28.0	-205.6		427.3	3898.3	388.3	3567.6	-39.0	-330.7

E.2.2 Project Justification Details in Iowa Area

Table E-9: Project Evaluation Details in Iowa Area

Transmission NUs	Miles	Constraints to be Relieved (MW)	Constraints to be aggravated (MW)	Potential New Constraints Caused by the NU (MW)
Franklin-Morgan Valley & Beverly-Sub 92 345 kV	198	J530 POI-Montezuma 345 kV: -154 J530 POI-Hills 345 kV: -143 J615 POI-Shaulis Rd 161 kV: -89 Adams 345-161-13.8 kV xfmr: -75 Wilmarth-Sheas Lk 345 kV: -233 Lakefield-Lakefield Gen 345 kV: -126 Field South-Crandal 345 kV: -66 Helena-Sheas Lk 345 kV: -259 Hayward-Glenworth 161 kV: -97 Emery-Floyd 161 kV: -95 Hazleton-Blackhawk 345 kV: -116 Hazleton-Washburn 161 kV: -130 Hazleton-Arnold 345 kV: -252 Hazleton-Mitchell Co 345 kV: -142 Hazleton-Hickory Crk 345 kV: -182 Hazleton-Blackhawk 345 kV: -322 Barton-Lime Crk 161 kV: -81 Raun-S3451 345 kV: -104 Grimes-Beaver Crk 345 kV: -172 Bondurant-Sycamore 345 kV: -56 Bondurant-Montezuma 345 kV: -147 Blackhawk-Midport 161 kV: -91 Blackhawk-DRFNDY 161 kV: -90 Blackhawk-Bremer Co 161 kV: -75 Midport-LUNDQST 161 kV: -90 LUNDQST-DR NE 161 kV: -123 DR NE-Washburn 161 kV: -121 Washburn-Shaulis Rd 161 kV: -96 Union Tap-Butler 161 kV: -53 Franklin-Butler 161 kV: -53 Floyd-Bremer Co 161 kV: -71	1) Increase contingency loading on Killdeer 345-161 kV xfmr (170 MVA), Walcott-Sub 92 345 kV (146 MVA), Oak Grove-Sub 93 345 kV (108 MVA). If these are DPP constraints, solutions will be identified in the final ACCC analysis. 2) Expensive	MVA Loading increase: 1) Killdeer 345-161 kV xfmr: 170 2) Walcott-Sub 92 345 kV: 146 3) Oak Grove-Sub 93 345 kV: 108

2022 Summer Shoulder Contingency Analysis Results

Transmission NUs	Miles	Constraints to be Relieved (MW)	Constraints to be aggravated (MW)	Potential New Constraints Caused by the NU (MW)
		Hills-Sub 92 345 kV: -251		
Webster-Franklin 345 kV	49.2	J530 POI-Montezuma 345 kV: -158 J530 POI-Hills 345 kV: -146 Adams 345-161-13.8 kV xfmr: -74 Wilmarth-Sheas Lk 345 kV: -236 Lakefield-Lakefield Gen 345 kV: -158 Helena-Scott Co 345 kV: -82 Helena-Sheas Lk 345 kV: -262 Emery-Colby 161 kV: -115 Emery-Floyd 161 kV: -91 Hayward-Freeborn 161 kV: -58 Hazleton-Mitchell Co 345 kV: -215 Colby 345-161 kV xfmr: -125 Raun-S3451 345 kV: -87 Grimes-Beaver Crk 345 kV: -286 Bondurant-Montezuma 345 kV: -174 Webster 345-161-13.8 kV xfmr: -216 Blackhawk-Bremer Co 161 kV: -94 Union Tap-Butler 161 kV: -105 Franklin-Butler 161 kV: -106 Floyd-Bremer Co 161 kV: -65	1) Hazleton-Arnold 345 kV: 87 2) Hazleton-Blackhawk 345 kV: 243	MVA Loading increase: 1) Hazleton-Arnold 345 kV: 76 2) Hazleton-Blackhawk 345 kV: 252 3) Arnold-Morgan Valley 345 kV: 80 4) Blackhawk 345-161 kV xfmr: 73
Ottumwa-Mt Pleasant 345 kV	67.2	J530 POI-Montezuma 345 kV: -432 J530 POI-Hills 345 kV: -442 Zackary 345-161 kV xfmr: -107 Adair-Zackary 161 kV: -113 Parnell-J438 POI 161 kV: -62 Parnell-Hills 161 kV: -54 Henry Co-Denmark 161 kV: -50 Henry Co-Jeff 161 kV: -61 Wapello-Jeff 161 kV: -71 Wapello-Appanoose 161 kV: -53 Ottumwa 345-161 kV xfmr: -226	Bondurant-Montezuma 345 kV: 78	MVA Loading increase: 1) Bondurant-Montezuma 345 kV: 61 Wapello 161-69 kV xfmr #2 is loaded at 134.1% flo Wapello 161-69 kV xfmr #1 Ottumwa-Montezuma 345 kV is loaded at 114.4% flo J530 POI-Hills 345 kV

2022 Summer Shoulder Contingency Analysis Results

Transmission NUs	Miles	Constraints to be Relieved (MW)	Constraints to be aggravated (MW)	Potential New Constraints Caused by the NU (MW)
		Raun-S3451 345 kV: -72 Grimes-Beaver Crk 345 kV: -107 Hills-Sub 92 345 kV: -183 Beaver Crk-Harmony 161 kV: -54		
Ida Co-Avoca 345 kV	75.6	Adams 345-161-13.8 kV xfmr: -75 Wilmarth-Sheas Lk 345 kV: -218 Helena-Sheas Lk 345 kV: -244 Raun-S3451 345 kV: -288 Grimes-Beaver Crk 345 kV: -294 Webster 345-161-13.8 kV xfmr: -130	None	Council Bluffs-Overland 161 kV is overloaded @109.8% flo Council Bluffs-Overland 345 kV

Table E-10: Projects Not Evaluated in Details in Iowa Area

NUs	Miles	Reasons for Not Testing
Hartford-Morgan Valley & Beverly-Sub 92 345 kV	168	1) Loadings on branches connected to New Hartford are lower with New Hartford-Morgan Valley & Beverly-Sub 92 NU than loadings on branches connected to Franklin with Franklin-Morgan Valley & Beverly-Sub 92 NU 1a) J415 POI-Franklin: 1312 MVA vs J415 POI-New Hartford: 1168 MVA 1b) Franklin-Morgan Valley: 581 MVA vs New Hartford-Morgan Valley: 486 MVA 1c) Franklin-Blackhawk: 733 MVA vs New Hartford-Morgan Valley: 683 MVA 2) Loadings on Beverly-Sub 92 are similar between two NUs 2a) Beverly-Sub 92 with Franklin-Morgan Valley & Beverly-Sub 92 NU: 480 MVA 2b) Beverly-Sub 92 with New Hartford-Morgan Valley & Beverly-Sub 92 NU: 461 MVA 3) Franklin-Morgan Valley option provides future flexibility to connect to Webster

Table E-11: Loss Reduction Comparison by Projects in Iowa Area

Area #	Area Name	Without Franklin-Morgan & Beverly-Sub92 NU		With Franklin-Morgan & Beverly-Sub92 NU		Loss Change	
		P losses (MW)	Q losses (MVAR)	P losses (MW)	Q losses (MVAR)	P losses (MW)	Q losses (MVAR)
357	AMIL	178.6	2532.0	181.4	2557.8	2.8	25.8
600	XEL	786.9	7122.9	773.6	7009.3	-13.3	-113.6
627	ALTW	437.1	3533.9	427.7	3661.3	-9.4	127.4
635	MEC	498.8	5198.0	467.0	4902.2	-31.8	-295.8
680	DPC	112.8	574.4	104.9	534.8	-7.9	-39.6
696	WPS	75.2	891.7	72.9	863.3	-2.3	-28.4
697	MGE	44.7	563.9	41.9	529.1	-2.8	-34.8
540	GMO	98.8	1148.6	96.3	1119.4	-2.5	-29.2
640	NPPD	249.6	2304.5	244.4	2263.5	-5.2	-41.0
645	OPPD	78.3	947.2	74.8	911.0	-3.5	-36.2
652	WAPA	427.3	3898.3	423.1	3872.1	-4.2	-26.2

E.3 Constraints Identified in Stage-2 Contingency Analysis with Initial Backbone NUs

Table E-12: Stage-2 System Intact Thermal Constraints

Table E-13: Stage-2 System Intact Voltage Constraints

Table E-14: Stage-2 Category P1 Thermal Constraints

Table E-15: Stage-2 Category P1 Worst Voltage Constraints

Table E-16: Stage-2 Category P2-P7 Thermal Constraints

Table E-17: Stage-2 Category P2-P7 Worst Voltage Constraints

Table E-18: Stage-2 Non-Converged Category P2-P7 Contingencies

Table E-19: Stage-2 Non-Converged Category P2-P7 Contingencies DCCC Results

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E.4 Step 4 - MWEX Project Justification

E.4.1 Comparison Results for Contingent Voltages

1) Voltage Comparison Based on Stage-2 Worst Voltage Results

Table E-20: Performance Comparison of MWEX NUs based on Stage-2 Worst Voltage Results

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2) Voltage Comparison under Three Critical Contingencies

Table E-21: Voltage Comparison under Montezuma-Ottumwa 345 kV Contingency

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
631193	HUNTLEY3	345	627	ALTW	0.9721	0.9488	0.0233
601002	ADAMS 3	345	600	XEL	0.9600	0.9380	0.0220
51144	J449 POI	345	627	ALTW	0.9637	0.9430	0.0207
631144	MITCHLCO3	345	627	ALTW	0.9609	0.9405	0.0204
601028	EAU CL 3	345	600	XEL	1.0064	0.9891	0.0173
603007	PRASWCP7	115	600	XEL	1.0466	1.0293	0.0173
657712	PRAIRIE7	115	600	XEL	1.0194	1.0026	0.0168
603025	NORDIC 7	115	600	XEL	1.0180	1.0012	0.0168
602034	SEVN ML5	161	600	XEL	1.0265	1.0112	0.0153
601044	BRIGGS RD 3	345	600	XEL	0.9902	0.9752	0.0150
602021	EAU CLA5	161	600	XEL	1.0380	1.0236	0.0144
602048	EAUCLAIRCAP7	161	600	XEL	1.0387	1.0243	0.0144
602038	JEFRSRD5	161	600	XEL	1.0369	1.0229	0.0140
602040	PRESTOT5	161	600	XEL	1.0376	1.0236	0.0140
602041	PRESTO 5	161	600	XEL	1.0374	1.0234	0.0140
602030	WHT 14 5	161	600	XEL	1.0334	1.0195	0.0139
602031	WHT 56 5	161	600	XEL	1.0363	1.0227	0.0136
603019	CASS CO7	115	600	XEL	1.0172	1.0040	0.0132
603021	REDRIVR7	115	600	XEL	1.0169	1.0037	0.0132
605488	GETWAY 8	69	600	XEL	1.0045	0.9914	0.0131
603018	SHEYNNE7	115	600	XEL	1.0202	1.0071	0.0131
603047	SHYSWCP7	115	600	XEL	1.0206	1.0075	0.0131
603020	MAPLE R7	115	600	XEL	1.0217	1.0087	0.0130
605347	HALLIE 5	161	600	XEL	1.0366	1.0236	0.0130
619406	GRE-LTLSAUK7	115	600	XEL	0.9729	0.9600	0.0129
619405	GRE-LTLSTP7	115	600	XEL	0.9729	0.9601	0.0128
602006	SHEYNNE4	230	600	XEL	0.9917	0.9790	0.0127
636610	SUB 91 3	345	635	MEC	0.9837	0.9712	0.0125

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
603027	DGLASCO7	115	600	XEL	0.9804	0.9681	0.0123
603254	DGLASCO CAP7	115	600	XEL	0.9804	0.9681	0.0123
615664	GRE-RIVRVRD3	345	600	XEL	0.9537	0.9414	0.0123
602016	REDCDR 5	161	600	XEL	1.0285	1.0163	0.0122
602044	GRAVL ISLE 5	161	600	XEL	1.0364	1.0242	0.0122
631138	LAKEFLD3	345	627	ALTW	0.9890	0.9771	0.0119
631197	LEDYARD3	345	627	ALTW	0.9766	0.9648	0.0118
601067	BISON 3	345	600	XEL	0.9676	0.9559	0.0117
615306	GRE-PL VLLY3	345	600	XEL	0.9695	0.9578	0.0117
601004	WILMART3	345	600	XEL	0.9587	0.9473	0.0114
601030	MEC-345	345	600	XEL	0.9587	0.9473	0.0114
601033	FIELD_N3	345	600	XEL	0.9690	0.9576	0.0114
636605	MECCORD3	345	635	MEC	1.0009	0.9899	0.0110
636611	SUB 91 5	161	635	MEC	0.9768	0.9659	0.0109
631141	ROCK CK3	345	627	ALTW	0.9957	0.9850	0.0107
601074	CRANDAL 3	345	600	XEL	0.9823	0.9717	0.0106
636687	QC ENRON 5	161	635	MEC	0.9749	0.9643	0.0106
601073	WOAD HILL 3	345	600	XEL	0.9824	0.9719	0.0105
601029	LKFLDXL3	345	600	XEL	0.9830	0.9725	0.0105
601068	TRIMONT WD 3	345	600	XEL	0.9835	0.9731	0.0104
602033	HYDROLN5	161	600	XEL	1.0298	1.0194	0.0104
605108	DGLAS C8	69	600	XEL	0.9997	0.9895	0.0102
636686	SUB 77 5	161	635	MEC	0.9721	0.9619	0.0102
636691	SB 71 5	161	635	MEC	0.9689	0.9588	0.0101
601032	FIELD_S3	345	600	XEL	0.9802	0.9702	0.0100
636650	SUB A 5	161	635	MEC	0.9670	0.9570	0.0100
636690	UTICA R5	161	635	MEC	0.9704	0.9604	0.0100
636693	SB 88 5	161	635	MEC	0.9657	0.9557	0.0100
605359	T SPEN 8	69	600	XEL	1.0149	1.0250	-0.0101
605360	SPENCER8	69	600	XEL	1.0150	1.0251	-0.0101
603273	PIP61 OWEN 8	115	600	XEL	1.0162	1.0272	-0.0110

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
605511	LAKE26 8	69	600	XEL	0.9764	0.9878	-0.0114
605507	TOWER RD	69	600	XEL	0.9715	0.9830	-0.0115
605504	WEBSTER8	69	600	XEL	0.9626	0.9742	-0.0116
605506	DANBURY8	69	600	XEL	0.9687	0.9803	-0.0116
605505	OAKLAN_8	69	600	XEL	0.9645	0.9762	-0.0117
680231	MEENON	69	600	XEL	0.9624	0.9741	-0.0117
601054	HAZEL CK3	345	600	XEL	0.9462	0.9580	-0.0118
605028	T CORNE8	69	600	XEL	1.0136	1.0260	-0.0124
603152	T-CRNRS7	115	600	XEL	1.0236	1.0411	-0.0175
603216	TCORNERSCAP7	115	600	XEL	1.0237	1.0412	-0.0175

Table E-22: Voltage Comparison under Montezuma-J530 POI 345 kV Contingency

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
601002	ADAMS 3	345	600	XEL	0.9553	0.9297	0.0256
631193	HUNTLEY3	345	627	ALTW	0.9656	0.9405	0.0251
51144	J449 POI	345	627	ALTW	0.9584	0.9342	0.0242
631144	MITCHLCO3	345	627	ALTW	0.9557	0.9318	0.0239
636610	SUB 91 3	345	635	MEC	1.0015	0.9842	0.0173
601028	EAU CL 3	345	600	XEL	1.0039	0.9871	0.0168
636611	SUB 91 5	161	635	MEC	0.9944	0.9790	0.0154
601044	BRIGGS RD 3	345	600	XEL	0.9877	0.9725	0.0152
636687	QC ENRON 5	161	635	MEC	0.9925	0.9774	0.0151
602034	SEVN ML5	161	600	XEL	1.0251	1.0100	0.0151
636686	SUB 77 5	161	635	MEC	0.9897	0.9751	0.0146
636691	SB 71 5	161	635	MEC	0.9864	0.9721	0.0143
636690	UTICA R5	161	635	MEC	0.9879	0.9736	0.0143
602021	EAU CLA5	161	600	XEL	1.0364	1.0222	0.0142
602048	EAUCLAIRCAP7	161	600	XEL	1.0372	1.0230	0.0142
636685	SB 74 5	161	635	MEC	0.9879	0.9737	0.0142
636693	SB 88 5	161	635	MEC	0.9833	0.9691	0.0142
636682	SB 58 5	161	635	MEC	0.9870	0.9729	0.0141
636683	SB 76 5	161	635	MEC	0.9870	0.9729	0.0141
636692	SB 70 5	161	635	MEC	0.9837	0.9696	0.0141
636650	SUB A 5	161	635	MEC	0.9844	0.9704	0.0140
636681	58TAP 5	161	635	MEC	0.9876	0.9736	0.0140
602040	PRESTOT5	161	600	XEL	1.0361	1.0222	0.0139
602041	PRESTO 5	161	600	XEL	1.0360	1.0221	0.0139
615306	GRE-PL VLLY3	345	600	XEL	0.9655	0.9516	0.0139
636605	MECCORD3	345	635	MEC	1.0160	1.0021	0.0139
602030	WHT 14 5	161	600	XEL	1.0321	1.0183	0.0138
602038	JEFRSRD5	161	600	XEL	1.0354	1.0216	0.0138
636680	SB 78 5	161	635	MEC	0.9875	0.9737	0.0138
636615	SB 56 3	345	635	MEC	0.9896	0.9759	0.0137

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
631141	ROCK CK3	345	627	ALTW	1.0100	0.9964	0.0136
636657	SB 48 5	161	635	MEC	0.9906	0.9772	0.0134
636679	ENT WAY 5	161	635	MEC	0.9875	0.9741	0.0134
602031	WHT 56 5	161	600	XEL	1.0348	1.0214	0.0134
636656	SB 47 5	161	635	MEC	0.9943	0.9813	0.0130
605347	HALLIE 5	161	600	XEL	1.0352	1.0223	0.0129
636661	SB 31 5	161	635	MEC	0.9889	0.9761	0.0128
636601	SB 39 5	161	635	MEC	0.9980	0.9853	0.0127
636660	SB 31T 5	161	635	MEC	0.9896	0.9769	0.0127
636662	SB 28 5	161	635	MEC	0.9888	0.9761	0.0127
636665	SB 17 5	161	635	MEC	0.9770	0.9644	0.0126
636667	SB 49 5	161	635	MEC	0.9734	0.9609	0.0125
631197	LEDYARD3	345	627	ALTW	0.9699	0.9574	0.0125
631062	ROCKCKW5	161	627	ALTW	0.9744	0.9620	0.0124
631063	ROCK CK5	161	627	ALTW	0.9745	0.9621	0.0124
636620	WALCOTT3	345	635	MEC	0.9848	0.9724	0.0124
602016	REDCDR 5	161	600	XEL	1.0271	1.0148	0.0123
636600	SB 39 3	345	635	MEC	1.0080	0.9957	0.0123
636616	SB 56 5	161	635	MEC	0.9883	0.9760	0.0123
636625	IPSCO 3	345	635	MEC	0.9832	0.9709	0.0123
602044	GRAVL ISLE 5	161	600	XEL	1.0351	1.0229	0.0122
636626	IPSCO 5	161	635	MEC	0.9886	0.9764	0.0122
631065	BVR CHN L1 5	161	627	ALTW	0.9689	0.9569	0.0120
631064	BVR CHN L2 5	161	627	ALTW	0.9688	0.9569	0.0119
631067	ALBANY 5	161	627	ALTW	0.9705	0.9587	0.0118
631138	LAKEFLD3	345	627	ALTW	0.9850	0.9732	0.0118
636668	SB 43 5	161	635	MEC	1.0004	0.9887	0.0117
636677	SB 53 5	161	635	MEC	0.9936	0.9821	0.0115
636675	SB 85 5	161	635	MEC	0.9940	0.9825	0.0115
636676	SB 52 5	161	635	MEC	0.9932	0.9817	0.0115
601004	WILMART3	345	600	XEL	0.9521	0.9407	0.0114

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
601030	MEC-345	345	600	XEL	0.9521	0.9407	0.0114
601033	FIELD_N3	345	600	XEL	0.9646	0.9534	0.0112
630177	SAVANNA8	69	627	ALTW	0.9871	0.9759	0.0112
631068	YORK 5	161	627	ALTW	0.9662	0.9551	0.0111
631033	ALBANY 6	138	627	ALTW	0.9796	0.9686	0.0110
636670	SB 18 5	161	635	MEC	1.0032	0.9924	0.0108
631066	SAVANNA5	161	627	ALTW	0.9674	0.9566	0.0108
631097	DEWITT 5	161	627	ALTW	0.9695	0.9588	0.0107
636636	OAKGROV5	161	635	MEC	1.0042	0.9936	0.0106
601029	LKFLDXL3	345	600	XEL	0.9786	0.9681	0.0105
601074	CRANDAL 3	345	600	XEL	0.9777	0.9672	0.0105
601068	TRIMONT WD 3	345	600	XEL	0.9792	0.9687	0.0105
601073	WOAD HILL 3	345	600	XEL	0.9778	0.9674	0.0104
602033	HYDROLN5	161	600	XEL	1.0287	1.0183	0.0104
636635	OAKGROV3	345	635	MEC	1.0032	0.9932	0.0100
601055	SCOTTCO3	345	600	XEL	0.9506	0.9606	-0.0100
605005	MAYNARD8	69	600	XEL	1.0017	1.0117	-0.0100
605006	MAYNARL8	69	600	XEL	1.0015	1.0116	-0.0101
605359	T SPEN 8	69	600	XEL	1.0143	1.0244	-0.0101
605360	SPENCER8	69	600	XEL	1.0144	1.0245	-0.0101
605007	CLARCTY8	69	600	XEL	0.9953	1.0054	-0.0101
605107	WESTPRT8	69	600	XEL	0.9881	0.9984	-0.0103
619123	GRE-OMMENTP8	69	600	XEL	0.9881	0.9985	-0.0104
605052	MNARDTP8	69	600	XEL	1.0009	1.0115	-0.0106
603273	PIP61 OWEN 8	115	600	XEL	1.0155	1.0266	-0.0111
605506	DANBURY8	69	600	XEL	0.9666	0.9784	-0.0118
605511	LAKE26 8	69	600	XEL	0.9742	0.9860	-0.0118
605507	TOWER RD	69	600	XEL	0.9693	0.9812	-0.0119
680231	MEENON	69	600	XEL	0.9603	0.9722	-0.0119
605505	OAKLAN_8	69	600	XEL	0.9624	0.9743	-0.0119
605504	WEBSTER8	69	600	XEL	0.9604	0.9724	-0.0120

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
619120	GRE-OMMEN 8	69	600	XEL	0.9875	0.9995	-0.0120
605028	T CORNE8	69	600	XEL	1.0130	1.0255	-0.0125
601054	HAZEL CK3	345	600	XEL	0.9402	0.9534	-0.0132
603007	PRASWCP7	115	600	XEL	1.0400	1.0533	-0.0133
605488	GETWAY 8	69	600	XEL	0.9979	1.0118	-0.0139
603152	T-CRNRS7	115	600	XEL	1.0231	1.0406	-0.0175
603216	TCORNERSCAP7	115	600	XEL	1.0231	1.0407	-0.0176
605051	FIESTAC8	69	600	XEL	0.9895	1.0130	-0.0235

Table E-23: Voltage Comparison under Square Butte Bipole Contingency

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
617055	GRE-ISLE 8	69	615	GRE	1.0347	1.0015	0.0332
618616	GRE-OPSTEAD8	69	615	GRE	1.0307	1.0006	0.0301
618620	GRE-OPSTDTP8	69	615	GRE	1.0307	1.0006	0.0301
603018	SHEYNNE7	115	600	XEL	1.0280	0.9991	0.0289
603047	SHYSWCP7	115	600	XEL	1.0284	0.9995	0.0289
603019	CASS CO7	115	600	XEL	1.0236	0.9952	0.0284
603021	REDRIVR7	115	600	XEL	1.0212	0.9938	0.0274
618615	GRE-ONAMIA 8	69	615	GRE	1.0290	1.0025	0.0265
603020	MAPLE R7	115	600	XEL	1.0238	0.9977	0.0261
618624	GRE-RUMRVTP8	69	615	GRE	1.0288	1.0032	0.0256
618617	GRE-GLEN 8	69	615	GRE	1.0246	0.9993	0.0253
631193	HUNTLEY3	345	627	ALTW	0.9664	0.9425	0.0239
601002	ADAMS 3	345	600	XEL	0.9632	0.9397	0.0235
51144	J449 POI	345	627	ALTW	0.9675	0.9454	0.0221
631144	MITCHLCO3	345	627	ALTW	0.9648	0.9428	0.0220
618622	GRE-VINELND8	69	615	GRE	1.0205	0.9988	0.0217
618618	GRE-SPIRITL8	69	615	GRE	1.0253	1.0048	0.0205
618619	GRE-SPRTLKS8	69	615	GRE	1.0263	1.0059	0.0204
616726	GRE-PINECTR8	69	615	GRE	1.0210	1.0014	0.0196
618623	GRE-VINLDTP8	69	615	GRE	1.0210	1.0014	0.0196
618621	GRE-PALISAD8	69	615	GRE	1.0222	1.0034	0.0188
618042	GRE-ROUNDLK8	69	615	GRE	1.0230	1.0043	0.0187
618079	GRE-BIGSANDT	69	615	GRE	1.0227	1.0040	0.0187
618080	GRE-BIGSAND8	69	615	GRE	1.0222	1.0035	0.0187
618041	GRE-WRIGHT 8	69	615	GRE	1.0295	1.0109	0.0186
601028	EAU CL 3	345	600	XEL	1.0092	0.9907	0.0185
618040	GRE-CRMWLLD8	69	615	GRE	1.0347	1.0162	0.0185
615494	GRE-CROMWLL8	69	615	GRE	1.0350	1.0166	0.0184
617035	GRE-OGILVIE8	69	615	GRE	1.0159	0.9975	0.0184

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
618046	GRE-BRANDEN8	69	615	GRE	1.0330	1.0146	0.0184
618047	GRE-GRANDLK8	69	615	GRE	1.0297	1.0113	0.0184
618043	GRE-GOWAN 8	69	615	GRE	1.0354	1.0171	0.0183
618051	GRE-GOWANSW8	69	615	GRE	1.0354	1.0171	0.0183
615518	GRE-4CORNRS8	69	615	GRE	1.0315	1.0132	0.0183
618045	GRE-LKHD GW8	69	615	GRE	1.0349	1.0166	0.0183
618048	GRE-SOLWAY 8	69	615	GRE	1.0312	1.0129	0.0183
618050	GRE-STURGEN8	69	615	GRE	1.0234	1.0052	0.0182
618052	GRE-MOOSERP8	69	615	GRE	1.0240	1.0058	0.0182
619926	GRE-MOOSEMN8	69	615	GRE	1.0239	1.0057	0.0182
619925	GRE-MOOSLK8	69	615	GRE	1.0248	1.0067	0.0181
618049	GRE-KETTLER8	69	615	GRE	1.0288	1.0108	0.0180
616735	GRE-PINCTRT8	69	615	GRE	1.0224	1.0048	0.0176
615566	GRE-WINGRIV4	230	608	MP	0.9321	0.9147	0.0174
615536	GRE-WILSONL8	69	615	GRE	1.0237	1.0065	0.0172
608642	VERNDLE7	115	608	MP	0.9505	0.9334	0.0171
615535	GRE-WILSNLK7	115	615	GRE	0.9784	0.9613	0.0171
615567	GRE-WINGRIV7	115	608	MP	0.9500	0.9329	0.0171
619701	GRE-ALDRICH7	115	608	MP	0.9513	0.9342	0.0171
619700	GRE-THMSTWN7	115	608	MP	0.9555	0.9387	0.0168
608639	MNP-STP7	115	608	MP	0.9541	0.9373	0.0168
608643	EAGLVLY7	115	608	MP	0.9469	0.9301	0.0168
602034	SEVN ML5	161	600	XEL	1.0279	1.0112	0.0167
616725	GRE-OAKLAWN8	69	615	GRE	1.0208	1.0041	0.0167
616734	GRE-OKLWNT8	69	615	GRE	1.0209	1.0042	0.0167
608647	LONG PR7	115	608	MP	0.9461	0.9297	0.0164
608644	DOGLAKE7	115	608	MP	0.9606	0.9442	0.0164
615341	GRE-HUBBARD4	230	608	MP	0.9488	0.9324	0.0164
619405	GRE-LTLSKTP7	115	600	XEL	0.9500	0.9337	0.0163
619703	GRE-REDEYE 7	115	608	MP	0.9519	0.9356	0.0163
608951	STRTRVR7	115	608	MP	0.9560	0.9398	0.0162

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
616715	GRE-FISHTTP7	115	608	MP	0.9645	0.9483	0.0162
616716	GRE-MOTLEY 7	115	608	MP	0.9634	0.9472	0.0162
616717	GRE-FISHTRP7	115	608	MP	0.9605	0.9443	0.0162
619406	GRE-LTLSAUK7	115	600	XEL	0.9499	0.9337	0.0162
619702	GRE-BLUEBRY7	115	608	MP	0.9541	0.9379	0.0162
619704	GRE-MENAHGA7	115	608	MP	0.9541	0.9379	0.0162
615342	GRE-HUBBARD7	115	608	MP	0.9584	0.9423	0.0161
617700	GRE-PALMLK7	115	608	MP	0.9627	0.9466	0.0161
617701	GRE-LONG LK7	115	608	MP	0.9615	0.9454	0.0161
617702	GRE-RDO 7	115	608	MP	0.9584	0.9423	0.0161
617703	GRE-RDO TAP7	115	608	MP	0.9592	0.9431	0.0161
617706	GRE-MANTRAP7	115	608	MP	0.9614	0.9453	0.0161
617707	GRE-POTATTP7	115	608	MP	0.9614	0.9453	0.0161
617708	GRE-POTATLK7	115	608	MP	0.9614	0.9453	0.0161
601044	BRIGGS RD 3	345	600	XEL	0.9948	0.9788	0.0160
608610	BADOURA4	230	608	MP	0.9557	0.9398	0.0159
608940	BAD1SUB7	115	608	MP	0.9720	0.9561	0.0159
608941	BAD2SUB7	115	608	MP	0.9720	0.9561	0.0159
608636	PEPINLK7	115	608	MP	0.9446	0.9288	0.0158
608640	BADOURA7	115	608	MP	0.9695	0.9537	0.0158
615421	GRE-BIRCHLK7	115	608	MP	0.9820	0.9662	0.0158
617725	GRE-TRIPPLK7	115	608	MP	0.9739	0.9581	0.0158
615428	GRE-RIVTON 8	69	615	GRE	1.0211	1.0053	0.0158
602021	EAU CLA5	161	600	XEL	1.0394	1.0237	0.0157
602048	EAUCLAIRCAP7	161	600	XEL	1.0402	1.0245	0.0157
616727	GRE-CROSSLK8	69	615	GRE	1.0165	1.0008	0.0157
616731	GRE-BREZYPT8	69	615	GRE	1.0162	1.0005	0.0157
616736	GRE-MISSION8	69	615	GRE	1.0165	1.0008	0.0157
617060	GRE-DENHAM 8	69	615	GRE	1.0271	1.0114	0.0157
616737	GRE-BRZYPTS8	69	615	GRE	1.0177	1.0021	0.0156
603027	DGLASCO7	115	600	XEL	0.9598	0.9442	0.0156

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
603254	DGLASCO CAP7	115	600	XEL	0.9598	0.9442	0.0156
616741	GRE-CRSSLKC8	69	615	GRE	1.0150	0.9994	0.0156
615416	GRE-PQUOTLK8	69	615	GRE	1.0211	1.0055	0.0156
616732	GRE-STONYBK8	69	615	GRE	1.0187	1.0031	0.0156
615422	GRE-BIRCHLK8	69	615	GRE	1.0225	1.0070	0.0155
616744	GRE-PLSNTLK8	69	615	GRE	1.0182	1.0027	0.0155
617028	GRE-MILACAD8	69	615	GRE	1.0342	1.0187	0.0155
617054	GRE-MILACA 8	69	615	GRE	1.0342	1.0187	0.0155
608655	SCRCYVL7	115	608	MP	0.9740	0.9585	0.0155
616733	GRE-WABEDO 8	69	615	GRE	1.0101	0.9946	0.0155
603221	WILSON CAP 7	115	600	XEL	0.9915	0.9761	0.0154
616728	GRE-EMILY 8	69	615	GRE	1.0126	0.9972	0.0154
616738	GRE-FFTYLKS8	69	615	GRE	1.0139	0.9985	0.0154
617704	GRE-SHINGOB7	115	608	MP	0.9703	0.9550	0.0153
617705	GRE-SHNGOTP7	115	608	MP	0.9703	0.9550	0.0153
608638	AKELEY7	115	608	MP	0.9701	0.9548	0.0153
608641	PINERVER7	115	608	MP	0.9727	0.9574	0.0153
616713	GRE-STHDALT7	115	608	MP	0.9747	0.9594	0.0153
620336	AUDUBON4	230	608	MP	0.9236	0.9083	0.0153
602040	PRESTOT5	161	600	XEL	1.0390	1.0237	0.0153
602038	JEFRSRD5	161	600	XEL	1.0382	1.0230	0.0152
616704	GRE-STHDALE7	115	608	MP	0.9743	0.9591	0.0152
603116	WILSON 7	115	600	XEL	0.9913	0.9761	0.0152
608645	BAXTER_7	115	608	MP	0.9806	0.9654	0.0152
613160	MORA	69	615	GRE	1.0040	0.9888	0.0152
616703	GRE-BAXTER 7	115	608	MP	0.9805	0.9653	0.0152
616729	GRE-OX LAKE8	69	615	GRE	1.0138	0.9986	0.0152
616743	GRE-LONGVLL8	69	615	GRE	1.0124	0.9972	0.0152
617056	GRE-MORA 8	69	615	GRE	1.0060	0.9908	0.0152
602030	WHT 14 5	161	600	XEL	1.0345	1.0193	0.0152
602041	PRESTO 5	161	600	XEL	1.0388	1.0236	0.0152

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
616742	GRE-LNGVLLT8	69	615	GRE	1.0128	0.9976	0.0152
616702	GRE-NOKAY 7	115	608	MP	0.9760	0.9609	0.0151
608651	MUDLAKE7	115	608	MP	0.9764	0.9614	0.0150
608652	BRAINRD7	115	608	MP	0.9766	0.9616	0.0150
616739	GRE-BLINDLK8	69	615	GRE	1.0153	1.0004	0.0149
608648	PEQUOT 7	115	608	MP	0.9746	0.9597	0.0149
602031	WHT 56 5	161	600	XEL	1.0375	1.0227	0.0148
608649	BLNCHRD7	115	608	MP	0.9597	0.9450	0.0147
616700	GRE-DEWING 7	115	608	MP	0.9731	0.9585	0.0146
615665	GRE-RIVRVRD7	115	615	GRE	0.9579	0.9434	0.0145
616710	GRE-BELLEVU7	115	608	MP	0.9626	0.9482	0.0144
617059	GRE-HARRY M8	69	615	GRE	1.0274	1.0131	0.0143
615664	GRE-RIVRVRD3	345	600	XEL	0.9328	0.9185	0.0143
616701	GRE-MERRFLD7	115	608	MP	0.9793	0.9650	0.0143
605347	HALLIE 5	161	600	XEL	1.0378	1.0236	0.0142
605000	GLEN LAKE8	69	600	XEL	1.0115	0.9974	0.0141
608657	RVT1BUS7	115	608	MP	0.9820	0.9679	0.0141
608658	60L TAP7	115	608	MP	0.9820	0.9679	0.0141
608659	RVT2BUS7	115	608	MP	0.9820	0.9679	0.0141
617027	GRE-GILMAN 8	69	615	GRE	1.0252	1.0111	0.0141
605133	HOLLYDL8	69	600	XEL	1.0358	1.0218	0.0140
605249	WSTGAT18	69	600	XEL	1.0188	1.0048	0.0140
608653	RIVERTN7	115	608	MP	0.9816	0.9676	0.0140
605108	DGLAS C8	69	600	XEL	0.9863	0.9723	0.0140
615573	GRE-MEDINA 8	69	615	GRE	1.0357	1.0218	0.0139
615579	GRE-BIGSWAN8	69	600	XEL	1.0355	1.0216	0.0139
619833	GRE-MEDINAD8	69	615	GRE	1.0357	1.0218	0.0139
608650	LITTLEF7	115	608	MP	0.9689	0.9551	0.0138
602016	REDCDR 5	161	600	XEL	1.0290	1.0152	0.0138
616714	GRE-LTTFLS7	115	608	MP	0.9685	0.9548	0.0137
618519	GRE-DASSEL 8	69	600	XEL	1.0297	1.0160	0.0137

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
605098	DASSEL 8	69	600	XEL	1.0300	1.0164	0.0136
615306	GRE-PL VLLY3	345	600	XEL	0.9717	0.9581	0.0136
617042	GRE-OAKPARK8	69	615	GRE	1.0232	1.0097	0.0135
617043	GRE-OAKPKTP8	69	615	GRE	1.0242	1.0107	0.0135
605107	WESTPRT8	69	600	XEL	0.9939	0.9804	0.0135
608612	RIVERTN4	230	608	MP	0.9702	0.9567	0.0135
619120	GRE-OMMEN 8	69	600	XEL	0.9950	0.9815	0.0135
619123	GRE-OMMENTP8	69	600	XEL	0.9940	0.9805	0.0135
617045	GRE-MAYHWSW8	69	615	GRE	1.0248	1.0113	0.0135
602044	GRAVL ISLE 5	161	600	XEL	1.0375	1.0241	0.0134
601004	WILMART3	345	600	XEL	0.9526	0.9392	0.0134
601030	MEC-345	345	600	XEL	0.9526	0.9392	0.0134
616730	GRE-THUN LK8	69	615	GRE	1.0135	1.0002	0.0133
617044	GRE-PIPE 2 8	69	615	GRE	1.0254	1.0121	0.0133
617041	GRE-PIPE2TP8	69	615	GRE	1.0255	1.0123	0.0132
617036	GRE-MAYHEW 8	69	615	GRE	1.0237	1.0106	0.0131
605106	VILLARD8	69	600	XEL	0.9949	0.9818	0.0131
617051	GRE-POPPLCR8	69	615	GRE	1.0269	1.0138	0.0131
617058	GRE-SANDSTN8	69	615	GRE	1.0268	1.0137	0.0131
617061	GRE-SAND TP8	69	615	GRE	1.0284	1.0153	0.0131
617063	GRE-SAND MP8	69	615	GRE	1.0284	1.0153	0.0131
615467	GRE-BEAR CK8	69	615	GRE	1.0279	1.0149	0.0130
617062	GRE-SAND SW8	69	615	GRE	1.0278	1.0148	0.0130
605096	COKATO 8	69	600	XEL	1.0209	1.0079	0.0130
636610	SUB 91 3	345	635	MEC	0.9962	0.9833	0.0129
605132	COKATOT8	69	600	XEL	1.0210	1.0081	0.0129
608617	MUDLAKE4	230	608	MP	0.9745	0.9617	0.0128
619121	GRE-LEVEN 8	69	600	XEL	0.9962	0.9835	0.0127
603118	E BLOOM7	115	600	XEL	0.9976	0.9850	0.0126
605066	KIMBALL8	69	600	XEL	1.0075	0.9949	0.0126
603269	WAKEFIELD 7	69	600	XEL	1.0354	1.0229	0.0125

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
631138	LAKEFLD3	345	627	ALTW	0.9844	0.9719	0.0125
605236	DELANO 8	69	600	XEL	0.9852	0.9727	0.0125
618520	GRE-KINGSTN8	69	600	XEL	1.0062	0.9937	0.0125
631197	LEDYARD3	345	627	ALTW	0.9732	0.9607	0.0125
605069	BLUEHRN8	69	600	XEL	1.0274	1.0150	0.0124
605004	LOWRY	69	600	XEL	0.9947	0.9823	0.0124
605105	GLENWD 8	69	600	XEL	0.9987	0.9863	0.0124
605235	WTRTWN 8	69	600	XEL	0.9977	0.9853	0.0124
605237	WACONIA8	69	600	XEL	0.9945	0.9821	0.0124
605233	LSTRPRA8	69	600	XEL	0.9940	0.9816	0.0124
605234	MCLEDCN8	69	600	XEL	0.9969	0.9845	0.0124
618426	GRE-NWGRMNY8	69	600	XEL	1.0000	0.9876	0.0124
619450	GRE-FAIRHVN8	69	600	XEL	1.0310	1.0186	0.0124
619451	GRE-FRHV TP8	69	600	XEL	1.0343	1.0219	0.0124
619457	GRE-FAIRCAP8	69	600	XEL	1.0342	1.0218	0.0124
605067	LUXBGTP8	69	600	XEL	1.0334	1.0211	0.0123
601033	FIELD_N3	345	600	XEL	0.9644	0.9521	0.0123
618427	GRE-HOLLYWD8	69	600	XEL	1.0011	0.9888	0.0123
605003	LOWRYTP8	69	600	XEL	0.9987	0.9864	0.0123
615320	GRE-BENTON 8	69	615	GRE	1.0336	1.0213	0.0123
617053	GRE-MINDEN 8	69	615	GRE	1.0336	1.0213	0.0123
618428	GRE-HOLLYTP8	69	600	XEL	1.0023	0.9900	0.0123
618434	GRE-GRMNYTP8	69	600	XEL	1.0000	0.9877	0.0123
619449	GRE-LUXMBRG8	69	600	XEL	1.0331	1.0208	0.0123
617000	GRE-LANGLTP7	115	600	XEL	0.9703	0.9581	0.0122
617912	GRE-SUNBURG8	69	615	GRE	1.0112	0.9990	0.0122
617917	GRE-PRWDTAP8	69	615	GRE	1.0072	0.9950	0.0122
605065	WATKINS8	69	600	XEL	1.0286	1.0164	0.0122
606008	RICH SPRING8	69	600	XEL	1.0252	1.0130	0.0122
608637	PLATTRV7	115	608	MP	0.9696	0.9574	0.0122
617001	GRE-LANGOLA7	115	600	XEL	0.9696	0.9574	0.0122

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
617916	GRE-PR WDS 8	69	615	GRE	1.0066	0.9944	0.0122
617923	GRE-SPICERT8	69	615	GRE	1.0239	1.0117	0.0122
618518	GRE-WATKINS8	69	600	XEL	1.0326	1.0204	0.0122
615060	GRE-ST BONI8	69	615	GRE	1.0067	0.9945	0.0122
601032	FIELD_S3	345	600	XEL	0.9746	0.9625	0.0121
605120	SOUTHVN8	69	600	XEL	1.0102	0.9981	0.0121
608654	AITKNMN7	115	608	MP	0.9792	0.9671	0.0121
617064	GRE-HINCKTP8	69	615	GRE	1.0195	1.0074	0.0121
617065	GRE-HINCKMP8	69	615	GRE	1.0174	1.0053	0.0121
617066	GRE-GRNDSTN8	69	615	GRE	1.0158	1.0037	0.0121
617913	GRE-HAWICK 8	69	615	GRE	1.0191	1.0070	0.0121
617914	GRE-KANDYOH8	69	615	GRE	1.0247	1.0126	0.0121
617915	GRE-SVEA 8	69	615	GRE	1.0280	1.0159	0.0121
617918	GRE-GREENLK8	69	615	GRE	1.0186	1.0065	0.0121
617919	GRE-SVEATAP8	69	615	GRE	1.0285	1.0164	0.0121
617924	GRE-SPICER 8	69	615	GRE	1.0237	1.0116	0.0121
618066	GRE-REMER 8	69	615	GRE	1.0126	1.0005	0.0121
618528	GRE-LKKORON8	69	615	GRE	1.0201	1.0080	0.0121
619122	GRE-GLENWD 8	69	600	XEL	0.9997	0.9876	0.0121
619977	GRE-WILLMAR8	69	615	GRE	1.0292	1.0171	0.0121
619985	WMU-WILMRMN8	69	615	GRE	1.0175	1.0054	0.0121
619987	WMU-WILMRSO8	69	615	GRE	1.0264	1.0143	0.0121
619989	WMU-WILMRTP8	69	615	GRE	1.0186	1.0065	0.0121
619990	WMU-WILMR E8	69	615	GRE	1.0190	1.0069	0.0121
603061	BLK DOG7	115	600	XEL	0.9824	0.9704	0.0120
605119	PAYNES 8	69	600	XEL	1.0216	1.0096	0.0120
616331	GRE-CABLE 8	69	615	GRE	1.0307	1.0187	0.0120
616361	GRE-CABLETP8	69	615	GRE	1.0314	1.0194	0.0120
617026	GRE-LONGSDG8	69	615	GRE	1.0333	1.0213	0.0120
617057	GRE-HNCKLEY8	69	615	GRE	1.0182	1.0062	0.0120
618529	GRE-LKKRNT8	69	615	GRE	1.0201	1.0081	0.0120

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
619439	GRE-ROSCOE 8	69	600	XEL	1.0180	1.0060	0.0120
619440	GRE-ROSCOTP8	69	600	XEL	1.0194	1.0074	0.0120
619986	WMU-WILMRW8	69	615	GRE	1.0198	1.0078	0.0120
619988	WMU-WILSWTP8	69	615	GRE	1.0216	1.0096	0.0120
618425	GRE-SHRMNT8	69	600	XEL	1.0109	0.9989	0.0120
606057	LK CAROLNTP8	69	600	XEL	1.0112	0.9993	0.0119
605097	WNSTDTP8	69	600	XEL	1.0164	1.0045	0.0119
605276	WINSTED8	69	600	XEL	1.0081	0.9962	0.0119
605277	MILRBND8	69	600	XEL	1.0081	0.9962	0.0119
619441	GRE-MUNSON 8	69	600	XEL	1.0195	1.0076	0.0119
619442	GRE-MUNSNTP8	69	600	XEL	1.0207	1.0088	0.0119
618424	GRE-SHERMAN8	69	600	XEL	1.0104	0.9985	0.0119
619853	GRE-LKCAROL8	69	600	XEL	1.0109	0.9990	0.0119
601029	LKFLDXL3	345	600	XEL	0.9786	0.9668	0.0118
601074	CRANDAL 3	345	600	XEL	0.9777	0.9659	0.0118
605071	FARM TP8	69	600	XEL	1.0225	1.0107	0.0118
605072	RICHMND8	69	600	XEL	1.0226	1.0108	0.0118
615040	GRE-LKFDGEN3	345	615	GRE	0.9786	0.9668	0.0118
618602	GRE-AITKIN 7	115	608	MP	0.9796	0.9678	0.0118
619425	GRE-W UNION8	69	600	XEL	0.9958	0.9840	0.0118
601073	WOAD HILL 3	345	600	XEL	0.9778	0.9661	0.0117
617221	GRE-JOHNSON8	69	600	XEL	1.0239	1.0122	0.0117
617238	GRE-PENELOE8	69	600	XEL	1.0280	1.0163	0.0117
617250	GRE-PENELOW8	69	600	XEL	1.0280	1.0163	0.0117
617910	GRE-PENNOCK8	69	615	GRE	1.0288	1.0171	0.0117
617911	GRE-PNNCKTP8	69	615	GRE	1.0294	1.0177	0.0117
617927	GRE-BRLINT8	69	615	GRE	1.0305	1.0188	0.0117
617928	GRE-WLMTRTP8	69	615	GRE	1.0303	1.0186	0.0117
619638	GRE-ST OLAF8	69	600	XEL	0.9986	0.9869	0.0117
619644	GRE-STOLAFT8	69	600	XEL	0.9986	0.9869	0.0117
619991	WMU-WTRTMT 8	69	615	GRE	1.0302	1.0185	0.0117

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
601068	TRIMONT WD 3	345	600	XEL	0.9791	0.9674	0.0117
603135	NINE MI7	115	600	XEL	0.9882	0.9765	0.0117
602033	HYDROLN5	161	600	XEL	1.0309	1.0192	0.0117
603002	WILMART7	115	600	XEL	0.9706	0.9590	0.0116
603179	MEC-115	115	600	XEL	0.9705	0.9589	0.0116
603186	SUMMIT	115	600	XEL	0.9703	0.9587	0.0116
605012	WILMART8	69	600	XEL	1.0255	1.0139	0.0116
605073	ANNDAL8	69	600	XEL	1.0143	1.0027	0.0116
605489	JOHNSON8	69	600	XEL	1.0241	1.0125	0.0116
617032	GRE-DUELM 8	69	615	GRE	1.0274	1.0158	0.0116
617049	GRE-DUELSW8	69	615	GRE	1.0279	1.0163	0.0116
617232	GRE-JHNSNTP8	69	600	XEL	1.0262	1.0146	0.0116
619432	GRE-CROW LK8	69	600	XEL	1.0020	0.9904	0.0116
619637	GRE-MATAWAN8	69	600	XEL	1.0016	0.9900	0.0116
619830	GRE-HOWARDL8	69	615	GRE	1.0182	1.0066	0.0116
619983	WMU-PRIAM 8	69	615	GRE	1.0306	1.0190	0.0116
605102	BELGRAD8	69	600	XEL	1.0025	0.9909	0.0116
605104	SEDAN 8	69	600	XEL	1.0022	0.9906	0.0116
50011	MEC CT2	115	600	XEL	0.9723	0.9608	0.0115
605140	MINNLK 8	69	600	XEL	1.0073	0.9958	0.0115
605169	MAPLTON8	69	600	XEL	1.0097	0.9982	0.0115
616360	GRE-BRGGLKS8	69	615	GRE	1.0299	1.0184	0.0115
617248	GRE-TRAVSWS8	69	600	XEL	1.0361	1.0246	0.0115
619635	GRE-HWY30TP8	69	600	XEL	1.0080	0.9965	0.0115
619636	GRE-DANVLL 8	69	600	XEL	1.0061	0.9946	0.0115
619827	GRE-DELANO 8	69	615	GRE	1.0328	1.0213	0.0115
619839	GRE-VICTOR 8	69	615	GRE	1.0187	1.0072	0.0115
619851	GRE-DELANOT8	69	615	GRE	1.0327	1.0212	0.0115
605227	VICTRTP8	69	600	XEL	1.0198	1.0084	0.0114
615366	GRE-BENSON 8	69	600	XEL	1.0255	1.0141	0.0114
616350	GRE-WESTEND8	69	615	GRE	1.0301	1.0187	0.0114

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
617047	GRE-PRINCTN8	69	615	GRE	1.0340	1.0226	0.0114
617241	GRE-STCLAIR8	69	600	XEL	1.0174	1.0060	0.0114
617246	GRE-TRAVERS8	69	600	XEL	1.0354	1.0240	0.0114
601067	BISON 3	345	600	XEL	0.9338	0.9224	0.0114
605171	SIBLYPK8	69	600	XEL	1.0103	0.9989	0.0114
605095	HOWRDLK8	69	600	XEL	1.0188	1.0074	0.0114
617236	GRE-NSWEDEN8	69	600	XEL	1.0248	1.0134	0.0114
617237	GRE-NWSWDTP8	69	600	XEL	1.0289	1.0175	0.0114
605094	WAVERLYT8	69	600	XEL	1.0220	1.0107	0.0113
606085	WAVERLY8	69	600	XEL	1.0220	1.0107	0.0113
613406	STPETER SUNR	69	600	XEL	1.0314	1.0201	0.0113
618723	GRE-JSNLDTP8	69	600	XEL	0.9965	0.9852	0.0113
605083	HENDRSN8	69	600	XEL	1.0024	0.9911	0.0113
605089	ARLNGTN8	69	600	XEL	0.9918	0.9805	0.0113
618741	GRE-JESNLND8	69	600	XEL	0.9963	0.9850	0.0113
605093	MONTRSE8	69	600	XEL	1.0250	1.0137	0.0113
617242	GRE-STCLRTP8	69	600	XEL	1.0189	1.0076	0.0113
617245	GRE-RUSHVR8	69	600	XEL	1.0206	1.0093	0.0113
619429	GRE-ZION TP8	69	600	XEL	1.0210	1.0097	0.0113
619453	GRE-ZION 8	69	600	XEL	1.0209	1.0096	0.0113
605188	EASTWD28	69	600	XEL	1.0315	1.0203	0.0112
613400	ST PETER	69	600	XEL	1.0297	1.0185	0.0112
613402	ST PETERBWY	69	600	XEL	1.0301	1.0189	0.0112
617240	GRE-DECORIA8	69	600	XEL	1.0216	1.0104	0.0112
617249	GRE-SS2826 8	69	600	XEL	1.0219	1.0107	0.0112
619444	GRE-BGFSHTP8	69	600	XEL	1.0220	1.0108	0.0112
619841	GRE-WOODLND8	69	615	GRE	1.0318	1.0206	0.0112
605103	BROOTEN8	69	600	XEL	1.0019	0.9907	0.0112
636611	SUB 91 5	161	635	MEC	0.9893	0.9781	0.0112
605223	KELSO SS 8	69	600	XEL	1.0165	1.0053	0.0112
619445	GRE-BIGFISH8	69	600	XEL	1.0207	1.0095	0.0112

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
619829	GRE-HIGHLND8	69	615	GRE	1.0150	1.0038	0.0112
605185	LKEMILY8	69	600	XEL	1.0314	1.0203	0.0111
619431	GRE-BANGRTP8	69	600	XEL	1.0035	0.9924	0.0111
619430	GRE-BANGOR 8	69	600	XEL	1.0017	0.9906	0.0111
615585	GRE-STNYCRK8	69	600	XEL	1.0283	1.0172	0.0111
618060	GRE-BOY RIV8	69	615	GRE	1.0106	0.9995	0.0111
605126	GROVELK8	69	600	XEL	1.0054	0.9944	0.0110
617243	GRE-EAGLELK8	69	600	XEL	1.0336	1.0226	0.0110
617200	GRE-NORTHPT7	115	600	XEL	0.9726	0.9616	0.0110
605245	DELANOT8	69	600	XEL	1.0403	1.0293	0.0110
617233	GRE-JAMESTW8	69	600	XEL	1.0324	1.0214	0.0110
617247	GRE-JMSTWTP8	69	600	XEL	1.0328	1.0218	0.0110
618061	GRE-SALEMSW8	69	615	GRE	1.0125	1.0015	0.0110
615070	GRE-MAPLE 1G	69	615	GRE	1.0175	1.0066	0.0109
618732	GRE-CLRY LK8	69	600	XEL	1.0195	1.0086	0.0109
636605	MECCORD3	345	635	MEC	1.0113	1.0004	0.0109
603005	DOVE 7	115	600	XEL	0.9680	0.9571	0.0109
636687	QC ENRON 5	161	635	MEC	0.9874	0.9765	0.0109
605255	GRNFLD 8	69	600	XEL	1.0424	1.0315	0.0109
616376	GRE-WBECKER8	69	615	GRE	1.0322	1.0213	0.0109
618521	GRE-LTCHFLD8	69	600	XEL	1.0350	1.0241	0.0109
619412	GRE-STSTPHT7	115	600	XEL	0.9722	0.9613	0.0109
619438	GRE-STSTPHN7	115	600	XEL	0.9721	0.9612	0.0109
605124	MAPLELK8	69	600	XEL	1.0175	1.0067	0.0108
613280	PRINNTN	69	615	GRE	1.0334	1.0226	0.0108
615404	GRE-LIBERTY8	69	615	GRE	1.0334	1.0226	0.0108
619831	GRE-MAPLELK8	69	615	GRE	1.0174	1.0066	0.0108
619835	GRE-SILVERC8	69	615	GRE	1.0194	1.0086	0.0108
619840	GRE-GOOSERP8	69	615	GRE	1.0175	1.0067	0.0108
605054	CYSTL T8	69	600	XEL	0.9797	0.9689	0.0108
605074	BECKER 8	69	600	XEL	1.0326	1.0218	0.0108

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
605166	EAGLELK8	69	600	XEL	1.0362	1.0254	0.0108
605569	JAMESTP8	69	600	XEL	1.0362	1.0254	0.0108
613110	LITCHFLD	69	615	GRE	1.0349	1.0241	0.0108
615066	GRE-LAKEMARG	69	615	GRE	1.0305	1.0197	0.0108
615441	GRE-LKMARN 8	69	600	XEL	1.0305	1.0197	0.0108
615443	GRE-LKMARN8	69	600	XEL	1.0305	1.0197	0.0108
615591	GRE-CROWRIV8	69	600	XEL	1.0473	1.0365	0.0108
615652	GRE-CHUBLAK8	69	600	XEL	1.0312	1.0204	0.0108
616061	GRE-SWIFTFL8	69	600	XEL	1.0179	1.0071	0.0108
618736	GRE-LKMRNTP8	69	600	XEL	1.0290	1.0182	0.0108
619828	GRE-GOOSELK8	69	615	GRE	1.0141	1.0033	0.0108
619847	GRE-HASTY 8	69	615	GRE	1.0242	1.0134	0.0108
605088	GAYLORD8	69	600	XEL	0.9788	0.9681	0.0107
605090	GRENISL8	69	600	XEL	1.0026	0.9919	0.0107
605099	LTCHMUT8	69	600	XEL	1.0350	1.0243	0.0107
619443	GRE-FARMING8	69	600	XEL	1.0218	1.0111	0.0107
618730	GRE-CLVLAND8	69	600	XEL	1.0296	1.0190	0.0106
602006	SHEYNNE4	230	600	XEL	0.9604	0.9498	0.0106
613260	PRINCETN	69	615	GRE	1.0159	1.0053	0.0106
613290	PRINPLNT	69	615	GRE	1.0157	1.0051	0.0106
616374	GRE-PRINCID8	69	615	GRE	1.0185	1.0079	0.0106
631141	ROCK CK3	345	627	ALTW	1.0061	0.9955	0.0106
605123	LINN ST8	69	600	XEL	1.0214	1.0109	0.0105
636686	SUB 77 5	161	635	MEC	0.9847	0.9742	0.0105
613270	PRINCTAP	69	615	GRE	1.0223	1.0118	0.0105
618526	GRE-LK JNN 8	69	615	GRE	1.0205	1.0100	0.0105
618601	GRE-KIMBRLY7	115	608	MP	0.9822	0.9717	0.0105
619848	GRE-WILLOW 8	69	615	GRE	1.0245	1.0140	0.0105
605592	BUFFALO8	69	600	XEL	1.0274	1.0170	0.0104
617920	GRE-LITCHTP8	69	615	GRE	1.0306	1.0202	0.0104
618525	GRE-LKJNN8SW8	69	615	GRE	1.0205	1.0101	0.0104

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
618715	GRE-ELKO 8	69	600	XEL	1.0217	1.0113	0.0104
618750	GRE-LKMRSTP8	69	600	XEL	1.0221	1.0117	0.0104
619842	GRE-LKJENTP8	69	615	GRE	1.0207	1.0103	0.0104
605122	PULASKI8	69	600	XEL	1.0330	1.0227	0.0103
616025	GRE-LKJOHNA8	69	600	XEL	1.0134	1.0031	0.0103
619826	GRE-CORCORN8	69	615	GRE	1.0179	1.0076	0.0103
619838	GRE-LAWNDAL8	69	615	GRE	1.0145	1.0042	0.0103
619844	GRE-BLACKLK8	69	615	GRE	1.0170	1.0067	0.0103
619845	GRE-BLKLKTP8	69	615	GRE	1.0175	1.0072	0.0103
605222	LSUEURE8	69	600	XEL	1.0157	1.0055	0.0102
616328	GRE-BALDWIN8	69	615	GRE	1.0205	1.0103	0.0102
617030	GRE-PINECTY8	69	615	GRE	1.0081	0.9979	0.0102
619832	GRE-MARY LK8	69	615	GRE	1.0302	1.0200	0.0102
619836	GRE-GRNFNSP8	69	615	GRE	1.0227	1.0125	0.0102
619837	GRE-CRCRNSW8	69	615	GRE	1.0221	1.0119	0.0102
636685	SB 74 5	161	635	MEC	0.9830	0.9728	0.0102
636690	UTICA R5	161	635	MEC	0.9829	0.9727	0.0102
636691	SB 71 5	161	635	MEC	0.9813	0.9711	0.0102
605221	LESUEUR8	69	600	XEL	1.0159	1.0058	0.0101
616062	GRE-GLCHRST8	69	600	XEL	1.0140	1.0039	0.0101
616347	GRE-THOMPLK8	69	615	GRE	1.0294	1.0193	0.0101
618415	GRE-WINTHRP8	69	600	XEL	0.9832	0.9731	0.0101
619846	GRE-ROCKFRD8	69	615	GRE	1.0290	1.0189	0.0101
630136	LECENR8	69	600	XEL	1.0222	1.0121	0.0101
636650	SUB A 5	161	635	MEC	0.9792	0.9691	0.0101
636683	SB 76 5	161	635	MEC	0.9821	0.9720	0.0101
636692	SB 70 5	161	635	MEC	0.9784	0.9683	0.0101
636693	SB 88 5	161	635	MEC	0.9780	0.9679	0.0101
601054	HAZEL CK3	345	600	XEL	0.9344	0.9451	-0.0107
605041	CROOKS 8	69	600	XEL	1.0060	1.0182	-0.0122
605056	RENVILL8	69	600	XEL	1.0053	1.0178	-0.0125

2022 Summer Shoulder Contingency Analysis Results

Bus Number	Bus Name	Base kV	Area Num	Area Name	Voltage (pu)		
					Alternative 1	Alternative 2	V Diff (Alt 1-Alt 2)
605042	EMMET R8	69	600	XEL	1.0053	1.0184	-0.0131
618421	GRE-HECTOR 8	69	600	XEL	1.0225	1.0362	-0.0137
618422	GRE-BROOKFD8	69	600	XEL	1.0217	1.0369	-0.0152
618423	GRE-BRKFDTP8	69	600	XEL	1.0218	1.0370	-0.0152
605058	HECTOR 8	69	600	XEL	1.0212	1.0379	-0.0167
605059	BUFLOLK8	69	600	XEL	1.0202	1.0371	-0.0169
618420	GRE-YELLWST8	69	600	XEL	1.0203	1.0372	-0.0169
618429	GRE-PRSTNLK8	69	600	XEL	1.0210	1.0380	-0.0170
605060	STEWART8	69	600	XEL	1.0230	1.0401	-0.0171
605040	BROWNT 8	69	600	XEL	1.0299	1.0471	-0.0172
605061	BROWNT8	69	600	XEL	1.0290	1.0462	-0.0172

E.4.2 Comparison Results of Potential Thermal Constraints

Table E-24: Potential Worst Thermal Constraints Comparison

Table E-24: Potential Worst Thermal Constraints Comparison												
Constraint				Constraint		Rating	Owner	Alternative 1		Alternative 2		Cont Type
								(MVA)	(%)	(MVA)	(%)	
51113 J438 POI 161 631085	PARNEL 5 161 1			Parnell-J438 POI 161 kV	274.0	MEC ITCM	269.9	98.5	NA	NA	P1	
65400 J506 POI 345 635200	RAUN 3 345 1			J506 POI-Raun 345 kV	864.0	MEC	1015.0	117.5	981.8	113.6	P1	
65400 J506 POI 345 635200	RAUN 3 345 1			J506 POI-Raun 345 kV	864.0	MEC	1015.5	117.5	982.2	113.7	P2-P7	
65400 J506 POI 345 635400	HIGHLND 3 345 1			J506 POI-Highland 345 kV	864.0	MEC	-840.3	97.3	NA	NA	P1	
65400 J506 POI 345 635400	HIGHLND 3 345 1			J506 POI-Highland 345 kV	864.0	MEC	-840.7	97.3	NA	NA	P2-P7	
75730 J530 POI 345 635730	MNTZUMA3 345 1			J530 POI-Montezuma 345 kV	1010.0	MEC	-1186.6	117.5	-1146.7	113.5	P1	
75730 J530 POI 345 635730	MNTZUMA3 345 1			J530 POI-Montezuma 345 kV	1010.0	MEC	-1237.4	122.5	-1198.3	118.6	P2-P7	
75730 J530 POI 345 636400	HILLS 3 345 1			J530 POI-Hills 345 kV	956.0	MEC	1019.0	106.6	977.8	102.3	P0	
75730 J530 POI 345 636400	HILLS 3 345 1			J530 POI-Hills 345 kV	1152.0	MEC	1396.6	121.2	1358.1	117.9	P1	
75730 J530 POI 345 636400	HILLS 3 345 1			J530 POI-Hills 345 kV	1152.0	MEC	1447.4	125.6	1409.7	122.4	P2-P7	
83021 J302&503 POI 230 86071	J607 POI 230 1			J302&J503 POI-J607 POI 230 kV	257.0	MDU	-284.2	110.6	-284.4	110.7	P2-P7	
83021 J302&503 POI 230 661042	HESKETT4 230 1			J302&J503 POI-Heskett 230 kV	257.0	MDU	341.5	132.9	341.4	132.8	P1	
83021 J302&503 POI 230 661042	HESKETT4 230 1			J302&J503 POI-Heskett 230 kV	257.0	MDU	478.6	186.2	478.7	186.3	P2-P7	
86071 J607 POI 230 661094	WISHEK 4 230 1			J607 POI-Wishek 230 kV	257.0	MDU	341.5	132.9	341.4	132.8	P1	
86071 J607 POI 230 661094	WISHEK 4 230 1			J607 POI-Wishek 230 kV	257.0	MDU	341.5	132.9	341.4	132.8	P2-P7	
86111 J611 161 541251	MARYVLE5 161 1			J611-Maryville 161 kV	199.0	MEC GMO	203.2	102.1	203.2	102.1	P0	
86111 J611 161 541251	MARYVLE5 161 1			J611-Maryville 161 kV	199.0	MEC GMO	247.3	124.3	247.5	124.4	P1	
86111 J611 161 541251	MARYVLE5 161 1			J611-Maryville 161 kV	199.0	MEC GMO	241.4	121.3	241.4	121.3	P2-P7	
300073 5GENTRY 161 300076	5FAIRPT 161 1			Gentry-Fairport 161 kV	227.0	AECI	224.4	98.9	224.3	98.8	P1	
300073 5GENTRY 161 300104	5NODWAY 161 1			Gentry-Nodaway 161 kV	227.0	AECI	-221.1	97.4	-220.9	97.3	P1	
300106 5NOVELY_SW 161 300364	2NOVLTY_SW 69.0 1			Novelty 161 -69 kV xfmr	56.0	AECI	66.4	118.6	66.8	119.3	P1	
300113 5SRIVER 161 300339	5EMERSN 161 1			South River-Emerson 161 kV	167.0	AECI	-180.5	108.1	-181.7	108.8	P1	
344000 7ZACKARY 345 344010	5ZACKARY 161 1			Zackary 345-161 kV xfmr	560.0	Ameren	581.1	103.8	587.4	104.9	P1	
344006 5ADAIR3 161 344010	5ZACKARY 161 1			Adair-Zackary 161 kV	557.0	Ameren	-579.7	104.1	-586.0	105.2	P1	
531451 MINGO 7 345 640325	REDWIL03 345 1			Red Willow-Mingo 345 kV	717.0	NPPD SUNC	-937.9	130.8	NA	NA	P0	
531451 MINGO 7 345 640325	REDWIL03 345 1			Red Willow-Mingo 345 kV	828.5	NPPD SUNC	-939.6	113.4	NA	NA	P1	
541199 ST JOE 3 345 640139	COOPER 3 345 1			St. Joseph-Cooper 345 kV	1195.0	NPPD GMO	-1179.9	98.7	-1178.1	98.6	P0	
541199 ST JOE 3 345 640139	COOPER 3 345 1			St. Joseph-Cooper 345 kV	1195.0	NPPD GMO	-1616.0	135.2	-1613.2	135.0	P1	
601002 ADAMS 3 345 631046	ADAMS 5 161 605739 ADAMS1 9 13.8 9			Adams 345-161-13.8 kV xfmr	334.6	ITCM	-353.2	105.6	NA	NA	P1	
601002 ADAMS 3 345 631046	ADAMS 5 161 605739 ADAMS1 9 13.8 9			Adams 345-161-13.8 kV xfmr	334.6	ITCM	-377.8	112.9	NA	NA	P2-P7	
601006 SPLT RK3 345 652864	SIOUXCY-LNX3 345 1			Split Rock-Sioux City 345 kV	717.0	XEL WAPA	706.5	98.5	692.3	96.6	P2-P7	
601015 BLUE LK3 345 601055	SCOTTC03 345 1			Blue Lake-Scott Co 345 kV	1515.8	XEL	-1488.4	98.2	NA	NA	P2-P7	
601029 LKFLDXL3 345 631138	LAKEFLD3 345 1			LakefieldXL-Lakefield 345 kV	956.1	ITCM XEL	NA	NA	-909.2	95.1	P1	
601029 LKFLDXL3 345 631138	LAKEFLD3 345 1			LakefieldXL-Lakefield 345 kV	956.1	ITCM XEL	-930.8	97.4	-997.2	104.3	P2-P7	
601031 BRKNGC03 345 652537	WHITE 3 345 2			Brookings Co-White 345 kV #2	788.8	XEL WAPA	NA	NA	905.2	114.8	P2-P7	
601032 FIELD_S3 345 601074	CRANDAL 3 345 1			Field South-Crandal 345 kV	1332.6	XEL	NA	NA	-1309.6	98.3	P1	
601032 FIELD_S3 345 601074	CRANDAL 3 345 1			Field South-Crandal 345 kV	1332.6	XEL	-1333.1	100.0	-1397.6	104.9	P2-P7	
601050 HELENA 3 345 601055	SCOTTC03 345 1			Helena-Scott Co 345 kV	1378.0	XEL	NA	NA	1311.9	95.2	P0	
602006 SHEYNNE4 230 657780	LAKEPK 4 230 1			Sheyenne-Lake Park 230 kV	294.0	XEL MPC	322.8	109.8	327.1	111.3	P0	
602006 SHEYNNE4 230 657780	LAKEPK 4 230 1			Sheyenne-Lake Park 230 kV	294.0	XEL MPC	329.9	112.2	334.1	113.6	P1	
602008 MINVALT4 230 652550	GRANITF4 230 1			Minn Valley-Granite Falls 230 kV	259.0	XEL WAPA	-320.8	123.9	-362.3	139.9	P1	
602008 MINVALT4 230 652550	GRANITF4 230 1			Minn Valley-Granite Falls 230 kV	259.0	XEL WAPA	-320.8	123.9	-362.3	139.9	P2-P7	
603018 SHEYNNE7 115 620203	MAPLTN 7 115 1			Sheyenne-Mapleton 115 kV	162.0	XEL OTP	-193.4	119.4	-193.9	119.7	P1	
603018 SHEYNNE7 115 620203	MAPLTN 7 115 1			Sheyenne-Mapleton 115 kV	162.0	XEL OTP	-186.2	114.9	-186.9	115.4	P2-P7	
605126 GROVELK8 69.0 616025	GRE-LKJOHNA869.0 1			Grove Lake-Lake Johnason 69 kV	38.6	XEL GRE	-39.9	103.4	-40.2	104.1	P1	
605126 GROVELK8 69.0 616025	GRE-LKJOHNA869.0 1			Grove Lake-Lake Johnason 69 kV	38.6	XEL GRE	-36.8	95.3	NA	NA	P2-P7	
613032 MURPHY CR 8 69.0 613042	MURPHY CR 5 161 1			Murphy Creek 161-69 kV xfmr	94.0	SMMPA	-110.0	117.0	NA	NA	P1	
613032 MURPHY CR 8 69.0 613042	MURPHY CR 5 161 1			Murphy Creek 161-69 kV xfmr	94.0	SMMPA	-110.0	117.0	NA	NA	P2-P7	
613040 AUSTIN 5 161 613042	MURPHY CR 5 161 1			Austin-Murphy Creek 161 kV	279.0	SMMPA	-272.5	97.7	NA	NA	P0	
613040 AUSTIN 5 161 613042	MURPHY CR 5 161 1			Austin-Murphy Creek 161 kV	307.0	SMMPA	-363.6	118.4	-320.0	104.2	P1	
613040 AUSTIN 5 161 613042	MURPHY CR 5 161 1			Austin-Murphy Creek 161 kV	307.0	SMMPA	-389.8	127.0	-344.0	112.1	P2-P7	
613042 MURPHY CR 5 161 631044	HAYWARD L2 5 161 1			Murphy Creek-Hatward 161 kV	276.0	SMMPA ITCM	-307.1	111.3	-276.6	100.2	P0	
613042 MURPHY CR 5 161 631044	HAYWARD L2 5 161 1			Murphy Creek-Hatward 161 kV	276.0	SMMPA ITCM	-400.6	145.1	-354.3	128.4	P1	

Table E-24: Potential Worst Thermal Constraints Comparison												
Constraint				Constraint		Rating	Owner	Alternative 1		Alternative 2		Cont Type
								(MVA)	(%)	(MVA)	(%)	
613042	MURPHY CR 5	161	631044	Murphy Creek-Hatward 161 kv		276.0	SMMPA ITCM	-427.5	154.9	-378.9	137.3	P2-P7
613330	RICE 5	161	630189	Rice 161-69 kv xfmr		106.0	SMMPA ITCM	138.8	130.9	138.9	131.0	P1
613330	RICE 5	161	630189	Rice 161-69 kv xfmr		106.0	SMMPA ITCM	138.8	130.9	138.9	131.0	P2-P7
613370	RUTLAND5	161	631042	Rutland-Fox Lake 161 kv		168.0	SMMPA ITCM	-167.7	99.8	-173.3	103.2	P0
613370	RUTLAND5	161	631042	Rutland-Fox Lake 161 kv		168.0	SMMPA ITCM	-237.7	141.5	-247.4	147.3	P1
613370	RUTLAND5	161	631042	Rutland-Fox Lake 161 kv		168.0	SMMPA ITCM	-257.4	153.2	-258.1	153.6	P2-P7
613370	RUTLAND5	161	631045	Rutland-Huntley 161 kv		168.0	SMMPA ITCM	188.9	112.4	198.0	117.9	P1
613370	RUTLAND5	161	631045	Rutland-Huntley 161 kv		168.0	SMMPA ITCM	209.7	124.8	210.5	125.3	P2-P7
615300	GRE-INMAN 4	230	615566	Wing River-Inman 230 kv		252.5	GRE	270.0	106.9	276.4	109.5	P0
615300	GRE-INMAN 4	230	615566	Wing River-Inman 230 kv		252.5	GRE	300.4	119.0	303.5	120.2	P1
620170	EFERGUS7	115	658102	Kerkhoven Tap-Kerkhoven 115 kv		103.0	GRE XEL	NA	NA	-97.9	95.0	P2-P7
620336	AUDUBON4	230	657780	Audubon-Lake Park 230 kv		294.0	OTP MPC	-308.4	104.9	-312.2	106.2	P0
620336	AUDUBON4	230	657780	Audubon-Lake Park 230 kv		294.0	OTP MPC	-315.5	107.3	-319.3	108.6	P1
630023	TRIBOJI J1	869.0	630687	Triboji-Fly Cloud 69 kv		48.0	ITCM	-47.4	98.8	-48.5	101.0	P2-P7
630104	WALTERS8	69.0	630120	Walters-Alden 69 kv		45.0	ITCM	51.0	113.3	43.0	95.6	P1
630120	ALDEN J8	69.0	630121	Alden-Alea West 69 kv		45.0	ITCM	50.5	112.2	NA	NA	P1
630189	RICE 8	69.0	630232	Rice-Cresco 69 kv		69.0	ITCM	67.3	97.5	67.5	97.8	P1
630189	RICE 8	69.0	630232	Rice-Cresco 69 kv		69.0	ITCM	67.5	97.8	67.6	98.0	P2-P7
631042	FOX LK 5	161	631210	Fox Lake-Jackson 161 kv #2		276.0	ITCM	-287.2	104.1	-287.2	104.1	P1
631042	FOX LK 5	161	631210	Fox Lake-Jackson 161 kv #2		276.0	ITCM	-287.2	104.1	-287.2	104.1	P2-P7
631044	HAYWARD L2	5	631174	Hayward-Glenworth 161 kv		325.0	ITCM	-325.4	100.1	NA	NA	P1
631048	EMERY 5	161	631199	Emery-Killdeer 345 kv		450.0	ITCM	600.3	133.4	562.7	125.0	P1
631048	EMERY 5	161	631202	Emery-Colby 161 kv		327.0	ITCM	-457.9	140.0	NA	NA	P1
631048	EMERY 5	161	636300	Emery-Floyd 161 kv		272.0	ITCM MEC	353.6	130.0	330.5	121.5	P1
631048	EMERY 5	161	636300	Emery-Floyd 161 kv		272.0	ITCM MEC	405.1	148.9	377.2	138.7	P2-P7
631050	HAZLTON L1	5	636200	Hazleton-Blackhawk 161 kv		325.0	ITCM MEC	-316.3	97.3	NA	NA	P1
631052	LANSING L1	5	681528	Lansing-Harmony 161 kv		153.0	ITCM DPC	NA	NA	-146.2	95.6	P1
631052	LANSING L1	5	681528	Lansing-Harmony 161 kv		153.0	ITCM DPC	-166.0	108.5	-167.1	109.2	P2-P7
631085	PARNEL 5	161	636401	Parnell-Hills 161 kv		325.0	MEC ITCM	310.9	95.7	NA	NA	P1
631106	HENRYC05	161	631108	Henry Co-Denmark 161 kv		169.0	ITCM	177.2	104.9	178.1	105.4	P2-P7
631106	HENRYC05	161	631111	Henry Co-Jeff 161 kv		197.0	ITCM	-206.0	104.6	-207.1	105.1	P2-P7
631110	WAPELL05	161	631111	Wapello-Jeff 161 kv		223.0	ITCM	213.9	95.9	213.7	95.8	P1
631110	WAPELL05	161	631111	Wapello-Jeff 161 kv		223.0	ITCM	255.8	114.7	256.3	114.9	P2-P7
631110	WAPELL05	161	631112	Wapello-Appanoose 161 kv		171.0	ITCM	200.7	117.4	204.1	119.4	P1
631110	WAPELL05	161	631112	Wapello-Appanoose 161 kv		171.0	ITCM	178.7	104.5	182.2	106.5	P2-P7
631115	OTTUMWA5	161	631143	Ottumwa 345-161 kv xfmr		400.0	ITCM	-549.5	137.4	-554.1	138.5	P1
631115	OTTUMWA5	161	631143	Ottumwa 345-161 kv xfmr		400.0	ITCM	-537.8	134.5	-542.4	135.6	P2-P7
631123	ADAMS L2	5	631127	Adams-Hayward 161 kv		216.0	ITCM	-289.1	133.8	-259.4	120.1	P1
631123	ADAMS L2	5	631127	Adams-Hayward 161 kv		216.0	ITCM	-289.1	133.8	-259.4	120.1	P2-P7
631127	HAYWARD L1	5	631180	Hayward-Freeborn 161 kv		446.0	ITCM	-470.8	105.6	NA	NA	P1
631127	HAYWARD L1	5	631180	Hayward-Freeborn 161 kv		446.0	ITCM	-479.8	107.6	NA	NA	P2-P7
631138	LAKEFLD3	345	635368	Lakefield-Obrien 345 kv		864.0	ITCM MEC	NA	NA	-834.8	96.6	P1
631138	LAKEFLD3	345	635368	Lakefield-Obrien 345 kv		864.0	ITCM MEC	NA	NA	-848.1	98.2	P2-P7
631139	HAZLTON3	345	631142	Hazleton-Arnold 345 kv		923.0	ITCM MEC	880.7	95.4	NA	NA	P2-P7
631139	HAZLTON3	345	631191	Hazleton-Hickory Crk 345 kv		1195.0	ITCM	NA	NA	1323.2	110.7	P2-P7
631148	MORGN VLLY	3	636420	Morgan Valley-Tiffin 345 kv		846.0	ITCM MEC	825.4	97.6	828.7	98.0	P1
631154	BARTON5	161	631160	Barton-Lime Creek 161 kv		240.0	ITCM	-235.5	98.1	NA	NA	P1
631154	BARTON5	161	631160	Barton-Lime Creek 161 kv		240.0	ITCM	-258.6	107.8	NA	NA	P2-P7
631183	CAYLER5	161	656570	Cayler-Wisdom 161 kv		215.0	ITCM CBPC	213.2	99.2	NA	NA	P1
631183	CAYLER5	161	656570	Cayler-Wisdom 161 kv		215.0	ITCM CBPC	220.5	102.6	205.7	95.7	P2-P7
631198	COLBY3	345	631202	Colby 345-161 kv xfmr		450.0	ITCM	468.9	104.2	NA	NA	P1
635200	RAUN 3	345	645451	Raun-S3451 345 kv		956.0	MEC OPPD	1002.5	104.9	986.9	103.2	P0
635200	RAUN 3	345	645451	Raun-S3451 345 kv		956.0	MEC OPPD	1236.0	129.3	1208.8	126.4	P1
635200	RAUN 3	345	645451	Raun-S3451 345 kv		956.0	MEC OPPD	1233.5	129.0	1206.4	126.2	P2-P7

Table E-24: Potential Worst Thermal Constraints Comparison										
Constraint	Constraint	Rating	Owner	Alternative 1		Alternative 2		Cont Type		
				(MVA)	(%)	(MVA)	(%)			
635600 GRIMES 3 345 635700 SYCAMOR3 345 1	Grimes-Sycamore 345 kV #1	1209.0	MEC	1218.8	100.8	1189.6	98.4	P1		
635600 GRIMES 3 345 635700 SYCAMOR3 345 1	Grimes-Sycamore 345 kV #1	1209.0	MEC	1155.8	95.6	NA	NA	P2-P7		
635600 GRIMES 3 345 635700 SYCAMOR3 345 2	Grimes-Sycamore 345 kV #2	1319.0	MEC	1395.8	105.8	1364.3	103.4	P2-P7		
635600 GRIMES 3 345 636003 BVR CRK 3 345 1	Grimes-Beaver Creek 345 kV	1222.0	MEC	-1233.4	100.9	-1179.9	96.6	P1		
635680 BOND RNT3 345 635700 SYCAMOR3 345 1	Bondurant-Sycamore 345 kV	1152.0	MEC	-1162.9	100.9	-1130.5	98.1	P1		
635680 BOND RNT3 345 635700 SYCAMOR3 345 1	Bondurant-Sycamore 345 kV	1152.0	MEC	-1242.3	107.8	-1208.9	104.9	P2-P7		
635680 BOND RNT3 345 635730 MNTZUMA3 345 1	Bondurant-Montezuma 345 kV	1067.0	MEC	1041.1	97.6	NA	NA	P0		
635680 BOND RNT3 345 635730 MNTZUMA3 345 1	Bondurant-Montezuma 345 kV	1067.0	MEC	1204.7	112.9	1146.4	107.4	P1		
635680 BOND RNT3 345 635730 MNTZUMA3 345 1	Bondurant-Montezuma 345 kV	1067.0	MEC	1254.9	117.6	1190.5	111.6	P2-P7		
636000 WEBSTER3 345 636001 WEBSTER5 161 636002 WEBS1XT9 13.8 1	Webster 345-161-13.8 kV xfmr	560.0	MEC	564.0	100.7	NA	NA	P1		
636000 WEBSTER3 345 636001 WEBSTER5 161 636002 WEBS1XT9 13.8 1	Webster 345-161-13.8 kV xfmr	560.0	MEC	598.7	106.9	561.0	100.2	P2-P7		
636001 WEBSTER5 161 636050 WRIGHT 5 161 1	Webster-Wright 161 kV	212.0	MEC	207.2	97.7	NA	NA	P2-P7		
636200 BLACKHAWK 5 161 656087 BREMER C05 161 1	Blackhawk-Bremer Co 161 kV	258.0	MEC CBPC	-275.1	106.6	-256.0	99.2	P1		
636200 BLACKHAWK 5 161 656087 BREMER C05 161 1	Blackhawk-Bremer Co 161 kV	258.0	MEC CBPC	-332.1	128.7	-267.5	103.7	P2-P7		
636225 UNIONTP5 161 636240 BUTLER 5 161 1	Union Tap-Butler 161 kV	410.0	MEC	-401.6	98.0	NA	NA	P1		
636230 FRANKLN5 161 636240 BUTLER 5 161 1	Franklin-Butler 161 kV	410.0	MEC	406.3	99.1	NA	NA	P1		
636300 FLOYD 5 161 656087 BREMER C05 161 1	Floyd-Bremer Co 161 kV	258.0	MEC CBPC	275.8	106.9	257.8	99.9	P1		
636300 FLOYD 5 161 656087 BREMER C05 161 1	Floyd-Bremer Co 161 kV	258.0	MEC CBPC	324.0	125.6	270.2	104.7	P2-P7		
652001 G13_001IST WATERTN7 115 1	G13_001IST-Watertown 115 kV	121.0	WAPA	118.5	97.9	117.5	97.1	P1		
652503 BLAIR 4 230 652550 GRANITF4 230 1	Blair-Granite Falls 230 kV	410.0	WAPA	NA	NA	443.2	108.1	P1		
652527 WHITL0K4 230 652821 SULLYBT-LNX3 230 1	Whitlock-Sully Buttes 230 kV	264.0	WAPA	263.6	99.8	261.0	98.9	P1		
652527 WHITL0K4 230 661038 GLENHAM4 230 1	Whitlock-Glenham 230 kV	260.0	MDU WAPA	-271.0	104.2	-268.5	103.3	P1		
652529 WATERTN3 345 652530 WATERTN4 230 652237 WATERT19 13.8 1	Watertown 345-230-13.8 kV xfmr	500.0	WAPA	NA	NA	510.2	102.0	P1		
652529 WATERTN3 345 652530 WATERTN4 230 652237 WATERT19 13.8 1	Watertown 345-230-13.8 kV xfmr	500.0	WAPA	NA	NA	510.2	102.0	P2-P7		
652530 WATERTN4 230 652582 APPLIEDORN 4 230 1	Watertown-Appledorn 230 kV	341.8	WAPA	NA	NA	374.5	109.6	P1		
652555 MORRIS 7 115 658102 GRANTC07 115 1	Morris-Grant Co 115 kV	103.0	MRES WAPA	98.7	95.8	102.6	99.6	P1		
652555 MORRIS 7 115 658102 GRANTC07 115 1	Morris-Grant Co 115 kV	103.0	MRES WAPA	100.4	97.5	102.8	99.8	P2-P7		
661042 HESKETT4 230 661053 MANDAN 4 230 1	Heskett-Mandan 230 kV	383.0	MDU	382.8	99.9	383.0	100.0	P2-P7		
661092 FOXTAIL4 230 661096 TTANKANORTH4 230 1	Foxtail-Tatanka North 230 kV	343.0	MDU	-430.6	125.5	-426.1	124.2	P1		
661092 FOXTAIL4 230 661096 TTANKANORTH4 230 1	Foxtail-Tatanka North 230 kV	343.0	MDU	-355.9	103.8	-349.8	102.0	P2-P7		
661092 FOXTAIL4 230 661098 ELLENDLMVP4 230 1	Foxtail-Ellendale 230 kV	419.0	MDU	539.2	128.7	534.8	127.6	P1		
661092 FOXTAIL4 230 661098 ELLENDLMVP4 230 1	Foxtail-Ellendale 230 kV	419.0	MDU	464.5	110.9	458.5	109.4	P2-P7		
661093 MERRCRT4 230 661094 WISHEK 4 230 1	Merricourt-Wishek 230 kV	257.0	MDU	-335.5	130.5	-335.5	130.5	P1		
661093 MERRCRT4 230 661094 WISHEK 4 230 1	Merricourt-Wishek 230 kV	257.0	MDU	-335.5	130.5	-335.5	130.5	P2-P7		
661093 MERRCRT4 230 661098 ELLENDLMVP4 230 1	Merricourt-Ellendale 230 kV	383.0	MDU	521.0	136.0	516.6	134.9	P1		
661093 MERRCRT4 230 661098 ELLENDLMVP4 230 1	Merricourt-Ellendale 230 kV	383.0	MDU	511.2	133.5	506.8	132.3	P2-P7		
680026 HARMNY 69.0 680175 CRESCO 69.0 1	Harmony-Cresco 69 kV	37.0	DPC	-36.7	99.2	-37.7	101.9	P1		

E.4.3 Detailed Estimated Cost of the Two Alternatives

Table E-25: Detailed Estimated Cost of Alternative 1

Alternative 1 (NRH-SRR)	1/1/2026 Cost (\$)	Comments
Construct a new South Rocky Run 345 kV switching station with an in-service date of 1/1/2026	\$53,500,000	a. Pre-certification costs for the entire project, b. 40 acres of real estate with 10 acres developed, c. a new control building, d. eight 345 kV circuit breakers (the substation is configured to eliminate P2 contingencies that result in the loss of two or more 345 kV lines), e. a 75 MVAR shunt capacitor, f. a line-connected shunt reactor (assumed to be 80 MVAR), and g. substation security. h. Also, redundant bus differential relays will need to be installed at the Rocky Run 345 kV Substation to eliminate the P5.5 contingency at Rocky Run 345 kV. i. The 12.848% tax gross-up is applied to 100% of the substation costs, which is the portion that ATC will own. The actual taxes at the time of construction will be applied. j. In addition to this final cost estimate, ATC also requires \$12.4M as an upfront charge to cover the O&M and other expenses of this substation over its 40 year life.
Construct a new North Rochester – South Rocky Run 345 kV line with an in-service date of 1/1/2026	\$621,000,000	a. Total line length assumed to be 180 miles. b. The 12.848% tax gross-up is applied to 50% of the line costs, which is the portion that ATC will own. The actual taxes at the time of construction will be applied. c. In addition to this final cost estimate, ATC also requires \$78.2M as an upfront charge to cover the O&M and other expenses of ATC's portion of this line over its 40 year life.
Construct a new 161/69 kV substation in Minnesota connecting the Wabaco – Alma 161 kV line to the Dairyland 69 kV system with an in-service date of 1/1/2026	\$5,500,000	a. This upgrade is required for the North Rochester – South Rocky Run 345 kV line to utilize the existing 161 kV position on CapX Alma transmission structures crossing the Mississippi. The final configuration will result in the 345 kV circuits being separated by the 161 kV circuit such that there won't a P7.1 contingency that causes an outage of both 345 kV circuits.
Install the following facilities at the North Rochester 345 kV Substation with an in-service date of 1/1/2026	\$6,000,000	a. 150 MVAR of shunt capacitance (modeled as two 75 MVAR mechanically switched capacitors) and related switchgear and (\$4 M) b. new breakers on the breaker-and-a-half 345 kV substation to connect the shunt capacitor(s) and the new 345 kV line. (\$2 M)
Install a 35% series capacitor at the Briggs Road 345 kV Substation on the Briggs Road – North Madison 345 kV line with an in-service date of 1/1/2026	\$18,000,000	
Install one breaker at the Briggs Road 345 kV Substation with an in-service date of 1/1/2026	\$2,000,000	a. This upgrade is required to eliminate the P2 contingency resulting in the simultaneous outage of both existing 345 kV lines.
Additional facilities, e.g. Forest Junction – Elkhart Lake 138 kV, are expected to be overloaded by this alternative and will need to be assigned Network Upgrades	NA	These estimates will be prepared if this alternative is selected as a final Network Upgrade and the final thermal analysis is performed.
Total (NRH-SRR)	\$706,000,000	

Table E-26: Detailed Estimated Cost of Alternative 2

Alternative 2 (J510-Sub 91)	1/1/2026 Cost (M\$)	MISO Comments
Brookings Substation upgrade	\$6.09	Add two line positions
Brookings – Huntley 345kV transmission line	\$393.54	~165 miles
Huntley Substation upgrade	\$6.09	Add two line positions
Huntley – Adams 345kV transmission line	\$232.71	~97 miles
Adams Substation upgrade	\$9.14	Add two line positions, move existing equipment
Adams – Hickory Creek 345kV transmission line	\$330.19	~139 miles
Hickory Creek Substation upgrade	\$9.14	Add two line positions, move existing transmission structures
Hickory Creek – Sub 91 345kV transmission line	\$205.91	~86 miles
Sub 91 Substation upgrade	\$9.14	Add two line positions, and additional bus work
Construction cost subtotal	\$1,201.35	
Contingency(+20%) (2018 dollars)	\$240.03	
AFUDC(+7.5%) (2018 dollars)	\$108.44	
Total (NRH-SRR)	\$1,549.81	

E.5 Constraints Identified in Stage-3 Contingency Analysis with Backbone and MWEX NUs

Table E-27: Stage-3 System Intact Thermal Constraints

Table E-28: Stage-3 System Intact Voltage Constraints

Table E-29: Stage-3 Category P1 Thermal Constraints

Table E-30: Stage-3 Category P1 Voltage Constraints

Table E-31: Stage-3 Category P2-P7 Thermal Constraints

Table E-32: Stage-3 Category P2-P7 Voltage Constraints

Table E-33: Stage-3 Non-Converged Category P2-P7 Contingencies

Table E-34: Stage-3 Non-Converged Category P2-P7 Contingencies DCCC Results

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Appendix

F

Local Planning Criteria Analysis Results

F.1 GRE Local Planning Criteria Analysis Results

F.1.1 J511 GRE Local Planning Criteria Analysis

Below is the J511 GRE local planning criteria analysis report.

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F.1.2 J596 GRE Local Planning Criteria Analysis

Below is the J596 GRE local planning criteria analysis report.

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F.2 OTP LPC Analysis

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F.3 MDU LPC Analysis

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F.4 DPC LPC Analysis

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F.5 Ameren LPC Analysis

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Affected System Contingency Analysis Results

G.1 CIPCO Company Affected System Analysis Results

Table G-1: 2022 SH CIPCO Affected System Analysis Results

Table G-2: 2022 SPK CIPCO Affected System Analysis Results

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G.2 MPC Affected System Study Results

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G.3 PJM Affected System Study Results

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G.4 SPP Affected System Study Results

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G.5 AECI Affected System Study Results

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Appendix

H

Transient Stability Results

H.1 2022 Summer Shoulder Stability Results Summary

Stability simulation was performed in the 2022 summer shoulder Stage-3 case with MISO Backbone NUs and MWEX voltage stability NUs modeled.

Stability study results summary is in Table H-1.

Table H-1: 2022 Summer Shoulder Stage-3 Stability Analysis Results Summary

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H.2 2022 Summer Shoulder Stability Plots

Plots of stability simulations for 2022 summer shoulder Stage-3 study case are in separate files which are listed below:

AppendixH2_2022SH_DPP 2016Aug-West_Stage-3_Study_Plots.zip

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MWEX Voltage Study Details

Below is the MWEX voltage stability study report provided by ATC.

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Short Circuit Analysis

- J.1 J302 Short Circuit Study Performed by Siemens PTI**
- J.2 J414 Short Circuit Study Performed by ITCM**
- J.3 J415 Short Circuit Study Performed by ITCM**
- J.4 J439 Short Circuit Study Performed by MEC**
- J.5 J457 Short Circuit Study Performed by Siemens PTI**
- J.6 J459, J613, J637, and J638 Short Circuit Study Performed by OTP**
- J.7 J476 Short Circuit Study Performed by MEC**
- J.8 J503 Short Circuit Study Performed by Siemens PTI**
- J.9 J511 Short Circuit Study Performed by GRE**
- J.10 J512, J569, J575, J577, J587, and J597 Short Circuit Studies Performed by Xcel**
- J.11 J541 and J598 Short Circuit Study Performed by Ameren**
- J.12 J555 Short Circuit Study Performed by MEC**
- J.13 J583 Short Circuit Study Performed by MEC**
- J.14 J590 Short Circuit Study Performed by MEC**
- J.15 J593 Short Circuit Study Performed by Siemens PTI**
- J.16 J594 Short Circuit Study Performed by ITCM**
- J.17 J596 Short Circuit Study Performed by GRE**

J.18 J599 Short Circuit Study Performed by Siemens PTI

J.19 J607 Short Circuit Study Performed by Siemens PTI

J.20 J611 Short Circuit Study Performed by MEC

J.21 J614 Short Circuit Study Performed by SMMPA

J.22 J615 Short Circuit Study Performed by MEC

Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J302 generating facility (44 GE 2.3 MW wind turbine generators) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

An ASPEN short circuit database including positive, negative and zero sequence parameters of the MDU system and its neighboring systems was provided by the MDU. This starting short circuit model is listed below:

- Short Circuit Model: "MDU 02012018.olr"

The J302 short circuit study model was developed as follows:

- Turned on generators which were nearby the POI but were off line in the starting short circuit model (Table 1-1).
- The J302 generating facility with its sequence parameters was added at the tap bus on Heskett to Wishek 230 kV line. J302 WTGs were aggregated and modeled as one induction generator with impedances set as follows:
 - o Positive sequence impedances (i.e. subtransient, transient and synchronous) (Z_1): $0.0+j0.2$ pu on 106.52 MVA base with the current limit A set to 445,647 A (five times of rated current)
 - o Zero sequence impedance (Z_0) is set to $0+j9999$ pu, and Negative sequence impedance (Z_2) is set to be the same as the Positive sequence impedance.
- The impedance and data for the J302 collection system was modeled as following:
 - o The positive and zero sequence impedance of the 34.5-0.69 kV GSU transformer impedance: $0.00759+j0.05699$ pu and $0.00687+j0.05154$ pu on 110 MVA base, respectively. Connection is Delta/Wye-ground.
 - o The 34.5 kV collection systems were modeled as a equivalent 34.5 kV line with the positive and zero sequence impedance at $0.00387+j0.00639$ pu on 100 MVA base.
 - o The main transformer is modeled as one 230-34.5 kV transformer. Positive and zero sequence impedances: $0.00242+j0.08496$ pu and $0.00228+j0.07996$ pu on 135 MVA base. Connection is Wye-ground/Wye-ground.
 - o Positive sequence impedance for the 230 kV Gentie: $0.0017+j0.01011$ pu; Zero sequence impedance: $0.0017+j0.01011$ pu.

Table 1-1: Generators Which Were Turn On

ASPEN Bus Name	ID	Starting Case Status	J302 Case Status
WATFORD-S 4.16kV	1	Offline	Online
CAPITOL G2 4.16kV	2	Offline	Online

ASPEN Bus Name	ID	Starting Case Status	J302 Case Status
LINTON GEN 3 13.8kV	1	Offline	Online
DICK DSU GEN 0.48kV	1	Offline	Online
GascWF singl 0.675kV	1	Offline	Online
LINTON GEN 2 13.8kV	1	Offline	Online
LINTON GEN 1 13.8kV	1	Offline	Online
St A-Hosp 12.5kV	1	Offline	Online
CAPITOL G4 4.16kV	1	Offline	Online
EDG-WF singl 0.675kV	1	Offline	Online
ST.JOE GEN 0.48kV	1	Offline	Online
Ellen WF gen 0.675kV	1	Offline	Online
SPROLE Gen 0.48kV	1	Offline	Online
GascWF gen 0.675kV	1	Offline	Online
TESORO G4 4.16kV	1	Offline	Online
BisWP Gen 4.16kV	1	Offline	Online
EII WF singl 0.675kV	1	Offline	Online
CAPITOL G3 4.16kV	2	Offline	Online
Bowdle WF G 0.675kV	1	Offline	Online
HESKET 3G 0.48kV	1	Offline	Online
CAPITOL G1 4.16kV	2	Offline	Online
RAY D 4.16kV	1	Offline	Online
ST.JOE GEN 0.48kV	2	Offline	Online
Bow WF singl 0.675kV	1	Offline	Online
MILCYCT9 13.8kV	1	Offline	Online
WILLCT 9 12.5kV	1	Offline	Online
COY MDU9 24.kV	1	Offline	Online
WILL DT9 12.5kV	1	Offline	Online
PARSHAL6 69.kV	1	Offline	Online

The following DPP queue projects are also added to the case:

- J457
- J503
- J593
- J599
- J607

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J302 project) and benchmark case (without the J302 project). The following fault simulation options were used in the short circuit analysis:

- The prefault voltage was set to “from a linear network solution”
- Current limited generator was set to “Enforce current limit A”

Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated on buses within the Study Project area. The short circuit results are summarized in Table 1.

The results of the short circuit analysis showed that the 3PH fault current at the J302&503 POI 230 kV bus to be 6,492 Amps with J302 (study case) and 6,023 Amps without J302 (benchmark case), and the SLG fault current at the J302&503 POI 230 kV bus to be 4,376 Amps with J302 and 4,241 Amps without J302.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 6,492 A (increased by 469 A) and SLG fault current is 4,376 A (increased by 136 A) at the J302&503 POI 230 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J302 generation project does not cause any Transmission Owner short circuit constraints.

Table 1: Three Phase (3PH) and Single-Line-to-Ground (SLG) Fault Currents with and without J302

ASPEN Bus Number	ASPEN Bus Name	ASPEN Base kV	LOCATION	AREA	ZONE	Benchmark Case without J302		Study Case with J302		Amps Change with J302 vs Benchmark		% Change with J302 vs Benchmark	
						3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault (A)	SLG Fault (A)	3PH Fault	SLG Fault
0	J302&503	230		2	1	5128.9	3721.4	5643.7	3881.4	514.8	160.0	10.0%	4.3%
0	J302&503 POI	230		2	1	6023.3	4240.5	6491.9	4376.2	468.6	135.7	7.8%	3.2%
0	J607 POI	230		2	1	6332.2	5999.1	6542.8	6114.0	210.6	114.9	3.3%	1.9%
1551	WISHEK 4	230	WISHEK 4	7	1	6309.7	5931.3	6491.2	6028.0	181.5	96.7	2.9%	1.6%
0	J607	230		2	1	5255.8	5428.1	5380.0	5508.4	124.2	80.3	2.4%	1.5%



Evaluation of Fault Currents and Interrupting Capabilities for Generation Interconnection Project J414

1-12-2018

Summary

Generator Interconnection Project J414 is a 120 MW wind powered generating facility with proposed interconnection to the 161 kV bus at ITC Midwest's Freeborn Substation. Three phase and single line to ground fault current values were evaluated before and after interconnection of the generating facility using Computer Aided Protection Engineering (CAPE) software by Electrocon International. The base CAPE model maintained and used by ITC Midwest's system protection group was updated to include planned BES future system topology representative of a 2019 ITCM system was used in the analysis.

Generating Facility Assumptions

The generating facility was modeled based on available information, and included:

- Two 60 MW equivalent generators
- Two 34.5kV to 690V equivalent transformers
- Two 34.5 kV collector system lines
- Two 34.5kV to 161 kV step up transformers
- 161 kV interconnection line

Figure 1 below shows a one-line diagram of the modeled generating facility and nearby system.

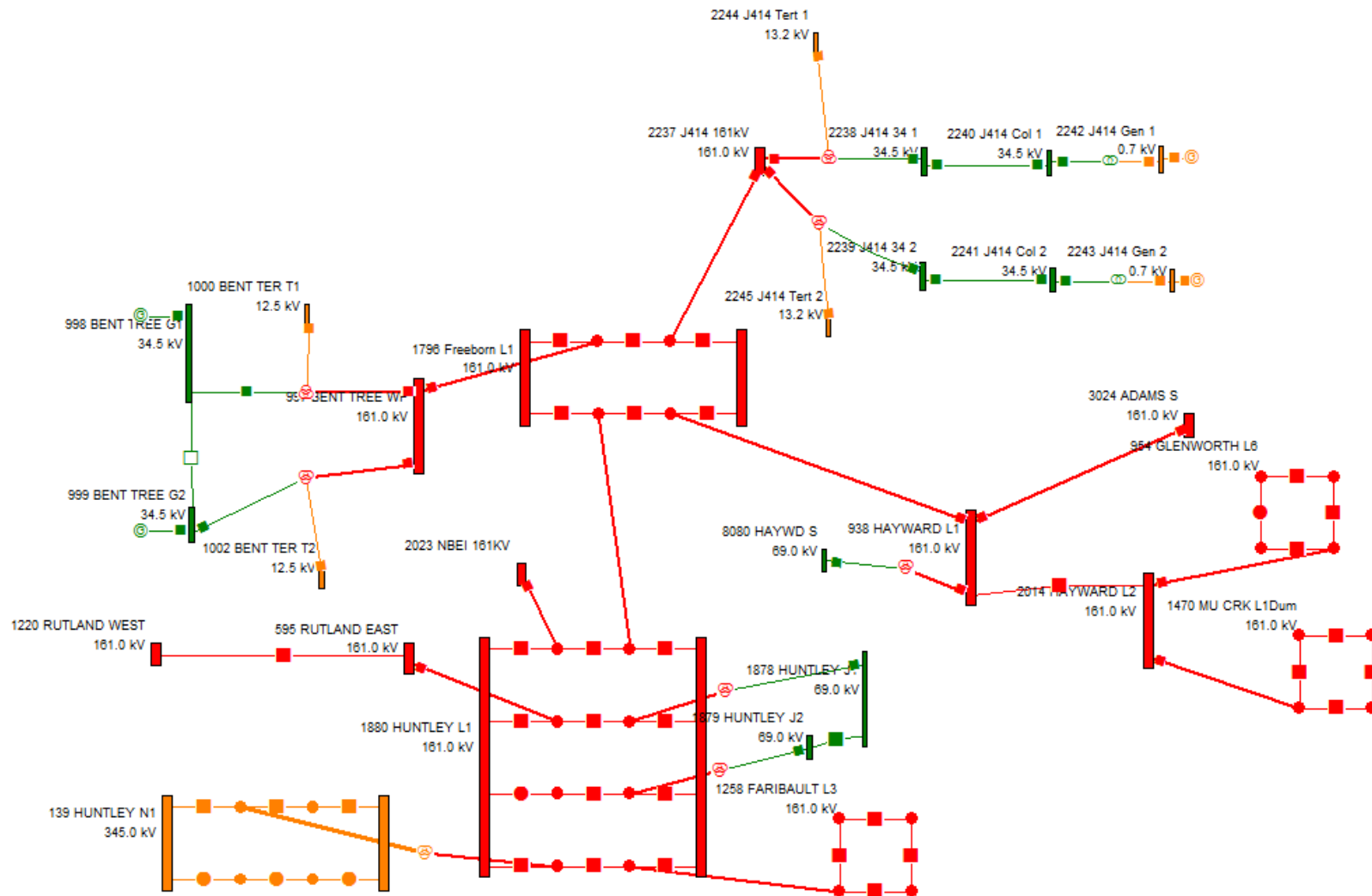


Figure1

Fault Currents

Table 1 below has the three phase and single line to ground fault currents before and after the addition of Project J414.

Substation	Owner	kV	SLG-pre (A)	3Ph-pre (A)	SLG-post (A)	3Ph-post (A)	Contribution SLG (A)	Contribution 3Ph (A)
MURPHY CREEK	SMMPA	161	9447.4	11600.5	9538.1	11758.9	90.7	158.4
GLENWORTH	ITCM	161	9552.3	12675.8	9814.3	13077.7	262	401.9
HAYWARD	ITCM	161	11609.9	14232.9	12308	14956.2	698.1	723.3
HAYWARD	ITCM	69	12889.3	11855.6	13054.4	11992.4	165.1	136.8
ADAMS	ITCM	161	17373	17787.6	17454.1	17913.4	81.1	125.8
FARIBAULT	ITCM	161	7263.5	8401.6	7310.9	8495.3	47.4	93.7
HUNTLEY	ITCM	345	8498.7	11441.6	8572.7	11594	74	152.4
HUNTLEY	ITCM	69	8811.7	8402.8	8843.9	8444.3	32.2	41.5
RUTLAND	SMMPA	161	7532.5	8648.2	7682.2	8903.2	149.7	255
N BLUE EARTH	XEL	161	8293.8	10566.2	8355.3	10702.5	61.5	136.3
FREEBORN	ITCM	161	8415.8	10526.5	10812.9	12071.6	2397.1	1545.1
HUNTLEY	ITCM	161	14231.4	14945.5	14443.7	15266.9	212.3	321.4
BENT TREE WF	Alliant	34.5	21674.2	19659.1	22028	19893.6	353.8	234.5
BENT TREE WF	Alliant	161	7747	9017	8963.7	9962	1216.7	945
WORTH COUNTY	ITCM	161	10488.2	13771	10567.9	13930	79.7	159

Table 1

Constraints

No ITCM constraints.

Conclusion

Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J414. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

CRITICAL ENERGY INFRASTRUCTURE INFORMATION NOTICE

The materials contained in this document include Critical Energy Infrastructure Information (CEII). All materials designated as CEII must be handled and protected per the requirements in FERC CEII Policy. There may be additional requirements for CEII materials in the future.

J415 Short Circuit Study Performed by MEC

The scope of this DPP short circuit facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J415, a proposed 200 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J415 was assumed to interconnect off the Quinn-Black Hawk 345 kV line. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in an ongoing DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 7,707 Amps without the Interconnection Customer included and 8,559 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, the preliminary generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation as well as for system contingencies.

As shown in the table, the changes in fault current at buses more than a couple buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J415

Bus Number	Bus Name	English Name	Base kV	Area Num	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
						Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
84151	J415 POI	J415 POI	345	635	MEC	5,738	7,098	1,360	7,707	8,559	852
636199	BLACKHAWK 3	Black Hawk	345	635	MEC	8,892	9,121	229	10,465	10,750	285
631206	QUINN3	Quinn	345	627	ITCM	6,042	7,094	1,052	7,938	8,670	732
631139	HAZLTON3	Hazleton	345	627	ITCM	11,979	12,078	99	13,650	13,830	180
636200	BLACKHAWK 5	Black Hawk	161	635	MEC	17,528	17,713	185	18,782	18,981	199
631199	KILLDEER3	Killdeer	345	627	ITCM	8,641	9,126	485	9,698	10,195	497
631144	MITCHLCO3	Mitchell County	345	627	ITCM	5,066	5,073	7	7,233	7,259	26
631142	ARNOLD 3	Duane Arnold	345	627	ITCM	11,156	11,173	17	12,588	12,636	48
631050	HAZLTON L1 5	Hazleton	161	627	ITCM	18,537	18,612	75	19,093	19,228	135
656087	BREMER CO5	Bremer County	161	652	CBPC	5,173	5,180	7	6,766	6,787	21
636202	MIDPORT5	Midport	161	635	MEC	12,193	12,257	64	14,937	15,052	115
636205	DRFNDRY5	Deere Foundry	161	635	MEC	12,725	12,778	53	15,722	15,837	115
638030	UNIONCF5	CFU Union	161	635	CFU	10,808	10,834	26	12,327	12,386	59
636240	BUTLER 5	Butler	161	635	MEC	5,189	5,824	635	7,437	7,453	16
636201	BLACKHAWK 8	Black Hawk	69	635	MEC	9,499	9,507	8	10,749	10,764	15
631198	COLBY3	Colby	345	627	ITCM	8,283	8,531	247	9,745	10,122	377
631048	EMERY 5	Emery	161	627	ITCM	25,610	25,898	288	22,676	23,022	346

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J439 Short Circuit Study Performed by MEC

The scope of this DPP short circuit facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J439, a proposed 500 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J439 was assumed to interconnect off the Clay County-Palo Alto County 345 kV line based on assumed interconnection points of higher queued generators off the original Obrien County-Kossuth County 345 kV line. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in an ongoing DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 10,253 Amps without the Interconnection Customer included and 11,796 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, the preliminary generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation as well as for system contingencies.

As shown in the table, the changes in fault current at buses more than a couple buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J439

Bus Number	Bus Name	English Name	Base kV	Area Num	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
						Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
84390	J439 POI	J439 POI	345	635	MEC	7,793	10,116	2,324	10,253	11,796	1,543
84391	J439 N ROCK1	J439 N ROCK1	345	635	IC	5,494	8,789	3,294	7,820	9,438	1,618
84392	J439 N ROCK2	J439 N ROCK2	345	635	IC	4,387	7,092	2,705	6,503	7,816	1,314
75368	J529_J590POI	Palo Alto (J529/J590 POI)	345	635	MEC	9,054	9,831	777	11,156	11,980	824
55368	J455 POI	Clay County (J455 POI)	345	635	MEC	10,171	10,914	743	11,984	12,861	877
635369	KOSSUTH 3	Kossuth County	345	635	MEC	9,274	9,594	320	12,495	12,995	500
635368	OBRIEN 3	Obrien County	345	635	MEC	11,723	12,141	418	13,878	14,563	685
631197	LEDYARD3	Ledyard	345	627	ITCM	8,669	8,785	116	11,896	12,147	251
631138	LAKEFLD3	Lakefield	345	627	ITCM	15,215	15,325	110	16,609	16,864	255
635400	HIGHLND 3	Highland	345	635	MEC	10,236	10,417	181	12,402	12,783	381
635410	OBRIEN B1 9	Obrien 34.5 kV Bus 1	34.5	635	MEC	19,917	19,948	31	19,523	19,572	49
635411	OBRIEN B2 9	Obrien 34.5 kV Bus 2	34.5	635	MEC	19,905	19,935	30	19,510	19,559	49

Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J457 generating facility (75 Vestas 2.0 MW wind turbine generators) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

An ASPEN short circuit database including positive, negative and zero sequence parameters of the MDU system and its neighboring systems was provided by the MDU. This starting short circuit model is listed below:

- Short Circuit Model: "MDU 02012018.olr"

The J457 short circuit study model was developed as follows:

- Turned on generators which were nearby the POI but were off line in the starting short circuit model (Table 1-1).
- The J457 generating facility with its sequence parameters was added at the existing Merricourt 230 kV line. J457 WTGs were aggregated and modeled as one induction generator with impedances set as follows:
 - o Positive sequence impedances (i.e. subtransient, transient and synchronous) (Z_1), and Negative sequence impedance (Z_2): 0.005 +j0.1669 pu on 174.5 MVA base with the current limit A set to 584,044 A (four times of rated current)
 - o Zero sequence impedance (Z_0) is set to 0+j9999 pu.
- The impedance and data for the J457 collection system was modeled as following:
 - o The positive and zero sequence impedance of the 34.5-0.69 kV GSU transformer impedance: 0.00927+j0.09733 pu and 0.00834+j0.08761 pu on 154.5 MVA base, respectively. Connection is Delta/Wye-ground.
 - o The 34.5 kV collection systems were modeled as an equivalent 34.5 kV line with the positive and zero sequence impedance at 0.00625+j0.00679 pu on 100 MVA base.
 - o The main transformer is modeled as one 230-34.5 kV transformer. Positive and zero sequence impedances: 0.00243+j0.08996 pu and 0.00218+j0.08097 pu on 100 MVA base. Connection is Wye-ground/Delta.
 - o Positive sequence impedance for the 230 kV Gentie: 0.00002+j0.00014 pu; Zero sequence impedance: 0.00002+j0.00014 pu.

Table 1-1: Generators Which Were Turn On

ASPEN Bus Name	ID	Starting Case Status	J457 Case Status
WATFORD-S 4.16kV	1	Offline	Online
CAPITOL G2 4.16kV	2	Offline	Online
LINTON GEN 3 13.8kV	1	Offline	Online

ASPEN Bus Name	ID	Starting Case Status	J457 Case Status
DICK DSU GEN 0.48kV	1	Offline	Online
GascWF singl 0.675kV	1	Offline	Online
LINTON GEN 2 13.8kV	1	Offline	Online
LINTON GEN 1 13.8kV	1	Offline	Online
St A-Hosp 12.5kV	1	Offline	Online
CAPITOL G4 4.16kV	1	Offline	Online
EDG-WF singl 0.675kV	1	Offline	Online
ST.JOE GEN 0.48kV	1	Offline	Online
Ellen WF gen 0.675kV	1	Offline	Online
SPROLE Gen 0.48kV	1	Offline	Online
GascWF gen 0.675kV	1	Offline	Online
TESORO G4 4.16kV	1	Offline	Online
BisWP Gen 4.16kV	1	Offline	Online
EII WF singl 0.675kV	1	Offline	Online
CAPITOL G3 4.16kV	2	Offline	Online
Bowdle WF G 0.675kV	1	Offline	Online
HESKET 3G 0.48kV	1	Offline	Online
CAPITOL G1 4.16kV	2	Offline	Online
RAY D 4.16kV	1	Offline	Online
ST.JOE GEN 0.48kV	2	Offline	Online
Bow WF singl 0.675kV	1	Offline	Online
MILCYCT9 13.8kV	1	Offline	Online
WILLCT 9 12.5kV	1	Offline	Online
COY MDU9 24.kV	1	Offline	Online
WILL DT9 12.5kV	1	Offline	Online
PARSHAL6 69.kV	1	Offline	Online

The following DPP queue projects are also added to the case:

- J302
- J503
- J593
- J599
- J607

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J457 project) and benchmark case (without the J457 project). The following fault simulation options were used in the short circuit analysis:

- The prefault voltage was set to “from a linear network solution”
- Current limited generator was set to “Enforce current limit A”

Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated on buses within the Study Project area. The short circuit results are summarized in Table 1.

The results of the short circuit analysis showed that the 3PH fault current at the Merricourt 230 kV bus to be 7,292 Amps with J457 (study case) and 6,268 Amps without J457 (benchmark case), and the SLG fault current at the Merricourt 230 kV bus to be 7,865 Amps with J457 and 6,231 Amps without J457.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 7,292 A (increased by 1,024 A) and SLG fault current is 7,865 A (increased by 1,634 A) at the Merricourt 230 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J457 generation project does not cause any Transmission Owner short circuit constraints.

Table 1: Three Phase (3PH) and Single-Line-to-Ground (SLG) Fault Currents with and without J457

ASPEN Bus Number	ASPEN Bus Name	ASPEN Base kV	LOCATION	AREA	ZONE	Benchmark Case without J457		Study Case with J457		Amps Change with J457 vs Benchmark		% Change with J457 vs Benchmark	
						3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault (A)	SLG Fault (A)	3PH Fault	SLG Fault
0	MERR	230		7	1	6267.9	6231.1	7291.8	7865.3	1023.9	1634.2	16.3%	26.2%
0	MERRICOURT	230		7	1	6266.2	6229.4	7289.4	7862.6	1023.2	1633.2	16.3%	26.2%
0	Tatanka TP	230	Ellendal	7	1	6283.7	6628.7	7151.6	7785.2	867.9	1156.5	13.8%	17.4%
0	TL91St238	230		1	1	6107.3	6577.6	6868.7	7491.9	761.4	914.3	12.5%	13.9%
0	Rough Rdr TP	230	Ellendal	7	1	6054.0	6669.3	6771.6	7501.6	717.6	832.3	11.9%	12.5%
0	TL91St112	230		1	1	5835.0	5152.5	6431.4	5767.6	596.4	615.1	10.2%	11.9%
1505	ELLENMD4	230	ELLENMD4	7	1	5867.7	5857.0	6422.2	6376.8	554.5	519.8	9.5%	8.9%
0	Tatanka SB	230	Ellendal	1	1	5119.3	5377.5	5569.0	5802.1	449.7	424.6	8.8%	7.9%
1551	WISHEK 4	230	WISHEK 4	7	1	6154.3	5749.7	6491.2	6028.0	336.9	278.3	5.5%	4.8%
0	J607 POI	230		2	1	6243.4	5876.9	6542.8	6114.0	299.4	237.1	4.8%	4.0%
1208	ELLENMD5	115	ELLENMD5	7	1	6815.0	7701.7	7096.4	8001.0	281.4	299.3	4.1%	3.9%
0	ELLENDALE 5	115	ELLENMD5	1	1	6810.2	7694.3	7091.1	7992.9	280.9	298.6	4.1%	3.9%
1602	OAKS 4	230	OAKS 4	28	1	4857.2	4153.8	5060.2	4278.7	203.0	124.9	4.2%	3.0%
0	J607	230		2	1	5192.2	5353.5	5380.0	5508.4	187.8	154.9	3.6%	2.9%
0	J302&503 POI	230		2	1	6350.7	4313.6	6491.9	4376.2	141.2	62.6	2.2%	1.5%
1506	FORMANO4	230	FORMANO4	28	1	5155.2	4411.4	5273.5	4479.7	118.3	68.3	2.3%	1.5%



Short Circuit Study Report

MISO J459, J613, J6337, and J438
Prepared for the Midwest ISO

February 16, 2018

Table of Contents

1. Executive Summary	3
2. J459	4
3. J613	5
4. J637	6
5. J638	7

1. Executive Summary

The Transmission Owner, Otter Tail, performed short circuit analysis for projects J459, J613, J637, and J638 on its facilities. When there was the potential for one project to impact another project, the Transmission Owner performed the short circuit analysis on the lower queued project before proceeding with the next study. If an earlier queue project withdraws, the Transmission Owner may need to restudy possible short-circuit impacts that could be introduced by the interconnecting project.

The Transmission Owner, Otter Tail, did not see any short circuit impacts on its facilities.

The Transmission Owner may also need to rerun any and all short circuit analysis if any of the assumptions change.

2. J459

A short-circuit analysis was performed to determine the impact of proposed projects on the adjacent substation fault current levels. The Transmission Owner added J459 to its existing ASPEN OneLiner system model (BANG13-0503.OLR) and all buses within one substation from the Big Stone South to Brookings County Line were evaluated. The Transmission Owner only owns the Big Stone South substation; therefore, it simulated the faults on the Big Stone South buses.

Results

With the proposed project J459 addition, the fault currents are roughly 15.3 kA at the Big Stone South 345 kV bus, 22.7 kA at the Big Stone South 230 kV bus, 13 kA at the Big Stone South 34.5 kV bus. Based on the short circuit analysis performed, the fault current ratings of the Transmission Owner's equipment at the Big Stone South Substation are not exceeded and there are no upgrades required. The Transmission Owner did not evaluate any impacts on the fault current levels at substations owned by other Transmission Owners.

The Transmission Owner will need to perform a general protective relay coordination review when the project approaches the backfeed date.

Assumptions

- The study included all generators with signed Generator Interconnection Agreements and prior queued generators in the area (Aug. 2015 Study Phase, Feb. 2016 Study Phase, and Aug. 2016 Study Phase),
- Detailed generation modeling values for the substation and feeders were not provided; therefore, Type 3 wind generator on a 0.2 pu impedance on generator base was assumed.

3. J613

A short-circuit analysis was performed to determine the impact of proposed projects on the adjacent substation fault current levels. The Transmission Owner added its existing ASPEN OneLiner system model (BANG13-0503.OLR) and all -buses within one substation from the Jamestown 115 kV substation had faults simulated on them.

Results

With the proposed projects additions, the fault currents are roughly 15.6 kA at the Jamestown 115 kV bus, 6.4 kA at the Jamestown 345 kV bus, 9.3 kA at the Jamestown 41.6 kV bus. Based on the short circuit analysis performed, the fault current ratings of the Transmission Owner's equipment in the area are not exceeded and there are no upgrades required. The Transmission Owner did not evaluate any impacts on the fault current levels at substations owned by other Transmission Owners.

The Transmission Owner will need to perform a general protective relay coordination review when the project approaches the backfeed date.

Assumptions

- The study included all generators with signed Generator Interconnection Agreements and prior queued generators in the area (Aug. 2015 Study Phase, Feb. 2016 Study Phase, and Aug. 2016 Study Phase),
- Detailed generation modeling values for the substation and feeders were not provided; therefore, Type 3 wind generators on a 0.2 pu impedance on generator base was assumed.

4. J637

A short-circuit analysis was performed to determine the impact of proposed projects on the adjacent substation fault current levels. The Transmission Owner added J637 to its existing ASPEN OneLiner system model (BANG13-0503.OLR) and all buses within one substation from the Big Stone South to Brookings County Line were evaluated. The Transmission Owner only owns only the Big Stone South substation; therefore, it simulated the faults on the Big Stone South buses.

Results

With the proposed projects additions, the fault currents are roughly 15.5 kA at the Big Stone South 345 kV bus, 22.9 kA at the Big Stone South 230 kV bus, 13 kA at the Big Stone South 34.5 kV bus. Based on the short circuit analysis performed, the fault current ratings of the Transmission Owner's equipment at the Big Stone South Substation are not exceeded and there are no upgrades required. The Transmission Owner did not evaluate any impacts on the fault current levels at substations owned by other Transmission Owners.

The Transmission Owner will need to perform a general protective relay coordination review when the project approaches the backfeed date.

Assumptions

- The study included all generators with signed Generator Interconnection Agreements and prior queued generators in the area (Aug. 2015 Study Phase, Feb. 2016 Study Phase, and Aug. 2016 Study Phase),
- Interconnection Customer's May 2016 one-lines, generator, and feeder values are accurate, and
- Other interconnections between this J637 and Big Stone South have same short circuit interrupting capabilities as Big Stone South.

5. J638

A short-circuit analysis was performed to determine the impact of proposed projects on the adjacent substation fault current levels. The Transmission Owner added J638 to its existing ASPEN OneLiner system model (BANG13-0503.OLR) and all -buses within one substation from the Big Stone South to Brooking County Line were evaluated. The Transmission Owner only owns the Big Stone South substation; therefore, it simulated the faults on the Big Stone South buses.

Results

With the proposed projects additions, the fault currents are roughly 15.7 kA at the Big Stone South 345 kV bus, 23.1 kA at the Big Stone South 230 kV bus, 13.2 kA at the Big Stone South 34.5 kV bus. Based on the short circuit analysis performed, the fault current ratings of the Big Stone South Substation are not exceeded and there are no upgrades required. The Transmission Owner did not evaluate any impacts on the fault current levels at substations owned by other Transmission Owners.

The Transmission Owner will need to perform a general protective relay coordination review when the project approaches the backfeed date.

Assumptions

- The study included all generators with signed Generator Interconnection Agreements and prior queued generators in the area (Aug. 2015 Study Phase, Feb. 2016 Study Phase, and Aug. 2016 Study Phase),
- Interconnection Customer's May 2016 one-lines, generator, and feeder are accurate, and
- Other interconnections between this J638 and Big Stone South have same short circuit interrupting capabilities as Big Stone South.

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J476 Short Circuit Study Performed by MEC

The scope of this DPP short circuit facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J476, a proposed 246 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J476 was assumed to interconnect off the Atchison County-Orient 345 kV line. Orient is the assumed interconnection point of a higher queued generator. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in an ongoing DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 13,807 Amps without the Interconnection Customer included and 14,136 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, the preliminary generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation as well as for system contingencies.

As shown in the table, the changes in fault current at buses more than a couple buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J476

Bus Number	Bus Name	English Name	Base kV	Area Num	Owner	SLG Fault Current Comparison		3 Ph Fault Current Comparison			
						Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
84760	J476 POI	J476 POI	345	635	MEC	9,817	11,576	1,759	13,807	14,136	329
635017	ATCHSN 3	Atchison	345	635	MEC	9,881	11,623	1,742	13,145	14,187	1,042
635570	ORIENT 3	Orient	345	635	MEC	13,355	13,457	102	16,962	17,096	134
640139	COOPER 3	Cooper	345	640	NPPD	21,796	22,468	672	21,232	21,908	676
635100	RLHILLS3	Rolling Hills	345	635	MEC	10,116	10,145	29	12,883	12,935	52
645458	S3458 3	Sub 3458	345	645	OPPD	17,249	17,349	100	18,134	18,292	158
640277	MOORE 3	Moore	345	640	NPPD	11,162	11,187	25	12,216	12,261	45
300039	7FAIRPT	Fairport	345	330	AECI	8,826	8,851	26	10,493	10,544	51
541199	ST JOE 3	St Joseph	345	540	GMO	12,089	12,124	35	14,500	14,570	70
640140	COOPER 5	Cooper	161	640	NPPD	11,164	11,217	53	9,887	9,939	53
635013	PNYCRK 3	Pony Creek	345	635	MEC	26,914	26,962	48	25,842	25,906	64
635630	BOONVIL3	Booneville	345	635	MEC	15,475	15,505	30	18,196	18,246	50
635635	MADISON3	Madison County	345	635	MEC	13,278	13,303	25	16,367	16,412	45

Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J503 generating facility (43 GE 2.3 MW wind turbine generators) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

An ASPEN short circuit database including positive, negative and zero sequence parameters of the MDU system and its neighboring systems was provided by the MDU. This starting short circuit model is listed below:

- Short Circuit Model: "MDU 02012018.olr"

The J503 short circuit study model was developed as follows:

- Turned on generators which were nearby the POI but were off line in the starting short circuit model (Table 1-1).
- The J503 generating facility with its sequence parameters was added at the tap bus on Heskett to Wishek 230 kV line. J503 WTGs were aggregated and modeled as one induction generator with impedances set as follows:
 - o Positive sequence impedances (i.e. subtransient, transient and synchronous) (Z_1): $0.0+j0.2$ pu on 104.1 MVA base with the current limit A set to 435,523 A (five times of rated current)
 - o Zero sequence impedance (Z_0) is set to $0+j9999$ pu, and Negative sequence impedance (Z_2) is set to be the same as the Positive sequence impedance.
- The impedance and data for the J503 collection system was modeled as following:
 - o The positive and zero sequence impedance of the 34.5-0.69 kV GSU transformer impedance: $0.00759+j0.05699$ pu on 107.5 MVA base. Connection is Delta/Wye-ground.
 - o The 34.5 kV collection systems were modeled as an equivalent 34.5 kV line with the positive and zero sequence impedance at $0.00386+j0.00638$ pu on 100 MVA base.
 - o The main transformer is modeled as one 230-34.5 kV transformer. Positive and zero sequence impedances: $0.00242+j0.08496$ pu and $0.00228+j0.07996$ pu on 135 MVA base. Connection is Wye-ground/Wye-ground.
 - o Positive sequence impedance for the 230 kV Gentie: $0.0017+j0.01011$ pu; Zero sequence impedance: $0.0017+j0.01011$ pu.

Table 1-1: Generators Which Were Turn On

ASPEN Bus Name	ID	Starting Case Status	J503 Case Status
WATFORD-S 4.16kV	1	Offline	Online
CAPITOL G2 4.16kV	2	Offline	Online

ASPEN Bus Name	ID	Starting Case Status	J503 Case Status
LINTON GEN 3 13.8kV	1	Offline	Online
DICK DSU GEN 0.48kV	1	Offline	Online
GascWF singl 0.675kV	1	Offline	Online
LINTON GEN 2 13.8kV	1	Offline	Online
LINTON GEN 1 13.8kV	1	Offline	Online
St A-Hosp 12.5kV	1	Offline	Online
CAPITOL G4 4.16kV	1	Offline	Online
EDG-WF singl 0.675kV	1	Offline	Online
ST.JOE GEN 0.48kV	1	Offline	Online
Ellen WF gen 0.675kV	1	Offline	Online
SPROLE Gen 0.48kV	1	Offline	Online
GascWF gen 0.675kV	1	Offline	Online
TESORO G4 4.16kV	1	Offline	Online
BisWP Gen 4.16kV	1	Offline	Online
EII WF singl 0.675kV	1	Offline	Online
CAPITOL G3 4.16kV	2	Offline	Online
Bowdle WF G 0.675kV	1	Offline	Online
HESKET 3G 0.48kV	1	Offline	Online
CAPITOL G1 4.16kV	2	Offline	Online
RAY D 4.16kV	1	Offline	Online
ST.JOE GEN 0.48kV	2	Offline	Online
Bow WF singl 0.675kV	1	Offline	Online
MILCYCT9 13.8kV	1	Offline	Online
WILLCT 9 12.5kV	1	Offline	Online
COY MDU9 24.kV	1	Offline	Online
WILL DT9 12.5kV	1	Offline	Online
PARSHAL6 69.kV	1	Offline	Online

The following DPP queue projects are also added to the case:

- J302
- J457
- J593
- J599
- J607

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J503 project) and benchmark case (without the J503 project). The following fault simulation options were used in the short circuit analysis:

- The prefault voltage was set to “from a linear network solution”
- Current limited generator was set to “Enforce current limit A”

Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated on buses within the Study Project area. The short circuit results are summarized in Table 1.

The results of short circuit analysis showed that the 3PH fault current at the J302&503 POI 230 kV bus to be 6,492 Amps with J503 (study case) and 6,036 Amps without J503 (benchmark case), and the SLG fault current at the J302&503 POI 230 kV bus to be 4,376 Amps with J503 and 4,244 Amps without J503.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 6,492 A (increased by 456 A) and SLG fault current is 4,376 A (increased by 132 A) at the J302&503 POI 230 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J503 generation project does not cause any Transmission Owner short circuit constraints.

Table 1: Three Phase (3PH) and Single-Line-to-Ground (SLG) Fault Currents with and without J503

ASPEN Bus Number	ASPEN Bus Name	ASPEN Base kV	LOCATION	AREA	ZONE	Benchmark Case without J503		Study Case with J503		Amps Change with J503 vs Benchmark		% Change with J503 vs Benchmark	
						3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault (A)	SLG Fault (A)	3PH Fault	SLG Fault
0	J302&503	230		2	1	5142.6	3725.8	5643.7	3881.4	501.1	155.6	9.7%	4.2%
0	J302&503 POI	230		2	1	6036.0	4244.3	6491.9	4376.2	455.9	131.9	7.6%	3.1%
0	J607 POI	230		2	1	6338.2	6002.4	6542.8	6114.0	204.6	111.6	3.2%	1.9%
1551	WISHEK 4	230	WISHEK 4	7	1	6314.9	5934.0	6491.2	6028.0	176.3	94.0	2.8%	1.6%
0	J607	230		2	1	5259.3	5430.4	5380.0	5508.4	120.7	78.0	2.3%	1.4%



Short Circuit Analysis Report

200 MW Windfarm in Oliver County, North Dakota

MISO # J511
Prepared for MISO

June 18, 2018
Great River Energy

Contact:
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Great River Energy

Contents

Introduction	3
Background	3
Methodology.....	3
Cases/Conditions	4
Results.....	7

Introduction

This report presents the results of the short circuit analysis for J511. The objective of the analysis is to determine the present fault current levels near the POI for J511 and determine the future fault current levels after the wind turbines become commercially operational. After the fault current levels are found, the rating of Great River Energy's (GRE's) breakers near the POI will be analyzed to ensure they are capable of interrupting any increase in fault current due to the contributions of J511.

Background

The first part of the study is to calculate the fault currents before and after the addition of J511's 200 MW of wind generation. Short circuit calculations are based on the impedances of the transmission lines, transformers, generators, and on the pre-fault voltages. This wind generation will be connected to the system via 16 mile transmission line into GRE's 230kV Stanton (S031) substation.

The second part of this study is to compare the fault currents to the interrupting ratings of nearby GRE equipment. Fault current will also be shown for GRE substations five buses away from Stanton as well as nearby foreign substations. These fault currents will be compared to interrupting ratings of GRE owned equipment but foreign owned equipment will not be analyzed.

Methodology

Aspen *One-liner* short circuit analysis software and the associated *Breaker Rating Module* are used to determine the symmetrical fault currents and circuit breaker interruption capabilities. Great River Energy has an Aspen model already built that includes Stanton and surrounding substations.

The *Breaker Rating Module* is a tool designed to check the rating of circuit breakers against the short circuit currents they need to interrupt. The breaker-rating logic simulates relevant faults to find the maximum short-circuit current that flows through the breaker. The momentary and breaking currents of breakers are then calculated using procedures recommended by ANSI/IEEE. Finally, the program compares these currents to the rated capabilities of the breakers and reports the findings. The breaker-rating logic uses a breaker connection model that is designed to work with traditional symmetrical-sequence networks where circuit breakers connections are not modeled explicitly. The connection model associates each breaker with the bus, branches and generators that it protects.

The generator, transformer, and the transmission line impedances for interconnection equipment were received from the customer. If the transmission line impedance was not available from the customer, it was assumed to be zero for worst case fault currents. All the buses are assumed to have a per unit pre-fault voltage equal to 1. GRE models wind turbines based on a method recommended by Aspen. The wind turbines used in this interconnection are type-3 generators. Wind generators with doubly-fed machines are also known as type-3 generators. Type-3 generators can be simulated in Aspen *One-liner* using the conventional generators model without current limits.

Short circuit studies with wind turbines modeled as conventional generators will yield a conservative estimate of the peak short-circuit current. Type-3 generators are ungrounded and contribute no zero-sequence current during a fault. When modeling type-3 generators in Aspen, the zero-sequence impedance is set to 9999 pu. The positive sequence impedance is set equal to the sub transient X_d'' impedance of the machine and the negative sequence impedance is set to be the same as the positive sequence impedance unless the negative-sequence impedance is known. All the current limits for the generator are set to zero.

Cases/Conditions

TABLE 1

Faults		Before J511			After J511		
Location	Fault Type	Short Circuit Current (A)	ANSI X/R	MVA	Short Circuit Current (A)	ANSI X/R	MVA
Stanton 230kV Bus	1LG	16,123	22.9	6540.9	17,245	21.9	6966.6
	3LG	18,638	30.9	7561.3	19,841	29.9	8015.6
Square Butte 230kV Bus	1LG	35,707	50.7	14271.4	35,843	50.3	14308.7
	3LG	30,813	59.5	12315.6	31,063	58.9	12400.7
Leland Olds 230kV Bus	1LG	23,474	56.1	9407.3	23,498	56.5	9410.4
	3LG	22,093	45.6	8853.6	22,119	46	8858
Coal Creek 230kV Bus	1LG	23,394	37.6	9574.8	23,742	36.7	9691
	3LG	19,474	35.2	7970.3	20,023	34.2	8172.7
McHenry 230kV Bus	1LG	5,480	8.2	2178.3	5,481	8.2	2174.6
	3LG	6,065	8.3	2410.7	6,081	8.3	2412.6
McHenry 115kV Bus	1LG	7,981	8.2	1571.9	7,976	8.2	1569
	3LG	8,522	8.8	1678.5	8,523	8.8	1676.7
Balta 230kV Bus	1LG	5,553	7.2	2189.1	5,552	7.2	2185.8
	3LG	6,764	8.1	2666.4	6,773	8	2666.8
Ramsey 230kV Bus	1LG	3,472	6.7	1362.1	3,470	6.7	1360.3
	3LG	4,169	7.3	1635.5	4,168	7.3	1634
Ramsey 115kV Bus	1LG	6,024	9.2	1188.7	6,021	9.2	1187.3
	3LG	5,764	9.1	1137.4	5,762	9.1	1136.3

Table 2

Before J511						
Location	Breaker	Duty (% of capacity)	Duty (See Note 1)	Breaker Capacity (A)	ANSI X/R [1]	Exceeds Breaker capacity?
Stanton 230kV	31RB1	53.3	21335.5	40000	30.9	No
	31RB2	46.7	18680	40000	31.4	No
	31RB3	53.3	21335.5	40000	30.9	No
	31RB4	53.3	21335.5	40000	30.9	No
	31RB6	53.3	21335.5	40000	30.9	No
	31RB7	53.3	21335.5	40000	30.9	No
	31RB8	53.3	21335.5	40000	30.9	No
Coal Creek 230kV Bus	61RB1	69.4	27747.4	40000	37.6	No
	61RB10	69.4	27747.4	40000	37.6	No
	61RB11	69.4	27747.1	40000	37.6	No
	61RB12	69.4	27747.4	40000	37.6	No
	61RB16	69.4	27747.4	40000	37.6	No
	61RB17	69.4	27747.1	40000	37.6	No
	61RB18	69.4	27747.4	40000	37.6	No
	61RB2	49.7	19861.5	40000	37.6	No
	61RB3	69.4	27747.4	40000	37.6	No
	61RB4	69.4	27747.4	40000	37.6	No
	61RB5	49.4	19754.5	40000	37.6	No
	61RB6	69.4	27747.4	40000	37.6	No
	61RB7	69.4	27747.4	40000	37.6	No
	61RB8	69.4	27747.1	40000	37.6	No
	61RB9	69.4	27747.4	40000	37.6	No

Table 3

After J511						
Location	Breaker	Duty (% of capacity)	Duty (See Note 1)	Breaker Capacity (A)	ANSI X/R [1]	Exceeds Breaker capacity?
Stanton 230kV	31RB1	56.4	22568.7	40000	29.9	No
	31RB2	49.8	19915.1	40000	30.2	No
	31RB3	56.4	22568.7	40000	29.9	No
	31RB4	56.4	22568.7	40000	29.9	No
	31RB6	56.4	22568.7	40000	29.9	No
	31RB7	56.4	22568.7	40000	29.9	No
	31RB8	56.4	22568.7	40000	29.9	No
Coal Creek 230kV Bus	61RB1	70.1	28042.9	40000	36.7	No
	61RB10	70.1	28042.9	40000	36.7	No
	61RB11	70.1	28042.7	40000	36.7	No
	61RB12	70.1	28042.9	40000	36.7	No
	61RB16	70.1	28042.9	40000	36.7	No
	61RB17	70.1	28042.7	40000	36.7	No
	61RB18	70.1	28042.9	40000	36.7	No
	61RB2	50.5	20180.9	40000	36.7	No
	61RB3	70.1	28042.9	40000	36.7	No
	61RB4	70.1	28042.9	40000	36.7	No
	61RB5	50.2	20074.6	40000	36.7	No
	61RB6	70.1	28042.9	40000	36.7	No
	61RB7	70.1	28042.9	40000	36.7	No
	61RB8	70.1	28042.7	40000	36.7	No
	61RB9	70.1	28042.9	40000	36.7	No

Results

Fault currents were calculated before and after the addition of J511's 200 MW of wind generation. Table 1 summarizes the fault currents at Stanton Substation and nearby substations.

Tables 2 and 3 compare the fault currents to the interrupting ratings of equipment at GRE's Stanton and Coal Creek Substations. Table 2 shows the data before the addition of J511 and Table 3 shows the data after the addition of J511.

The data in Table 3 shows that none of the circuit breaker interrupting capabilities at Stanton or Coal Creek Substations will be exceeded after the addition of J511.



Interconnection Facility Study Report

Generation Interconnection MISO projects
J512, J569, J575, J577, J587 and J597

March 15, 2018

Submitted by Tim Bickford, PE

Xcel Energy Services, Inc.

1.0 J512

A short circuit analysis was performed to determine the impact of the proposed 250MW wind generation MISO project J512 interconnecting on the 115 kV Transmission Line between Nobles County and Fenton substations. The proposed generation was added to the existing Transmission Owner's CAPE model and faults were studied on all busses within four busses from the point of interconnection.

Based on the results of the study, the fault current ratings of the Transmission Owner's equipment in the area will not be exceeded. With the proposed J512 generation modeled the fault currents in the area are below 10 kA for 69 kV busses and approximately 12 kA for 115 kV and 345 kV busses. All of the Transmission Owner equipment is rated at 31.5kA or greater.

There are no short circuit upgrades required in this area due to the proposed additional generation.

2.0 J569

A short circuit analysis was performed to determine the impact of the proposed 100MW wind generation MISO project J569 interconnecting at the 161 kV Rock County substation. The proposed generation was added to the existing Transmission Owner's CAPE model and faults were studied on all busses within four busses from the point of interconnection.

Based on the results of the study, the fault current ratings of the Transmission Owner's equipment in the area will not be exceeded. With the proposed J569 generation modeled the fault currents in the area are below 15 kA for 161 kV and 115 kV busses with the exception of the Anson 115 kV bus which was approximately 37 kA. All of the Transmission Owner equipment is rated at 40 kA or greater and the 115 kV equipment at Anson is rated at 63 kA.

There are no short circuit upgrades required in this area due to the proposed additional generation.

3.0 J575, J577, J587, J597

The short circuit analysis for J575 (100 MW wind), J577 (102.8 MW Wind), J587 (200 MW wind) and J597 (300MW wind) was performed to determine the impact of the cumulative proposed 702.8 MW wind generation interconnecting at the 345 kV Brookings County substation and on the 345 kV Transmission Line between Brookings County and Hawks Nest Lake substations. These MISO generation interconnections were studied using the same model due to the proximity of the interconnections. The

proposed generation was added to the existing Transmission Owner's CAPE model and faults were studied on all busses within four busses from the point of interconnections.

Based on the results of the study, the fault current ratings of the Transmission Owner's equipment in the area will not be exceeded. With the proposed J575, J577, J587 and J597 generation modeled the fault currents in the area are below 30 kA for 115 kV and 345 kV busses. All of the Transmission Owner equipment is rated at 40 kA or greater.

There are no short circuit upgrades required in this area due to the proposed additional generation.

C. Short Circuit Analysis

Short circuit analysis was performed with and without the study generation in service using Aspen on a model maintained by Ameren specifically for calculating fault current. A comparison of fault current calculations provided the necessary information to determine if the interrupting capability of existing circuit breakers at substations within three buses of the J541 and J598 interconnections would be exceeded. Per MISO criteria, the study generation is not responsible for circuit breaker interrupting capability upgrades if the incremental increase is less than 5.0%. Results of the short circuit analysis are shown in tables IV-N-1 and IV-N-2 below:

J541 & J598 Short Circuit Results

			Without Projects		With Projects		Delta	
Substation	Voltage	T.O.	3-Phase amps	L-G amps	3-Phase amps	L-G amps	3-Phase amps	L-G amps
J598 POI	345 kV	Ameren	8,055	6,075	9,944	9,542	1,889	3,467
J541 POI	345 KV	Ameren	8,120	5,924	10,406	9,946	2,286	4,022
Zachary	345 KV	Ameren	8,770	6,468	10,393	8,701	1,623	2,233
Zachary	161 kV	Ameren	13,305	9,976	14,254	11,673	949	1,697
Adair	161 kV	Ameren	12,952	9,484	13,698	10,706	746	1,222
Maywood	345 kV	Ameren	15,722	11,906	16,210	12,293	488	387
Ottumwa	345 KV	MEC	11,257	12,309	12,146	13,232	889	923
Ottumwa	161 kV	MEC	34,542	36,514	35,486	37,736	944	1,222
Bridgeport	161 kV	MEC	20,613	16,831	20,903	17,031	290	200
Wapello	161 kV	MEC	13,885	12,502	14,033	12,597	148	95
Appanoose	161 kV	AECI	6,469	4,978	6,530	5,021	61	43
Novelty	161 kV	AECI	6,252	4,732	6,337	4,810	85	78

No circuit breaker upgrades are required on the Ameren System based on the expected fault contribution of the J541 and J598 generation. Breaker ratings for MEC and AECI would be required in order to determine if ratings are exceed for their facilities in this study. However, since the incremental increase in fault current is greater than 5% for only the Ottumwa 345 kV bus, this would be the only facility where the customers would be required to provide mitigation if necessary.

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J475 and J555 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J475, a proposed 200 MW wind farm, and J555, a proposed 140 MW expansion, and nearby substations both with and without the Interconnection Customer interconnected. J475 and J555 were assumed to interconnect at the MidAmerican and ITC Midwest jointly-owned Montezuma Substation. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in DPP study cycles of the MISO generation interconnection process, where the system impact studies are not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 13,008 Amps without the Interconnection Customer included and 14,334 Amps with the full 340 MW from the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results, especially the 34.5 kV results that should be calculated by the Interconnection Customer. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican breakers become overdutied as a result of the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

**Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J475 and J555**

Bus Number	Bus Name	English Name	Base kV	Area Num	Owner	SLG Fault Current Comparison					3 Ph Fault Current Comparison				
						Base SLG w/o wind farms J475/J555	SLG with new J475 wind farm	SLG Difference w/ J475 wind farm vs Base	SLG with J475 & J555 wind farm	SLG Difference w/ J475 & J555 wind farm vs Base	Base 3PH w/o wind farms J475/J555	3PH with new J475 wind farm	3PH Difference w/ J475 wind farm vs Base	3PH with J475 & J555 wind farm	3PH Difference w/ J475 & J555 wind farm vs Base
635730	MNTZUMA3	Montezuma	345	635	MEC	9,020	10,083	1,063	11,078	2,058	13,008	13,705	697	14,334	1,326
		J475 Collector Bus	34.5			19,371	32,795	13,424	32,856	13,485	16,744	27,713	10,969	27,756	11,012
		J555 Collector Bus	34.5			19,371	19,371	-	29,325	9,954	16,744	16,744	-	24,307	7,563
635680	BONDRNT3	Bondurant	345	635	MEC	17,016	17,131	115	17,223	207	19,055	19,188	133	19,299	244
631143	OTTUMWA3	Ottumwa	345	627	ITCM	9,031	9,193	162	9,322	291	10,891	11,055	164	11,194	303
	J530 POI	J530 POI	345	635	MEC	10,293	10,542	249	10,742	449	12,902	13,146	244	13,354	452
635700	SYCAMOR3	Sycamore	345	635	MEC	19,005	19,070	65	19,122	117	20,106	20,193	87	20,265	159
635690	GDMEC	GDMEC	345	635	MEC	18,566	18,638	72	18,697	131	18,624	18,711	87	18,783	159
344000	7ADAIR	Adair/Zachary 345	345	356	AMMO	8,920	8,934	15	8,946	26	12,465	12,494	29	12,517	52
631115	OTTUMWA5	Ottumwa	161	627	ITCM	27,848	27,929	81	27,993	145	23,967	24,035	68	24,091	124
636400	HILLS 3	Hills	345	635	MEC	16,822	16,902	80	16,966	144	19,937	20,054	117	20,151	214
		J530 Collector Bus	34.5			30,409	30,480	71	30,537	128	26,522	26,574	52	26,617	95
635600	GRIMES 3	Grimes	345	635	MEC	18,245	18,292	47	18,330	85	20,460	20,535	75	20,597	137
635701	SYCAMOR5	Sycamore	161	635	MEC	33,003	33,067	64	33,119	116	32,925	33,003	78	33,068	143
635650	SE POLK3	SE Polk	345	635	MEC	17,992	18,057	65	18,110	118	18,356	18,438	82	18,505	149
345435	7MAYWOOD	Maywood	345	356	AMMO	8,222	8,226	4	8,228	6	11,420	11,429	9	11,436	16
344006	5ADAIR3	Adair/Zachary 161	161	356	AMMO	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
631110	WAPELLO5	Wapello	161	627	ITCM	10,727	10,735	8	10,741	14	11,096	11,108	12	11,117	21
631118	LUCAS 5	Lucas Co	161	627	ITCM	3,221	3,222	1	3,222	1	3,339	3,340	1	3,341	2
631116	BRDGPRT5	Bridgeport	161	627	ITCM	15,772	15,794	22	15,811	39	17,111	17,144	33	17,170	59
631134	TRICNTY5	Tri County	161	627	ITCM	15,937	15,960	23	15,978	41	17,358	17,391	33	17,418	60
635880	PLSNT CRNR 5	Pleasant Corner	161	635	MEC	9,837	9,846	9	9,853	15	12,403	12,420	17	12,434	31
636420	TIFFIN 3	Tiffin	345	635	MEC	11,251	11,275	24	11,295	44	14,815	14,864	49	14,905	90
636620	WALCOTT3	Walcott	345	635	MEC	11,076	11,083	7	11,088	12	15,648	15,667	19	15,682	34
636640	LOUISA 3	Louisa	345	635	MEC	19,061	19,085	24	19,105	44	19,077	19,111	34	19,139	62
636645	SUB T 3	Sub T Haskins	345	635	MEC	11,232	11,252	20	11,269	37	15,712	15,759	47	15,797	85
636401	HILLS 5	Hills	161	635	MEC	19,245	19,278	33	19,304	59	19,685	19,723	38	19,755	70
631034	ROSEHLW5	Rose Hollow	161	627	ITCM	5,802	5,803	2	5,804	3	6,551	6,554	3	6,556	5

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J583 Short Circuit Study Performed by MEC

The scope of this DPP short circuit facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J583, a proposed 200 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J583 was assumed to interconnect at an expanded Fallow Ave Substation. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in an ongoing DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 10,216 Amps without the Interconnection Customer included and 10,844 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, the preliminary generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation as well as for system contingencies.

As shown in the table, the changes in fault current at buses more than a couple buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J583

Bus Number	Bus Name	English Name	Base kV	Area Num	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
						Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
635590	FALLOW 3	Fallow Ave	345	635	MEC	8,704	9,323	619	10,216	10,844	628
85831	J583	J583 Sub	345	635	IC	8,344	8,980	636	9,927	10,556	629
635580	ARBR HL 3	Arbor Hill	345	635	MEC	9,239	9,501	263	11,441	11,827	386
635014	OVLND 3	Overland Trail	345	635	MEC	23,173	23,244	71	24,238	24,378	140
635589	ECLIPSE3	Eclipse	345	635	MEC	7,028	7,410	382	8,409	8,818	409
635600	GRIMES 3	Grimes	345	635	MEC	19,284	19,361	77	21,843	22,015	172
635000	CBLUFFS3	CBEC	345	635	MEC	30,982	31,058	76	28,226	28,363	137
635015	OVLND 5	Overland Trail	161	635	MEC	24,540	24,558	18	26,188	26,236	48
636003	BVR CRK 3	Beaver Creek	345	635	MEC	9,622	9,626	4	12,476	12,492	16
635700	SYCAMOR3	Sycamore	345	635	MEC	19,678	19,734	56	21,259	21,393	134
635630	BOONVIL3	Booneville	345	635	MEC	15,478	15,505	27	18,170	18,246	76
635601	GRIMES 5	Grimes	161	635	MEC	22,606	22,636	30	24,177	24,239	62
635599	GRIM1XT9	Grimes 9T1 Tertiary	13.8	635	MEC	-	-	-	32,723	32,731	8
635013	PNYCRK 3	Pony Creek	345	635	MEC	26,905	26,962	57	25,792	25,906	114
635016	STHLND 3	Southland	345	635	MEC	26,877	26,933	56	25,775	25,889	114
645456	S3456 3	Sub 3456	345	645	OPPD	26,457	26,485	28	27,851	27,924	73
635001	CBLUFFS5	CBEC	161	635	MEC	37,185	37,220	35	35,674	35,750	76
635026	CBLF2XT9	CBEC 9T2 Tertiary	13.8	635	MEC	-	-	-	35,119	35,125	6

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J529 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J529, a proposed 250 MW wind farm along with J590, a proposed 90 MW expansion at the same interconnection point, and nearby substations both with and without the Interconnection Customer interconnected. J529/J590 was assumed to interconnect off the Obrien County-Kossuth County 345 kV line. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in a DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 10,000 Amps without the Interconnection Customer included and 11,705 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than a couple buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J529/J590

Bus Number	Bus Name	English Name	Base kV	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
					Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
		J529J590 POI	345		7,660	9,703	2,043	10,000	11,705	1,705
		J529J590 Collector	34.5		23,730	37,957	14,227	20,331	32,060	11,730
55368	J455 POI	J455 POI	345	MEC	9,827	10,320	493	11,388	11,993	606
635369	KOSSUTH 3	Kossuth Co	345	MEC	8,502	9,587	1,085	11,804	12,852	1,048
635368	OBRIEN 3	O'Brien Co	345	MEC	11,489	11,804	315	13,740	14,192	452
631197	LEDYARD3	Ledyard	345	ITCM	8,258	8,705	448	11,552	12,112	560
636000	WEBSTER3	Webster	345	MEC	9,825	10,033	208	11,901	12,150	249
631138	LAKEFLD3	ITCM Lakefield	345	ITCM	15,041	15,234	192	16,469	16,736	267
635400	HIGHLND 3	Highland	345	MEC	10,112	10,283	171	12,330	12,588	258
631193	HUNTLEY3	Huntley	345	ITCM	7,823	8,019	196	10,432	10,725	293
631198	COLBY3	Colby	345	ITCM	7,974	8,039	65	9,386	9,484	98
636010	LEHIGH 3	Lehigh	345	MEC	9,802	9,910	109	12,385	12,541	156
636001	WEBSTER5	Webster	161	MEC	17,205	17,352	147	17,701	17,831	130
601034	NOBLES 3	Nobles Co	345	XCEL	5,670	5,687	16	8,543	8,587	43
601029	LKFLDXL3	Xcel Lakefield	345	XCEL	20,018	20,173	156	14,337	14,438	101
631041	LAKEFLD5	ITCM Lakefield	161	ITCM	17,185	17,286	101	18,941	19,081	140
635200	RAUN 3	Raun	345	MEC	27,244	27,345	101	25,743	25,839	95

Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J593 generating facility (112 Vestas V110 2.0 MW wind turbine generators) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

An ASPEN short circuit database including positive, negative and zero sequence parameters of the MDU system and its neighboring systems was provided by the MDU. This starting short circuit model is listed below:

- Short Circuit Model: "MDU 02012018.olr"

The J593 short circuit study model was developed as follows:

- Turned on generators which were nearby the POI but were off line in the starting short circuit model (Table 1-1).
- The J593 generating facility with its sequence parameters was added at the existing Tioga 230 kV line. J593 WTGs were aggregated and modeled as one induction generator with impedances set as follows:
 - o Positive sequence impedances (i.e. subtransient, transient and synchronous) (Z_1), and Negative sequence impedance (Z_2): 0.005 +j0.1669 pu on 235.76 MVA base with the current limit A set to 789,079 A (four times of rated current)
 - o Zero sequence impedance (Z_0) is set to 0+j9999 pu.
- The impedance and data for the J593 collection system was modeled as following:
 - o The positive and zero sequence impedance of the 34.5-0.69 kV GSU transformer impedance: 0.00799+ j0.07758 pu and 0.008+j0.07256 pu on 235.2 MVA base, respectively. Connection is Delta/Wye-ground.
 - o The 34.5 kV collection systems were modeled as an equivalent 34.5 kV line with the positive and zero sequence impedance at 0.00297+j0.00474 pu and 0.01517+j 0.00215 on 100 MVA base.
 - o The main transformer is modeled as one 230-34.5 kV transformer. Positive and zero sequence impedances: 0.00189+j0.07597 pu and 0.00189+j0.07597 pu on 144 MVA base. Connection is Wye-ground/Delta.
 - o Positive sequence impedance for the 230 kV Gentie: 0.00456+j0.02933 pu; Zero sequence impedance: 0.0274+j0.0835 pu.

Table 1-1: Generators Which Were Turn On

ASPEN Bus Name	ID	Starting Case Status	J593 Case Status
WATFORD-S 4.16kV	1	Offline	Online
CAPITOL G2 4.16kV	2	Offline	Online

ASPEN Bus Name	ID	Starting Case Status	J593 Case Status
LINTON GEN 3 13.8kV	1	Offline	Online
DICK DSU GEN 0.48kV	1	Offline	Online
GascWF singl 0.675kV	1	Offline	Online
LINTON GEN 2 13.8kV	1	Offline	Online
LINTON GEN 1 13.8kV	1	Offline	Online
St A-Hosp 12.5kV	1	Offline	Online
CAPITOL G4 4.16kV	1	Offline	Online
EDG-WF singl 0.675kV	1	Offline	Online
ST.JOE GEN 0.48kV	1	Offline	Online
Ellen WF gen 0.675kV	1	Offline	Online
SPROLE Gen 0.48kV	1	Offline	Online
GascWF gen 0.675kV	1	Offline	Online
TESORO G4 4.16kV	1	Offline	Online
BisWP Gen 4.16kV	1	Offline	Online
EII WF singl 0.675kV	1	Offline	Online
CAPITOL G3 4.16kV	2	Offline	Online
Bowdle WF G 0.675kV	1	Offline	Online
HESKET 3G 0.48kV	1	Offline	Online
CAPITOL G1 4.16kV	2	Offline	Online
RAY D 4.16kV	1	Offline	Online
ST.JOE GEN 0.48kV	2	Offline	Online
Bow WF singl 0.675kV	1	Offline	Online
MILCYCT9 13.8kV	1	Offline	Online
WILLCT 9 12.5kV	1	Offline	Online
COY MDU9 24.kV	1	Offline	Online
WILL DT9 12.5kV	1	Offline	Online
PARSHAL6 69.kV	1	Offline	Online

The following DPP queue projects are also added to the case:

- J302
- J457
- J503
- J599
- J607

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J593 project) and benchmark case (without the J593 project). The following fault simulation options were used in the short circuit analysis:

- The prefault voltage was set to “from a linear network solution”
- Current limited generator was set to “Enforce current limit A”

Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated on buses within the Study Project area. The short circuit results are summarized in Table 1.

The results of short circuit analysis showed that the 3PH fault current at the Tioga 230 kV bus to be 6,933 Amps with J593 (study case) and 5,609 Amps without J593 (benchmark case), and the SLG fault current at the Tioga 230 kV bus to be 7,086 Amps with J593 and 5,589 Amps without J593.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 6,933 A (increased by 1,324 A) and SLG fault current is 7,086 A (increased by 1,497 A) at the Tioga 230 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J593 generation project does not cause any Transmission Owner short circuit constraints.

Table 1: Three Phase (3PH) and Single-Line-to-Ground (SLG) Fault Currents with and without J593

ASPEN Bus Number	ASPEN Bus Name	ASPEN Base kV	LOCATION	AREA	ZONE	Benchmark Case without J593		Study Case with J593		Amps Change with J593 vs Benchmark		% Change with J593 vs Benchmark	
						3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault (A)	SLG Fault (A)	3PH Fault	SLG Fault
1552	TIOGA 4	230	TIOGA 4	6	1	5609.3	5589.0	6933.0	7085.7	1323.7	1496.7	23.6%	26.8%
0	NESET 4	230	Tioga	1	1	5600.0	5576.4	6918.4	7063.3	1318.4	1486.9	23.5%	26.7%
1132	TIOGA E5	115	TIOGA E5	6	1	7189.7	8158.8	7922.1	9043.1	732.4	884.3	10.2%	10.8%
0	TIOGA CAP E5	115	TIOGA CA	1	1	7180.2	8146.6	7910.5	9028.0	730.3	881.4	10.2%	10.8%
0	NESET	115	Tioga	1	1	7166.3	8128.5	7895.7	9008.4	729.4	879.9	10.2%	10.8%
1228	TIOGA W5	115	TIOGA W5	6	1	6385.6	6267.4	6910.6	6727.1	525.0	459.7	8.2%	7.3%
0	MW-N Tioga T	115	Tioga	1	1	6030.1	5882.4	6532.2	6319.8	502.1	437.4	8.3%	7.4%
0	WHEELock 4	230	WILL	1	1	5281.5	5088.0	5776.3	5463.2	494.8	375.2	9.4%	7.4%
0	PV1	115		1	1	5456.0	4968.5	5860.5	5272.1	404.5	303.6	7.4%	6.1%
0	PV2	115	Sherwood	1	1	5024.0	4325.4	5362.4	4550.0	338.4	224.6	6.7%	5.2%
0	BLAISDELL 4	230	TIOGA	1	1	3986.9	4005.2	4305.6	4242.6	318.7	237.4	8.0%	5.9%
1598	WILLIST4	230	WILLIST4	1	1	8479.4	8659.3	8775.9	8871.9	296.5	212.6	3.5%	2.5%
0	JUDSON	230		0	0	8528.4	8718.5	8800.7	8911.5	272.3	193.0	3.2%	2.2%
1553	BDV 4	230	BDV 4	32	1	5552.8	5605.2	5758.7	5752.0	205.9	146.8	3.7%	2.6%
0	BD PS-LTC	230	BD PS-LT	1	1	5851.1	5990.4	6040.5	6128.8	189.4	138.4	3.2%	2.3%
0	RAY 5	115	RAY	6	1	4730.9	3525.4	4907.5	3605.4	176.6	80.0	3.7%	2.3%
0	MW-PVALLEY5	115	Tioga	1	1	3441.8	2704.9	3590.8	2784.2	149.0	79.3	4.3%	2.9%
1510	LOGAN 4	230	LOGAN 4	1	1	4937.3	4520.7	5080.8	4599.5	143.5	78.8	2.9%	1.7%
0	JUDSON	345		0	0	6284.3	6202.6	6424.2	6295.0	139.9	92.4	2.2%	1.5%
1709	CHARLCR3	345	CHARLCR3	21	1	8875.5	9509.4	9012.5	9612.4	137.0	103.0	1.5%	1.1%
1230	WILLSTN5	115	WILLSTN5	6	1	8465.6	7577.9	8598.5	7650.3	132.9	72.4	1.6%	1.0%
0	WILLSTN CAP	115	WILLSTN	1	1	8452.4	7567.3	8584.9	7639.5	132.5	72.2	1.6%	1.0%
0	WILL CAP-A	115	WILL CAP	1	1	8439.2	7556.8	8571.3	7628.8	132.1	72.0	1.6%	1.0%
0	WILL CAP-B	115	WILL CAP	1	1	8439.2	7556.8	8571.3	7628.8	132.1	72.0	1.6%	1.0%
1596	CHARCRK4	230	CHARCRK4	1	1	10033.4	11202.6	10157.6	11303.4	124.2	100.8	1.2%	0.9%
1284	LOGAN 5	115	LOGAN 5	1	1	7757.3	8031.9	7879.5	8116.8	122.2	84.9	1.6%	1.1%
0	WHEELock 5	115	WILL	1	1	4004.5	4804.2	4125.5	4928.9	121.0	124.7	3.0%	2.6%
1554	BDX 4	230	BDX 4	32	1	11529.6	13945.0	11647.9	14060.5	118.3	115.5	1.0%	0.8%
0	BLAIS CAP	115	TIOGA	1	1	3561.5	4552.4	3675.3	4677.8	113.8	125.4	3.2%	2.8%
0	BLAISDELL 5	115	TIOGA	1	1	3561.8	4553.8	3675.6	4679.3	113.8	125.5	3.2%	2.8%
1262	STANLEY5	115	STANLEY5	6	1	3396.7	2384.7	3504.0	2427.6	107.3	42.9	3.2%	1.8%
1540	LELANDO4	230	LELANDO4	21	1	21254.9	22873.4	21362.2	22953.4	107.3	80.0	0.5%	0.3%



Evaluation of Fault Currents and Interrupting Capabilities for Generation Interconnection Project J594

1-12-2018

Summary

Generator Interconnection Project J594 is a 300 MW wind powered generating facility with proposed interconnection to the 161 kV bus at ITC Midwest's Jackson North Substation. Three phase and single line to ground fault current values were evaluated before and after interconnection of the generating facility using Computer Aided Protection Engineering (CAPE) software by Electrocon International. The base CAPE model maintained and used by ITC Midwest's system protection group was updated to include planned BES future system topology representative of a 2019 ITCM system was used in the analysis.

Generating Facility Assumptions

The generating facility was modeled based on available information, and included:

- One 300 MW equivalent generator
- One 34.5kV to 690V equivalent transformer
- One 34.5 kV collector system line
- Two 34.5kV to 161 kV step up transformers
- 161 kV interconnection line

Figure 1 below shows a one-line diagram of the modeled generating facility and nearby system.

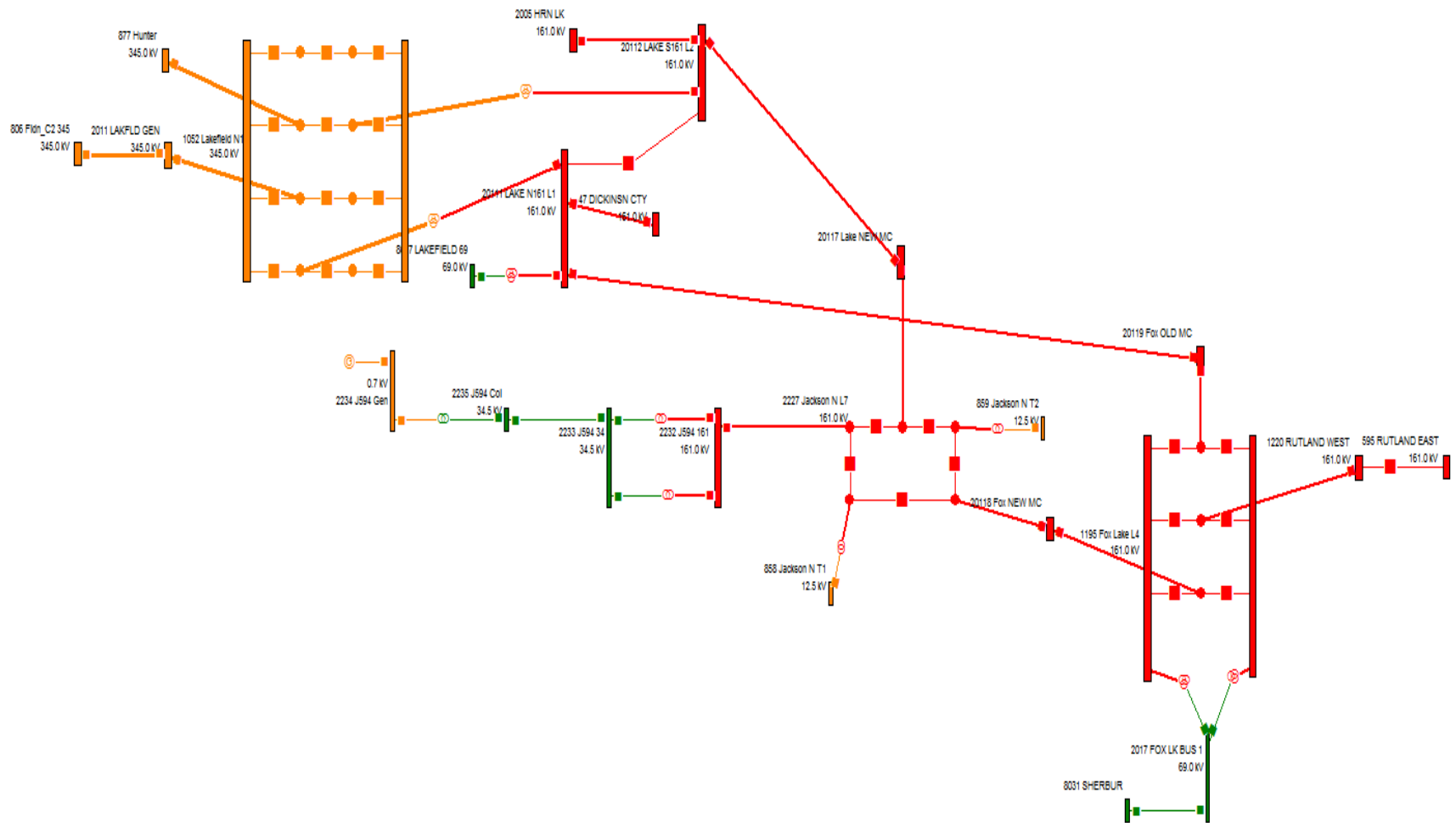


Figure1

Fault Currents

Table 1 below has the three phase and single line to ground fault currents before and after the addition of Project J594.

Substation	Owner	kV	SLG-pre (A)	3Ph-pre (A)	SLG-post (A)	3Ph-post (A)	Contribution SLG (A)	Contribution 3Ph (A)
LAKEFIELD	ITCM	345	16387.6	18976	16845.9	19553.6	458.3	577.6
LAKEFIELD	ITCM	69	6248.7	5608.2	6289.4	5651.5	40.7	43.3
RUTLAND	SMMPA	161	7532.5	8648.2	7682.2	8903.2	149.7	255
FOX LAKE	ITCM	69	9402.4	8491.2	9571	8665.2	168.6	174
RUTLAND	ITCM	161	7532.5	8648.2	7682.2	8903.2	149.7	255
DICKINSON COUNTY	ITCM	161	7720.1	9052.7	7771.8	9151.8	51.7	99.1
HERON LAKE	ITCM	161	5826.7	7115.7	5896.6	7254.3	69.9	138.6
LAKEFIELD	ITCM	161	19128.9	20945.6	20518.9	22736.7	1390	1791.1
FOX LAKE	ITCM	161	9617.9	10667	10312.3	11592.3	694.4	925.3
LAKEFIELD	ITCM	161	19128.9	20945.6	20518.9	22736.7	1390	1791.1
JACKSON NORTH	MRES	12.5	6947.6	6821.5	7000.8	6903.4	53.2	81.9
LAKEFIELD	ITCM	161	15207.4	17886.6	16555.9	19627	1348.5	1740.4
JACKSON NORTH	ITCM	161	6932	9602	9976.4	12278.3	3044.4	2676.3
LAKEFIELD GEN	XEL	345	13914.1	17917.7	14014.5	18084.8	100.4	167.1
FIELDON	XEL	345	11430.8	20379.9	11447.2	20439.4	16.4	59.5
SHERBURN	SCREC	69	7688.8	7756.6	7797.38	7898.43	108.58	141.83
HUNTER WF	EDF	345	16266.5	18851.7	16717.1	19421.1	450.6	569.4

Table 1



Constraints

No ITCM constraints.

Conclusion

Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J594. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.



Short Circuit Analysis Report

100 MW Windfarm in Stevens County, Minnesota

MISO # J596
Prepared for MISO

June 21, 2018
Great River Energy

Contact:
Krystal DeBruine, Transmission Project Manager
Great River Energy

Contents

Introduction	3
Background	3
Methodology.....	3
Cases/Conditions	4
Results.....	7

Introduction

This report presents the results of the short circuit analysis for J596. The objective of the analysis is to determine the present fault current levels near the POI for J596 and determine the future fault current levels after the wind turbines become commercially operational. After the fault current levels are found, the rating of Great River Energy's (GRE's) breakers near the POI will be analyzed to ensure they are capable of interrupting any increase in fault current due to the contributions of J596. The report also includes the fault current values at non-GRE owned substations. However, it does not include an assessment of the ratings of the equipment in those substations.

Background

The first part of the study is to calculate the fault currents before and after the addition of J596's 100 MW of wind generation. Short circuit calculations are based on the impedances of the transmission lines, transformers, generators, and on the pre-fault voltages. This wind generation will be connected to the system via a tap bus on GRE's existing AG-MB transmission line between Morris and Walden Substations.

The second part of this study is to compare the fault currents to the interrupting ratings of GRE equipment up to five buses away from the point of interconnection. Fault currents will also be calculated for nearby GRE and foreign substations. These fault currents will be compared to interrupting ratings of GRE owned equipment but foreign owned equipment will not be analyzed.

This study was completed under the assumption of an in-service date of December 2018. Nearby interconnections J493 and J526 were not yet modeled as data was not available at the time of the study. However, after considering low fault currents in the area and large margins between interrupting capacity and calculated duty, it was determined that a short circuit study for J596 could proceed.

Methodology

Aspen *One-liner* short circuit analysis software and the associated *Breaker Rating Module* are used to determine the symmetrical fault currents and circuit breaker interruption capabilities. Great River Energy has an Aspen model already built that includes 115kV Morris-Walden line and surrounding substations.

The *Breaker Rating Module* is a tool designed to check the rating of circuit breakers against the short circuit currents they need to interrupt. The breaker-rating logic simulates relevant faults to find the maximum short-circuit current that flows through the breaker. The momentary and breaking currents of breakers are then calculated using procedures recommended by ANSI/IEEE. Finally, the program compares these currents to the rated capabilities of the breakers and reports the findings. The breaker-rating logic uses a breaker connection model that is designed to work with traditional symmetrical-sequence networks where circuit breakers connections are not modeled explicitly. The connection model associates each breaker with the bus, branches and generators that it protects.

The generator, transformer, and the transmission line impedances for interconnection equipment were received from the customer. If the transmission line impedance was not available from the customer, it was assumed to be zero for worst case fault currents. All the buses are assumed to have a per unit pre-fault voltage equal to 1. GRE models wind turbines based on a method recommended by Aspen. The wind turbines used in this interconnection are type-3 generators. Wind generators with doubly-fed machines are also known as type-3 generators. Type-3 generators can be simulated in Aspen *One-liner* using the conventional generators model without current limits.

Short circuit studies with wind turbines modeled as conventional generators will yield a conservative estimate of the peak short-circuit current. Type-3 generators are ungrounded and contribute no zero-sequence current during a fault. When modeling type-3 generators in Aspen, the zero-sequence impedance is set to 9999 pu. The positive sequence impedance is set equal to the sub transient X_d'' impedance of the machine and the negative sequence impedance is set to be the same as the positive sequence impedance unless the negative-sequence impedance is known. All the current limits for the generator are set to zero.

Cases/Conditions

TABLE 1

Faults		Before J596			After J596		
Location	Fault Type	Short Circuit Current (A)	ANSI X/R	MVA	Short Circuit Current (A)	ANSI X/R	MVA
Morris-Walden Tap 115kV Bus	1LG	4,019	4.9	815.0	5,626	5.5	1122.7
	3LG	5,239	4.7	1,062.5	6,413	6.2	1279.8
Morris 230kV Bus	1LG	3,557	10.2	1,419.5	3,703	10.4	1467.8
	3LG	4,546	6.7	1,814.1	4,662	7.2	1847.7
Morris 115kV Bus	1LG	6,408	7.7	1,291.0	7,319	8	1454.7
	3LG	7,011	5.7	1,412.5	7,889	6.2	1567.9
Morris 41.6kV Bus	1LG	3,789	12.9	280.3	3,795	13.5	277.5
	3LG	3,594	10.6	265.8	3,622	11.1	264.8
Walden 115kV Bus	1LG	3,147	4.4	642.7	3,769	4.5	758.2
	3LG	4,380	4.4	894.6	4,977	5	1001.3
Walden 41.6kV Bus	1LG	2,555	14.4	211.9	2,565	15.5	209.6
	3LG	2,334	13.2	193.6	2,361	14.5	192.9
Benson 115kV Bus	1LG	2,762	4.8	559.7	2,895	4.8	580.2
	3LG	3,807	4.1	771.6	3,983	4.2	798.3
Benson 69kV Bus	1LG	2,956	8.0	364.1	3,014	8.2	369.2
	3LG	3,159	6.8	389.2	3,205	7.2	392.5
Benson 41.6kV Bus	1LG	3,139	12.3	238.4	3,138	12.8	235.6
	3LG	2,848	10.4	216.3	2,859	10.8	214.6

Table 2

Before J596						
Location	Breaker	Duty (% of capacity)	Duty (A)	Breaker Capacity (A)	ANSI X/R [1]	Exceeds Breaker capacity?
Walden 115 kV	1512	14.5	4343	30000	4.4	No
Walden 41.6 kV	415	14.1	2393.5	17000	14.4	No
Graceville 115 kV	1515	11.7	3688.7	31500	2.8	No
	1535	17.9	3575.8	20000	2.7	No
Graceville 41.6 kV	555	19.4	3289.7	17000	8.6	No
	565	19.4	3289.6	17000	8.6	No
	575	14.8	2521.1	17000	10.2	No
	595	14.8	2521	17000	10.2	No
Benson 115 kV	1514	15.2	3806.9	25000	4.1	No
	1515	9.2	3671.1	40000	4.1	No
	1525	9.5	3806.9	40000	4.1	No
	1535	5.2	2075.2	40000	3.9	No
	1555	9.2	3671.1	40000	4.1	No
	1565	9.5	3806.9	40000	4.1	No
Benson 69 kV	735	6	2417.6	40000	8.7	No
Benson 41.6 kV	425	18.5	3137	17000	12.3	No
	745	7.8	3137	40000	12.3	
	765	7.8	3138.4	40000	12.3	
	785	7.8	3138.6	40000	12.3	
Willmar 230 kV	13RB2	10.6	4234.4	40000	7.6	No
Willmar 69 kV	13NB2	24.8	9904.7	40000	14.3	No
	13NB3	24.8	9904.7	40000	14.3	No
	13NB5	24.8	9904.7	40000	14.3	No
Panther 230 kV	127RB15	10.1	4049.9	40000	7.1	No
	127RB16	8.8	3530.5	40000	7.4	No

Table 3

After J596						
Location	Breaker	Duty (% of capacity)	Duty (A)	Breaker Capacity (A)	ANSI X/R [1]	Exceeds Breaker capacity?
Walden 115 kV	1512	16.5	4940.6	30000	4.9	No
Walden 41.6 kV	415	14.1	2404.3	17000	15.5	No
Graceville 115 kV	1515	12	3784.5	31500	2.8	No
	1535	18.3	3659.5	20000	2.7	No
Graceville 41.6 kV	555	19.3	3276.7	17000	8.6	No
	565	19.3	3276.6	17000	8.6	No
	575	14.7	2505.9	17000	10.3	No
	595	14.7	2505.8	17000	10.3	No
Benson 115 kV	1514	15.9	3983.2	25000	4.2	No
	1515	9.6	3849.4	40000	4.2	No
	1525	10	3983.2	40000	4.2	No
	1535	5.2	2071.8	40000	3.9	No
	1555	9.6	3849.4	40000	4.2	No
	1565	10	3983.2	40000	4.2	No
Benson 69 kV	735	6.1	2447.3	40000	9.1	No
Benson 41.6 kV	425	18.4	3135.6	17000	12.8	No
	745	7.8	3135.6	40000	12.8	No
	765	7.8	3137	40000	12.8	No
	785	7.8	3137.2	40000	12.8	No
Willmar 230 kV	13RB2	10.6	4235.8	40000	7.6	No
Willmar 69 kV	13NB2	24.7	9884.5	40000	14.3	No
	13NB3	24.7	9884.5	40000	14.3	No
	13NB5	24.7	9884.5	40000	14.3	No
Panther 230 kV	127RB15	10.1	4048	40000	7.1	No
	127RB16	8.8	3528	40000	7.3	No

¹ The ANSI X/R ratio of the network at breaker location directly determines the rate of ac and dc decay of the short circuit current that flows through the breaker. Therefore calculating X/R is an important step in determining breaker interrupting and momentary duty. ANSI/IEEE standard C37-010-1999 recommends applying a multiplying factor to fault currents calculated using the symmetrical sequence network model. The multiplying factor is calculated from an equation derived from X/R ratio curves in ANSI/IEEE standard C37-010-1999. Duty is equal to the worst case short circuit current through breaker times the multiplying factor.

Table 4

Min Fault Current (Tap Bus-Morris 115kV Open, All Local Generation Offline)

Location	Fault Type	Short Circuit Current (A)	ANSI X/R	MVA
Morris-Walden Tap 115kV Bus	1LG	1,310	3.90	261.1
	3LG	1,075	3.82	214.3

Results

Fault currents were calculated before and after the addition of J596's 100 MW of wind generation. Table 1 summarizes the fault currents at the point of interconnection and nearby substations.

Tables 2 and 3 compare the fault currents to the interrupting ratings of equipment at GRE's substations within five buses of the interconnection. Table 2 shows the data before the addition of J596 and Table 3 shows the data after the addition of J596.

The data in Table 3 shows that none of the circuit breaker interrupting capabilities at these substations will be exceeded after the addition of J596. Equipment at non-GRE owned substations was not evaluated.

Table 4 summarizes the minimum fault currents at the POI calculated for an N-1 contingency and local generation, including Alberta Wind offline. Based on the results shown in Table 4, the short-circuit ratio at the POI for Project J596 (100 MW wind) could be as low as 2.14.

Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J599 generating facility (100 Vestas V110 2.0 MW wind turbine generators) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

An ASPEN short circuit database including positive, negative and zero sequence parameters of the MDU system and its neighboring systems was provided by the MDU. This starting short circuit model is listed below:

- Short Circuit Model: "MDU 02012018.olr"

The J599 short circuit study model was developed as follows:

- Turned on generators which were nearby the POI but were off line in the starting short circuit model (Table 1-1).
- The J599 generating facility with its sequence parameters was added at the existing Glenham 230 kV line. J599 WTGs were aggregated and modeled as one induction generator with impedances set as follows:
 - o Positive sequence impedances (i.e. subtransient, transient and synchronous) (Z_1), and Negative sequence impedance (Z_2): 0.005 +j0.1669 pu on 209.3 MVA base with the current limit A set to 700,518 A (four times of rated current)
 - o Zero sequence impedance (Z_0) is set to 0+j9999 pu.
- The impedance and data for the J599 collection system was modeled as following:
 - o The positive and zero sequence impedance of the 34.5-0.69 kV GSU transformer impedance: 0.008+ j0.077589 pu and 0.008+j0.07256 pu on 210 MVA base, respectively. Connection is Delta/Wye-ground.
 - o The 34.5 kV collection systems were modeled as an equivalent 34.5 kV line with the positive and zero sequence impedance at 0.0031+j0.00495 pu on 100 MVA base.
 - o The main transformers is modeled as two 230-34.5 kV transformer. Positive and zero sequence impedances: 0.00223+j0.07596 pu on 72 MVA base. Connection is Wye-ground/Delta.
 - o Positive sequence impedance for the 230 kV Gentie: 0.00404+j0.02598 pu; Zero sequence impedance: 0.00404+j0.02598 pu.

Table 1-1: Generators Which Were Turn On

ASPEN Bus Name	ID	Starting Case Status	J599 Case Status
WATFORD-S 4.16kV	1	Offline	Online
CAPITOL G2 4.16kV	2	Offline	Online
LINTON GEN 3 13.8kV	1	Offline	Online

ASPEN Bus Name	ID	Starting Case Status	J599 Case Status
DICK DSU GEN 0.48kV	1	Offline	Online
GascWF singl 0.675kV	1	Offline	Online
LINTON GEN 2 13.8kV	1	Offline	Online
LINTON GEN 1 13.8kV	1	Offline	Online
St A-Hosp 12.5kV	1	Offline	Online
CAPITOL G4 4.16kV	1	Offline	Online
EDG-WF singl 0.675kV	1	Offline	Online
ST.JOE GEN 0.48kV	1	Offline	Online
Ellen WF gen 0.675kV	1	Offline	Online
SPROLE Gen 0.48kV	1	Offline	Online
GascWF gen 0.675kV	1	Offline	Online
TESORO G4 4.16kV	1	Offline	Online
BisWP Gen 4.16kV	1	Offline	Online
EII WF singl 0.675kV	1	Offline	Online
CAPITOL G3 4.16kV	2	Offline	Online
Bowdle WF G 0.675kV	1	Offline	Online
HESKET 3G 0.48kV	1	Offline	Online
CAPITOL G1 4.16kV	2	Offline	Online
RAY D 4.16kV	1	Offline	Online
ST.JOE GEN 0.48kV	2	Offline	Online
Bow WF singl 0.675kV	1	Offline	Online
MILCYCT9 13.8kV	1	Offline	Online
WILLCT 9 12.5kV	1	Offline	Online
COY MDU9 24.kV	1	Offline	Online
WILL DT9 12.5kV	1	Offline	Online
PARSHAL6 69.kV	1	Offline	Online

The following DPP queue projects are also added to the case:

- J302
- J457
- J503
- J593
- J607

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J599 project) and benchmark case (without the J599 project). The following fault simulation options were used in the short circuit analysis:

- The prefault voltage was set to “from a linear network solution”
- Current limited generator was set to “Enforce current limit A”

Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated on buses within the Study Project area. The short circuit results are summarized in Table 1.

The results of short circuit analysis showed that the 3PH fault current at the Glenham 230 kV bus to be 5,099 Amps with J599 (study case) and 3,843 Amps without J599 (benchmark case), and the SLG fault current at the Glenham 230 kV bus to be 5,751 Amps with J599 and 3,984 Amps without J599.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 5,099 A (increased by 1,256 A) and SLG fault current is 5,751 A (increased by 1,767 A) at the Glenham 230 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J599 generation project does not cause any Transmission Owner short circuit constraints.

Table 1: Three Phase (3PH) and Single-Line-to-Ground (SLG) Fault Currents with and without J599

ASPEN Bus Number	ASPEN Bus Name	ASPEN Base kV	LOCATION	AREA	ZONE	Benchmark Case without J599		Study Case with J599		Amps Change with J599 vs Benchmark		% Change with J599 vs Benchmark	
						3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault (A)	SLG Fault (A)	3PH Fault	SLG Fault
1526	GLENHAM4	230	GLENHAM4	7	1	3842.9	3984.2	5098.6	5751.1	1255.7	1766.9	32.7%	44.3%
0	GLENHAM-4	230	GLENHAM-	1	1	3838.1	3980.2	5089.8	5741.2	1251.7	1761.0	32.6%	44.2%
652499	Camp County	230		1	1	3873.7	4075.9	4706.5	4948.5	832.8	872.6	21.5%	21.4%
659408	Camp-Wind	230		1	1	3869.2	4072.5	4699.5	4942.7	830.3	870.2	21.5%	21.4%
1546	WHITLKT4	230	WHITLKT4	7	1	4557.3	4034.7	5066.8	4406.5	509.5	371.8	11.2%	9.2%
0	GLENHAM-5	115	GLENHAM-	1	1	4057.4	4848.7	4498.2	5460.5	440.8	611.8	10.9%	12.6%
1212	GLENHAM5	115	GLENHAM5	7	1	4054.5	4844.9	4494.7	5455.5	440.2	610.6	10.9%	12.6%
1515	SULLYB 4	230	SULLYB 4	31	1	6281.0	5561.6	6617.9	5774.7	336.9	213.1	5.4%	3.8%
1543	OAHE 4	230	OAHE 4	31	1	12773.8	14218.5	12998.9	14406.2	225.1	187.7	1.8%	1.3%
0	BIS WAPA E	230	BIS WAPA	31	1	14923.3	13473.2	15101.1	13557.7	177.8	84.5	1.2%	0.6%
1501	BIS WAPA W	230	BIS WAPA	31	1	14959.8	13505.6	15136.9	13589.8	177.1	84.2	1.2%	0.6%
1220	MOBRIDG5	115	MOBRIDG5	7	1	2626.7	2536.6	2792.8	2662.8	166.1	126.2	6.3%	5.0%
0	BOWDLE WF TP	115		7	1	2379.6	2282.8	2514.4	2380.9	134.8	98.1	5.7%	4.3%
1823	F THOM1T	345	F THOM1T	1	1	14441.4	11353.1	14550.3	11395.5	108.9	42.4	0.8%	0.4%
1841	F THOM2T	345	F THOM2T	1	1	14441.4	11353.1	14550.3	11395.5	108.9	42.4	0.8%	0.4%

Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J607 generating facility (75 Vestas V110 2.0 MW wind turbine generators) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

An ASPEN short circuit database including positive, negative and zero sequence parameters of the MDU system and its neighboring systems was provided by the MDU. This starting short circuit model is listed below:

- Short Circuit Model: "MDU 02012018.olr"

The J607 short circuit study model was developed as follows:

- Turned on generators which were nearby the POI but were off line in the starting short circuit model (Table 1-1).
- The J607 generating facility with its sequence parameters was added at the tap bus on Heskett to Wishek 230 kV line. J607 WTGs were aggregated and modeled as one induction generator with impedances set as follows:
 - o Positive sequence impedances (i.e. subtransient, transient and synchronous) (Z_1), and Negative sequence impedance (Z_2): 0.005 +j0.1669 pu on 157.9 MVA base with the current limit A set to 528,485 A (four times of rated current)
 - o Zero sequence impedance (Z_0) is set to 0+j9999 pu.
- The impedance and data for the J607 collection system was modeled as following:
 - o The positive and zero sequence impedance of the 34.5-0.69 kV GSU transformer impedance: 0.008+j0.077589 pu and 0.008+j0.07256 pu on 157.5 MVA base, respectively. Connection is Delta/Wye-ground.
 - o The 34.5 kV collection systems were modeled as an equivalent 34.5 kV line with the positive and zero sequence impedance at 0.00313+j0.00499 pu and 0.01582+j0.00224 pu on 100 MVA base, respectively.
 - o The main transformer is modeled as one 230-34.5 kV transformer. Positive and zero sequence impedances: 0.00205+j0.07597 pu and 0.00175+j0.06497 pu on 96 MVA base. Connection is Wye-ground/Delta.
 - o Positive sequence impedance for the 230 kV Gentie: 0.00196+j0.01263 pu; Zero sequence impedance: 0.0118+j0.03596 pu.

Table 1-1: Synchronous Generators Which Were Turn On

ASPEN Bus Name	ID	Starting Case Status	J607 Case Status
WATFORD-S 4.16kV	1	Offline	Online
CAPITOL G2 4.16kV	2	Offline	Online

ASPEN Bus Name	ID	Starting Case Status	J607 Case Status
LINTON GEN 3 13.8kV	1	Offline	Online
DICK DSU GEN 0.48kV	1	Offline	Online
GascWF singl 0.675kV	1	Offline	Online
LINTON GEN 2 13.8kV	1	Offline	Online
LINTON GEN 1 13.8kV	1	Offline	Online
St A-Hosp 12.5kV	1	Offline	Online
CAPITOL G4 4.16kV	1	Offline	Online
EDG-WF singl 0.675kV	1	Offline	Online
ST.JOE GEN 0.48kV	1	Offline	Online
Ellen WF gen 0.675kV	1	Offline	Online
SPROLE Gen 0.48kV	1	Offline	Online
GascWF gen 0.675kV	1	Offline	Online
TESORO G4 4.16kV	1	Offline	Online
BisWP Gen 4.16kV	1	Offline	Online
EII WF singl 0.675kV	1	Offline	Online
CAPITOL G3 4.16kV	2	Offline	Online
Bowdle WF G 0.675kV	1	Offline	Online
HESKET 3G 0.48kV	1	Offline	Online
CAPITOL G1 4.16kV	2	Offline	Online
RAY D 4.16kV	1	Offline	Online
ST.JOE GEN 0.48kV	2	Offline	Online
Bow WF singl 0.675kV	1	Offline	Online
MILCYCT9 13.8kV	1	Offline	Online
WILLCT 9 12.5kV	1	Offline	Online
COY MDU9 24.kV	1	Offline	Online
WILL DT9 12.5kV	1	Offline	Online
PARSHAL6 69.kV	1	Offline	Online

The following DPP queue projects are also added to the case:

- J302
- J457
- J503
- J593
- J599

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J607 project) and benchmark case (without the J607 project). The following fault simulation options were used in the short circuit analysis:

- The prefault voltage was set to “from a linear network solution”
- Current limited generator was set to “Enforce current limit A”

Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated on buses within the Study Project area. The short circuit results are summarized in Table 1.

The results of short circuit analysis showed that the 3PH fault current at the J607 POI 230 kV bus to be 6,543 Amps with J607 (study case) and 5,558 Amps without J607 (benchmark case), and the SLG fault current at the J607 POI 230 kV bus to be 6,114 Amps with J607 and 4,318 Amps without J607.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 6,543 A (increased by 985 A) and SLG fault current is 6,114 A (increased by 1,796 A) at the J607 POI 230 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J607 generation project does not cause any Transmission Owner short circuit constraints.

Table 1: Three Phase (3PH) and Single-Line-to-Ground (SLG) Fault Currents with and without J607

ASPEN Bus Number	ASPEN Bus Name	ASPEN Base kV	LOCATION	AREA	ZONE	Benchmark Case without J607		Study Case with J607		Amps Change with J607 vs Benchmark		% Change with J607 vs Benchmark	
						3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault (A)	SLG Fault (A)	3PH Fault	SLG Fault
0	J607 POI	230		2	1	5557.5	4318.0	6542.8	6114.0	985.3	1796.0	17.7%	41.6%
1551	WISHEK 4	230	WISHEK 4	7	1	5621.6	4578.5	6491.2	6028.0	869.6	1449.5	15.5%	31.7%
0	TL91St112	230		1	1	5944.3	5206.9	6431.4	5767.6	487.1	560.7	8.2%	10.8%
0	J302&503 POI	230		2	1	6014.5	3798.0	6491.9	4376.2	477.4	578.2	7.9%	15.2%
0	MERR	230		7	1	6936.0	7505.3	7291.8	7865.3	355.8	360.0	5.1%	4.8%
0	MERRICOURT	230		7	1	6933.9	7502.8	7289.4	7862.6	355.5	359.8	5.1%	4.8%
0	J457	230		7	1	6917.4	7487.7	7270.8	7845.4	353.4	357.7	5.1%	4.8%
0	J302&503	230		2	1	5314.2	3429.2	5643.7	3881.4	329.5	452.2	6.2%	13.2%
0	Tatanka TP	230	Ellendal	7	1	6854.3	7507.3	7151.6	7785.2	297.3	277.9	4.3%	3.7%
0	TL91St238	230		1	1	6610.1	7259.8	6868.7	7491.9	258.6	232.1	3.9%	3.2%
0	Rough Rdr TP	230	Ellendal	7	1	6528.8	7282.2	6771.6	7501.6	242.8	219.4	3.7%	3.0%
1299	WISHEK 5	115	WISHEK 5	7	1	4916.0	5296.9	5115.5	5663.9	199.5	367.0	4.1%	6.9%
0	WISHEK-5	115	WISHEK 5	7	1	4913.3	5293.2	5112.5	5659.6	199.2	366.4	4.1%	6.9%
1505	ELLENMD4	230	ELLENMD4	7	1	6236.4	6242.0	6422.2	6376.8	185.8	134.8	3.0%	2.2%
1527	HESKETT4	230	HESKETT4	2	1	15780.8	14418.3	15900.1	14524.6	119.3	106.3	0.8%	0.7%
0	MANDAN	230	Mandan	1	1	16564.8	15349.3	16681.4	15450.5	116.6	101.2	0.7%	0.7%

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J611 Short Circuit Study Performed by MEC

The scope of this DPP short circuit facilities study is a review of the available fault current at the proposed 161 kV interconnection substation for MISO generation queue request J611, a proposed 110 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J611 was assumed to interconnect off the Clarinda-Maryville 161 kV line. Maryville Substation and additional buses in Missouri are not owned by MidAmerican and are not evaluated here. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in an ongoing DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 161 kV interconnection substation bus to be 5,338 Amps without the Interconnection Customer included and 6,207 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, the preliminary generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation as well as for system contingencies.

As shown in the table, the changes in fault current at buses more than a couple buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J611

Bus Number	Bus Name	English Name	Base kV	Area Num	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
						Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with with new wind farm	3PH Difference w/ wind farm vs Base
86111	J611 POI	J611 POI	161	635	MEC	4,107	5,040	933	5,338	6,207	869
86112	J611	J611 IC Sub	161	635	IC	3,154	4,268	1,115	4,381	5,289	909
635034	CLARINDA 5	Clarinda	161	635	MEC	5,678	6,046	368	6,144	6,647	503
541251	MARYVLE5	Maryville (GMO)	161	540	GMO	4,642	4,945	303	6,171	6,585	414
635037	BROOKS 5	Brooks	161	635	MEC	4,236	4,300	64	5,321	5,478	157
635032	HASTING5	Hastings	161	635	MEC	5,898	5,954	56	6,963	7,091	128
635035	CLARINDA 8	Clarinda	69	635	MEC	9,565	9,809	244	7,707	7,960	253
635056	ADAMS 5	Adams County	161	635	MEC	4,948	4,999	51	5,870	5,999	129
635031	BUNGE 5	Bunge	161	635	MEC	19,104	19,129	25	22,761	22,838	77
635033	HASTING8	Hastings	69	635	MEC	9,510	9,561	51	9,177	9,272	95
635065	REDOAK 8	Red Oak	69	635	MEC	2,420	2,427	7	3,736	3,766	30
635050	SHENAND8	Shenandoah	69	635	MEC	2,502	2,509	7	3,873	3,901	28
629170	VILISCAJCT8	Villisca Jct	69	627	CIPCO	2,116	2,124	8	2,001	2,018	17
652560	CRESTON5	Creston	161	652	WAPA	9,378	9,431	54	9,825	9,949	124



Short Circuit Study for J614

66 MW Wind Generation Interconnection

Rice Substation

Howard County, Iowa

MISO# J614

March 9, 2017

Prepared by:
Dairyland Power Cooperative

1. Overview of the Project

J614 is a proposed 66 MW wind generation interconnection to Southern Minnesota Municipal Power's (SMP) Rice substation. A circuit position will need to be added at Rice Substation to accommodate this proposed interconnection. In addition to the circuit position being required, the additional generation may require a capacity increase to the 161/69 kV transformer.

2. Short-Circuit Fault Study

The purpose of the short circuit study is to review the short circuit withstand capabilities of the equipment at Rice Substation. The customer requested two separate scenarios be studied. The first scenario was to include the interconnection at Rice Substation with the existing 161/69 kV transformer, no increase in transformer capacity. The second scenario was to replace the existing 161/69 kV transformer with a larger capacity (200 MVA) transformer. The customer intends to install wind machines per documentation received by Dairyland Power Cooperative (DPC) on January 23, 2017. DPC used the information contained in file "GI_J614_Attachment A Revised 20161011" titled "Large Generating Facility Data" supplied by SMP. A summary of the short circuit impacts to the Rice Substation is attached.

In summary, the short circuit study concludes the fault current rating of the existing Rice Substation equipment has the capabilities to withstand the additional generation. Due to the increased short circuit current, it is recommended that a protection coordination review be performed on all 161 kV and 69 kV terminals in the Rice Substation and on any other affected stations remote to Rice Substation. The scope and cost associated with this protection coordination review is captured in a facility study being performed by others.

Short Circuit Study - Rice Substation

Scenario 1: Rice Substation with Existing Transformer

Bkr Owner	Bus Name	BKR 1 Name	Remote End	Line	Base kV	Fault on Line (kA)			Fault on Bus (kA)			Int. Capability (kA)	
						SLG	TPH	DLG	SLG	TPH	DLG	BKR	Over Rating?
SMMPA	Rice	6670	Saratoga Tap	Q-30	161	1.9	1.6	1.8	4.5	5.0	4.7	40	No
SMMPA	Rice	6680	JADE AVE	SMMPA	161	5.4	5.8	5.5	1.3	1.0	1.3	40	No
SMMPA	Rice	0507	Rice 69	AT1	161	5.7	5.7	5.6	1.0	1.0	1.0	40	No
SMMPA	Rice	050	Rice 161	AT1	69	2.0	3.6	3.1	5.0	4.4	4.7	31.5	No
SMMPA	Rice	051	Saratoga Tap	N-81	69	6.4	6.0	6.0	1.0	1.9	1.6	31.5	No
SMMPA	Rice	052	Riceville	SMMPA	69	7.2	7.0	6.8	0.5	0.8	0.7	31.5	No
SMMPA	Rice	053	Cresco IPW	SMMPA	69	6.9	6.6	6.5	0.6	1.1	1.0	31.5	No

Scenario 2: Rice Substation with Increased Capacity Transformer (200 MVA)

Bkr Owner	Bus Name	BKR 1 Name	Remote End	Line	Base kV	Fault on Line (kA)			Fault on Bus (kA)			Int. Capability (kA)	
						SLG	TPH	DLG	SLG	TPH	DLG	BKR	Over Rating?
SMMPA	Rice	6670	Saratoga Tap	Q-30	161	2.4	1.9	2.2	4.5	5.0	4.7	40	No
SMMPA	Rice	6680	JADE AVE	SMMPA	161	6.1	6.0	5.9	1.3	1.0	1.3	40	No
SMMPA	Rice	0507	Rice 69	AT1	161	5.7	5.7	5.6	1.6	1.3	1.4	40	No
SMMPA	Rice	050	Rice 161	AT1	69	2.0	3.6	3.1	9.2	7.9	8.8	31.5	No
SMMPA	Rice	051	Saratoga Tap	N-81	69	10.6	9.4	10.0	1.0	1.9	1.6	31.5	No
SMMPA	Rice	052	Riceville	SMMPA	69	11.5	10.4	10.8	0.5	0.8	0.7	31.5	No
SMMPA	Rice	053	Cresco IPW	SMMPA	69	11.2	10.0	10.5	0.6	1.1	1.0	31.5	No

Values current as of 1-25-17

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J615 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 161 kV interconnection substation for MISO generation queue request J615, a proposed 70 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J615 was assumed to interconnect at a new interconnection substation off the Electriform-Shaulis Rd 161 kV line. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in a DPP study cycle of the MISO generation interconnection process, where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 161 kV interconnection substation bus to be 11,802 Amps without the Interconnection Customer included and 12,568 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results, especially the 34.5 kV results that should be calculated by the Interconnection Customer. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican breakers become overdutied as a result of the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J615

Bus Number	Bus Name	English Name	Base kV	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
					Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
		J615 POI Sub	161	MEC	9,996	11,198	1,203	11,802	12,568	766
		J615 Collector Bus	34.5		4,484	10,114	5,630	5,699	13,283	7,584
636220	EL FARM5	Electrifarm	161	MEC	12,693	13,433	740	13,171	13,761	590
636215	SHAULISRD5	Shaulis Rd	161	MEC	9,434	10,254	820	11,612	12,205	593
636208	DR ENG 5	Deere Engine	161	MEC	12,774	13,250	476	13,966	14,460	494
636222	ELFARMG1	Electrifarm Unit 1	13.2	MEC	10,179	10,184	5	30,960	31,039	79
636223	ELFARMG2	Electrifarm Unit 2	13.2	MEC	Isolated	Isolated	NA	33,434	33,578	144
636221	EL FARM8	Electrifarm	69	MEC	17,119	17,332	213	14,935	15,123	188
636219	ELFARMXT9	Electrifarm 161-69 kV Tertiary	13.8	MEC	Isolated	Isolated	NA	20,484	20,547	63
636210	WASHBRN5	Washburn	161	MEC	10,696	11,123	427	13,399	13,825	426
636207	WTWEST 5	Waterloo West	161	MEC	12,369	12,659	290	14,690	15,076	386
638040	CFUINDP5	CFU Industrial	161	CFU	11,767	12,116	349	13,135	13,540	405
636224	ELFARMG3	Electrifarm Unit 3	13.2	MEC	Isolated	Isolated	NA	34,513	34,620	107
636261	RIDGWAY8	Ridgeway	69	MEC	11,035	11,110	75	13,742	13,882	140
636262	MURPHY 8	Murphy	69	MEC	6,786	6,817	NA	8,237	8,292	56
631051	HAZLTON L2 5	Hazleton	161	ITCM	18,365	18,467	102	18,893	19,039	146
631086	DYSART 5	Dysart	161	ITCM	7,380	7,429	49	10,021	10,091	70
636204	DR NE 5	Deere NE	161	MEC	9,331	9,533	202	12,299	12,585	286
636211	WASHBRN8	Washburn	69	MEC	8,763	8,813	50	8,314	8,370	56

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2022 Cost Allocation Results

K.1 Distribution Factor (DF) and MW Contribution Results for Cost Allocation in 2022

Table K-1: Distribution Factor and MW Contribution on Constraints for Franklin-Morgan Valley & Beverly-Sub 92 345 kV Backbone NU Cost Allocation

Table K-2: Distribution Factor and MW Contribution on Constraints for J510-Hazel Creek-Scott County 345 kV Backbone NU Cost Allocation

Table K-3: Distribution Factor and MW Contribution on Constraints for Big Stone-Alexandria 345 kV Backbone NU Cost Allocation

Table K-4: Distribution Factor and MW Contribution on Constraints for Webster-Franklin 345 kV Backbone NU Cost Allocation

Table K-5: Distribution Factor and MW Contribution on Constraints for Franklin Reactive Backbone NU Cost Allocation

Table K-6: Distribution Factor and MW Contribution on MWEX Constraints for NROC-Rocky Run NU Cost Allocation

Table K-7: Distribution Factor and MW Contribution on Constraints for Other Thermal NU Cost Allocation

Table K-8: Distribution Factor and MW Contribution on Voltage Constraints for Other Reactive Power NU Cost Allocation

Table K-9: Distribution Factor and MW Contribution on Constraints under the GRE P7 CUDC bipole contingency for NU Cost Allocation

Table K-10: SPP Suggested Mitigations within DPP 2016 August West Area Study

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K.2 Cost Allocation Details

Table K-11: Network Upgrades Cost Allocation in 2022

Table K-11: Network Upgrades Cost Allocation in 2022

[illegible]

Table K-11: Network Upgrades Cost Allocation in 2022																																		
Monitored Element	English Name	Cost	J302	J414	J415	J439	J457	J459	J476	J503	J511	J512	J541	J555	J560	J575	J577	J583	J587	J590	J593	J594	J596	J597	J598	J599	J607	J611	J613	J614	J615	J637	J638	Upgrade for
Total Cost Per Project for Actual NRIS Elections for each Project	Total Cost Per Project for Actual NRIS Elections for each Project	\$3,249,061,765	\$113,421,255	\$68,972,590	\$169,633,709	\$174,124,689	\$116,742,354	\$129,344,863	\$112,129,358	\$110,751,215	\$383,833,137	\$68,881,675	\$122,290,754	\$51,444,402	\$19,904,197	\$45,145,758	\$45,147,689	\$64,119,460	\$95,748,419	\$23,403,718	\$236,531,173	\$151,964,370	\$41,301,577	\$142,365,440	\$86,582,891	\$166,732,343	\$148,084,441	\$14,562,588	\$61,481,795	\$14,777,729	\$65,413,612	\$62,844,052	\$141,380,512	

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